CHAPTER

23

COMMUNICATIONS

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-EFFECTIVE PAGES		
1 thru 4	Jun 15/2026	
23-HOW TO USE THE FIM		
1	Feb 15/2013	
2	Feb 15/2013	
3	Feb 15/2013	
4	Feb 15/2013	
5	Feb 15/2013	
6	Feb 15/2013	
23-FAULT CODE INDEX		
101	Feb 15/2013	
102	Feb 15/2013	
103	Jun 15/2015	
104	Feb 15/2023	
105	Feb 15/2023	
106	Feb 15/2023	
107	Feb 15/2023	
108	BLANK	
23-MAINT MSG INDEX		
101	Sep 15/2023	
102	Sep 15/2023	
103	Feb 15/2017	
104	Feb 15/2026	
105	Jun 15/2014	
106	Jun 15/2014	
107	Feb 15/2023	
108	BLANK	
23-11 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Oct 15/2024	
204	Jun 15/2020	
205	Feb 15/2026	
206	Oct 15/2024	
207	Feb 15/2026	
208	Oct 15/2024	
209	Feb 15/2026	
210	Feb 15/2026	
23-11 TASKS (cont)		
211	Feb 15/2026	
212	Feb 15/2026	
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Feb 15/2026	
218	Feb 15/2025	
219	Oct 15/2024	
220	Oct 15/2024	
221	Feb 15/2025	
222	Feb 15/2026	
223	Feb 15/2026	
224	Feb 15/2026	
225	Feb 15/2026	
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227	Feb 15/2026	
228	Oct 15/2025	
229	Feb 15/2026	
230	Feb 15/2025	
23-12 TASKS		
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202	Jun 15/2024	
203	Jun 15/2024	
204	Jun 15/2024	
205	Jun 15/2024	
206	Feb 15/2026	
207	Feb 15/2026	
208	Oct 15/2024	
209	Feb 15/2026	
210	Feb 15/2026	
211	Feb 15/2026	
212	Feb 15/2025	
213	Oct 15/2024	
214	Oct 15/2024	
215	Feb 15/2026	
23-12 TASKS (cont)		
216	Feb 15/2026	
217	Feb 15/2026	
218	Feb 15/2026	
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220	Feb 15/2026	
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225	Feb 15/2025	
226	Feb 15/2026	
227	Feb 15/2026	
228	Feb 15/2026	
229	Feb 15/2025	
230	Feb 15/2025	
23-24 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Oct 15/2018	
204	Feb 15/2020	
205	Feb 15/2020	
206	Feb 15/2023	
207	Feb 15/2026	
208	Feb 15/2026	
23-27 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2026	
204	Feb 15/2026	
205	Feb 15/2026	
206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
210	Feb 15/2026	
211	Feb 15/2026	
212	Feb 15/2026	

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23-EFFECTIVE PAGES

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CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-27 TASKS (cont)		
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Feb 15/2026	
218	Feb 15/2026	
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226	Feb 15/2026	
227	Feb 15/2026	
228	BLANK	
23-28 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2026	
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206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Oct 15/2025	
210	Feb 15/2026	
211	Feb 15/2026	
212	Oct 15/2024	
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
23-31 TASKS		
201	Oct 15/2015	
202	Oct 15/2025	
203	Oct 15/2025	
204	Feb 15/2026	
205	Oct 15/2025	
206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
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212	Feb 15/2026	
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226	Feb 15/2026	
227	Feb 15/2026	
228	Feb 15/2026	
229	Feb 15/2026	
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231	Feb 15/2026	
232	Feb 15/2026	
233	Jun 15/2025	
234	Jun 15/2025	
235	Jun 15/2025	
236	BLANK	
23-32 TASKS		
201	Oct 15/2024	
202	Oct 15/2024	
203	Oct 15/2024	
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216	Oct 15/2024	
217	Oct 15/2024	
218	Oct 15/2024	
219	Oct 15/2024	
220	BLANK	
23-34 TASKS		
201	Oct 15/2024	
202	Oct 15/2024	
203	Oct 15/2024	
204	Oct 15/2024	
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211	Oct 15/2024	
212	Oct 15/2024	
213	Oct 15/2024	
214	Oct 15/2024	
215	Oct 15/2024	
216	BLANK	
23-41 TASKS		
201	Oct 15/2015	

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COMMUNICATIONS

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23-41 TASKS (cont)		
202	Oct 15/2025	
203	Oct 15/2025	
204	Oct 15/2015	
205	Oct 15/2015	
206	Oct 15/2025	
207	Jun 15/2016	
208	Feb 15/2026	
209	Oct 15/2025	
210	Feb 15/2026	
211	Feb 15/2026	
212	Feb 15/2026	
213	Feb 15/2026	
214	Oct 15/2022	
23-42 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2017	
204	Feb 15/2026	
205	Oct 15/2015	
206	Jun 15/2016	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
210	Feb 15/2026	
211	Jun 15/2022	
212	Jun 15/2022	
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Feb 15/2026	
218	Feb 15/2026	
219	Feb 15/2026	
220	Feb 15/2026	
221	Feb 15/2026	
222	Feb 15/2026	

Subject/Page	Date	COC
23-42 TASKS (cont)		
223	Feb 15/2026	
224	Feb 15/2026	
225	Feb 15/2026	
226	Feb 15/2026	
227	Feb 15/2026	
228	Feb 15/2026	
229	Feb 15/2026	
230	Feb 15/2026	
231	Feb 15/2026	
232	Feb 15/2026	
233	Feb 15/2026	
234	Feb 15/2026	
235	Feb 15/2026	
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237	Feb 15/2026	
238	Feb 15/2026	
239	Feb 15/2026	
240	Feb 15/2026	
241	Feb 15/2026	
242	BLANK	
23-43 TASKS		
201	Oct 15/2022	
202	Oct 15/2022	
203	Feb 15/2026	
204	Oct 15/2015	
205	Jun 15/2016	
206	Jun 15/2016	
23-51 TASKS		
201	Feb 15/2026	
202	Oct 15/2025	
203	Feb 15/2022	
204	Oct 15/2025	
205	Feb 15/2026	
206	Feb 15/2026	
207	Feb 15/2023	
208	Oct 15/2025	
209	Oct 15/2025	

Subject/Page	Date	COC
23-51 TASKS (cont)		
210	Oct 15/2025	
211	Oct 15/2025	
212	Feb 15/2023	
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Oct 15/2025	
218	BLANK	
23-71 TASKS		
201	Feb 15/2026	
202	Jun 15/2021	
203	Feb 15/2026	
204	Feb 15/2026	
205	Feb 15/2026	
206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
210	Feb 15/2026	
211	Feb 15/2026	
212	Feb 15/2026	
213	Feb 15/2026	
214	Feb 15/2026	
215	Oct 15/2024	
216	Feb 15/2026	
217	Feb 15/2026	
218	Sep 15/2023	
219	Feb 15/2026	
220	Feb 15/2026	
221	Feb 15/2026	
222	Feb 15/2026	
223	Feb 15/2026	
224	Feb 15/2026	
225	Feb 15/2026	
226	Feb 15/2026	

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Subject/Page	Date	COC
23-71 TASKS (cont)		
227	Sep 15/2023	
228	Sep 15/2023	
229	Feb 15/2026	
230	Feb 15/2026	
231	Feb 15/2026	
232	Feb 15/2026	
23-75 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2026	
204	Feb 15/2026	
205	Feb 15/2026	
206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
210	Feb 15/2026	
211	Feb 15/2026	
212	BLANK	
23-82 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2026	
204	Feb 15/2026	

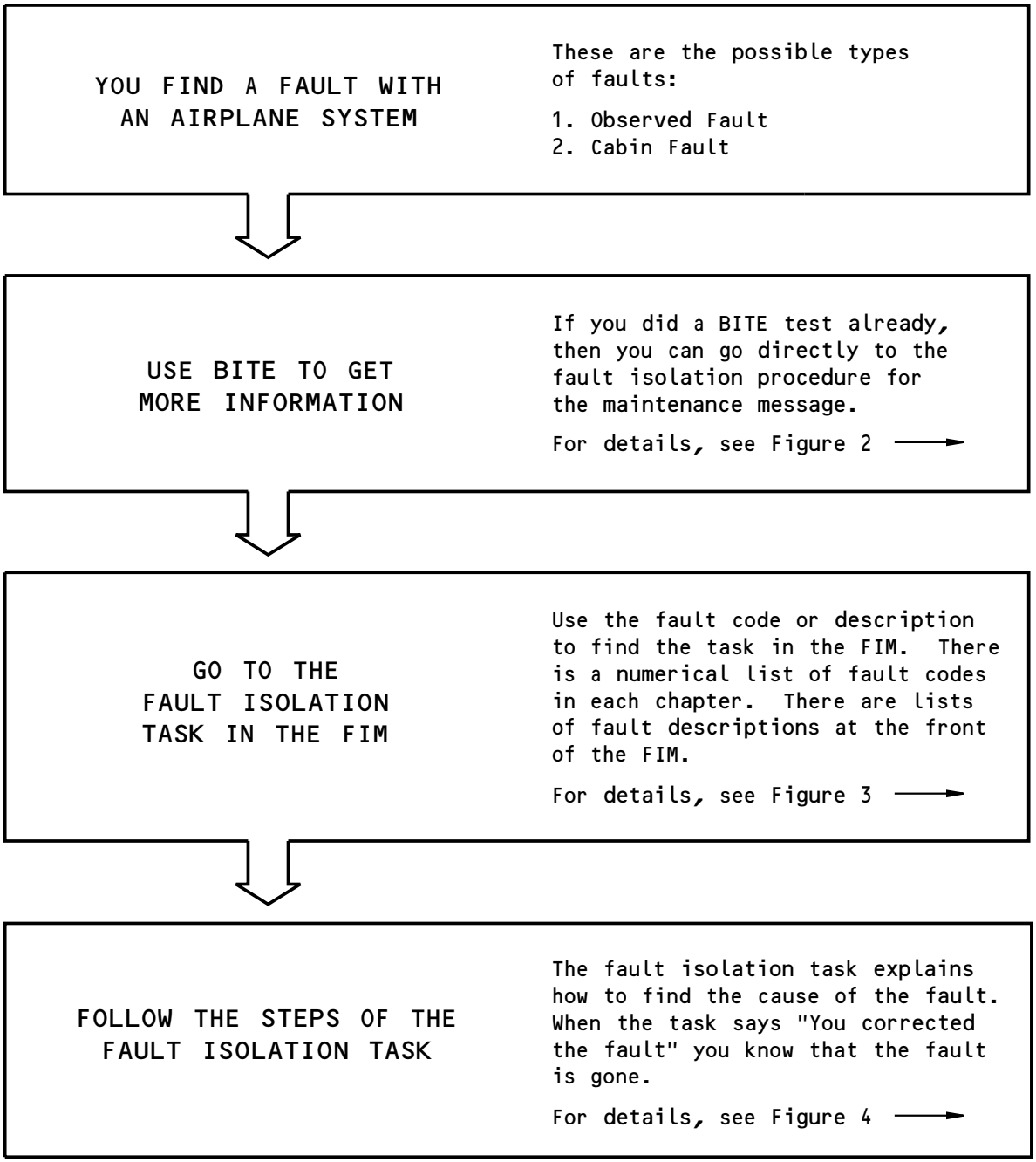
Subject/Page	Date	COC

Subject/Page	Date	COC

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23-EFFECTIVE PAGES

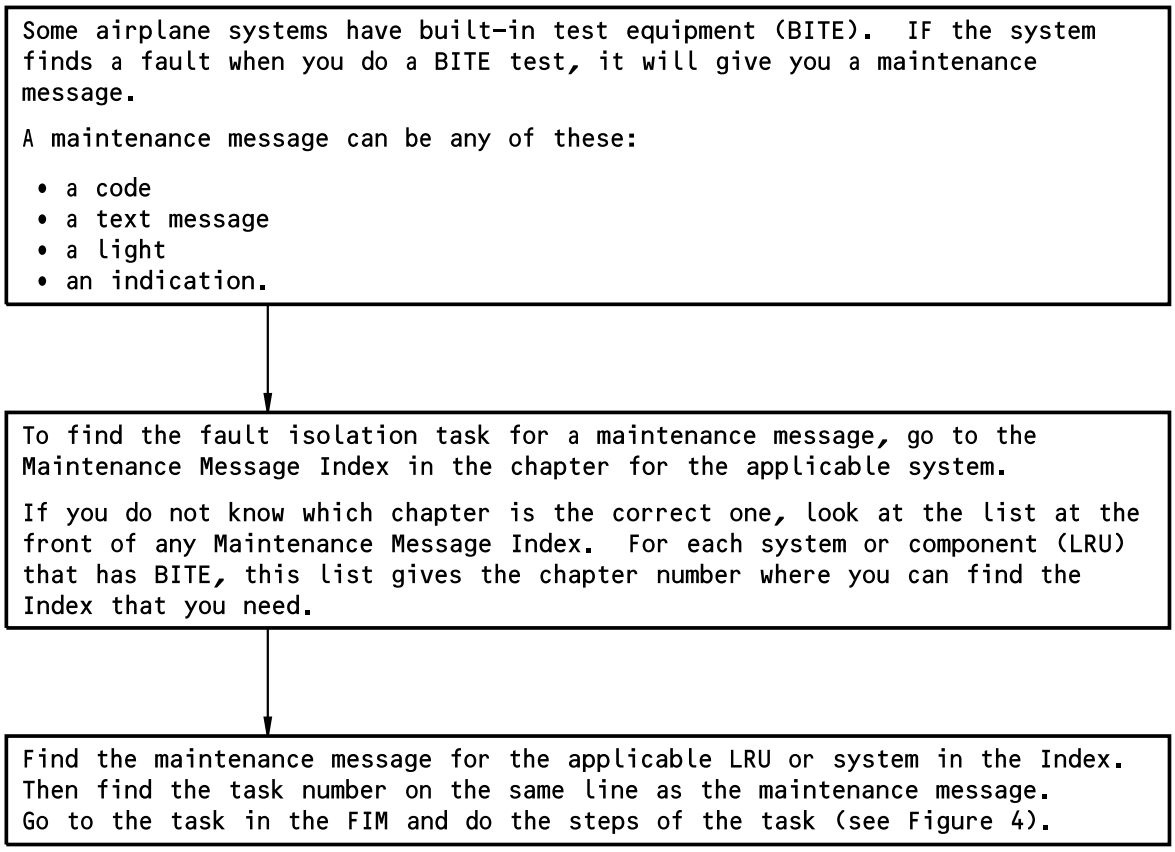
ECCN 9E991 BOEING PROPRIETARY - See title page for details



G04902 S0000148576_V1

Basic Fault Isolation Process
Figure 1

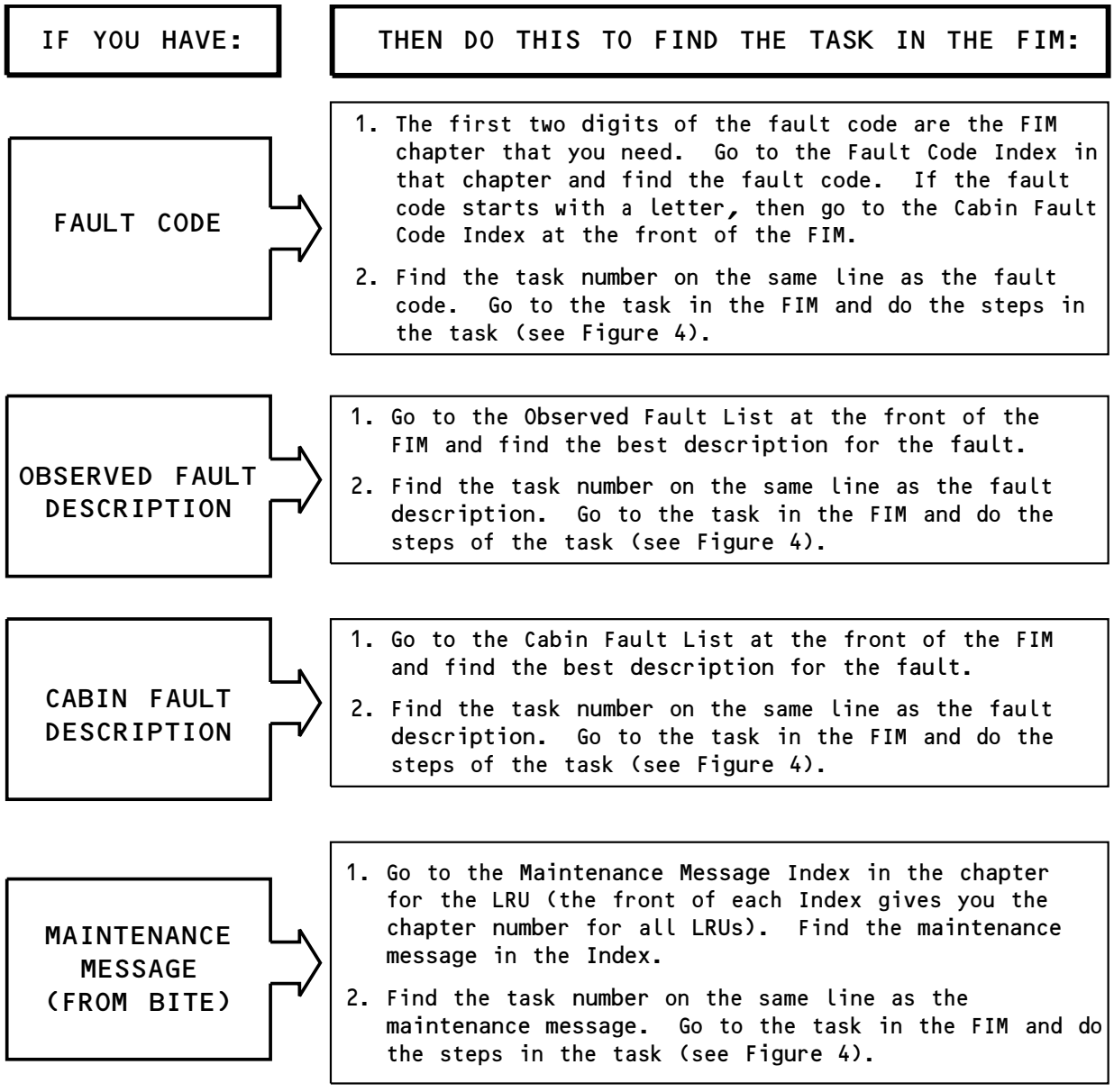
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G04950 S0000148578_V1

Getting Fault Information from BITE
Figure 2

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G04979 S0000148579_V2

Finding the Fault Isolation Task in the FIM
Figure 3

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ASSUMED CONDITIONS AT START OF TASK

- External electrical power is ON
- Hydraulic power and pneumatic power are OFF
- Engines are shut down
- No equipment in the system is deactivated

POSSIBLE CAUSES

- The list of possible causes has the most likely cause first and the least likely cause last.
- You can use the maintenance records of your airline to determine if the fault occurred before. Compare the list of possible causes to the past maintenance actions. This will help prevent repetition of the same maintenance actions.

INITIAL EVALUATION PARAGRAPH

- The primary purpose of the Initial Evaluation paragraph at the start of the task is to help you find out if you can detect the fault right now:
  - If you cannot detect the fault right now, then the task cannot isolate the fault and the Initial Evaluation paragraph will say that there was an intermittent fault.
  - If you have an intermittent fault, you must use your judgement (and follow your airline's policy) to decide which maintenance action to take. Then monitor the airplane to see if the fault happens again on subsequent flights.
- The Initial Evaluation paragraph can also help you find out which Fault Isolation Procedure to use to isolate and correct the fault.

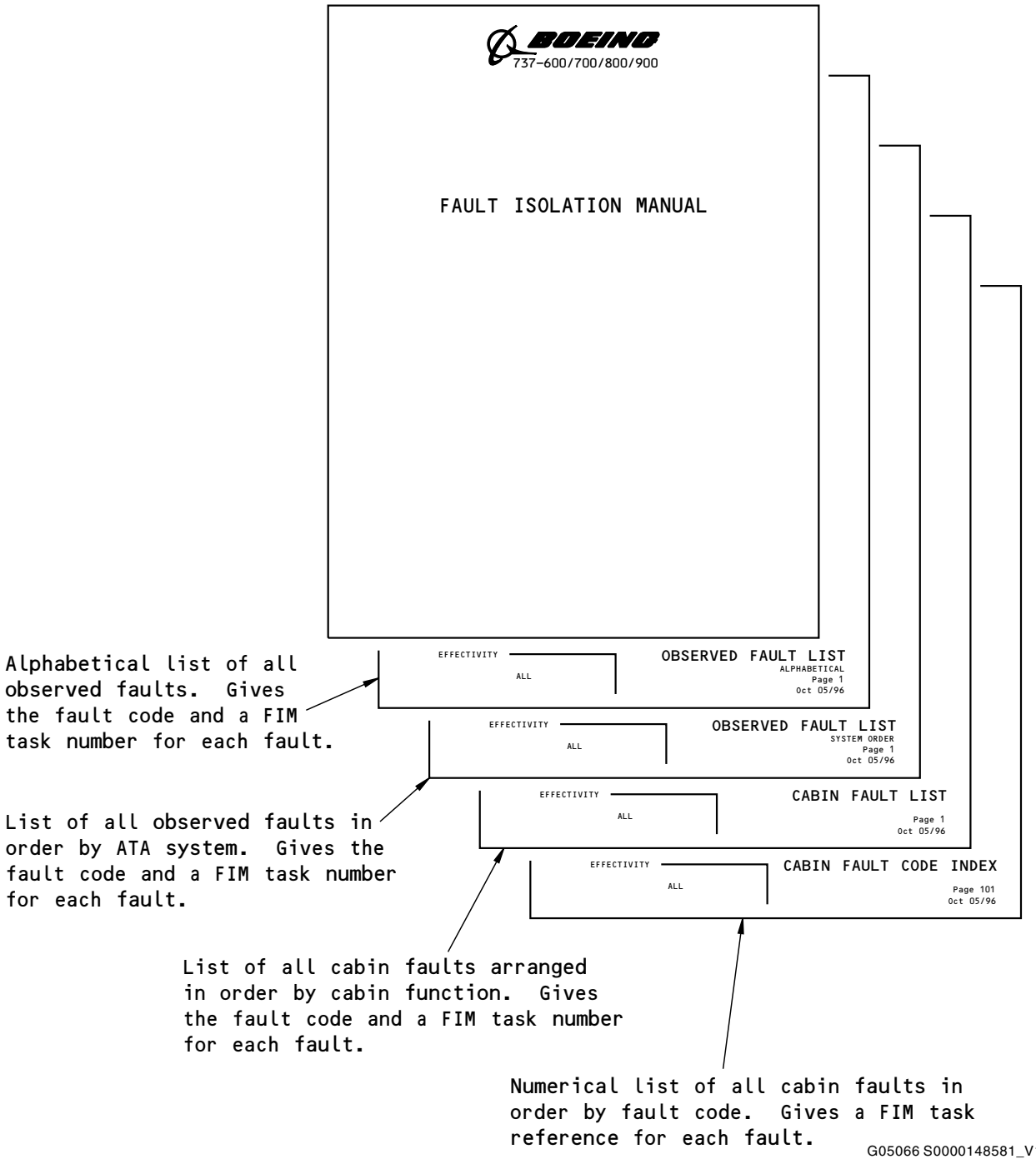
FAULT ISOLATION STEPS

- The FIM task steps are presented in a specified order. The "If... then" statements will guide you along a logical path. But if you do not plan to follow the FIM task exactly, make sure that you read it before you start to isolate the fault. Some FIM procedures start with important steps that have an effect on the other steps in the procedure.
- When you are at the endpoint of the path, the step says "...you corrected the fault." Complete the step and exit the procedure.

G05009 S0000148580_V3

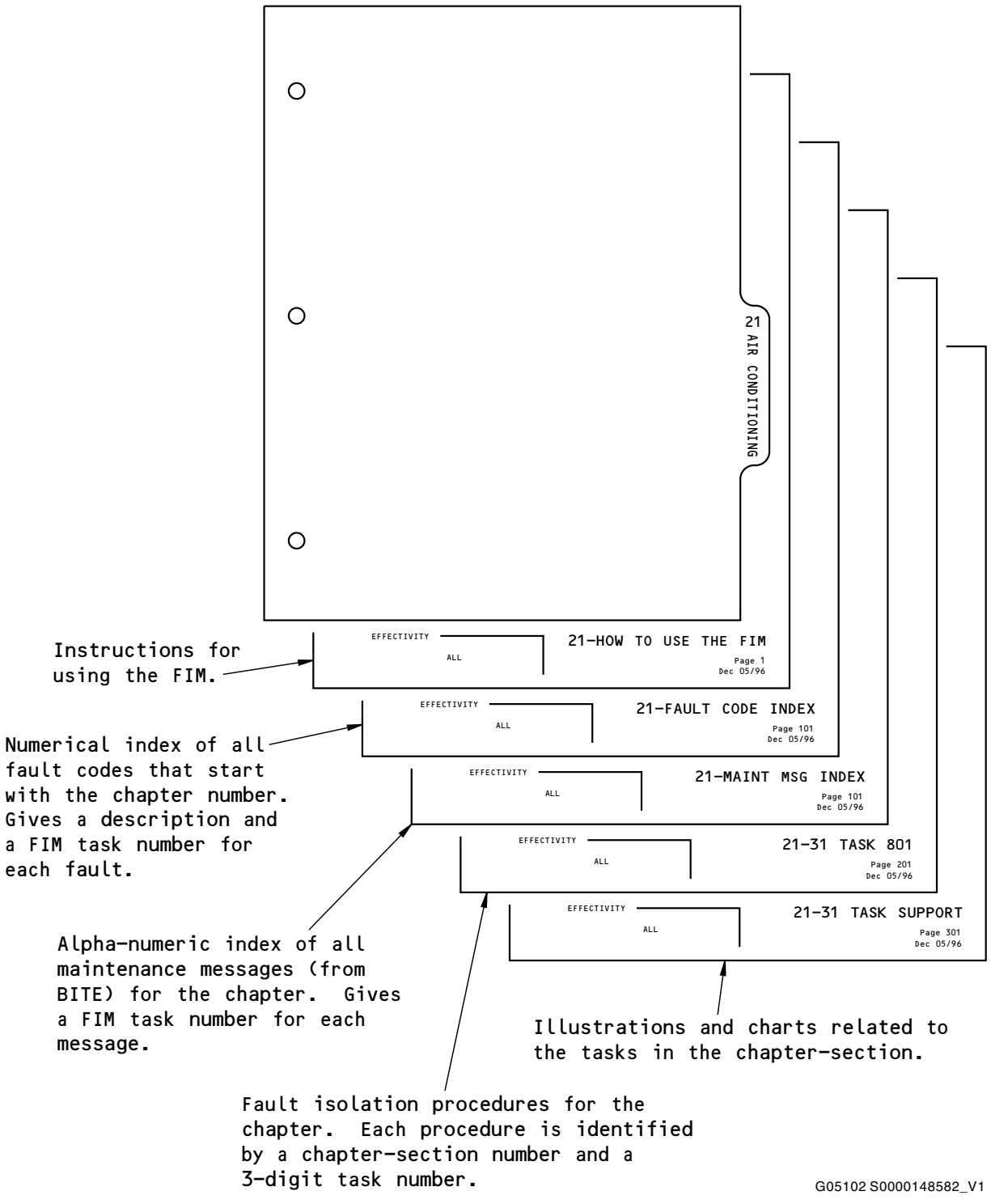
Doing the Fault Isolation Task
Figure 4

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Subjects at Front of FIM
Figure 5

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Subjects in Each FIM Chapter
Figure 6

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
231 010 00	HF 1 control panel problem.	23-11 TASK 805
231 020 00	HF 2 control panel problem.	23-11 TASK 805
231 030 00	VHF 1 control panel problem.	23-12 TASK 814
231 040 00	VHF 2 control panel problem.	23-12 TASK 814
231 050 00	VHF 3 control panel problem.	23-12 TASK 814
231 060 31	HF 1 radio: reception and transmission problem at one station - captain's.	23-11 TASK 806
231 060 32	HF 1 radio: reception and transmission problem at one station - first officer's.	23-11 TASK 806
231 060 33	HF 1 radio: reception and transmission problem at one station - first observer's.	23-11 TASK 806
231 060 48	HF 1 radio: reception and transmission problem at all stations.	23-11 TASK 806
231 070 31	HF 1 radio: reception problem at one station - captain's.	23-11 TASK 806
231 070 32	HF 1 radio: reception problem at one station - first officer's.	23-11 TASK 806
231 070 33	HF 1 radio: reception problem at one station - first observer's.	23-11 TASK 806
231 070 48	HF 1 radio: reception problem at all stations.	23-11 TASK 806
231 080 31	HF 1 radio: Transmission problem at one station - captain's.	23-11 TASK 806
231 080 32	HF 1 radio: Transmission problem at one station - first officer's.	23-11 TASK 806
231 080 33	HF 1 radio: Transmission problem at one station - first observer's.	23-11 TASK 806
231 080 48	HF 1 radio: Transmission problem at all stations.	23-11 TASK 806
231 180 31	HF 2 radio: reception and transmission problem at one station - captain's.	23-11 TASK 806
231 180 32	HF 2 radio: reception and transmission problem at one station - first officer's.	23-11 TASK 806
231 180 33	HF 2 radio: reception and transmission problem at one station - first observer's.	23-11 TASK 806
231 180 48	HF 2 radio: reception and transmission problem at all stations.	23-11 TASK 806
231 190 31	HF 2 radio: reception problem at one station - captain's.	23-11 TASK 806
231 190 32	HF 2 radio: reception problem at one station - first officer's.	23-11 TASK 806
231 190 33	HF 2 radio: reception problem at one station - first observer's.	23-11 TASK 806
231 190 48	HF 2 radio: reception problem at all stations.	23-11 TASK 806
231 200 31	HF 2 radio: Transmission problem at one station - captain's.	23-11 TASK 806
231 200 32	HF 2 radio: Transmission problem at one station - first officer's.	23-11 TASK 806
231 200 33	HF 2 radio: Transmission problem at one station - first observer's.	23-11 TASK 806
231 200 48	HF 2 radio: Transmission problem at all stations.	23-11 TASK 806

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
231 300 31	Radio tuning panel problem - captain's.	23-12 TASK 815
231 300 32	Radio tuning panel problem - first officer's.	23-12 TASK 815
231 300 33	Radio tuning panel problem - first observer's.	23-12 TASK 815
231 310 31	VHF 1 radio: reception and transmission problem at one station - captain's.	23-12 TASK 816
231 310 32	VHF 1 radio: reception and transmission problem at one station - first officer's.	23-12 TASK 816
231 310 33	VHF 1 radio: reception and transmission problem at one station - first observer's.	23-12 TASK 816
231 310 48	VHF 1 radio: reception and transmission problem at all stations.	23-12 TASK 816
231 320 31	VHF 1 radio: reception problem at one station - captain's.	23-12 TASK 816
231 320 32	VHF 1 radio: reception problem at one station - first officer's.	23-12 TASK 816
231 320 33	VHF 1 radio: reception problem at one station - first observer's.	23-12 TASK 816
231 320 48	VHF 1 radio: reception problem at all stations.	23-12 TASK 816
231 330 31	VHF 1 radio: Transmission problem at one station - captain's.	23-12 TASK 816
231 330 32	VHF 1 radio: Transmission problem at one station - first officer's.	23-12 TASK 816
231 330 33	VHF 1 radio: Transmission problem at one station - first observer's.	23-12 TASK 816
231 330 48	VHF 1 radio: Transmission problem at all stations.	23-12 TASK 816
231 430 31	VHF 2 radio: reception and transmission problem at one station - captain's.	23-12 TASK 816
231 430 32	VHF 2 radio: reception and transmission problem at one station - first officer's.	23-12 TASK 816
231 430 33	VHF 2 radio: reception and transmission problem at one station - first observer's.	23-12 TASK 816
231 430 48	VHF 2 radio: reception and transmission problem at all stations.	23-12 TASK 816
231 440 31	VHF 2 radio: reception problem at one station - captain's.	23-12 TASK 816
231 440 32	VHF 2 radio: reception problem at one station - first officer's.	23-12 TASK 816
231 440 33	VHF 2 radio: reception problem at one station - first observer's.	23-12 TASK 816
231 440 48	VHF 2 radio: reception problem at all stations.	23-12 TASK 816
231 450 31	VHF 2 radio: Transmission problem at one station - captain's.	23-12 TASK 816
231 450 32	VHF 2 radio: Transmission problem at one station - first officer's.	23-12 TASK 816
231 450 33	VHF 2 radio: Transmission problem at one station - first observer's.	23-12 TASK 816
231 450 48	VHF 2 radio: Transmission problem at all stations.	23-12 TASK 816
231 550 31	VHF 3 radio: reception and transmission problem at one station - captain's.	23-12 TASK 816

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
231 550 32	VHF 3 radio: reception and transmission problem at one station - first officer's.	23-12 TASK 816
231 550 33	VHF 3 radio: reception and transmission problem at one station - first observer's.	23-12 TASK 816
231 550 48	VHF 3 radio: reception and transmission problem at all stations.	23-12 TASK 816
231 560 31	VHF 3 radio: reception problem at one station - captain's.	23-12 TASK 816
231 560 32	VHF 3 radio: reception problem at one station - first officer's.	23-12 TASK 816
231 560 33	VHF 3 radio: reception problem at one station - first observer's.	23-12 TASK 816
231 560 48	VHF 3 radio: reception problem at all stations.	23-12 TASK 816
231 570 31	VHF 3 radio: Transmission problem at one station - captain's.	23-12 TASK 816
231 570 32	VHF 3 radio: Transmission problem at one station - first officer's.	23-12 TASK 816
231 570 33	VHF 3 radio: Transmission problem at one station - first observer's.	23-12 TASK 816
231 570 48	VHF 3 radio: Transmission problem at all stations.	23-12 TASK 816
232 020 00	SELCAL: Call switch does not reset for HF 1, PTT reset OK.	23-28 TASK 801
232 030 00	SELCAL: Call switch does not reset for HF 2, PTT reset OK.	23-28 TASK 801
232 040 00	SELCAL: Call switch does not reset for VHF 1, PTT reset OK.	23-28 TASK 801
232 050 00	SELCAL: Call switch does not reset for VHF 2, PTT reset OK.	23-28 TASK 801
232 060 00	SELCAL: Call switch does not reset for VHF 3, PTT reset OK.	23-28 TASK 801
232 080 00	SELCAL: call light does not come on for calls on HF 1.	23-28 TASK 802
232 090 00	SELCAL: call light does not come on for calls on HF 2.	23-28 TASK 802
232 100 00	SELCAL: call light does not come on for calls on VHF 1.	23-28 TASK 802
232 110 00	SELCAL: call light does not come on for calls on VHF 2.	23-28 TASK 802
232 120 00	SELCAL: call light does not come on for calls on VHF 3.	23-28 TASK 802
232 140 00	SELCAL: Chime does not operate for calls on HF 1, SELCAL call light comes on.	23-28 TASK 803
232 150 00	SELCAL: Chime does not operate for calls on HF 2, SELCAL call light comes on.	23-28 TASK 803
232 160 00	SELCAL: Chime does not operate for calls on VHF 1, SELCAL call light comes on.	23-28 TASK 803
232 170 00	SELCAL: Chime does not operate for calls on VHF 2, SELCAL call light comes on.	23-28 TASK 803
232 180 00	SELCAL: Chime does not operate for calls on VHF 3, SELCAL call light comes on.	23-28 TASK 803
232 200 00	SELCAL: does not operate for HF 1.	23-28 TASK 804
232 210 00	SELCAL: does not operate for HF 2.	23-28 TASK 804
232 220 00	SELCAL: does not operate for VHF 1.	23-28 TASK 804
232 230 00	SELCAL: does not operate for VHF 2.	23-28 TASK 804

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
232 240 00	SELCAL: does not operate for VHF 3.	23-28 TASK 804
232 250 00	SELCAL: does not operate for any radio.	23-28 TASK 804
232 260 00	SELCAL: all call lights on.	23-28 TASK 805
232 310 00	ACARS: does not operate correctly.	23-27 TASK 801
232 315 00	ACARS: CDU scratchpad shows APM WARN.	23-27 TASK 825
232 325 00	ACARS: communication problem.	23-27 TASK 814
232 326 00	ACARS: CDU interface problem.	23-27 TASK 815
232 327 00	ACARS: input problem.	23-27 TASK 816
232 328 00	ACARS: call annunciation problem.	23-27 TASK 818
232 329 00	ACARS: call annunciation problem.	23-27 TASK 820
232 401 00	Emergency Locator Transmitter (ELT): Self Test Fails.	23-24 TASK 801
232 410 00	Emergency Locator Transmitter (ELT): no ELT signal on VHF when ELT switch set to ON during test.	23-24 TASK 801
232 411 00	Emergency Locator Transmitter (ELT): ELT light is on.	23-24 TASK 807
232 719 00	ACARS: CMU cannot print data to the Multi-function Printer.	23-27 TASK 812
232 720 00	ACARS: MU cannot print data to the Multi-function Printer.	23-27 TASK 812
233 010 00	Passenger address system - does not operate from flight compartment.	23-31 TASK 801
233 020 00	Passenger address system - distorted from flight compartment.	23-31 TASK 801
233 030 00	Passenger address system - intermittent from flight compartment.	23-31 TASK 801
233 040 00	Passenger address system - volume problem from flight compartment.	23-31 TASK 802
234 010 00	Service interphone: does not connect to flight interphone.	23-41 TASK 801
234 020 48	Service interphone: does not operate - all jacks.	23-41 TASK 801
234 030 48	Service interphone: distorted - all jacks.	23-41 TASK 803
234 040 48	Service interphone: intermittent - all jacks.	23-41 TASK 803
234 050 48	Service interphone: volume problem - all jacks.	23-41 TASK 803
234 060 00	Service interphone: does not operate - jack at EE rack.	23-41 TASK 802
234 070 00	Service interphone: distorted - jack at EE rack.	23-41 TASK 802
234 080 00	Service interphone: intermittent - jack at EE rack.	23-41 TASK 802
234 090 00	Service interphone: volume problem - jack at EE rack.	23-41 TASK 802
234 100 00	Service interphone: does not operate - jack adjacent to APU.	23-41 TASK 802
234 110 00	Service interphone: intermittent - jack adjacent to APU.	23-41 TASK 802
234 120 00	Service interphone: distorted - jack adjacent to APU.	23-41 TASK 802
234 130 00	Service interphone: volume problem - jack adjacent to APU.	23-41 TASK 802

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
234 140 00	Service interphone: does not operate - jack at aft entry light panel.	23-41 TASK 802
234 150 00	Service interphone: distorted - jack at aft entry light panel.	23-41 TASK 802
234 160 00	Service interphone: intermittent - jack at aft entry light panel.	23-41 TASK 802
234 170 00	Service interphone: volume problem - jack at aft entry light panel.	23-41 TASK 802
234 180 00	Service interphone: does not operate - jack at external power receptacle panel.	23-41 TASK 802
234 190 00	Service interphone: distorted - jack at external power receptacle panel.	23-41 TASK 802
234 200 00	Service interphone: intermittent - jack at external power receptacle panel.	23-41 TASK 802
234 210 00	Service interphone: volume problem - jack at external power receptacle panel.	23-41 TASK 802
234 220 00	Service interphone: does not operate - jack at left wheel well.	23-41 TASK 802
234 230 00	Service interphone: distorted - jack at left wheel well.	23-41 TASK 802
234 240 00	Service interphone: intermittent - jack at left wheel well.	23-41 TASK 802
234 250 00	Service interphone: volume problem - jack at left wheel well.	23-41 TASK 802
234 260 00	Service interphone: does not operate - jack at right wheel well.	23-41 TASK 802
234 270 00	Service interphone: distorted - jack at right wheel well.	23-41 TASK 802
234 280 00	Service interphone: intermittent - jack at right wheel well.	23-41 TASK 802
234 290 00	Service interphone: volume problem - jack at right wheel well.	23-41 TASK 802
234 300 00	Service interphone: does not operate - jack at right wing refueling slat.	23-41 TASK 802
234 310 00	Service interphone: distorted - jack at right wing refueling slat.	23-41 TASK 802
234 320 00	Service interphone: intermittent - jack at right wing refueling slat.	23-41 TASK 802
234 330 00	Service interphone: volume problem - jack at right wing refueling slat.	23-41 TASK 802
234 340 00	Call horn: does not sound.	23-43 TASK 801
234 350 00	Call horn: operates continuously.	23-43 TASK 802
234 360 00	Call system: does not operate, ground crew to flight compartment.	23-43 TASK 803
234 370 00	Call system: call light does not come on, ground crew to flight compartment.	23-43 TASK 804
234 371 00	Call system: does not operate, attendant to flight compartment.	23-42 TASK 805
234 380 00	Call system: call light does not come on, attendant to flight compartment.	23-42 TASK 806
234 390 00	Call system: Chime does not sound in the flight compartment.	23-42 TASK 804

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
234 395 00	Call system: attendant call chime does not operate in passenger cabin.	23-42 TASK 803
234 400 00	Call system: does not operate, flight compartment to attendant.	23-42 TASK 802
234 410 00	Service interphone: does not operate - jack in the electronic equipment bay.	23-41 TASK 802
234 420 00	Service interphone: intermittent - jack in the electronic equipment bay.	23-41 TASK 802
234 430 00	Service interphone: distorted - jack in the electronic equipment bay.	23-41 TASK 802
234 440 00	Service interphone: volume problem - jack in the electronic equipment bay.	23-41 TASK 802
234 811 00	Attendant control panel: amber light on ACP switch assembly does not come on during operational test.	23-42 TASK 841
235 010 31	Boom microphone/headset: does not operate - captain's.	23-51 TASK 801
235 010 32	Boom microphone/headset: does not operate - first officer's.	23-51 TASK 801
235 020 31	Boom microphone/headset: damaged - captain's.	23-51 TASK 802
235 020 32	Boom microphone/headset: damaged - first officer's.	23-51 TASK 802
235 030 31	Flight interphone: does not operate - captain's.	23-51 TASK 804
235 030 32	Flight interphone: does not operate - first officer's.	23-51 TASK 804
235 030 33	Flight interphone: does not operate - first observer's.	23-51 TASK 804
235 030 48	Flight interphone: does not operate - all stations.	23-51 TASK 803
235 040 31	Headphone: does not operate - captain's.	23-51 TASK 801
235 040 32	Headphone: does not operate - first officer's.	23-51 TASK 801
235 040 33	Headphone: does not operate - first observer's.	23-51 TASK 801
235 040 34	Headphone: does not operate - second observer's.	23-51 TASK 801
235 050 31	Hand microphone: does not operate - captain's.	23-51 TASK 801
235 050 32	Hand microphone: does not operate - first officer's.	23-51 TASK 801
235 050 33	Hand microphone: does not operate - first observer's.	23-51 TASK 801
235 060 31	Hand microphone: damaged - captain's.	23-51 TASK 802
235 060 32	Hand microphone: damaged - first officer's.	23-51 TASK 802
235 060 33	Hand microphone: damaged - first observer's.	23-51 TASK 802
235 070 31	Headphone: damaged - captain's.	23-51 TASK 802
235 070 32	Headphone: damaged - first officer's.	23-51 TASK 802
235 070 33	Headphone: damaged - first observer's.	23-51 TASK 802
235 070 34	Headphone: damaged - second observer's.	23-51 TASK 802
235 071 31	Oxygen mask: communication problem - captain's.	23-51 TASK 808
235 071 32	Oxygen mask: communication problem - first officer's.	23-51 TASK 808

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
235 071 33	Oxygen mask: communication problem - first observer's.	23-51 TASK 808
235 080 31	Speaker in the flight compartment: does not operate - captain's.	23-51 TASK 805
235 080 32	Speaker in the flight compartment: does not operate - first officer's.	23-51 TASK 806
235 101 31	Audio control panel indicator light problem - captain's.	23-51 TASK 807
235 101 32	Audio control panel indicator light problem - first officer's.	23-51 TASK 807
235 101 33	Audio control panel indicator light problem - first observer's.	23-51 TASK 807
235 102 31	Audio control panel volume control problem - captain's.	23-51 TASK 807
235 102 32	Audio control panel volume control problem - first officer's.	23-51 TASK 807
235 102 33	Audio control panel volume control problem - first observer's.	23-51 TASK 807
235 103 31	Audio control panel selector switch problem - captain's.	23-51 TASK 807
235 103 32	Audio control panel selector switch problem - first officer's.	23-51 TASK 807
235 103 33	Audio control panel selector switch problem - first observer's.	23-51 TASK 807
237 010 00	Voice recorder: Signal problem at monitor jack.	23-71 TASK 801
237 015 00	Voice recorder: RIPS maintenance discrete check fails.	23-71 TASK 806
237 020 00	Voice recorder: STATUS light or TEST light does not come on when TEST switch is pushed.	23-71 TASK 804
237 050 00	Voice recorder: system operation problem (airplanes with AUTO/ON switch).	23-71 TASK 805
237 501 01	Video Surveillance System: no video for one camera - CAM 1.	23-75 TASK 807
237 501 02	Video Surveillance System: no video for one camera - CAM 2.	23-75 TASK 807
237 501 03	Video Surveillance System: no video for one camera - CAM 3.	23-75 TASK 807
237 503 00	Video Surveillance System: no video at all three cameras.	23-75 TASK 809
237 510 00	Video Surveillance System: Distorted video from one camera.	23-75 TASK 808
238 302 00	Flight Compartment: Outlet Unit Does Not Operate.	23-82 TASK 802
238 303 00	Flight Compartment: Outlet Unit Has No Power or USB Problems.	23-82 TASK 803

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LRU/SYSTEM	SHORT NAME	CHAPTER
Air Data Inertial Reference System	ADIRS	34
Air Traffic Controller Transponder - 1 (Left)	ATC XPDR - 1 (L)	34
Air Traffic Controller Transponder - 2 (Right)	ATC XPDR - 2 (R)	34
Airborne Vibration Monitor System Signal Conditioner	AVM SIG COND	77
Antiskid Control Unit	ANTISKID	32
Attendant Control Panel	ACP	23
Automatic Direction Finder Receiver - 1	ADF RECVR - 1	34
Automatic Direction Finder Receiver - 2	ADF RECVR - 2	34
Autothrottle Computer	A/T COMPUTER	22
Auxiliary Power Unit	APU	49
Auxiliary Power Unit Generator Control Unit	APU GCU	24
Bus Power Control Unit	BPCU	24
Cabin Pressure Controller	CAB PRESS CON	21
Cabin Temperature Controller	CAB TEMP CONT	21
Cargo Electronic Unit - Lower Aft	CEU - LWR AFT	26
Cargo Electronic Unit - Lower Forward	CEU - LWR FWD	26
Cargo Electronic Unit - Main Aft	CEU - MAIN AFT	26
Cargo Electronic Unit - Main Forward	CEU - MAIN FWD	26
Cargo Fire Control Panel	CFCP	26
Common Display System	CDS	31
Compartment Overheat Detection Control Module	WING/BODY OHT	26
Digital Flight Control System	DFCS	22
Distance Measurement Equipment Interrogator	DME INTRROGTR	34
Electrical Meters, Battery, and Galley Power Module	P5-13	24
Electronic Engine Controller - 1	ENGINE - 1	73
Electronic Engine Controller - 2	ENGINE - 2	73
Emergency Locator Transmitter	ELT	23
Engine Accessory Unit	EAU	78
Engine Accessory Unit/TR DEPLOY ENG 1	EAU/TR DPLOY-ENG 1	78
Engine Accessory Unit/TR DEPLOY ENG 2	EAU/TR DPLOY-ENG 2	78
Engine Accessory Unit/TR STOW ENG 1	EAU/TR STOW-ENG 1	78
Engine Accessory Unit/TR STOW ENG 2	EAU/TR STOW-ENG 2	78
Engine and Auxiliary Power Unit Fire Detection Control Module	ENG/APU FIRE	26
Enhanced Digital Flight Control Computer-A	EDFCC-A	22
Enhanced Digital Flight Control System	EDFCS	22

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LRU/SYSTEM	SHORT NAME	CHAPTER
Flap/Slat Electronics Unit	FSEU	27
Flight Data Acquisition Unit	FDAU	31
Flight Management Computer System	FMCS	34
Fuel Quantity Indicating System	FQIS	28
Generator Control Unit - 1	GCU - 1	24
Generator Control Unit - 2	GCU - 2	24
Ground Proximity Computer	GROUND PROX	34
High Frequency Transceiver	HF XCVR	23
Low Limit (35 Degree F) Controller - Left	35 DEG CONT L	21
Low Limit (35 Degree F) Controller - Right	35 DEG CONT R	21
Multi-Mode Receiver	MMR	34
Nitrogen Generation System BITE Display Unit	NGS	47
Pack Flow Temperature Controller	PFTC	21
Pack/Zone Temperature Controller - Left	PACK/ZN CON - L	21
Pack/Zone Temperature Controller - Right	PACK/ZN CON - R	21
Proximity Switch Electronics Unit	PSEU	32
Radio Altimeter Receiver/Transmitter	RADIO ALTIMTR	34
Stall Management Yaw Damper Computer - 1	SMYD - 1	27
Stall Management Yaw Damper Computer - 2	SMYD - 2	27
Supplemental Proximity Switch Electronics Unit	SPSEU	32
Traffic Alert and Collision Avoidance System Computer	TCAS COMPUTER	34
VHF Omnidirectional Ranging Marker Beacon Receiver	VOR/MKR RCVR	34
Very High Frequency Transceiver	VHF XCVR	23
Waste Tank Logic Control Module	WASTE TANK	38
Weather Radar Receiver/Transmitter	WEATHER RADAR	34
Window Heat Control Unit - Left Forward	WHCU - L FWD	30
Window Heat Control Unit - Left Side	WHCU - L SIDE	30
Window Heat Control Unit - Right Forward	WHCU - R FWD	30
Window Heat Control Unit - Right Side	WHCU - R SIDE	30
Window Heat Control Unit 1 - Left Forward and Right Side	WHCU1 - L FWD/R SIDE	30
Window Heat Control Unit 2 - Right Forward and Left Side	WHCU2 - R FWD/L SIDE	30

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11001 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11002 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11003 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11004 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11005 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11006 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11007 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11008 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 809
ACP	23-11010 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11011 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11012 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11013 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11014 INTERNAL ACP FAULT DETECTED IN FORWARD AND AFT ACPS	23-42 TASK 809
ACP	23-11015 <FORWARD | AFT> WORK LIGHT FAULT DETECTED	23-42 TASK 811
ACP	23-11016 <FORWARD | AFT> MDCD LIGHT FAULT DETECTED	23-42 TASK 812
ACP	23-11017 28 VDC POWER FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 814
ACP	23-11018 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11019 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11020 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11021 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11022 COMMUNICATION WITH AFT ACP LOST	23-42 TASK 815
ACP	23-11023 CABIN TEMPERATURE FAULT DETECTED	23-42 TASK 816
ACP	23-11024 <FORWARD | AFT> THRESHOLD LIGHT FAULT DETECTED	23-42 TASK 813

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11025 INTERNAL ACP FAULT DETECTED	23-42 TASK 807
ACP	23-11026 LAVATORY INOPERATIVE	23-42 TASK 817
ACP	23-11027 CLEAN CHECK SENSORS	23-42 TASK 817
ACP	23-11028 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11029 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11030 LIGHT <COLUMN>-<ADDRESS> EXTRA RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11031 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 810
ACP	23-11032 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 810
ACP	23-11033 LIGHT <COLUMN>-<ADDRESS> INCORRECT STANDARD SCENES DETECTED	23-42 TASK 810
ACP	23-11034 LIGHT <COLUMN>-<ADDRESS> INCORRECT CUSTOM SCENES DETECTED	23-42 TASK 810
ACP	23-11035 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 1	23-42 TASK 819
ACP	23-11036 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 2	23-42 TASK 819
ACP	23-11037 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 3	23-42 TASK 819
ACP	23-11038 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 4	23-42 TASK 819
ACP	23-11039 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 1-<X>	23-42 TASK 818
ACP	23-11040 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 2-<X>	23-42 TASK 818
ACP	23-11041 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 3-<X>	23-42 TASK 818
ACP	23-11042 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 4-<X>	23-42 TASK 818
ACP	23-11043 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ADDRESSING	23-42 TASK 820
ACP	23-11044 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ADDRESSING	23-42 TASK 844
ACP	23-11045 LIGHT <COLUMN>-<ADDRESS> EXTRA RESPONSE DURING ADDRESSING	23-42 TASK 821
ACP	23-11046 LIGHT <COLUMN>-<ADDRESS> INCORRECT LIGHT TYPE DETECTED	23-42 TASK 822

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11047 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ZONING	23-42 TASK 823
ACP	23-11048 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ZONING	23-42 TASK 823
ACP	23-11049 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 824
ACP	23-11050 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11051 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11052 LIGHT <COLUMN>-<ADDRESS> FAILED STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11053 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11054 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11055 LIGHT <COLUMN>-<ADDRESS> FAILED CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11056 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 1	23-42 TASK 825
ACP	23-11057 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 2	23-42 TASK 825
ACP	23-11058 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 3	23-42 TASK 825
ACP	23-11059 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 4	23-42 TASK 825
ACP	23-11060 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 1	23-42 TASK 826
ACP	23-11061 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 2	23-42 TASK 826
ACP	23-11062 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 3	23-42 TASK 826
ACP	23-11063 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 4	23-42 TASK 826
ACP	23-11064 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11065 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11066 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11067 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11068 COMMUNICATIONS WITH AFT ACP LOST	23-42 TASK 828

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11069 LSAP CONFIGURATION INVALID	23-42 TASK 829
ACP	23-11070 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11071 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11072 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11073 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11074 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11075 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11076 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11077 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11078 INCOMPATIBLE DISK DETECTED	23-42 TASK 832
ACP	23-11079 INCOMPATIBLE DISK DETECTED	23-42 TASK 832
ACP	23-11080 COMMUNICATIONS WITH ADL LOST	23-42 TASK 833
ACP	33-11001 Light <column>-<address> Component ID Error	33-20 TASK 817
ACP	33-11002 Light <column>-<address> Calibration Data CRC Error	33-20 TASK 817
ACP	33-11003 Light <column>-<address> Firmware Version Disagree Error	33-20 TASK 817
ACP	33-11004 Light <column>-<address> Power Supply Error	33-20 TASK 817
ACP	33-11005 Light <column>-<address> Temperature Sensor Error	33-20 TASK 817
ACP	33-11006 Light <column>-<address> RAM Check Error	33-20 TASK 817
ACP	33-11007 Light <column>-<address> Slave Token Error	33-20 TASK 817
ACP	33-11008 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11009 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11010 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11011 Light <column>-<address> LED Wrap Data	33-20 TASK 817
ACP	33-11012 Light <column>-<address> Zone/Address Data CRC Error	33-20 TASK 818
ACP	33-11013 Light <column>-<address> Zone Address Disagree	33-20 TASK 818
ACP	33-11014 Light <column>-<address> Standard Scene CRC Error	33-20 TASK 818
ACP	33-11015 Light <column>-<address> Custom Scene CRC Error	33-20 TASK 818
ACP	33-11016 Light <column>-<address> Watchdog Timer Error	33-20 TASK 817

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	33-11017 Light <column>-<address> Master Token Timeout Error	33-20 TASK 819
ACP	33-11018 Light <column>-<address> Master Token Release Error	33-20 TASK 819
ACP	33-11019 Light <column>-<address> Loss of communication	33-20 TASK 819
ELT	LED flashes one time	23-24 TASK 802
ELT	LED flashes seven times	23-24 TASK 804
ELT	LED flashes three or four times	23-24 TASK 803
HF XCVR	CONTROL FAIL	23-11 TASK 802
HF XCVR	CONTROL INPUT FAIL	23-11 TASK 802
HF XCVR	KEY INTERLOCK	23-11 TASK 804
HF XCVR	LRU FAIL	23-11 TASK 803
HF XCVR	LRU STATUS	23-11 TASK 803
VHF XCVR	ANTENNA FAIL	23-12 TASK 813
VHF XCVR	CONTROL FAIL	23-12 TASK 812
VHF XCVR	LRU STATUS	23-12 TASK 818

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801. HF Communication System - BITE Procedure

A. General

(1) You do the High Frequency (HF) Communication System BITE Procedure from the:

OMR 701-799, 802

(a) HF Control Panel on the Aft Electronic Panel, P8, in the Flight Deck.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Front Panel of the HF Transceiver.

OMR 701-799, 809; OMR 802 PRE SB 737-23-1334; OMR 807 PRE SB 737-23-1330

(2) No. 1 and No. 2 HF Transceivers are located in the Main Equipment Center on the Electronic Equipment Rack, E4.

OMR 807 POST SB 737-23-1330

(3) No. 1 and No. 2 (if installed) HF Transceivers are located in the Main Equipment Center on the Electronic Equipment Rack, E5.

OMR 805, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 802 POST SB 737-23-1334

(4) No. 1 and No. 2 HF Transceivers are located in the Aft Cargo Compartment on the Electronic Equipment Rack, E6.

OMR ALL

(5) The HF Communication System BITE Test does a self check for existing Internal and External Faults.

(a) Results of the BITE Test are shown with Fault Lights on the HF Transceiver Front Panel.

OMR ALL; WITH DUAL COLLINS HF TRANSCEIVERS

B. BITE Procedure - COLLINS DUAL HF Transceivers

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do these steps to do the BITE Procedure for the HF Communication System:

(a) Push and release the TEST Switch on the applicable HF Transceiver-1 (2) Front Panel.

(b) Make sure that these conditions occur:

1) All three LEDs on the HF Transceiver Front Panel come ON Red for one to three seconds.

2) The LRU STATUS LED comes ON Green and the KEY INTERLOCK and CONTROL FAIL LEDs stay Red for one to three seconds.

3) All three LEDs go OFF for one to seven seconds.

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

4) The LRU STATUS LED comes ON Green for about 30 seconds and the KEY INTERLOCK and CONTROL FAIL LEDs stay OFF.

(c) If the LRU STATUS LED comes ON Green for about 30 seconds and the KEY INTERLOCK and CONTROL FAIL LEDs stay OFF, then the BITE Test passed.

(d) If the Red LRU STATUS, KEY INTERLOCK, or CONTROL FAIL LED is ON, then refer to the table at the end of this task to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show.

NOTE: LRU STATUS is the same as LRU FAIL.

CONTROL FAIL is the same as CONTROL INPUT FAIL.

OMR 701-799, 802

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(2) Do the BITE Procedure for the HF Communication System as follows:

(a) Push and hold the SQL/LAMP TEST Switch on the applicable HF Transceiver 1 (2) Front Panel to do the Lamp Test.

1) Make sure that the LRU FAIL, KEY INTERLOCK, and CONTROL INPUT FAIL Lights come ON and stay ON.

(b) Release the SQL/LAMP TEST Switch.

1) Make sure that the LRU FAIL, KEY INTERLOCK, and CONTROL INPUT FAIL Lights go OFF.

(c) Connect a Headset/Boom Microphone to the Captain's Jack Panel.

(d) Do these steps to prepare the Captain's Audio Control Panel (ACP), on the P8 Panel, for HF System Operation:

1) Push and release the applicable HF-1 or HF-2 Switch.

a) Make sure that the Switch Light comes ON.

2) Push and release the applicable HF-1 or HF-2 Volume Control.

a) Make sure that the Volume Control Indicator Light comes ON.

3) Set the Volume Control to the middle position.

4) Set the MASK/BOOM Switch (if installed) to the BOOM position.

(e) On the P8 Panel, do these steps at the applicable Radio Tuning Panel (RTP)-1 (2):

1) Push the applicable HF-1 (2) Switch Light.

a) Make sure that the Switch Light comes ON.

2) Push the AM Switch Light for AM or USB Mode of operation.

a) Make sure that the Switch Light is ON for AM, or OFF for USB.

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OMR 701-799, 802 (Continued)

3) Rotate the SENS Control fully clockwise.

4) Set the STANDBY Frequency Window to an approved Test Frequency.

5) Push the Display Transfer Switch.

a) Make sure that the STANDBY Frequency moves to the ACTIVE Frequency Window.

(f) Push and release the Captain's Push-To-Talk (PTT) Switch.

1) Make sure that you hear a 1 kHz Tune-in-progress Tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 2 to 4 seconds typical, 7 seconds maximum.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected (about 1 second), in which case the tune tone may be only a momentary beep or may not be audible. But at the first tuning after a cold start, the tune tone is always audible regardless of whether this frequency is stored (average 2 to 4 seconds, 7 seconds maximum).

NOTE: Data for the last 100 tuned frequencies is stored in memory. When you replace the HF Antenna or HF Transceiver, the Frequency Memory is reset.

(g) If the LRU FAIL, KEY INTERLOCK, and CONTROL INPUT FAIL Lights on the HF Transceiver Front Panel are OFF, then the BITE Test passed.

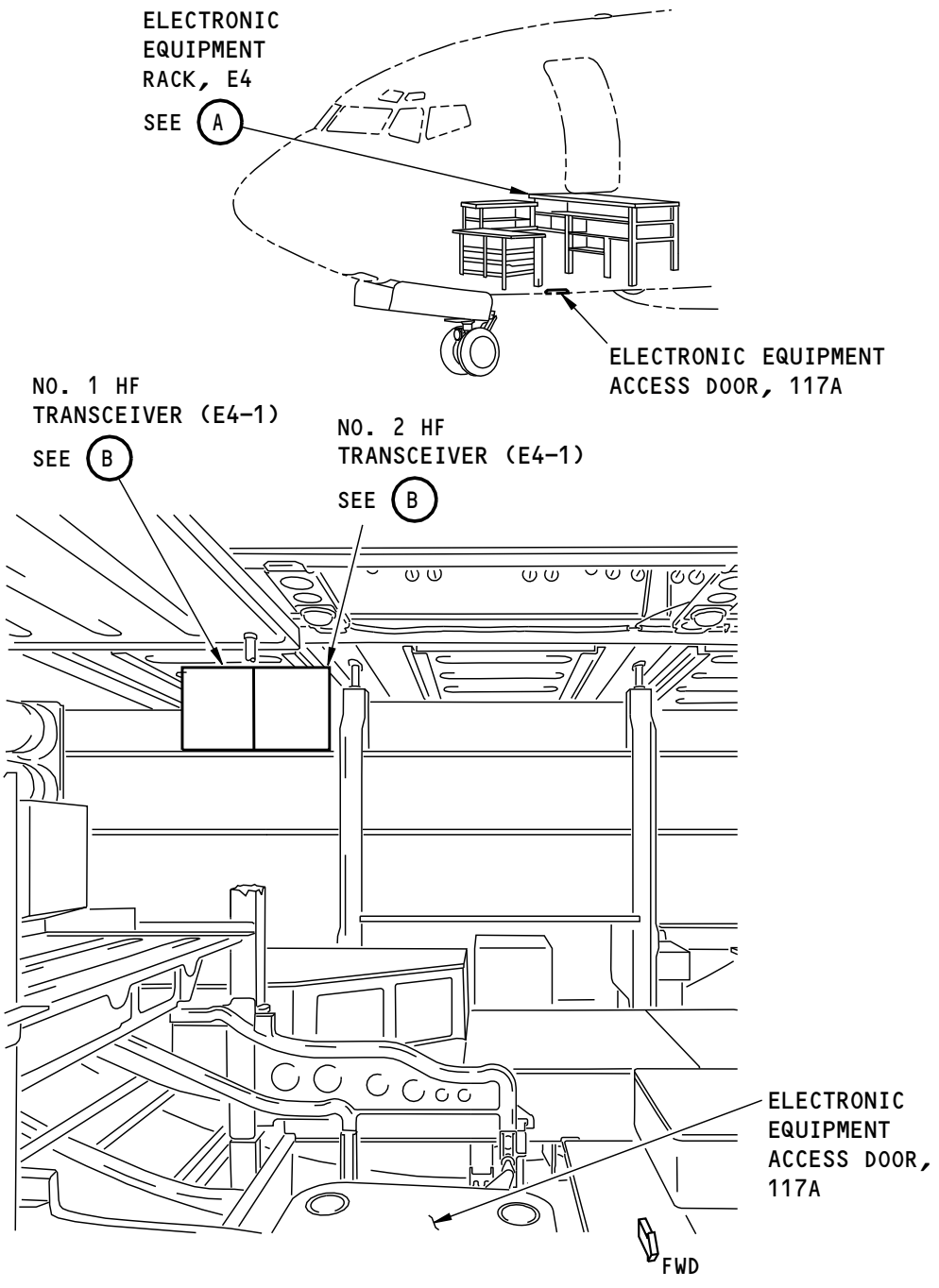
(h) If a LRU FAIL, KEY INTERLOCK, or CONTROL INPUT FAIL Light is ON, then refer to the table at the end of this task to find the applicable FIM Tasks for the maintenance messages that show.

OMR ALL

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
HF XCVR	CONTROL FAIL	23-11 TASK 802
HF XCVR	CONTROL INPUT FAIL	23-11 TASK 802
HF XCVR	KEY INTERLOCK	23-11 TASK 804
HF XCVR	LRU FAIL	23-11 TASK 803
HF XCVR	LRU STATUS	23-11 TASK 803

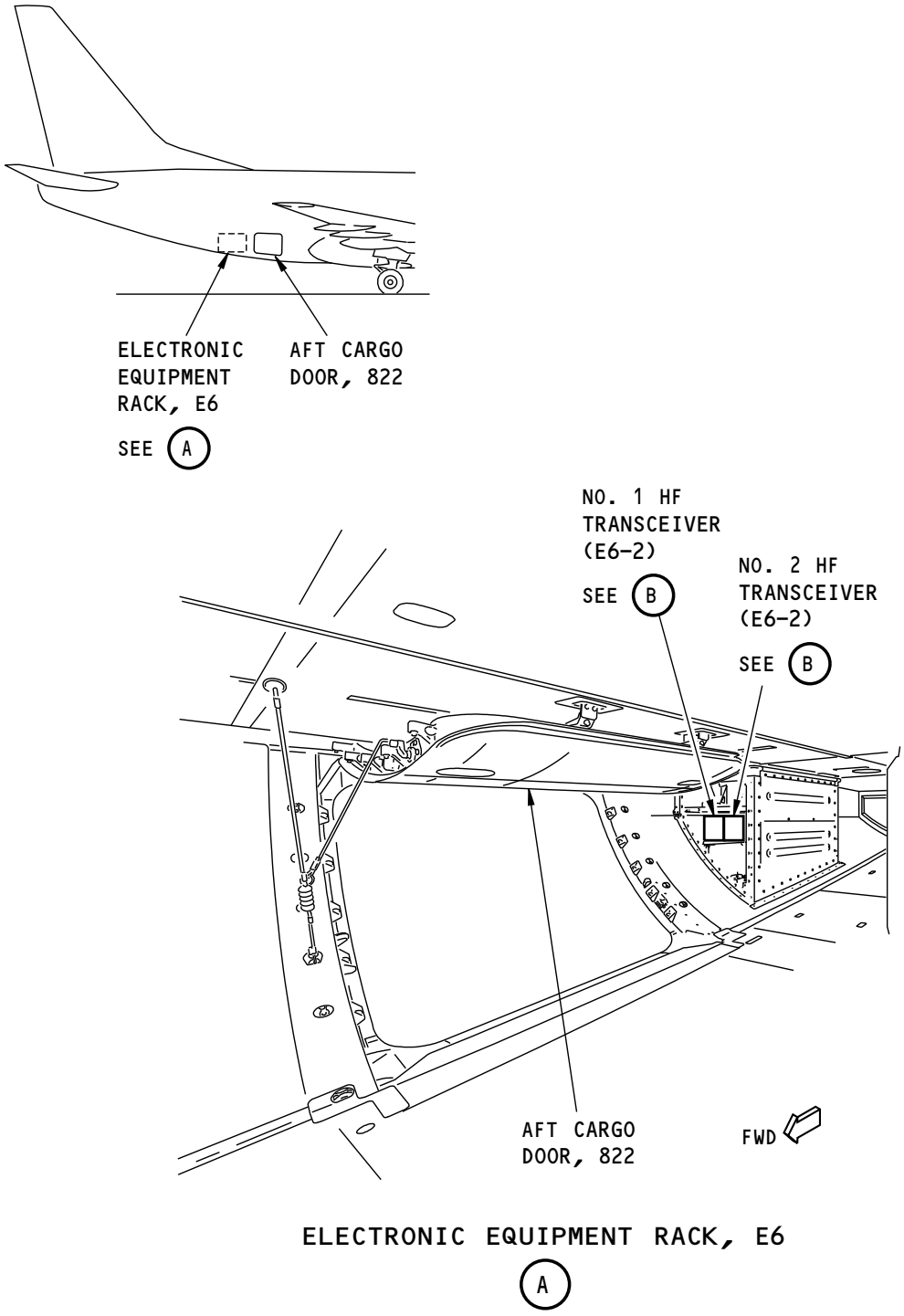
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



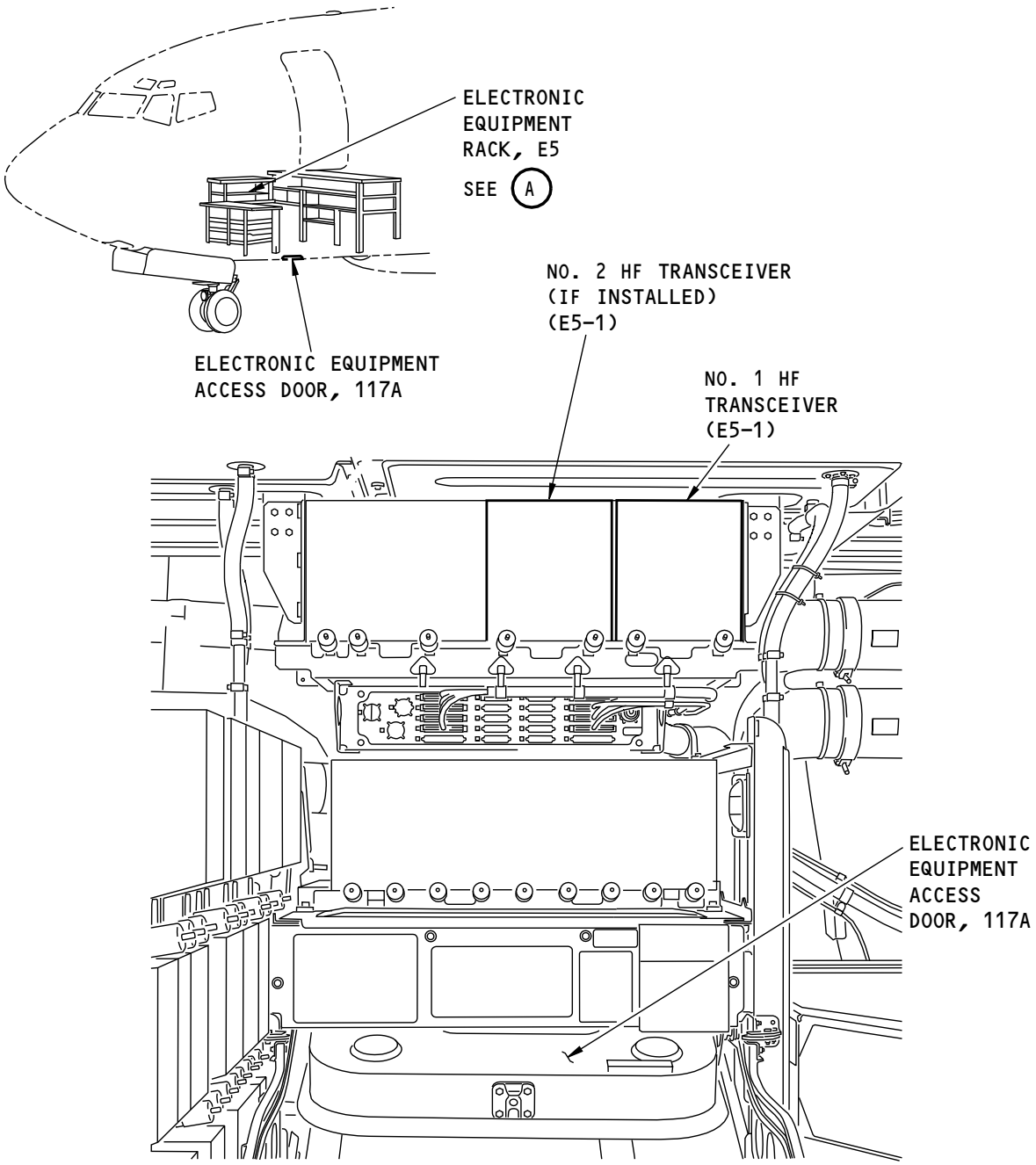
HF Transceiver - Component Location
Figure 201/23-11-00-990-831

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HF Transceiver - Component Location
Figure 202/23-11-00-990-827

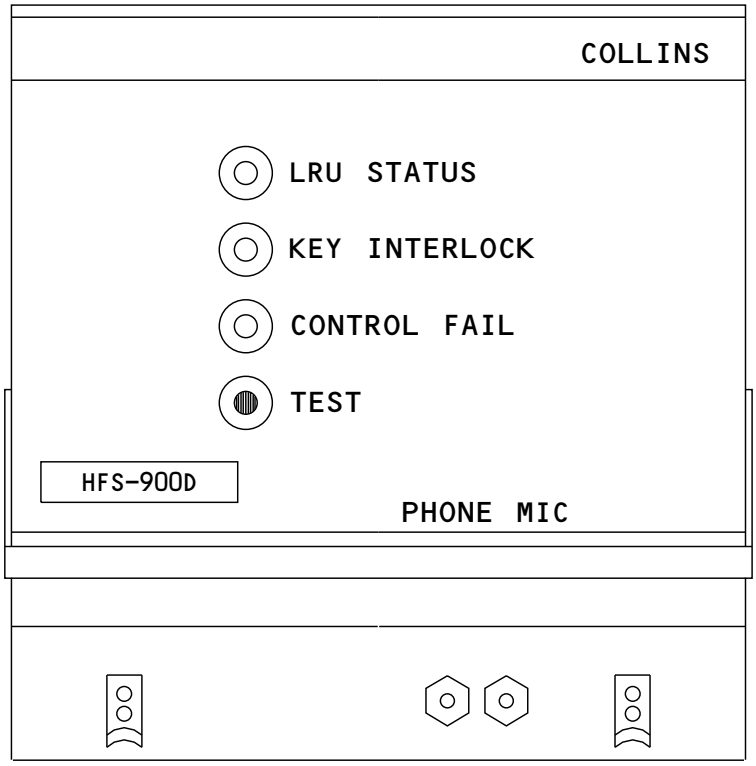
ECCN 9E991 BOEING PROPRIETARY - See title page for details



K01207 S0000146851_V1

HF Transceiver - Component Location
Figure 203/23-11-00-990-826

ECCN 9E991 BOEING PROPRIETARY - See title page for details

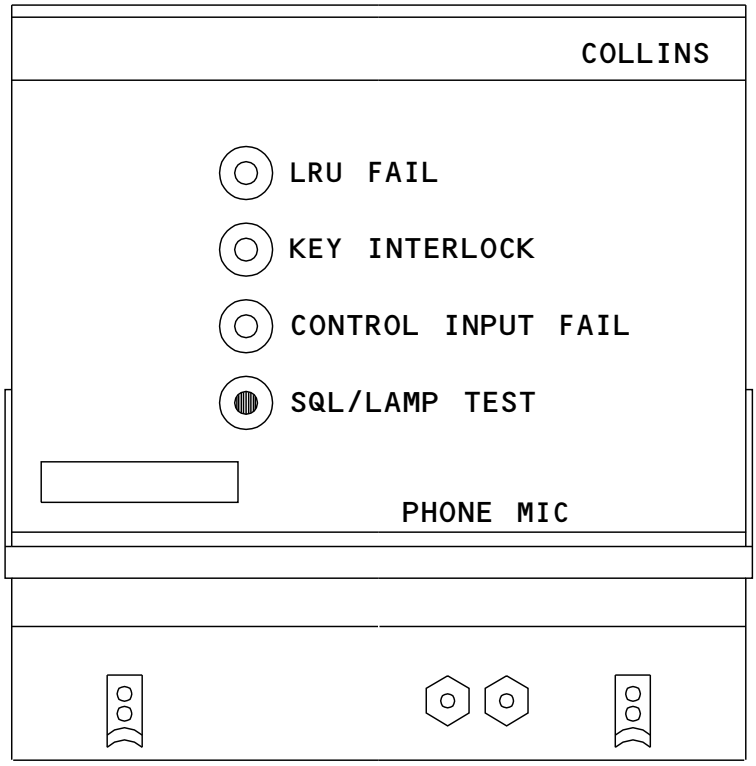


HF TRANSCEIVER

L36671 S0000146937_V1

HF Transceiver Front Panel
Figure 204/23-11-00-990-820

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HF TRANSCEIVER

N56070 S0000146940_V1

HF Transceiver Front Panel
Figure 205/23-11-00-990-817

ECCN 9E991 BOEING PROPRIETARY - See title page for details

802. HF Transceiver CONTROL INPUT FAIL Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) CONTROL INPUT FAIL

OMR 701-799, 802

(2) The HF transceiver receives no input from the HF control panel.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) The HF transceiver receives no input from the radio tuning panel.

OMR ALL

B. Possible Causes

OMR 701-799, 802

(1) HF control panel, P8-11 (HF-1) or P8-12 (HF-2).

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2).

OMR ALL

(3) Wiring problem.

(4) HF transceiver, M226 (HF-1) or M439 (HF-2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name

OMR ALL

C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

D. Related Data

(1) (SSM 23-11-11, 23-11-21)

(2) (WDM 23-11-11, 23-11-21)

E. Initial Evaluation

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If the maintenance message does not show on the front panel of the transceiver, then there was an intermittent fault.

(b) If the maintenance message shows on the front panel of the transceiver, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

(2) Do these steps at the HF control panel, P8-11 (HF-1), on the aft electronic panel, P8:

(a) Set the mode selector switch to the AM or USB position.

(b) Set the HF frequency display to an approved test HF frequency.

(c) Make sure that the frequency display shows the set frequency.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Do these steps at the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2), on the aft electronic panel, P8:

(a) Push the applicable HF switch light (HF 1 or HF 2).

1) Make sure that the switch light comes on.

(b) Set the STANDBY frequency window to an approved test frequency.

(c) Push the display transfer switch.

(d) Make sure that the ACTIVE frequency display shows the set frequency.

OMR ALL

F. Fault Isolation Procedure - Frequency Is Not Shown

OMR 701-799, 802

(1) Do this check for 115V AC at the HF control panel:

(a) Remove the HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To remove it, do this task: HF Control Panel - Removal, AMM TASK 23-11-31-000-801.

(b) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 701-799, 802			
D	2	C00857	COMMUNICATIONS HF 2

(c) Do a check for 115V AC between pin 3 of connector D419 (HF-1) or D773 (HF-2), and structure ground.

(d) Open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 701-799, 802			
D	2	C00857	COMMUNICATIONS HF 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802 (Continued)

(e) If there was 115V AC at pin 3 of connector D419 (HF-1) or D773 (HF-2), then do these steps:

1) Install a new HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To install it, do this task: HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

2) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message CONTROL INPUT FAIL does not show on the front panel of the transceiver, then you corrected the fault.

(f) If there was no 115V AC at pin 3 of connector D419 (HF-1) or D773 (HF-2), then continue.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Do this check for 28V DC at the radio tuning panel:

(a) Remove the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2). To remove it, do this task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(b) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

(c) Do a check for 28V DC at pin 40 of connector D203 (RTP-1) or D209 (RTP-2), to structure ground.

(d) Open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

(e) If there was 28V DC at pin 40 of connector D203 (RTP-1) or D209 (RTP-2), then do these steps:

1) Install a new radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2). To install it, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

2) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

(f) If there was no 28V DC at pin 40 of connector D203 (RTP-1) or D209 (RTP-2), then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

(3) Do this check for 115V AC at the circuit breaker:

(a) Do a check for 115V AC between the load terminal of HF-1 circuit breaker C839, or HF-2 circuit breaker C857, and structure ground.

(b) If there is no 115V AC at the load terminal of HF-1 circuit breaker C839, or HF-2 circuit breaker C857, then do these steps:

1) Replace one of these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 701-799, 802			
D	2	C00857	COMMUNICATIONS HF 2

2) Re-install the HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To install it, do this task: HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

(c) If there is 115V AC at the load terminal of HF-1 circuit breaker C839, or HF-2 circuit breaker C857, then do these steps:

1) Repair the wiring between pin 3 of connector D419 (HF-1) or D773 (HF-2), at the HF control panel, P8-11 (HF-1) or P8-12 (HF-2), and the load terminal of HF-1 circuit breaker C839, or HF-2 circuit breaker C857, at the circuit breaker panel, P18-2 (HF-1) or P6-1 (HF-2).

2) Re-install the HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To install it, do this task: HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Do this check for 28V DC at the circuit breaker:

(a) Do a check for 28V DC between the load terminal of VHF-1 circuit breaker C165, or VHF-2 circuit breaker C166, and structure ground.

(b) If there is no 28V DC at the load terminal of VHF-1 circuit breaker C165, or VHF-2 circuit breaker C166, then do these steps:

1) Replace one of these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

2) Re-install the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2). To install it, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

(c) If there is 28V DC at the load terminal of VHF-1 circuit breaker C165, or VHF-2 circuit breaker C166, then do these steps:

1) Repair the wiring between pin 40 of connector, D203 (RTP-1) or D209 (RTP-2), at the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2), and the load terminal of VHF-1 circuit breaker, C165, or VHF-2 circuit breaker, C166, at the circuit breaker panel, P18-2 (VHF-1) or P6-1 (VHF-2)

2) Re-install the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2). To install it, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

OMR ALL

G. Fault Isolation Procedure - Frequency Is Shown

OMR 701-799, 802

(1) Replace the HF control panel, P8-11 (HF-1) or P8-12 (HF-2).

These are the tasks:

HF Control Panel - Removal, AMM TASK 23-11-31-000-801,

HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Replace the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2).

These are the tasks:

Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801,

Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

OMR ALL

(3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

(b) If the maintenance message shows on the front panel of the transceiver, then continue.

(4) Replace the HF transceiver, M226 (HF-1) or M439 (HF-2).

These are the tasks:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HF Transceiver - Removal, AMM TASK 23-11-21-000-801,

HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

2) If the maintenance message shows on the front panel of the transceiver, then continue.

OMR 701-799, 802

(5) Do this check of the wiring:

(a) Remove the HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To remove it, do this task: HF Control Panel - Removal, AMM TASK 23-11-31-000-801.

(b) Remove the HF transceiver, M226 (HF-1) or M439 (HF-2). To remove it, do this task: HF Transceiver - Removal, AMM TASK 23-11-21-000-801.

(c) Do a wiring check between these pins of connector, D419 (HF-1) or D773 (HF-2), at the aft electronic panel, P8, and these pins of transceiver tray connector, D345B (HF-1) or D623B (HF-2), on the electronic equipment shelf:

	HF CONTROL PANEL	
	PANEL CONNECTOR	**XCVR CONNECTOR**
HF-1 (P8-11)	**D419**	**D345B**
	pin 9	pin G3
	pin 10	pin H3

	HF CONTROL PANEL	
	PANEL CONNECTOR	**XCVR CONNECTOR**
HF-2 (P8-12)	**D773**	**D623B**
	pin 9	pin G3
	pin 10	pin H3

(d) Repair any airplane wiring problems you find.

(e) Re-install the HF control panel, P8-11 (HF-1) or P8-12 (HF-2). To install it, do this task: HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

(f) Re-install the HF transceiver, M226 (HF-1) or M439 (HF-2). To install it, do this task: HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(g) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message CONTROL INPUT FAIL does not show on the front panel of the transceiver, then you corrected the fault.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(6) Do this check of the wiring:

(a) Remove the radio tuning panels, P8-71 (RTP-1) and P8-72 (RTP-2). To remove them, do this task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(b) Remove the HF transceiver, M226 (HF-1) or M439 (HF-2). To remove it, do this task: HF Transceiver - Removal, AMM TASK 23-11-21-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(c) Do a wiring check between these pins of connector, D203 (RTP-1) and D209 (RTP-2), at the aft electronic panel, P8, and these pins of tranceiver tray connector, D345B (HF-1) or D623B (HF-2), at the electronic equipment shelf:

	RTP CONNECTOR		RTP XCVR CONNECTOR
RTP-1 (P8-71)	**D203**		**D345B**
	pin 2	. . . . .	pin E3
	pin 3	. . . . .	pin F3

	RTP CONNECTOR		RTP XCVR CONNECTOR
RTP-2 (P8-72)	**D209**		**D345B**
	pin 2	. . . . .	pin G3
	pin 3	. . . . .	pin H3

RTP-1 (P8-71)	**D203**		**D623B**
	pin 2	. . . . .	pin G3
	pin 3	. . . . .	pin H3

RTP-2 (P8-72)	**D209**		**D623B**
	pin 2	. . . . .	pin E3
	pin 3	. . . . .	pin F3

(d) Repair any airplane wiring problems you find.

(e) Re-install the radio tuning panels, P8-71 (RTP-1) and P8-72 (RTP-2). To install them, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

(f) Re-install the HF transceiver, M226 (HF-1) or M439 (HF-2). To install it, do this task: HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(g) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message does not show on the front panel of the transceiver, then you corrected the fault.

OMR ALL

——— END OF TASK ———

803. HF Transceiver LRU FAIL Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) LRU FAIL

(2) The HF Transceiver has an internal fault.

B. Possible Causes

(1) HF-1 (2) Transceiver, M226 (M439)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	2	C00857	COMMUNICATIONS HF 2

D. Related Data

(1) WDM 23-11-11

(2) WDM 23-11-21

(3) SSM 23-11-11

(4) SSM 23-11-21)

E. Initial Evaluation

(1) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If the maintenance message LRU FAIL does not show on the HF Transceiver Front Panel, then there was an intermittent fault.

(b) If the maintenance message LRU FAIL shows on the HF Transceiver Front Panel, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the applicable HF Transceiver-1 (2), M226 (M439). These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message LRU FAIL does not show on the HF Transceiver Front Panel, then you corrected the problem.

——— END OF TASK ———

804. KEY INTERLOCK or COUPLER FAIL Fault - Fault Isolation

A. Description

(1) This task is for maintenance message:

(a) KEY INTERLOCK

(2) The HF transceiver detects an antenna coupler failure.

B. Possible Causes

(1) HF antenna coupler, M227 (HF-1) or M440 (HF-2)

(2) HF transceiver, M226 (HF-1) or M439 (HF-2)

(3) Wiring

(4) HF antenna, M228

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

D. Related Data

(1) SSM 23-11-11, 23-11-21

(2) WDM 23-11-11, 23-11-21

E. Initial Evaluation

(1) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If the maintenance message does not show on the front panel of the transceiver, then there was an intermittent fault.

(b) If the maintenance message shows on the front panel of the transceiver, then continue.

F. Fault Isolation Procedure

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

(2) Replace the HF antenna coupler, M227 (HF-1) or M440 (HF-2). These are the tasks:

- HF Antenna Coupler - Removal, AMM TASK 23-11-61-000-801
- HF Antenna Coupler - Installation, AMM TASK 23-11-61-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message shows on the front panel of the transceiver, then continue.

(3) Replace the HF transceiver, M226 (HF-1) or M439 (HF-2). These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

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1) If the maintenance message does not show on the front panel of the transceiver, then you corrected the problem.

2) If the maintenance message shows on the front panel of the transceiver, then continue.

(4) Do this check of the coaxial cable:

(a) Remove the HF transceiver, M226 (HF-1) or M439 (HF-2). This is the task: HF Transceiver - Removal, AMM TASK 23-11-21-000-801.

(b) Remove the applicable access panels:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

NOTE: The HF-1 antenna coupler is installed on the left side of the vertical stabilizer. The HF-2 antenna coupler is installed on the right side of the vertical stabilizer.

(c) Disconnect the coaxial cable connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2).

(d) Do a time or frequency domain reflectometry check of the coaxial cable between connector, D345B (HF-1) or D623B (HF-2), at the HF transceiver, M226 (HF-1) or M439 (HF-2), and connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2) (WDM 23-11-11, -21), do this task: AMM TASK 20-10-72-210-801 or AMM TASK 20-10-72-200-801.

(e) Repair any problems that you find.

(f) Re-connect the coaxial cable connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2).

(g) Install the applicable access panels:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

(h) Re-install the HF transceiver, M226 (HF-1) or M439 (HF-2). This is the task: HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(i) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

1) If the maintenance message shows on the front panel of the transceiver, then continue.

(5) Do a check of the electrical bond of the HF antenna, do this task: HF Antenna - Electrical Bond Check, AMM TASK 23-11-51-760-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801

1) If the maintenance message does not show on the front panel of the transceiver, then you corrected the problem.

——— END OF TASK ———

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OMR 701-799, 802

805. HF Control Panel Problem - Fault Isolation

A. Description

(1) The HF control panel does not operate correctly.

B. Possible Causes

(1) HF control panel, P8-11 (HF-1) or P8-12 (HF-2).

(2) HF transceiver, M226 (HF-1) or M439 (HF-2).

(3) Wiring problem.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 701-799, 802			
D	2	C00857	COMMUNICATIONS HF 2

D. Related Data

(1) (SSM 23-11-11, 23-11-21)

(2) (WDM 23-11-11, 23-11-21)

E. Initial Evaluation

(1) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If there is a maintenance message shown on the front panel of the transceiver, then do the fault isolation procedure for the maintenance message.

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE PERSONNEL STAY A MINIMUM OF 6 FEET AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(2) Do not operate the HF system while a fuel operation is done on the airplane.

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OMR 701-799, 802 (Continued)

(3) Do this reception/transmission test of the HF communication system:

NOTE: HF communication can be degraded while the airplane is on the ground due to external interference or signal blockage. Before you identify a HF communication system fault, make sure you try to transmit and receive at several frequencies across the HF frequency band. Make sure the airplane is not in or near any large metal structures. In some cases, you can move the airplane to correct the problem.

(a) Connect a headset/boom microphone, M428 (captain's), to the boom microphone jack, D6027 (captain's).

(b) Do these steps at the audio control panel, P8-6 (captain's ACP):

1) Push and release the applicable HF-1 or HF-2 switch.
   a) Make sure that the switch light comes on.
2) Push and release the applicable HF-1 or HF-2 volume control.
   a) Make sure that the volume control indicator light comes on.
3) Set the volume control to the middle position.
4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Do these steps at the applicable HF control panel, P8-11 (HF-1) or P8-12 (HF-2):

1) Set the mode selector switch to the AM or USB position.
2) Turn the RF SENS control fully clockwise.
3) Set the HF frequency display to an approved test frequency.

(d) Push and release the captain's push-to-talk (PTT) switch.

1) Make sure you hear a 1 kHz tune-in-progress tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 1 to 7 seconds.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected, in which case the tune tone may be only a momentary beep.

(e) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold the PTT switch while you speak.
   a) Make sure you hear the sidetone in the headset while you speak.
2) Release the PTT switch while you listen.
   a) Make sure the quality of the transmitted and received voice is satisfactory.
   b) Make sure the sound of the received voice decreases and increases when you turn the RF SENS control counterclockwise and clockwise on the HF control panel.
   c) Make sure the sound of the received voice changes when you turn the applicable HF-1 or HF-2 volume control on the ACP, P8-6 (captain's), with no change in voice quality.

(f) If more than one mode of operation (AM and USB) is available at your location, do this step:

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OMR 701-799, 802 (Continued)

1) Set the mode selector switch to the other mode of operation and make a voice transmission.

(4) If the HF control panel operates satisfactorily, then there was an intermittent fault.

(5) If the HF control panel does not operate satisfactorily, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the HF control panel, P8-11 (HF-1) or P8-12 (HF-2).

These are the tasks:

HF Control Panel - Removal, AMM TASK 23-11-31-000-801,

HF Control Panel - Installation, AMM TASK 23-11-31-400-801.

(2) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(3) Do a voice communication test with a radio tower operator.

(a) Make sure that the quality of the transmitted and received voice is satisfactory.

(b) If the HF control panel operates satisfactorily, then you corrected the fault.

(c) If the HF control panel does not operate satisfactorily, then continue.

(4) Replace the HF transceiver, M226 (HF-1) or M439 (HF-2).

These are the tasks:

HF Transceiver - Removal, AMM TASK 23-11-21-000-801,

HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(5) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(6) Do a voice communication test with a radio tower operator.

(a) Make sure that the quality of the transmitted and received voice is satisfactory.

(b) If the HF control panel operates satisfactorily, then you corrected the fault.

OMR ALL

——— END OF TASK ———

806. HF Communication System Receive/Transmit Problem - Fault Isolation

A. Description

(1) The HF communication system does not operate correctly.

(a) Poor reception and/or poor transmission at any or all audio control panel (ACP) locations for the HF-1 or HF-2 communication system.

(b) HF operation while the airplane is in flight is intermittent, but is satisfactory when the airplane is on the ground.

B. Possible Causes

(1) HF-1 transceiver, M226.

(2) HF-2 transceiver, M4396

(3) HF-1 antenna coupler, M227

(4) HF-2 antenna coupler, M440.

(5) HF antenna, M228

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OMR 701-799, 802

(6) HF-2 control panel, P8-12

OMR ALL

(7) Radio tuning panel, P8-71 (RTP-1)

(8) Radio tuning panel, P8-72 (RTP-2)

(9) Radio tuning panel, P8-72 (RTP-3)

(10) Audio control panel, P8-6 (captain's)

(11) Audio control panel, P8-7 (first officer's)

(12) Audio control panel, P8-6 (first observer's)

(13) Remote electronics unit, M1353 (REU)

(14) Wiring.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
OMR ALL			
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) (SSM 23-11-11)

(2) (SSM 23-11-21

(3) WDM 23-11-11

(4) WDM 23-11-21

E. Initial Evaluation

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

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(WARNING PRECEDES)

WARNING

MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

(2) Do this reception/transmission test of the HF communication system:

NOTE: HF communication can be degraded while the airplane is on the ground due to external interference or signal blockage. Before you identify a HF communication system fault, make sure you try to transmit and receive at several frequencies across the HF frequency band. Make sure the airplane is not in or near any large metal structures. In some cases, you can move the airplane to correct the problem.

(a) Connect a headset/boom microphone, M428 (captain's), to the boom microphone jack, D6027 (captain's).

(b) Do these steps at the audio control panel, P8-6 (captain's ACP):

1) Push and release the applicable HF-1 or HF-2 switch.

a) Make sure that the switch light comes on.

2) Push and release the applicable HF-1 or HF-2 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

OMR 701-799, 802

(c) Do these steps at the applicable HF control panel, P8-11 (HF-1) or P8-12 (HF-2):

1) Set the mode selector switch to the AM or USB position.

2) Turn the RF SENS control fully clockwise.

3) Set the HF frequency display to an approved test frequency.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Do these steps at the radio tuning panel, P8-71 (RTP-1):

1) Push the applicable HF switch light (HF 1 or HF 2).

a) Make sure that the switch light comes on.

2) Push the AM switch light for AM or USB mode of operation.

a) Make sure that the switch light is on for AM, or off for USB.

3) Turn the SENS control clockwise to the maximum position.

4) Set the STANDBY frequency window to an approved test frequency.

5) Push the display transfer switch.

a) Make sure that the STANDBY frequency moves to the ACTIVE frequency window.

OMR ALL

(e) Push and release the captain's push-to-talk (PTT) switch.

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1) Make sure that you hear a 1 kHz tune-in-progress tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 2 to 4 seconds typical, 7 seconds maximum. A continuous tone indicates a failed HF coupler and as long as this failure exists, it will be annunciated on the front of the HF transceiver.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected (about 1 second), in which case the tune tone may be only a momentary beep or may not be audible. But at the first tuning after a cold start, the tune tone is always audible regardless of whether this frequency is stored (average 2 to 4 seconds, 7 seconds maximum).

NOTE: Data for the last 100 tuned frequencies is stored in memory. When either HF antenna or HF transceiver is replaced, the frequency memory is reset.

(f) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold the PTT switch while you speak.

NOTE: If no audio sidetone is heard, the HF transceiver has failed and as long as this failure exists, it will be annunciated on the front of the HF transceiver.

a) Make sure that you hear the sidetone in the headset while you speak.

b) Make sure that the HF transceiver blower fan operates when you transmit.

c) Make sure that the FAIL lights on the front panel of the HF transceiver are not on.

2) Release the PTT switch while you listen.

a) Make sure that the quality of the transmitted and received voice is satisfactory.

OMR 701-799, 802

b) Make sure that the sound of the received voice decreases and increases when you turn the RF SENS control counterclockwise and clockwise on the HF control panel.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

c) Make sure that the sound of the received voice decreases and increases when you turn the SENS control counterclockwise and clockwise on the RTP.

OMR ALL

d) Make sure that the sound of the received voice changes when you turn the applicable HF-1 or HF-2 volume control on the ACP, P8-6 (captain's), with no change in voice quality.

(g) If more than one mode of operation (AM and USB) is available at your location, do this step:

OMR 701-799, 802

1) Set the mode selector switch to the other mode of operation and make a voice transmission.

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) Push the AM switch light to the other mode of operation and make a voice transmission.

OMR ALL

(h) At each remaining ACP, set the applicable HF-1 or HF-2 switch and make a voice transmission.

1) Make sure that the quality of the transmitted and received voice is satisfactory.

2) Make sure that the sound of the received voice changes when you turn the applicable HF-1 or HF-2 volume control on the applicable ACP with no change in voice quality.

(3) If the HF communication system operates satisfactorily, then there was an intermittent fault.

(4) If the HF communication system does not operate satisfactorily at one location, then do the Fault Isolation Procedure - Problem At One Location below.

(5) If the HF communication system does not operate satisfactorily at all locations, then do the Fault Isolation Procedure - Problem At All Locations below.

(6) If the HF communication system does not operate satisfactorily or is intermittent while in flight, but operates satisfactorily on the ground, do the Fault Isolation Procedure - Intermittent Operation In Flight, Ground Operation OK below.

F. Fault Isolation Procedure - Problem At One Location

(1) Replace the audio control panel, P8-6 (captain's ACP), P8-7 (first officer's ACP) or P5-15 (first observer's ACP). These are the tasks:

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801

(a) Do a voice communication test with a radio tower operator using the applicable ACP.

1) Make sure that the quality of the transmitted and received voice is satisfactory.

(b) If the HF communication system operates satisfactorily, then you corrected the problem.

(c) If the HF communication system does not operate satisfactorily, then continue.

(2) Replace the remote electronics unit, M1353 (REU). These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do a voice communication test with a radio tower operator using the applicable ACP.

1) Make sure that the quality of the transmitted and received voice is satisfactory.

(b) If the HF communication system operates satisfactorily, then you corrected the problem.

G. Fault Isolation Procedure - Problem At All Locations

(1) Do this exchange check of the HF transceiver, M226 (HF-1) or M439 (HF-2):

(a) Put a tag that reads SUSPECT on the HF transceiver, M226 (HF-1) or M439 (HF-2), with poor reception or transmission.

(b) Put a tag that reads OK on the other HF transceiver, M226 (HF-1) or M439 (HF-2).

(c) Exchange the locations of the HF transceivers, M226 (HF-1) and M439 (HF-2). These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

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(d) Do a BITE test of the HF transceivers, M226 (HF-1) and M439 (HF-2). This is the task: HF Communication System - BITE Procedure, 23-11 TASK 801

(e) Do a voice communication test with a radio tower operator.

1) Make sure the quality of the transmitted and received voice is satisfactory.

(f) If the poor reception or transmission moves with the HF transceiver tagged SUSPECT, then do these steps:

1) Replace the HF transceiver, M226 (HF-1) or M439 (HF-2), tagged SUSPECT. These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

NOTE: If it is your airlines' policy, you must install the HF transceiver with the OK tag in its initial location.

2) Do a BITE test of the HF transceiver that replaced the HF transceiver tagged SUSPECT. This is the task: HF Communication System - BITE Procedure, 23-11 TASK 801.

a) Do the Repair Confirmation at the end of this task.

3) Remove the tags from the HF transceivers, M226 (HF-1) and M439 (HF-2).

(g) If the poor reception or transmission goes away after you exchange the HF transceivers, M226 (HF-1) and M439 (HF-2), then do these steps to complete the task:

NOTE: There was an intermittent fault in either the equipment rack connector, D345B (HF-1) or D623B (HF-2), or in the HF transceiver, M226 (HF-1) or M439 (HF-2), with the SUSPECT tag.

1) If it is your airlines' policy, you must install the HF transceivers, M226 (HF-1) and M439 (HF-2), in their initial locations. These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

2) Remove the tags from the HF transceivers, M226 (HF-1) and M439 (HF-2).

(h) If the poor reception or transmission stays with the HF-1 or HF-2 communication system after you exchange the HF transceivers, M226 (HF-1) and M439 (HF-2), then do these steps and continue:

1) If it is your airlines' policy, you must install the HF transceivers, M226 (HF-1) and M439 (HF-2), in their initial locations. These are the tasks:

- HF Transceiver - Removal, AMM TASK 23-11-21-000-801
- HF Transceiver - Installation, AMM TASK 23-11-21-400-801

2) Remove the tags from the HF transceivers, M226 (HF-1) and M439 (HF-2).

OMR 701-799, 802

(2) Replace the HF control panel, P8-11 (HF-1). These are the tasks:

- HF Control Panel - Removal, AMM TASK 23-11-31-000-801
- HF Control Panel - Installation, AMM TASK 23-11-31-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(b) Do the Repair Confirmation at the end of this task.

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Replace the radio tuning panel, P8-71 (RTP-1) or P8-72 (RTP-2). These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(b) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802

(4) Do this exchange check of the HF control panel, P8-11 (HF-1) or P8-12 (HF-2):

(a) Put a tag that reads SUSPECT on the HF control panel, P8-11 (HF-1) or P8-12 (HF-2), with poor reception or transmission.

(b) Put a tag that reads OK on the other HF control panel, P8-11 (HF-1) or P8-12 (HF-2).

(c) Exchange the locations of the HF control panels, P8-11 (HF-1) and P8-12 (HF-2). These are the tasks:

- HF Control Panel - Removal, AMM TASK 23-11-31-000-801
- HF Control Panel - Installation, AMM TASK 23-11-31-400-801

(d) Do a BITE test of the HF transceivers, M226 (HF-1) and M439 (HF-2). This is the task: HF Communication System - BITE Procedure, 23-11 TASK 801

(e) Do a voice communication test with a radio tower operator.

1) Make sure that the quality of the transmitted and received voice is satisfactory.

(f) If the poor reception or transmission moves with the HF control panel tagged SUSPECT, then do these steps:

1) Replace the HF control panel, P8-11 (HF-1) or P8-12 (HF-2), tagged SUSPECT. These are the tasks:

- HF Control Panel - Removal, AMM TASK 23-11-31-000-801
- HF Control Panel - Installation, AMM TASK 23-11-31-400-801

NOTE: If it is your airlines' policy, you must install the HF control panel with the OK tag in its initial location.

2) Do a BITE test of the HF transceiver system for the replaced HF control panel tagged SUSPECT. This is the task: HF Communication System - BITE Procedure, 23-11 TASK 801.

3) Do the Repair Confirmation at the end of this task.

a) Remove the tags from the HF control panels, P8-11 (HF-1) and P8-12 (HF-2).

(g) If the poor reception or transmission goes away after you exchange the HF control panels, P8-11 (HF-1) and P8-12 (HF-2), then do these steps to complete the task:

NOTE: There was an intermittent fault in either the HF control panel connector, D419 (HF-1) or D773 (HF-2), or in the HF control panel, P8-11 (HF-1) or P8-12 (HF-2), with the SUSPECT tag.

1) If it is your airlines' policy, you must install the HF control panels, P8-11 (HF-1) and P8-12 (HF-2), in their initial locations. These are the tasks:

- HF Control Panel - Removal, AMM TASK 23-11-31-000-801
- HF Control Panel - Installation, AMM TASK 23-11-31-400-801

2) Remove the tags from the HF control panels, P8-11 (HF-1) and P8-12 (HF-2).

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OMR 701-799, 802 (Continued)

(h) If the poor reception or transmission stays with the HF-1 or HF-2 communication system after you exchange the HF control panels, P8-11 (HF-1) and P8-12 (HF-2), then continue:

1) If it is your airlines' policy, you must install the HF control panels, P8-11 (HF-1) and P8-12 (HF-2), in their initial locations. These are the tasks:

- HF Control Panel - Removal, AMM TASK 23-11-31-000-801
- HF Control Panel - Installation, AMM TASK 23-11-31-400-801

2) Remove the tags from the HF control panels, P8-11 (HF-1) and P8-12 (HF-2).

OMR ALL

(5) Replace the remote electronics unit, M1353 (REU). These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the HF antenna coupler, M227 (HF-1) or M440 (HF-2). These are the tasks:

- HF Antenna Coupler - Removal, AMM TASK 23-11-61-000-801
- HF Antenna Coupler - Installation, AMM TASK 23-11-61-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(b) Do the Repair Confirmation at the end of this task.

(7) Do this check of the coaxial cable:

(a) Remove the HF transceiver, M226 (HF-1) or M439 (HF-2). This is the task: HF Transceiver - Removal, AMM TASK 23-11-21-000-801.

(b) Remove the applicable access panel:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge

or open this access panel:

Number	Name/Location
322AR	Vertical Fin, Fixed Leading Edge

NOTE: The HF-1 antenna coupler is installed on the left side of the vertical stabilizer. The HF-2 antenna coupler is installed on the right side of the vertical stabilizer.

(c) Disconnect the coaxial cable connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2).

(d) Do a time or frequency domain reflectometry check of the coaxial cable between connector, D345B (HF-1) or D623B (HF-2), at the HF transceiver, M226 (HF-1) or M439 (HF-2), and connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2) (WDM 23-11-11, -21), do this task: AMM TASK 20-10-72-210-801 or AMM TASK 20-10-72-200-801.

(e) Repair any problems that you find.

(f) Re-connect the coaxial cable connector, D337 (HF-1) or D625 (HF-2), at the HF antenna coupler, M227 (HF-1) or M440 (HF-2).

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(g) Install the applicable access panel:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge

or close this access panel:

Number	Name/Location
322AR	Vertical Fin, Fixed Leading Edge

(h) Re-install the HF transceiver, M226 (HF-1) or M439 (HF-2). This is the task: HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(i) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

(j) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(k) Do the Repair Confirmation at the end of this task.

(8) Do a check of the electrical bond of the HF antenna, do this task: HF Antenna - Electrical Bond Check, AMM TASK 23-11-51-760-801.

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(b) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - Intermittent Operation In Flight, Ground Operation OK below.

NOTE: This fault indicates that a HF antenna coupler has a pressurization leak. Make sure that the replacement HF coupler is fully pressurized with dry nitrogen in accordance with the applicable Component Maintenance Manual (CMM).

(1) Replace the applicable HF antenna coupler, M227 (HF-1) or M440 (HF-2). These are the tasks:

- HF Antenna Coupler - Removal, AMM TASK 23-11-61-000-801
- HF Antenna Coupler - Installation, AMM TASK 23-11-61-400-801

(a) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(b) Monitor HF system performance on subsequent flights.

(c) If the HF communication system operates satisfactorily while in flight, then you corrected the problem.

I. Repair Confirmation

(1) Do a voice communication test with a radio tower operator.

(a) Make sure that the quality of the transmitted and received voice is satisfactory.

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(2) If the HF communication system operates satisfactorily, then you corrected the problem.

(3) If the HF communication system does not operate satisfactorily, then continue the Fault Isolation procedure at the subsequent step.

——— END OF TASK ———

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801. VHF Communication Transceiver System - BITE Procedure

A. General

(1) You do the Very High Frequency (VHF) Communication System Built-In-Test Equipment (BITE) Test at the Front Panel of the VHF Communication Transceiver.

(2) The No. 1 VHF Communication Transceiver is located on the E1-3 Shelf and the No. 2 VHF Communication Transceiver is located on the E1-5 Shelf in the Main Equipment Center.

(3) The No. 3 VHF Communication Transceiver is located on the E3-3 Shelf in the Main Equipment Center.

(4) The VHF Communication System BITE Test does a self check for existing internal and external faults.

(a) Results of the BITE Test are displayed by fault lights on the Front Panel of the VHF Communication Transceiver.

(5) The VHF Communication Transceiver flight fault memory stores fault data from previous flight legs. You can view faults that occurred on previous flight legs to help isolate an intermittent fault that is not found by the BITE Self-Test.

B. BITE Procedure

(1) Do these steps to do the BITE procedure for the VHF Communication System:

OMR 809; OMR 805, 807 PRE SB 737-23-1330

NOTE: Before you do the BITE Test, make sure that VHF-3 is in the voice mode (a voice frequency shows in the active (left) display of the Radio Tuning Panel (RTP)).

OMR ALL

(a) Push and release the TEST Switch on the Front Panel of the applicable VHF Communication Transceiver.

1) Make sure these conditions occur:

a) All three Light Emitting Diode (LED)s on the VHF Transceiver Front Panel turn red for about two seconds.

b) The LRU STATUS LED turns green and the CONTROL FAIL and ANTENNA FAIL LEDs remain red for about two seconds.

OMR 701-799, 802, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 805, 807 POST SB 737-23-1330

c) All three LEDs go OFF for about two seconds.

OMR 809; OMR 805, 807 PRE SB 737-23-1330

d) All three LEDs go OFF for about five seconds.

OMR ALL

(b) If the LRU STATUS LED comes ON green for about 30 seconds and the CONTROL FAIL and ANTENNA FAIL LEDs remain OFF, then the BITE Test passed.

(c) If the red LRU STATUS, CONTROL FAIL, or ANTENNA FAIL LED is ON, then refer to the table at the end of this task to find the Fault Isolation Task for the applicable maintenance message.

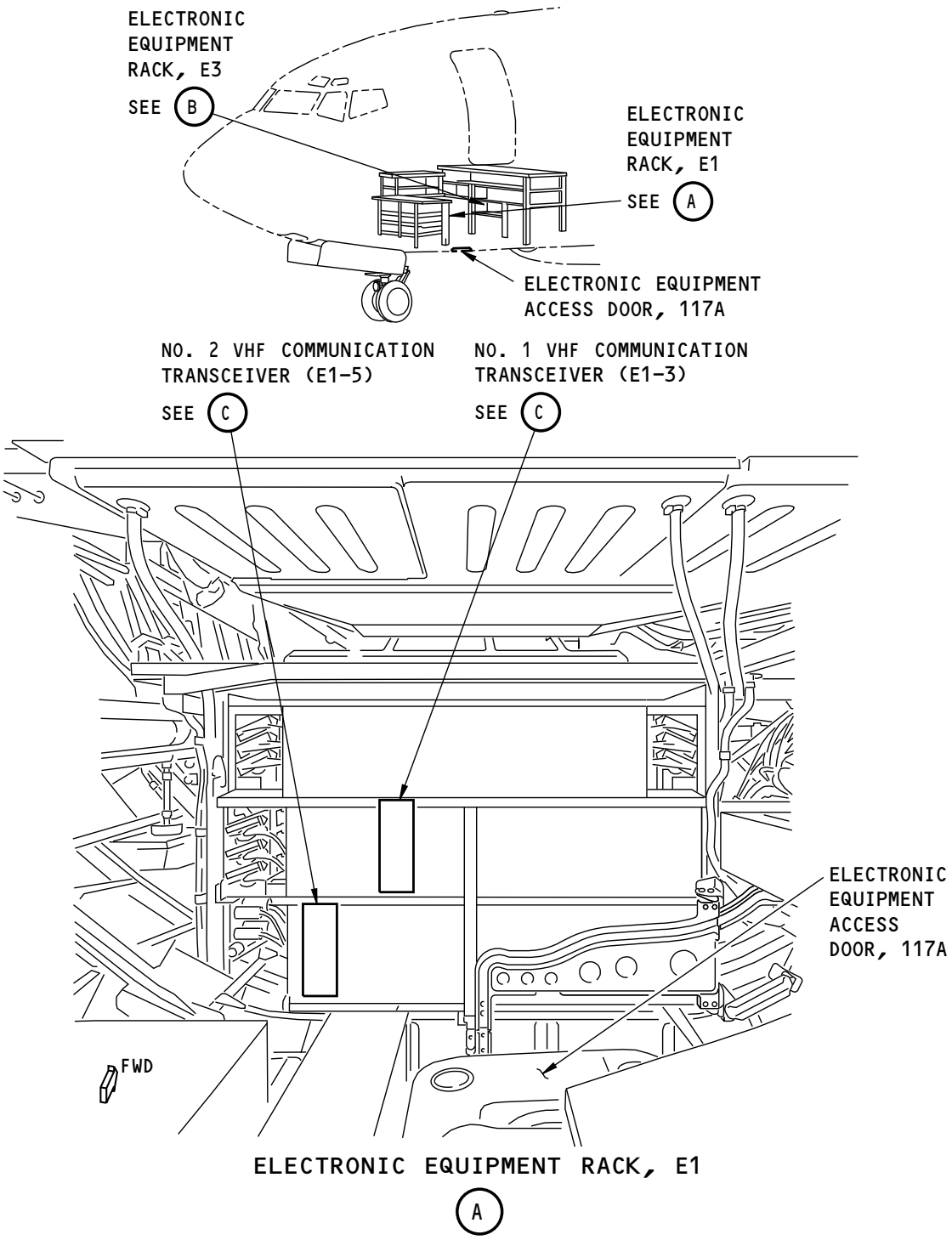
LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
VHF XCVR	ANTENNA FAIL	23-12 TASK 813
VHF XCVR	CONTROL FAIL	23-12 TASK 812

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
VHF XCVR	LRU STATUS	23-12 TASK 818

——— END OF TASK ———

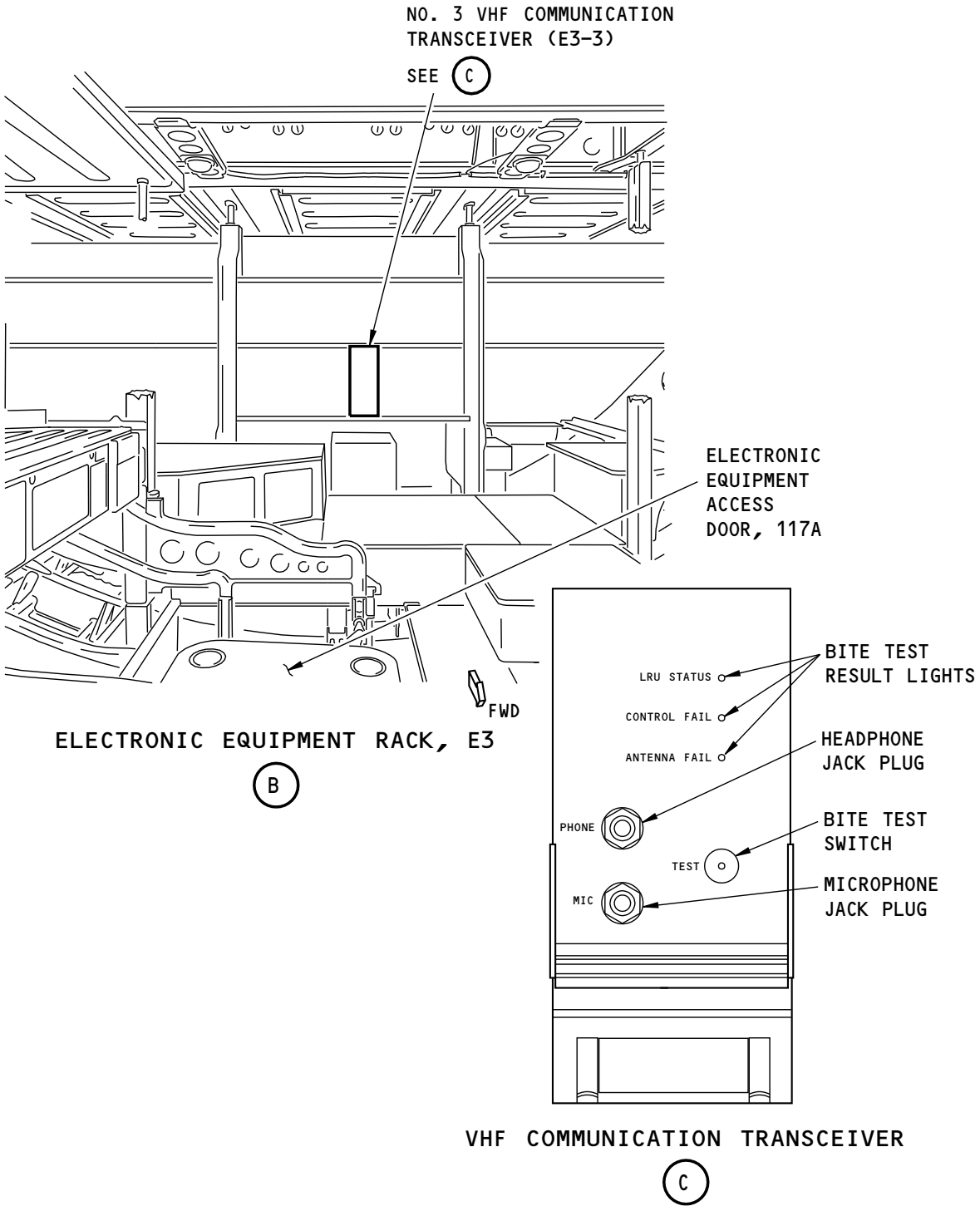
ECCN 9E991 BOEING PROPRIETARY - See title page for details



H01994 S0000146942_V1

VHF Communication System Installation
Figure 201/23-12-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VHF Communication System Installation
Figure 201/23-12-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

811. VHF Transceiver VHF DATA RADIO FAILED Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) VHF DATA RADIO FAILED

(2) The VHF communication transceiver has an internal fault.

B. Possible Causes

(1) VHF communication transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

D. Related Data

(1) (SSM 23-12-11,-21, -31)

(2) (WDM 23-12-11,-21,-31)

E. Initial Evaluation

(1) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If VHF DATA RADIO FAILED does not show on the LCD screen of the transceiver, then there was an intermittent fault.

(b) If VHF DATA RADIO FAILED shows on the LCD screen of the transceiver, then continue.

F. Fault Isolation Procedure

(1) Replace the VHF communication transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3).

These are the tasks:

VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801,

VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.

(a) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

1) If VHF DATA RADIO FAILED does not show on the LCD screen of the transceiver, then you corrected the fault.

END OF TASK

812. VHF Transceiver CONTROL Problem or TUNING PORT B MISSING INPUT Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) CONTROL FAIL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

(2) The VHF Communication Transceiver receives no input from the VHF Communication Control Panel.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) The VHF Communication Transceiver receives no input from the RTP.

OMR ALL

B. Possible Causes

OMR 701-799, 802

(1) VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3)

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) RTP, P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3)

OMR ALL

(3) VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3)

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

D. Related Data

(1) SSM 23-12-11,-21, -31

(2) WDM 23-12-11,-21,-31

E. Initial Evaluation

(1) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the maintenance message or fail light does not show on the transceiver, then there was an intermittent fault.

(b) If the maintenance message or fail light shows on the transceiver, then continue.

OMR 701-799, 802

(2) Set the TFR Switch on the applicable VHF Communication Control Panel to the left position.

(a) Make sure the active frequency light, above the left frequency indicator, is ON.

(b) If the active frequency light is not ON, then do the Fault Isolation Procedure - Active Frequency Light Not On/Frequency Is Not Shown below.

(c) If the active frequency light is ON, then do the Fault Isolation Procedure - Active Frequency Light Is On/Frequency Is Shown below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Set the radio tuning panels for the applicable VHF Communication System.

(a) Set the STANDBY frequency to an approved test frequency.

(b) If the frequency is not shown, then do the Fault Isolation Procedure - Frequency Is Not Shown below.

(c) If the frequency is shown, then do the Fault Isolation Procedure - Frequency Is Shown below.

OMR 701-799, 802

F. Fault Isolation Procedure - Active Frequency Light Not On

(1) Do this check for 28 VDC at the VHF Control Panel:

(a) Remove the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801.

(b) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

(c) Do a check for 28V DC at pin 13 of connector D203 (VHF-1), D209 (VHF-2), or D549 (VHF-3), to Structure Ground.

(d) Open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

(e) If there is 28V DC at pin 13 of connector D203 (VHF-1), D209 (VHF-2), or D549 (VHF-3), then do these steps:

1) Install a new VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801.

2) Do the Repair Confirmation at the end of this task.

(f) If there is not 28V DC at pin 13 of connector D203 (VHF-1), D209 (VHF-2), or D549 (VHF-3), then continue.

(2) Do this check for 28 VDC at the circuit breaker.

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OMR 701-799, 802 (Continued)

(a) Do a check for 28V DC from the load terminal of VHF COMM 1 circuit breaker, C165, VHF COMM 2 circuit breaker, C166, or VHF COMM 3 circuit breaker, C471, to Structure Ground.

(b) If there is not 28V DC at the load terminal of VHF COMM 1 circuit breaker, C165, VHF COMM 2 circuit breaker, C166, or VHF COMM 3 circuit breaker, C471, then do these steps:

1) Replace the applicable circuit breaker, VHF COMM 1, C165, VHF COMM 2, C166, or VHF COMM 3, C471.

2) Re-install the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801.

3) Do the Repair Confirmation at the end of this task.

(c) If there is 28V DC at the load terminal of VHF COMM 1 circuit breaker, C165, VHF COMM 2 circuit breaker, C166, or VHF COMM 3 circuit breaker, C471, then do these steps:

1) Repair the wiring between pin 13 of connector D203 (VHF-1), D209 (VHF-2), or D549 (VHF-3), at the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3), to the load terminal of VHF COMM 1 circuit breaker, C165, VHF COMM 2 circuit breaker, C166, or VHF COMM 3 circuit breaker, C471, at the Circuit Breaker Panel, P18-1 (VHF-1), P6-1 (VHF-2), or P18-2 (VHF-3).

2) Re-install the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801.

3) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Active Frequency Light Is On or Frequency Is Shown

(1) Replace the VHF communications transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring:

(a) Remove the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801.

(b) Remove the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801.

(c) Do a wiring check between these pins of connector D203 (VHF-1), D209 (VHF-2), or D549 (VHF-3), at the Aft Electronic Panel, P8, and these pins of connector D199B (VHF-1), D201B (VHF-2), or D539B (VHF-3), at the Shelf, E1-3 (VHF-1), E1-5 (VHF-2), or E3-3 (VHF-3), in the Electronic Equipment (EE) Compartment:

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OMR 701-799, 802 (Continued)

	VHF CNTRL PNL	
	PNL CONNECTOR	**XCVR CONNECTOR**
VHF-1 (P8-2)	**D203**	**D199B**
	pin 21	pin A7
	pin 22	pin B7
VHF-2 (P8-3)	**D209**	**D201B**
	pin 21	pin A7
	pin 22	pin B7
VHF-3 (P8-8)	**D549**	**D539B**
	pin 21	pin A7
	pin 22	pin B7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the airplane wiring.
2) Re-install the VHF Communication Control Panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3). This is the task: VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801.
3) Re-install the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.
4) Do the Repair Confirmation at the end of this task.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

H. Fault Isolation Procedure - Frequency Is Not Shown

(1) Replace the RTP, P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3). These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check for 28 VDC at the RTP:

(a) Remove the RTP, P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3). This is the task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(b) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

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(c) Do a check for 28V DC at pin 40 of connector D203 (RTP-1), D209 (RTP-2), or D549 (RTP-3) to Structure Ground.

1) If there is not 28V DC at pin 40 of connector D203 (RTP-1), D209 (RTP-2), or D549 (RTP-3), then do this check for 28 VDC at the circuit breaker:

2) Do a check for 28V DC from the load terminal of VHF 1 circuit breaker, C165, VHF 2 circuit breaker, C166, or VHF 3 circuit breaker, C471, to Structure Ground.

3) If there is not 28V DC at the load terminal, then do these steps:

a) Replace the applicable circuit breaker, VHF 1, C165, VHF 2, C166, or VHF 3, C471.

b) Re-install the RTP. This is the task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

c) Do the Repair Confirmation at the end of this task.

I. Fault Isolation Procedure - Frequency Is Shown

(1) Replace the RTP, P8-71 (RTP-1) for VHF-1, P8-72 (RTP-2) for VHF-2, or P8-73 (RTP-3) for VHF-3. These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the VHF communications transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

(a) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring:

(a) Remove the applicable VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801.

(b) Remove the applicable radio tuning panels (see table below): P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3). This is the task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(c) Do a wiring check between these pins of the applicable VHF Communication Transceiver in the EE Compartment, and the specified RTPs at the Aft Electronic Panel, P8, (WDM 23-12-11,-21,-31):

VHF COMM TRANSCEIVER	TRANS CONNECTOR		RTP CONNECTOR	RTP
VHF-1 (M149)	D199B	-----	D203	RTP-1 (P8-71)
	pin A11	-----	pin 2	
	pin B11		pin 3	
	D199B	-----	D209	RTP-2 (P8-72)
	pin A7	-----	pin 2	
	pin B7		pin 3	

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(Continued)

VHF COMM TRANSCEIVER	TRANS CONNECTOR		RTP CONNECTOR	RTP
VHF-2 (M150)	D201B pin A7 pin B7	----- -----	D203 pin 2 pin 3	RTP-1 (P8-71)
	D201B pin A11 pin B11	----- -----	D209 pin 2 pin 3	RTP-2 (P8-72)
VHF-3 (M411)	D539B pin A7 pin B7	----- -----	D549 pin 2 pin 3	RTP-3 (P8-73)

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the VHF Communication Transceiver. This is the task: VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.

3) Re-install the RTPs. This is the task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

4) Do the Repair Confirmation at the end of this task.

OMR ALL

J. Repair Confirmation

(1) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the maintenance message or fail light does not show on the transceiver, then you corrected the problem.

(b) If the maintenance message or fail light shows on the transceiver, then continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

813. VHF Transceiver ANTENNA FAIL or RF FAULT Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) ANTENNA FAIL

(2) The VHF Communication Transceiver BITE detects an Radio Frequency (RF) fault.

B. Possible Causes

(1) VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3).

(2) VHF Communication Antenna, M57 (VHF-1), M58 (VHF-2), or M225 (VHF-3).

(3) Coaxial Cable

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

D. Related Data

(1) SSM 23-12-11,-21, -31

(2) WDM 23-12-11,-21,-31

E. Initial Evaluation

(1) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the red ANTENNA FAIL LED is not ON, on the front panel of the transceiver, then there was an intermittent fault.

(b) If the red ANTENNA FAIL LED is ON, on the front panel of the transceiver, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the VHF Communication Antenna, M57 (VHF-1), M58 (VHF-2), or M225 (VHF-3), and the RF coaxial cable:

(a) Remove the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801.

(b) Do a time or frequency domain reflectometry check of the coaxial cable from the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3), to the VHF Communication Antenna, M57 (VHF-1), M58 (VHF-2), or M225 (VHF-3) (WDM 23-12-11,-21,-31), do this task: AMM TASK 20-10-72-210-801 or AMM TASK 20-10-72-200-801.

(c) If you find a problem with the coaxial cable, then do this step:

1) Repair or replace the coaxial cable.

(d) If you find a problem with the VHF Communication Antenna, then do this step:

1) Replace the VHF Communication Antenna, M57 (VHF-1), M58 (VHF-2), or M225 (VHF-3). These are the tasks:

- VHF Communication Antenna - Removal, AMM TASK 23-12-11-000-801
- VHF Communication Antenna - Installation, AMM TASK 23-12-11-400-801

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(e) Re-install the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.

(f) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the red ANTENNA FAIL LED is not ON, on the front panel of the transceiver, then you corrected the problem.

(b) If the red ANTENNA FAIL LED is ON, on the front panel of the transceiver, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 701-799, 802

814. VHF Communication Control Panel Problem - Fault Isolation

A. Description

(1) The VHF Communication Control Panel does not operate correctly.

B. Possible Causes

(1) VHF-1 (2, 3) Communication Control Panel, P8-2 (P8-3, P8-8)

(2) VHF-1 (2, 3) Communication Transceiver, M149 (M150, M411)

(3) Wiring problem.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

D. Related Data

(1) WDM 23-12-11, WDM 23-12-21, WDM 23-12-31

(2) SSM 23-12-11, SSM 23-12-21, SSM 23-12-31

E. Initial Evaluation

(1) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If you find a maintenance messages on the front panel of the Transceiver, refer to the table at the end of the BITE Task to find the applicable Fault Isolation Tasks for the maintenance messages that show.

(b) If you do not find a maintenance message on the front panel of the Transceiver, then continue.

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OMR 701-799, 802 (Continued)

(2) Do this Reception/Transmission Test of the VHF Communication System:

NOTE: VHF communication can be degraded while the airplane is on the ground due to external interference or signal blockage. Before you identify a VHF communication system fault, make sure you try to transmit and receive at several frequencies across the VHF frequency band. Make sure the airplane is not in or near any large metal structures. In some cases, you can move the airplane to correct the problem.

(a) Connect a headset/boom microphone, M428 (captain's), to the boom microphone jack, D6027 (captain's).

(b) Do these steps at the audio control panel, P8-6 (captain's ACP):

1) Push and release the applicable VHF-1, VHF-2, or VHF-3 switch.
   a) Make sure the switch light comes ON.
2) Push and release the applicable VHF-1, VHF-2, or VHF-3 volume control.
   a) Make sure the volume control indicator light comes ON.
3) Set the volume control to the middle position.
4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Do these steps at the VHF communication control panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3):

1) Use the right frequency select knobs to set the right frequency display to an approved test frequency.
2) Use the left frequency select knobs to set the left frequency display to an approved test frequency.
3) Set the TFR switch to the left position.
   a) Make sure the light above the left frequency display comes ON.
   b) Make sure the light above the right frequency display is not ON.
4) Set the TFR switch to the right position.
   a) Make sure the light above the right frequency display comes ON.
   b) Make sure the light above the left frequency display is not ON.

(d) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold a push-to-talk (PTT) switch while you speak.
   a) Make sure you hear the sidetone in the headset while you speak.
2) Release the PTT switch while you listen.
   a) Make sure the quality of the transmitted and received voice is satisfactory.
   b) Make sure the sound of the received voice changes when you turn the applicable VHF-1, VHF-2, or VHF-3 volume control on the ACP, P8-6 (captain's), with no change in voice quality.

(3) If the VHF Communication Control Panel operates correctly, then there was an intermittent fault.

(4) If the VHF Communication Control Panel does not operate correctly, then do the Fault Isolation Procedure below.

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OMR 701-799, 802 (Continued)

F. Fault Isolation Procedure

(1) Replace the applicable VHF-1 (2, 3) Communication Control Panel, P8-2 (P8-3, P8-8). These are the tasks:

- VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801
- VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the VHF-1 (2, 3) Communication Transceiver, M149 (M150, M411). These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

G. Repair Confirmation

(1) Do a Voice Communication Test with a Radio Tower Operator.

(a) If the VHF Communication Control Panel operates correctly, then you corrected the problem.

(b) If the VHF Communication Control Panel does not operate correctly, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

815. Radio Tuning Panel Problem - Fault Isolation

A. Description

(1) The radio tuning panel (RTP) does not operate correctly.

B. Possible Causes

(1) Radio tuning panel (RTP), P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3).

(2) Wiring problem.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

D. Related Data

(1) (SSM 23-12-11,-21, -31)

(2) (SSM 23-11-11,-21)

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(3) (WDM 23-12-11,-21,-31)

(4) (WDM 23-11-11, -21)

E. Initial Evaluation

(1) AIRPLANES WITH GABLES P/N G7404-XX RADIO TUNING PANELS;

Do these steps for a frequency lock-up problem:

(a) Push the VHF-1 switch on all the RTPs.

(b) Make sure that you can change the ACTIVE frequency for VHF-1 at each RTP.

1) Make sure that the ACTIVE frequencies at the other RTP(s) also change to the new frequency.

(c) If you can change the ACTIVE frequency at one RTP, but the other RTP(s) will not accept a frequency change, then do these steps:

1) Push the OFF switch on the RTP that operates satisfactorily.

a) Make sure the RTP is in the off condition.

2) Set the other RTP(s) to new ACTIVE frequencies to see if they will now accept a frequency change.

3) If the RTP(s) with the frequency lock-up problem now operate satisfactorily, then do the Fault Isolation Procedure - RTP Frequency Lock-up Problem below.

(2) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If there is a maintenance message shown on the front panel of the transceiver, then do the fault isolation procedure for the maintenance message.

(3) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

(a) If there is a maintenance message shown on the front panel of the transceiver, then do the fault isolation procedure for the maintenance message.



WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.



WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(4) Do a VHF communication test (AMM TASK 23-12-00-730-801) and/or an HF communication test (AMM TASK 23-11-00-730-801) to confirm the problem.

(a) If the radio tuning panel (RTP) operates satisfactorily, then there was an intermittent fault.

(b) If the radio tuning panel (RTP) does not operate satisfactorily, then do the Fault Isolation Procedure - RTP Problem below.

F. Fault Isolation Procedure - RTP Problem

(1) Replace the radio tuning panel (RTP), P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3).

These are the tasks:

Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801,

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

(a) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(b) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(c) Do a VHF communication test and/or an HF communication test with a radio tower operator.

1) Make sure the quality of the transmitted and received voice is satisfactory.

(d) If the radio tuning panel (RTP) operates satisfactorily, then you corrected the fault.

(e) If the radio tuning panel (RTP) does not operate satisfactorily, then continue.

(2) Do this check of the wiring:

(a) Remove the applicable radio tuning panel (RTP) (see table below). To remove it, do this task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(b) Remove the applicable VHF or HF communication transceiver (see table below).

1) To remove the VHF communication transceiver, do this task: VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801.

2) To remove the HF transceiver, do this task: HF Transceiver - Removal, AMM TASK 23-11-21-000-801.

(c) Do a wiring check between these pins of the radio tuning panel (RTP) at the aft electronic panel, P8, and the applicable transceiver in the electronic equipment compartment (WDM 23-12-11,-21,-31) (WDM 23-11-11, -21):

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

Table 201/23-12-00-993-802

RADIO TUNING PANEL (RTP)	RTP CONNECTOR		XCVR CONNECTOR	TRANSCEIVER
RTP-1 (P8-71)	D203		D199B	VHF-1 (M149)
	pin 2	-----	pin A11	
	pin 3	-----	pin B11	
	pin 8	-----	pin A7	
	pin 9	-----	pin B7	
	D203		D345B	HF-1 (M226) (if installed)
	pin 2	-----	pin E3	
	pin 3	-----	pin F3	
	pin 8	-----	pin G3	
	pin 9	-----	pin H3	
	D203		D201B	VHF-2 (M150)
	pin 2	-----	pin A7	
	pin 3	-----	pin B7	
	pin 8	-----	pin A11	
	pin 9	-----	pin B11	
	D203		D623B	HF-2 (M439) (if installed)
	pin 2	-----	pin G3	
	pin 3	-----	pin H3	
	pin 8	-----	pin E3	
	pin 9	-----	pin F3	
	D203		D539B	VHF-3 (M411) (if installed)
	pin 5	-----	pin A7	
	pin 6	-----	pin B7	

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

Table 201/23-12-00-993-802 (Continued)

RADIO TUNING PANEL (RTP)	RTP CONNECTOR		XCVR CONNECTOR	TRANSCEIVER
RTP-2 (P8-72)	D209		D201B	VHF-2 (M150)
	pin 2	-----	pin A11	
	pin 3	-----	pin B11	
	pin 5	-----	pin A7	
	pin 6	-----	pin B7	
	D209		D623B	HF-2 (M439) (if installed)
	pin 2	-----	pin E3	
	pin 3	-----	pin F3	
	pin 5	-----	pin G3	
	pin 6	-----	pin H3	
	D209		D199B	VHF-1 (M149)
	pin 2	-----	pin A7	
	pin 3	-----	pin B7	
	pin 5	-----	pin A11	
	pin 6	-----	pin B11	
	D209		D345B	HF-1 (M226) (if installed)
	pin 2	-----	pin G3	
	pin 3	-----	pin H3	
	pin 5	-----	pin E3	
	pin 6	-----	pin F3	
	D209		D539B	VHF-3 (M411) (if installed)
	pin 8	-----	pin A7	
	pin 9	-----	pin B7	

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

Table 201/23-12-00-993-802 (Continued)

RADIO TUNING PANEL (RTP)	RTP CONNECTOR		XCVR CONNECTOR	TRANSCEIVER
RTP-3 (P8-73) (if installed)	D549		D539B	VHF-3 (M411) (if installed)
	pin 2	-----	pin A7	
	pin 3	-----	pin B7	
	D549		D623B	HF-2 (M439) (if installed)
	pin 5	-----	pin E3	
	pin 6	-----	pin F3	
	pin 8	-----	pin G3	
	pin 9	-----	pin H3	
	D549		D199B	VHF-1 (M149)
	pin 5	-----	pin A7	
	pin 6	-----	pin B7	
	pin 8	-----	pin A11	
	pin 9	-----	pin B11	
	D549		D345B	HF-1 (M226) (if installed)
	pin 5	-----	pin G3	
	pin 6	-----	pin H3	
	pin 8	-----	pin E3	
	pin 9	-----	pin F3	
	D549		D201B	VHF-2 (M150)
	pin 5	-----	pin A11	
	pin 6	-----	pin B11	
	pin 8	-----	pin A7	
	pin 9	-----	pin B7	

(d) Repair any airplane wiring problems you find.

(e) Re-install the applicable radio tuning panel (RTP). To install it, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

(f) Re-install the applicable VHF or HF communication transceiver:

1) To install the VHF communication transceiver, do this task: VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.

2) To install the HF transceiver, do this task: HF Transceiver - Installation, AMM TASK 23-11-21-400-801.

(g) Do this task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(h) Do this task: HF Communication System - BITE Procedure, 23-11 TASK 801.

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WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(i) Do a VHF communication test and/or an HF communication test with a radio tower operator.

1) Make sure the quality of the transmitted and received voice is satisfactory.

(j) If the radio tuning panel (RTP) operates satsifactorily, then you corrected the fault.

G. Fault Isolation Procedure - RTP Frequency Lock-up Problem

(1) Replace the radio tuning panel, P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3), that you set to off in the Initial Evaluation.

NOTE: If an RTP has a cross-tuning failure, it may tune frequencies satisfactorily while it prevents other RTPs from tuning frequencies satisfactorily.

(a) Set each RTP to a new ACTIVE frequency to make sure that each RTP will accept a frequency change.

1) If the RTPs operate satisfactorily, then you corrected the fault.

2) If the RTPs do not operate satisfactorily, then continue the procedure.

(2) Do this check of the wiring:

(a) Remove the RTPs. To remove the RTPs, do this task: Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801.

(b) Do a wiring check between these pins of RTP connectors D203, D209 and D549 at P8 (WDM 23-12-11,-21,-31 OR -41,):

D203 (RTP-1)	D209 (RTP-2)
pin 2	pin 5
pin 3	pin 6
pin 5	pin 8
pin 6	pin 9
pin 8	pin 2
pin 9	pin 3

D203 (RTP-1)	D549 (RTP-3)
pin 2	pin 8
pin 3	pin 9
pin 5	pin 2
pin 6	pin 3
pin 8	pin 5
pin 9	pin 6

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

D209 (RTP-2)	D549 (RTP-3)
pin 2	pin 5
pin 3	pin 6
pin 5	pin 8
pin 6	pin 9
pin 8	pin 2
pin 9	pin 3

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

(d) Install the RTPs. To install the RTPs, do this task: Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801.

(e) Set each RTP to a new ACTIVE frequency to make sure that each RTP will accept a frequency change.

1) If the RTP operates satisfactorily, then you corrected the fault.

OMR ALL

——— END OF TASK ———

816. VHF Communication System - Receive/Transmit Problem

A. Description

(1) The VHF-1, VHF-2, or VHF-3 communication system does not operate correctly.

(2) Poor reception and/or poor transmission at the audio control panel, P8-6 (captain), P8-7 (first officer) or P5-15 (first observer) on the VHF-1, VHF-2, or VHF-3 communication system.

B. Possible Causes

(1) VHF communication transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3)

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(2) VHF communication control panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3)

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(3) Radio tuning panels (RTP): P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3)

OMR ALL

(4) Audio control panel, P8-6 (captain), P8-7 (first officer) or P5-15 (first observer)

(5) A push-to-talk (PTT) switch is stuck in transmit

(6) Remote electronics unit, M1353

(7) VHF communication antenna, M57 (VHF-1), M58 (VHF-2), or M225 (VHF-3)

(8) RF coaxial cable problem

(9) Wiring

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-12-11,-21, -31

(2) SSM 23-51-11,-21,-31

(3) WDM 23-12-11,-21,-31

(4) WDM 23-51-11,-21,-31

E. Initial Evaluation

(1) Make sure that none of the push-to-talk (PTT) switches are stuck in transmit (AMM TASK 23-51-00-710-801).

NOTE: A stuck PTT can be a cause of a temporary VHF transmit failure.

(a) Do these steps to do a fast check for a PTT that is stuck in transmit:

1) Set all the audio control panels to the Flight Interphone (or Interphone) transmit position.

2) Without pushing a PTT switch, speak into the microphones at the different crew stations.

3) If the station transmits, then it has a stuck PTT switch.

(2) Do a BITE test of the VHF communication transceivers. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the BITE test finds a fault, then do the fault isolation procedure for the maintenance message.

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(3) Do this reception/transmission test of the VHF communication system:

NOTE: VHF communications can be degraded while the airplane is on the ground due to external interference or signal blockage. Before you identify a VHF communication system fault, make sure that you try to transmit and receive at several frequencies across the VHF frequency band. Make sure that the airplane is not in or near any large metal structures. In some cases, you can move the airplane to correct the problem.

(a) Use a boom microphone/headphone to do the test.

(b) At the audio control panel, P8-6 (captain), P8-7 (first officer) or P5-15 (first observer), do these steps:

1) Push and release the applicable microphone selector switch (VHF-1, VHF-2, or VHF-3).

a) Make sure that the applicable switch light comes on.

2) Adjust the applicable volume control switch to the middle position.

3) Make sure that the BOOM/MASKS switch is set to the BOOM position.

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(c) At the VHF communication control panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3), do these steps:

1) Use the right frequency select knobs to set the right frequency display to an approved test frequency.

2) Use the left frequency select knobs to set the left frequency display to an approved test frequency.

3) Set the TFR FREQ SEL switch to the left position.

a) Make sure that the active frequency light is on above the left frequency indicator.

b) Make sure that the active frequency light above the right frequency display is not on.

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(d) Set the radio tuning panel for the applicable VHF communication system.

1) Set the STANDBY frequency to an approved test frequency.

2) Push the transfer switch.

3) Make sure that the STANDBY and ACTIVE frequencies are exchanged.

OMR ALL

(e) Do these steps to do a voice communication test with a radio operator:

1) Push and hold a push-to-talk (PTT) switch while you speak into the microphone.

a) Make sure that you hear the sidetone in the headset while you speak.

2) Release the PTT switch while you listen.

a) Make sure that the quality of the transmitted and received voice is satisfactory.

b) Make sure that the volume of the received voice changes when you turn the volume control on the audio control panel, P8-6 (captain), P8-7 (first officer) or P5-15 (first observer) with no change in voice quality.

(4) If the VHF communication system operates satisfactorily, then there was an intermittent fault.

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(5) If the VHF communication system does not operate satisfactorily, then do a reception/transmission test of the VHF communication system at each audio control panel.

(a) If the problem occurs at all audio control panels, then do Problem at All Audio Control Panels - Fault Isolation Procedure below.

(b) If the problem occurs at only one audio control panel, then do Problem at Only One Audio Control Panel - Fault Isolation Procedure below.

F. Problem at All Audio Control Panels - Fault Isolation Procedure

(1) Do this exchange check of the VHF communication transceiver, M149 (VHF-1), M150 (VHF-2) or M411 (VHF-3):

(a) Put a tag that reads SUSPECT on the VHF communication transceiver with poor reception or transmission.

(b) Put a tag that reads OK on the other VHF communication transceiver.

(c) Exchange the locations of the two VHF communication transceivers. These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

(d) Do a BITE test of each VHF communication transceiver. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(e) Speak with the radio operator on each VHF communication system.

1) Make sure that the transmitted and received voice signals are satisfactory.

(f) If the poor reception or transmission moves with the VHF communication transceiver tagged SUSPECT, then do these steps:

1) Replace the VHF communication transceiver tagged SUSPECT. These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

NOTE: If it is your airlines' policy, you must install the VHF communication transceiver with the OK tag in its initial location.

2) Do a BITE test of the applicable VHF communication transceiver. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

3) Speak with a radio operator on each VHF communication system.

a) Make sure that the transmitted and received voice is satisfactory.

4) If the VHF communication system operates correctly, then you corrected the problem.

a) Remove the tags from the VHF communication transceivers.

(g) If the poor reception or transmission goes away after you exchange the VHF communication transceivers, then do these steps to complete the task:

NOTE: There was an intermittent fault in either the equipment rack connector, or in the VHF transceiver with the SUSPECT tag.

1) If it is your airlines' policy, you must install the VHF communication transceivers in their initial locations. These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

2) Remove the tags from the VHF communication transceivers.

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(h) If the poor reception or transmission stays with the applicable communication system after you exchanged the VHF communication transceivers, then continue:

1) If it is your airlines' policy, you must install the VHF communication transceivers in their initial locations. These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

2) Remove the tags from the VHF communication transceivers.

OMR 701-799, 802

(2) Do this exchange check of the VHF communication control panel, P8-2 (VHF-1), P8-3 (VHF-2), or P8-8 (VHF-3):

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Do this exchange check of the radio tuning panels, P8-71 (RTP-1), P8-72 (RTP-2), or P8-73 (RTP-3):

OMR ALL

(4) Do the following steps:

(a) Put a tag that reads SUSPECT on the panel that has the reception or transmission problem.

(b) Put a tag that reads OK on the other panel.

OMR 701-799, 802

(c) Exchange the locations of the VHF communication control panels. These are the tasks:

- VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801
- VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Exchange the locations of the radio tuning panels. These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

OMR ALL

(e) Do a BITE test of the VHF communication transceivers. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(f) Do a voice communication test with a radio operator on each VHF communication system.

1) Make sure that the transmitted and received voice signals are satisfactory.

(g) If the fault moves with the panel tagged SUSPECT, then do these steps:

OMR 701-799, 802

1) Replace the VHF communication control panel tagged SUSPECT. These are the tasks:

- VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801
- VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801

NOTE: If it is your airlines' policy, you must install the VHF communication control panel with the OK tag in its initial location.

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) Replace the radio tuning panel tagged SUSPECT. These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

NOTE: If it is your airlines' policy, you must install the radio tuning panel with the OK tag in its initial location.

OMR ALL

3) Do a BITE test of the applicable VHF communication transceiver. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

4) Do a voice communication test on each VHF communication system.

a) Make sure that the transmitted and received voice signals are satisfactory.

5) If the VHF communication system operates correctly, then you corrected the problem.

a) Remove the tags from the panels.

(h) If the fault goes away after you exchange the panels, then do these steps to complete the task:

NOTE: There was an intermittent fault in the panel connector, or in the panel tagged SUSPECT.

OMR 701-799, 802

1) If it is your airlines' policy, you must install the VHF communication control panels in their initial locations. These are the tasks:

- VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801
- VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) If it is your airlines' policy, you must install the radio tuning panels in their initial locations. These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

OMR ALL

3) Remove the tags from the panels.

(i) If the poor transmission or reception stays with the communication system after you exchange the panels, then continue the procedure.

OMR 701-799, 802

1) If it is your airlines' policy, you must install the VHF communication control panels in their initial locations. These are the tasks:

- VHF Communication Control Panel - Removal, AMM TASK 23-12-31-000-801
- VHF Communication Control Panel - Installation, AMM TASK 23-12-31-400-801

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) If it is your airlines' policy, you must install the radio tuning panels in their initial locations. These are the tasks:

- Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801
- Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

OMR ALL

3) Remove the tags from the panels.

(5) Replace the remote electronics unit, M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) Do this check of the VHF communication antenna, VHF-1, VHF-2 or VHF-3 and the RF coaxial cable:

(a) Remove the VHF communication transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). This is the task: VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801.

(b) Do a check of the coaxial cable from the VHF communication transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3) to the VHF communication antenna, VHF-1, VHF-2, or VHF-3 (WDM 23-12-11,-21,-31). Do this task: AMM TASK 20-10-72-210-801 or AMM TASK 20-10-72-200-801.

(c) If you find a problem with the coaxial cable, then do this step:

1) Repair or replace the coaxial cable.

(d) If you find a problem with the VHF communication antenna, then do this step:

1) Replace the VHF communication antenna, VHF-1, VHF-2 or VHF-3. These are the tasks:

- VHF Communication Antenna - Removal, AMM TASK 23-12-11-000-801
- VHF Communication Antenna - Installation, AMM TASK 23-12-11-400-801

(e) Re-install the VHF communication transceiver. This is the task: VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801.

(f) Do a BITE test of the VHF communication transceiver. This is the task: VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(g) Do the Repair Confirmation at the end of this task.

G. Problem at Only One Audio Control Panel - Fault Isolation Procedure

(1) Replace the applicable audio control panel, P8-6 (captain), P8-7 (first officer) or P5-15 (first observer). These are the tasks:

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the remote electronics unit, M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do the Repair Confirmation at the end of this task.

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H. Repair Confirmation

(1) Do a radio check with the radio operator on the applicable VHF communication system.

(a) Make sure that the transmitted and received voice is satisfactory.

(2) If the VHF communication system operates correctly, then you corrected the problem.

(3) If the VHF communication system does not operate correctly, then continue the Fault Isolation procedure at the subsequent step.

END OF TASK

818. VHF Transceiver Internal Fault Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) LRU STATUS

(2) The VHF Communication Transceiver has an internal fault.

B. Possible Causes

(1) VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

D. Related Data

(1) SSM 23-12-11,-21, -31

E. Initial Evaluation

(1) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

(a) If the red LRU STATUS LED is not ON, on the front panel of the transceiver, then there was an intermittent fault.

(b) If the red LRU STATUS LED is ON, on the front panel of the transceiver, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the VHF Communication Transceiver, M149 (VHF-1), M150 (VHF-2), or M411 (VHF-3). These are the tasks:

- VHF Communication Transceiver - Removal, AMM TASK 23-12-21-020-801
- VHF Communication Transceiver - Installation, AMM TASK 23-12-21-420-801

(a) Do the VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801.

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1) If the red LRU STATUS LED is not ON, on the front panel of the transceiver, then you corrected the problem.

——— END OF TASK ———

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801. Emergency Locator Transmitter (ELT) BITE Procedure

A. General

(1) You do the Emergency Locator Transmitter (ELT) BITE Procedure from the front panel of the ELT. The ELT is above an access panel in the AFT Passenger Cabin Ceiling.

(2) The ELT BITE Test does an operational check of the Emergency Locator Transmitter System.

(a) The Light Emitting Diode (LED) on the ELT front panel will come ON when the ELT detects a failure in the ELT 406 MHz Transmitter, the Antenna, the Battery, or the G-Switch Loop. The LED can also come ON to indicate that the interface to the Navigation System is not connected.

(3) Make sure that the airplane has electrical power.

(a) If it is necessary, do this task: Supply Electrical Power, AMM TASK 24-22-00-860-811.

(4) Make sure that you do the ELT Test in the first five minutes of the hour (UTC).

NOTE: The ELT will transmit the 121.5/243.0 emergency signals during the ELT Test.

(a) If you do the ELT Test outside of the first five minutes of the hour (UTC), then do this:

1) Speak to the applicable emergency authority (Air Traffic Control Tower or Flight Service Station) to tell them that there will be an ELT Test Transmission on the Emergency Frequency.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Open the access panel to get access to the ELT Unit if applicable (Lowered Ceiling Removal, AMM TASK 25-21-71-000-801 or Lowered Ceiling Removal, AMM TASK 25-21-71-000-815).

OMR ALL

B. BITE Procedure

(1) Do the BITE Procedure for the ELT as follows (Figure 201):

(a) Make sure that the ELT ARM/ON Switch on the ELT Remote Control Panel is in the ARM position and its guard is CLOSED.

(b) Set the ELT ON/OFF Switch on the front panel of the ELT Unit to the ON position for one second, then move it back to the OFF position.

(c) Make sure that the LED on the ELT Unit comes ON momentarily and then goes OFF after the switch is moved to the OFF position.

1) If the LED flashes after it goes OFF, then count the number of flashes.

NOTE: The number of flashes is the maintenance message for the fault.

a) If the LED flashes one time, three times, four times or seven times, there was a self-test failure.

NOTE: If the LED flashes five times, this is not a failure. It is an indication that the interface to the Navigation System is not connected and the Navigation Interface Unit (NIU)/PPIU is not installed. Refer to WDM 23-24-11 to determine the airplane configuration.

b) Refer to the table at the end of this task to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show.

(2) Speak to the applicable emergency authority to tell them that the ELT Test is completed if applicable.

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OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) CLOSE the access panel for the ELT Unit if applicable (Lowered Ceiling Installation, AMM TASK 25-21-71-400-801 or Lowered Ceiling Installation, AMM TASK 25-21-71-400-815).

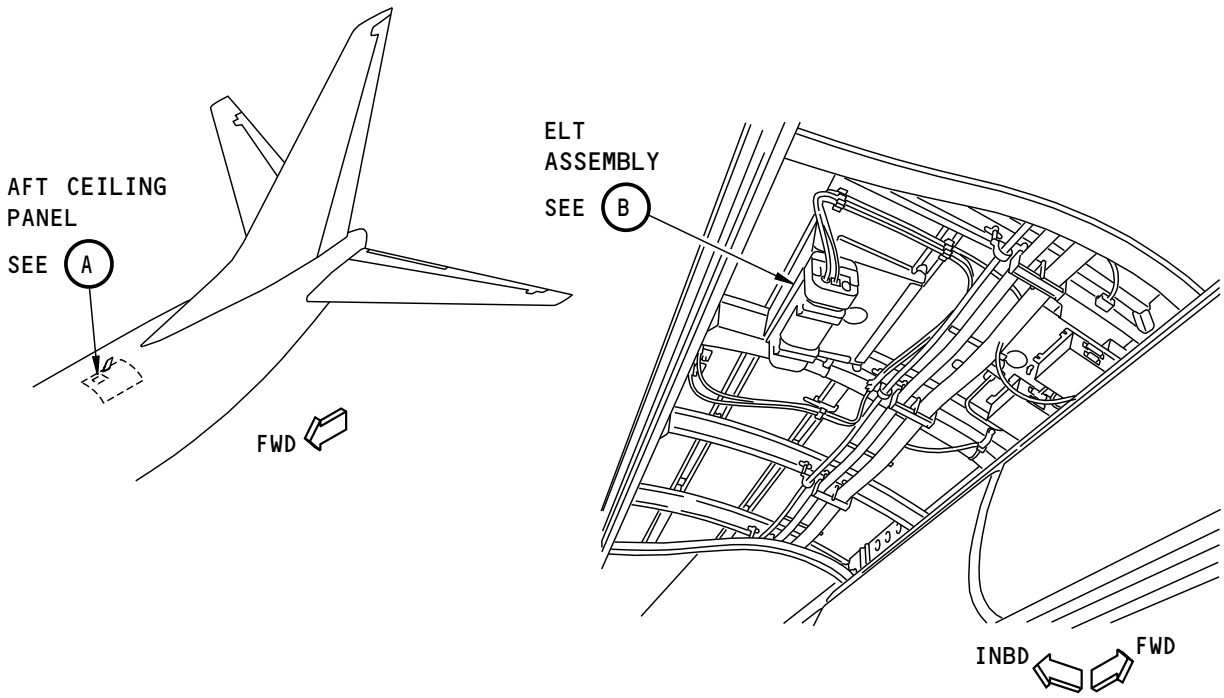
OMR ALL

(4) If the electrical power is not necessary, do this task: Remove Electrical Power, AMM TASK 24-22-00-860-812.

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ELT	LED flashes one time	23-24 TASK 802
ELT	LED flashes seven times	23-24 TASK 804
ELT	LED flashes three or four times	23-24 TASK 803

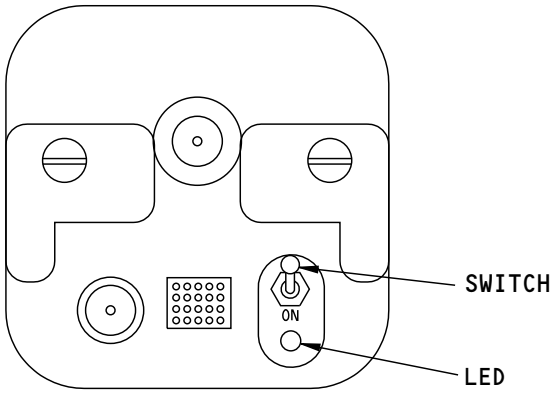
——— END OF TASK ———

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AFT CEILING PANEL
(REMOVED)

A



EMERGENCY LOCATOR
TRANSMITTER (ELT)

B

H34067 S0000146958_V1

Emergency Locator Transmitter (ELT)
Figure 201/23-24-00-990-803

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802. ELT G-Switch Loop Problem - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) LED flashes one time.

(2) The ELT BITE test does an operational check of the emergency locator transmitter system. The LED on the ELT front panel will flash one time when the ELT detects a failure in the G-switch loop.

B. Possible Cause

(1) G-switch loop

C. Related Data

(1) (WDM 23-24-11)

(2) (SSM 23-24-11)

D. Initial Evaluation

(1) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.

(a) If there is one flash, then do the Fault Isolation Procedure below.

(b) If there is no flash, then there was an intermittent fault.

E. Fault Isolation Procedure

(1) Do this check of the G-switch loop on the emergency locator transmitter (ELT), M1523:

(a) Remove the bottom cover of the ELT (AMM TASK 23-24-00-000-801-001).

NOTE: The ELT is above an access panel in the aft passenger cabin ceiling.

(b) Make sure the G-switch loop that connects between pin 5 and pin 8 is installed correctly.

1) If the G-switch loop is missing or not installed correctly, then install the G-switch loop.

(c) Re-install the bottom cover of the ELT.

(d) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.

(e) If there is no flash, then you corrected the fault.

——— END OF TASK ———

803. ELT 406 MHz Transmitter or Bad Antenna Problem - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) LED flashes three or four times.

1) ELT Model B406-1: LED flashes four times

2) ELT Model B406-4: LED flashes three times

(2) The ELT BITE test does an operational check of the emergency locator transmitter system. The LED on the ELT front panel will flash three or four times when the ELT detects a failure in the ELT 406 MHz transmitter or the antenna.

B. Related Data

(1) (WDM 23-24-11)

(2) (SSM 23-24-11)

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C. Possible Causes

(1) ELT

(2) ELT antenna

(3) Wiring

D. Equipment

(1) PE6029-50 or PE6066-50 or equivalent (50 ohm resistor termination - TNC female, 0.5 watt minimum power dissipation), Pasternack Enterprises, LLC, P.O. Box 16759, Irvine, CA, 92623-6759

E. Initial Evaluation

(1) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.

(a) If there are three or four flashes, then do the Fault Isolation Procedure below.

(b) If there are no flashes, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this check of the emergency locator transmitter (ELT):

(a) Remove the bottom cover of the ELT (AMM TASK 23-24-00-000-801-001).

NOTE: The ELT is above an access panel in the aft passenger cabin ceiling.

(b) Disconnect coaxial cable connector D10585 from the antenna.

NOTE: The ELT antenna mounted on the top of the fuselage near the transmitter (STA 796.5, LBL 4.5).

(c) Connect 50 ohms resistor termination to the coaxial cable connector D10585.

NOTE: Terminator must be 50 ohm resistance, TNC female and minimum 0.5 watt power dissipation.

(d) Turn the ELT to the ON position for about one second, and then back to the OFF position.

1) If the flashes come on again, then do this step:

a) Replace the ELT.

These are the tasks:

Emergency Locator Transmitter Removal, AMM TASK 23-24-00-000-801-001,

Emergency Locator Transmitter Installation, AMM TASK 23-24-00-400-801-001.

b) Remove the 50 ohms resistor termination from connector D10585.

c) Connect coaxial cable connector D10585 to the antenna.

d) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.

e) If there are no flashes, then you corrected the fault.

2) If there are no flashes, then remove the 50 ohms resistor termination from coaxial cable connector D10585 and continue.

(2) Do this check of the wiring:

(a) Disconnect coaxial cable connector D10583 from the ELT.

(b) Do a check of the coaxial cable from connector D10583 to connector D10585. To do the check, do this task: Coaxial Cable Inspection, AMM TASK 20-10-72-210-801.

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D10583		D10585
pin A1	. . . . .	pin A1

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connector D10583.
3) Re-connect the connector D10585.
4) Install the bottom cover of the ELT.
5) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.
6) If there are no flashes, then you corrected the fault.

(d) If you do not find a problem with the wiring, then do these steps and continue:

1) Re-connect the connector D10583.
2) Re-connect the connector D10585.
3) Install the bottom cover of the ELT.
4) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.
5) If there are no flashes, then you corrected the fault.
6) If the flashes come on again, then continue.

(3) Replace the ELT antenna.

These are the tasks:

ELT Antenna - Removal, AMM TASK 23-24-01-000-802,

ELT Antenna - Installation, AMM TASK 23-24-01-400-802.

(a) If there are no flashes, then you corrected the fault.

——— END OF TASK ———

804. ELT Battery Problem - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) LED flashes seven times.

(2) The ELT BITE test does an operational check of the emergency locator transmitter system. The LED on the ELT front panel will flash seven times when the ELT detects a failure in the battery.

B. Related Data

(1) (WDM 23-24-11)

(2) (SSM 23-24-11)

C. Possible Cause

(1) ELT battery

D. Initial Evaluation

(1) Do this task: ELT System - System Test, AMM TASK 23-24-00-730-804-001.

(a) If there are seven flashes, then do the Fault Isolation Procedure below.

(b) If there are no flashes, then there was an intermittent fault.

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E. Fault Isolation Procedure

(1) Replace the ELT battery assembly. To replace it, do this task: Emergency Locator Transmitter Battery Replacement, AMM TASK 23-24-00-900-801-001.

(a) If there are no flashes, then you corrected the fault.

——— END OF TASK ———

807. ELT On - Fault Isolation

A. Initial Evaluation

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) The emergency locator transmitter (ELT) is on. Set the ELT ARM/ON switch on the overhead panel, P5, to the ON position for approximately one second and move it back to the ARM position (ELT System - Operational Test, AMM TASK 23-24-00-710-801-001).

(a) If the ELT annunciator goes OFF, then no maintenance action is necessary.

(b) If the ELT annunciator stays ON, then do the Fault Isolation Procedure below.

OMR ALL

B. Fault Isolation Procedure

(1) Disconnect connector D3003 from the ELT, M1523.

(a) If the ELT annunciator goes OFF, then do the steps that follow:

1) Replace the ELT, M1523.

These are the tasks:

Emergency Locator Transmitter Removal, AMM TASK 23-24-00-000-801-001

Emergency Locator Transmitter Installation, AMM TASK 23-24-00-400-801-001

a) If the ELT annunciator goes OFF, then you corrected the fault.

b) If the ELT annunciator stays ON, then continue the procedure.

(b) If the ELT annunciator stays ON, then do these steps:

1) Connect the connector D3003 back to the ELT, M1523.

2) Continue the procedure.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Do a wiring check of the ELT ARM/ON switch on the ELT control panel, P5–80 (P5) (WDM 23-24-11).

(a) If you find a problem with the switch, then replace the ELT control panel.

These are the tasks:

Emergency Locator Transmitter (ELT) Control Panel Removal, AMM TASK 23-24-02-000-801

Emergency Locator Transmitter (ELT) Control Panel Installation, AMM TASK 23-24-02-400-801

1) If the ELT annunciator goes OFF, then you corrected the fault.

2) If the ELT annunciator stays ON, then continue the procedure.

(b) If you do not find a problem with the switch, then continue the procedure.

(3) Do a wiring check between the ELT ARM/ON switch on the ELT control panel, P5–80 (P5) and the ELT, M1523 (WDM 23-24-11).

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(a) Repair the problems that you find.

(b) If the ELT annunciator goes OFF, then you corrected the fault.

——— END OF TASK ———

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801. ACARS System Problem - Fault Isolation

A. Description

(1) The Aircraft Communications Addressing And Reporting System (ACARS) has one of these problems:

(a) The ACARS communications management unit cannot communicate with an ACARS ground station (NO COMM shows in the CDU scratchpad display).

(b) The ACARS menu selection prompt (<ACARS) is missing from line 2 on one of the CDU Main Menu displays.

(c) The ACARS menu selection prompt (<ACARS) is missing from line 2 on both of the CDU Main Menu displays.

(d) The ACARS does not respond to CDU key pushes.

(e) The ground station reports that downlink messages from the ACARS system are incomplete.

(f) The ACARS communications management unit cannot print data on the multi-input printer.

(g) The flight compartment chime does not sound for an ACARS uplink message.

OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(h) The VHF-C lamp on the SELCAL control panel does not come on for an ACARS uplink message.

(i) Pushing the VHF-C switchlight on the SELCAL control panel does not acknowledge the ACARS uplink message (VHF-C CALL lamp does not go out).

OMR 815, 816, 818-826, 901-999

(j) The VHF-C lamp on the audio control panels does not come on for an ACARS uplink message.

(k) Pushing the VHF-C switchlight on the audio control panel does not acknowledge the ACARS uplink message (VHF-C CALL lamp does not go out).

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

B. Corrective Action

(1) If the CMU cannot communicate with an ACARS ground station, do this task: ACARS Communication Problem - Fault Isolation, 23-27 TASK 814

(2) If the <ACARS prompt is missing from one, or both, of the CDU Main Menu Displays, do this task: ACARS/CDU Interface Problem - Fault Isolation, 23-27 TASK 815

(3) If the ACARS does not respond to CDU key pushes, do this task: ACARS/CDU Interface Problem - Fault Isolation, 23-27 TASK 815

(4) If the ground station reports that downlink messages from the ACARS system are incomplete, do this task: ACARS Input Problem - Fault Isolation, 23-27 TASK 816

(5) If the ACARS communications management unit cannot print on the multi-input printer, do this task: ACARS Print Problem - Fault Isolation, 23-27 TASK 812

(6) If the flight compartment chime does not sound for an ACARS uplink message, do this task: ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 818 or ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 820

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OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(7) If the VHF-C lamp on the SELCAL control panel does not come on for an ACARS uplink message, do this task: ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 818

(8) If pushing the VHF-C switchlight on the SELCAL control panel does not acknowledge the ACARS uplink message (VHF-C CALL lamp does not go out), do this task: ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 818

OMR 815, 816, 818-826, 901-999

(9) If the VHF-C lamp on the audio control panels does not come on for an ACARS uplink message, do this task: ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 820

(10) If pushing the VHF-C switchlight on the audio control panel does not acknowledge the ACARS uplink message (VHF-C CALL lamp does not go out), do this task: ACARS Call Annunciation Problem - Fault Isolation, 23-27 TASK 820

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

C. Fault Isolation Procedure - ACARS Management Unit Problem

(1) Replace the ACARS communications management unit (CMU) 1, M2127. These are the tasks:

- ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801
- ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

(a) If the installation test for the ACARS CMU is satisfactory, then you corrected the problem.

——— END OF TASK ———

812. ACARS Print Problem - Fault Isolation

A. Description

(1) The ACARSCommunications Management Unit (CMU) cannot print data on the Multi-input Printer.

B. Possible Causes

(1) ACARS Communications Management Unit (CMU-1), M2127

(2) Multi-input Printer, M2195

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	5	C01216	PRINTER

OMR 802 POST SB 737-23-1334

E	7	C00744	ACARS MU AC

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

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These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

D. Related Data

(1) WDM 23-27-32

(2) WDM 31-33-01

(3) SSM 23-27-32

(4) SSM 31-33-01

E. Initial Evaluation

(1) Do a check of the Multi-input Printer as follows:

(a) Open these circuit breakers, wait at least 10 seconds, then close the circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	5	C01216	PRINTER
OMR 802 POST SB 737-23-1334			
E	7	C00744	ACARS MU AC
OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334			
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(b) On the CDU-1, push the MENU Key.

(c) Go to the ACARS-PART NUMBER Page.

1) Refer to ACARS - System Test, AMM TASK 23-27-00-730-806-010 for detailed instructions.

(d) Push the Line Select Key (LSK) adjacent to the *PRINT prompt.

1) If the ACARS-PART NUMBER Page prints, then there was an intermittent fault.

2) If the page does not print, push and hold the PRINTER TEST push-button for 1 second, then release the push-button.

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a) The printer does a self-test and prints a pattern on the printer paper.

b) The FAIL or FAULT Light stays OFF.

<1> If the Printer Self-Test result is not satisfactory, replace the printer. These are the tasks:

- Printer Removal, AMM TASK 31-33-01-000-801
- Printer Installation, AMM TASK 31-33-01-400-801

<2> If the Printer Self-Test result is satisfactory, then do the Fault Isolation Procedure below.

3) If the page does not print, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure - MU or NON Honeywell CMU

(1) On the Control Display Unit (CDU), access the Digital Flight Data Acquisition Unit (DFDAU) Menu and print a known printable Screen Page.

(a) If the page from the DFDAU printed, examine the wiring as follows:

1) Examine the ARINC 429 Bus wiring from pins J12 and K12 of CMU connector D10727B to the common point TB3201 (WDM 31-33-01).

2) If you find a problem with the wiring, then repair the wiring.

a) Do the Repair Confirmation below.

(b) If the page from the DFDAU does not print, examine the wiring as follows:

1) Examine the ARINC 429 Bus wiring from pins E and F, of printer connector D2649 to the common point TB3201 (WDM 31-33-01).

2) If you find a problem with the wiring, then repair the wiring.

a) Do the Repair Confirmation below.

(2) Replace the Printer. These are the tasks:

- Printer Removal, AMM TASK 31-33-01-000-801
- Printer Installation, AMM TASK 31-33-01-400-801

(a) Do the Repair Confirmation below.

G. Repair Confirmation

(1) On CDU-1, push the MENU key.

(2) Go to the ACARS-PART NUMBER Page.

(a) Refer to ACARS - System Test, AMM TASK 23-27-00-730-806-010 for detailed instructions.

(3) Push the LSK adjacent to the PRINT* prompt.

(a) If the ACARS-PART NUMBER Page prints on the printer, then you have corrected the problem.

(b) If the ACARS-PART NUMBER Page does not print, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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814. ACARS Communication Problem - Fault Isolation

A. General

(1) There are several conditions that may result in a NO COMM condition. Only a few of these conditions are the result of an airplane system failure.

(2) This may include a VHF failure, a wiring problem, or an incorrect configuration setting.

(3) At times, a NO COMM condition is normal:

(a) When the airplane is out of range of ground stations

(b) RF shadowing (VHF)

(c) VHF mode (POA/AOA) transitions

(4) Under these “normal” conditions, the NO COMM goes away by itself when the communication link with the ground network is reestablished.

(5) In addition, the ACARS keeps messages queued up during the NO COMM condition and transmits them when communication is again available.

B. Description

(1) The Aircraft Communications Addressing and Reporting System (ACARS) communications management unit (CMU) cannot communicate with an ACARS ground station (NO COMM shows in the CDU scratchpad display).

C. Possible Causes

(1) ACARS Communications Management Unit (CMU-1), M2127

(2) VHF-3 Communications Transceiver (VHF-3), M411

OMR 805 POST SB 737-23-1330

(3) HF-1 Communications Transceiver (HF-1), M226

(4) HF-2 Communications Transceiver (HF-2), M439

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(5) Airplane Programming Module (APM), M2502

(6) APM settings

(7) #1 Radio Tuning Panel (RTP-1), P8-71

(8) #2 Radio Tuning Panel (RTP-2), P8-72

(9) #3 Radio Tuning Panel (RTP-3), P8-73

(10) Wiring

D. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2
E	8	C01483	CMU-1 AC

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(Continued)

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

E. Related Data

(1) SSM 23-27-35

OMR 805 POST SB 737-23-1330

(2) SSM 23-27-36

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(3) WDM 23-27-35

OMR 805 POST SB 737-23-1330

(4) WDM 23-27-36

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

F. Fault Isolation Procedure

(1) On the CMU front panel, push and release the RESET switch.

(a) Make sure the SYSTEM FAIL indicator stays off.

(b) Make sure the LRU PASS/FAIL indicator is red for 2 seconds, then is green for 2 seconds, then is off for at least 2 seconds and no more than 5 seconds.

(c) If the test passes, the LRU PASS/FAIL indicator is green for 30 seconds, then is off.

(d) If the test fails, the LRU PASS/FAIL indicator is red for 30 seconds, then is off.

1) If these indications did occur, go to the VHF Link Test step.

2) If these indications did not occur, replace the ACARS CMU. These are the tasks:

ACARS Communications Management Unit (CMU) Removal, AMM
TASK 23-27-33-020-801

ACARS Communications Management Unit (CMU) Installation, AMM
TASK 23-27-33-420-801

(2) On the left CDU, access the ACARS-IDENT BLK PGM page 1.

(a) If the airplane registration number that shows adjacent to the REG NUM prompt on the CDU screen agrees with the airplane registration number on the captain's instrument panel placard, then continue.

(b) If the registration numbers do not agree, do these steps:

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1) Do this task: ACARS Airplane Personality Module (APM) Software Installation, AMM TASK 23-27-35-700-802

2) If the APM will not accept data, replace the APM, M2502. These are the tasks:

ACARS Airplane Personality Module (APM) Removal, AMM TASK 23-27-35-020-801

ACARS Airplane Personality Module (APM) Installation, AMM TASK 23-27-35-420-801

(c) If the airline ID code adjacent to the AIRLINE ID prompt on the CDU screen is correct, continue to the captain's radio tuning panel step.

(d) If the airline ID code is incorrect or is missing, do these steps:

1) Do this task: ACARS Airplane Personality Module (APM) Software Installation, AMM TASK 23-27-35-700-802

2) Also do subtask 23-27-00-730-081-009 of ACARS - System Test, AMM TASK 23-27-00-730-806-010.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 805, 807 POST SB 737-23-1330

(3) On the captain's radio tuning panel (RTP), select VHF-3.

(a) If "ACARS" or "DATA" shows in the active (left) display, go to the VHF Link Test step.

(b) If "ACARS" or "DATA" shows in the standby (right) display, do these steps:

1) Push the transfer switch on the captain's RTP to move "ACARS" or "DATA" to the active display.

2) Continue with the VHF Link Test step.

(c) If "ACARS" or "DATA" does not show in the standby (right) display, do these steps:

1) Use the frequency select knob to select a frequency in the standby (right) display that is outside the range of the normal VHF voice frequency range.

2) If "ACARS" or "DATA" now shows in the standby (right) display, push the transfer switch on the captain's RTP to move "ACARS" or "DATA" to the active (left) display, then continue with the VHF Link Test step.

3) If "ACARS" or "DATA" does not show in the standby (right) display, do these steps:

a) On the first officers RTP, select VHF-3.

b) Use the frequency select knob to select a frequency in the standby (right) display that is outside the range of the normal VHF voice frequency range.

c) If "ACARS" or "DATA" now shows in the standby (right) display, replace the captains RTP. These are the tasks:

Radio Tuning Panel (RTP) - Removal, AMM TASK 23-12-41-000-801

Radio Tuning Panel (RTP) - Installation, AMM TASK 23-12-41-400-801

d) If "ACARS" or "DATA" still does not show in the standby (right) display, examine the wiring between the CMU and the RTPs (WDM 23-27-35).

e) If no wiring faults are found, replace the ACARS CMU. These are the tasks:

ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801

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OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 805, 807 POST SB 737-23-1330 (Continued)

ACARS Communications Management Unit (CMU) Installation, AMM
TASK 23-27-33-420-801

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(4) Do the VHF Link Test subtask (AMM 23-27-00-730-115) of the ACARS System Test (AMM TASK 23-27-00-730-806-010).

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(a) If the VHF Link Test completes with a "PASS" indication, then there was an intermittent fault.

OMR 805 POST SB 737-23-1330

(b) If the VHF Link Test completes with a "PASS" indication, continue with the HF Link Test step.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(c) If the VHF Link Test completes with a "FAIL" indication, then do these steps:

1) Do a voice communications test over VHF-3 (subtask 23-12-00-730-004 of VHF Communication System - System Test, AMM TASK 23-12-00-730-801).

a) If the voice communications test results are not satisfactory:

<1> Make sure there no obstructions between the VHF-3 antenna and the ACARS VHF ground station.

<2> If the fault continues, refer to PAGEBLOCK 23-12-00/201 for further fault isolation.

b) If the voice communications test results are satisfactory:

<1> Make sure there no obstructions between the VHF-3 antenna and the ACARS VHF ground station.

<2> Make sure the ACARS VHF ground station and network is operational.

<3> Check for messages on the ACARS network from the airplane. If the airplane is not in the ground station database, messages will not be acknowledged.

<4> If the fault continues, replace VHF-3 transceiver, M411. These are the tasks:

VHF Communication Transceiver - Removal, AMM
TASK 23-12-21-020-801

VHF Communication Transceiver - Installation, AMM
TASK 23-12-21-420-801

<5> If the fault continues, replace the ACARS CMU, M2127. These are the tasks:

ACARS Communications Management Unit (CMU) Removal, AMM
TASK 23-27-33-020-801

ACARS Communications Management Unit (CMU) Installation, AMM
TASK 23-27-33-420-801

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<6> If the fault continues, examine the wiring between the ACARS CMU and the #3 VHF Transceiver WDM 23-27-37. Repair the problems that you find.

2) If the VHF frequency is not correct for the area that the airplane is located in, do these steps:

a) Open the ACARS DATA FREQUENCY page.

b) Push the LSK adjacent to the region where the airplane is located.

<1> The new region selected is indicated by an arrow.

c) Access the ACARS COMM STATUS page.

d) Wait approximately 2 minutes, then go back to the VHF Link Test step.

OMR 805 POST SB 737-23-1330

(5) Do the HF Link Test subtask (AMM 23-27-00-730-116) of the ACARS System Test (AMM TASK 23-27-00-730-806-010).

(a) If the HF Link Test completes with a "PASS" indication for both HF-1 and HF-2, then there was an intermittent fault.

(b) If the HF Link Test completes with a "FAIL" indication for HF-1, then do these steps:

1) Do the HF Communication - System Test over the HF-1 communication system (HF Communication System - System Test, AMM TASK 23-11-00-730-801).

a) If the test results are not satisfactory, go to HF Communication System Receive/Transmit Problem - Fault Isolation, 23-11 TASK 806 for further fault isolation.

b) If the test results are satisfactory, access the ACARS-PERIPHERALS screen, page 2.

<1> If "INACTIVE" shows under the CONNECT prompt adjacent to the HFDR 1 prompt, replace HF-1 transceiver. These are the tasks:

HF Transceiver - Removal, AMM TASK 23-11-21-000-801

HF Transceiver - Installation, AMM TASK 23-11-21-400-801

<2> If “PRESENT” shows under the CONNECT prompt and “OK” shows under the STAT prompt adjacent to the HFDR 1 prompt, then continue.

c) Access the ACARS-LINK STATUS page.

<1> If “IN COMM” shows under the STATE prompt for HF, the ACARS CMU and HF-1 are operational. There may be conditions external to the airplane that caused the HF Link Test to fail.

<2> If “LOGD OFF” shows under the STATE prompt for HF, the ACARS CMU and HF-1 are communicating with each other, but HF-L has not established an HF link to the ACARS network. This may be caused by conditions external to the airplane.

(c) If the HF Link Test completes with a "FAIL" indication for HF-2, then do these steps:

1) Do the HF Communication - System Test over the HF-2 communication system (HF Communication System - System Test, AMM TASK 23-11-00-730-801).

a) If the test results are not satisfactory, go to HF Communication System Receive/Transmit Problem - Fault Isolation, 23-11 TASK 806 for further fault isolation.

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OMR 805 POST SB 737-23-1330 (Continued)

b) If the test results are satisfactory, access the ACARS-PERIPHERALS screen, page 2.

<1> If "INACTIVE" shows under the CONNECT prompt adjacent to the HFDR 2 prompt, replace HF-2 transceiver. These are the tasks:

HF Transceiver - Removal, AMM TASK 23-11-21-000-801

HF Transceiver - Installation, AMM TASK 23-11-21-400-801

<2> If “PRESENT” shows under the CONNECT prompt and “OK” shows under the STAT prompt adjacent to the HFDR 2 prompt, then continue.

c) Access the ACARS-LINK STATUS page.

<1> If “IN COMM” shows under the STATE prompt for HF, the ACARS CMU and HFDR-2 are operational. There may be conditions external to the airplane that caused the HF Link Test to fail.

<2> If “LOGD OFF” shows under the STATE prompt for HF, the ACARS CMU and HF-2 are communicating with each other, but HF-2 has not established an HF link to the ACARS network. This may be caused by conditions external to the airplane.

——— END OF TASK ———

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

815. ACARS/CDU Interface Problem - Fault Isolation

A. Description

(1) The Aircraft Communications Addressing and Reporting System (ACARS) menu selection prompt (<ACARS) is missing from line 2 on one of the CDU Main Menu displays.

(2) The ACARS does not respond to CDU key pushes.

B. Possible Causes

(1) ACARS Communications Management Unit (CMU-1), M2127

(2) Left Flight Management Control Display Unit, P9-65

(3) Right Flight Management Control Display Unit, P9-66

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	7	C01238	FMCS MCDU 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	15	C01239	FMCS MCDU 2
E	8	C01483	CMU-1 AC

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This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

D. Related Data

(1) SSM 23-27-32, 23-27-33

(2) WDM 23-27-32, 23-27-33

E. Fault Isolation Procedure

(1) On CDU-1 and CDU-2, push the MENU function key.

(2) If the <ACARS prompt shows on the two CDU screens, then there was an intermittent fault.

(3) If the <ACARS prompt does not show on one of the two CDU screens, do these steps:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(b) Wait 30 sceonds.

(c) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

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OMR 805 POST SB 737-23-1330 (Continued)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(d) If the fault continues, install the ACARS software. This is the task: ACARS Software Installation with a Portable Data Loader (PDL), AMM TASK 23-27-33-470-801 or ACARS Communications Management Unit Software Installation with an Airborne Data Loader (ADL), AMM TASK 23-27-33-470-802 or ACARS Communications Management Unit Software Installation with an Enhanced Airborne Data Loader (eADL), AMM TASK 23-27-33-470-805

(e) If the fault continues after the software is installed, replace the ACARS CMU. These are the tasks:

- ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801
- ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

(4) If the <ACARS prompt shows on only one of the CDU screens, then do these steps:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(b) If the fault occurred on CDU-1, open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	7	C01238	FMCS MCDU 1

(c) If the fault occurred on CDU-2, open this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	15	C01239	FMCS MCDU 2

(d) Wait 30 seconds.

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(e) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(f) If the fault occurred on CDU-1, close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	7	C01238	FMCS MCDU 1

(g) If the fault occurred on CDU-2, close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	15	C01239	FMCS MCDU 2

(h) If the <ACARS prompt now shows on the two CDU screens, then there was an intermittent fault.

(i) If the <ACARS prompt continues to show on only one of the CDU screens, do these steps:

1) Interchange the positions of CDU-1 and CDU-2. These are the tasks:
   - FMCS Control Display Unit (CDU) Removal, AMM TASK 34-61-01-000-802
   - FMCS Control Display Unit (CDU) - Installation, AMM TASK 34-61-01-400-802

2) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

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OMR 805 POST SB 737-23-1330 (Continued)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

3) Wait 30 sceonds.

4) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

5) If the fault moved to the other CDU position (1 to 2 or 2 to 1), replace the CDU that is now missing the <ACARS prompt. These are the tasks:
   - FMCS Control Display Unit (CDU) Removal, AMM TASK 34-61-01-000-802
   - FMCS Control Display Unit (CDU) - Installation, AMM TASK 34-61-01-400-802

6) If the <ACARS prompt did not show on CDU-1 and still does not show on CDU-1, do these steps:
   a) Access the ACARS-SYSTEM CONFIG page on CDU-2.
   b) Push the LSK adjacent to the * PRINT prompt.
   c) If the ACARS-SYSTEM CONFIG page prints on the multifunction printer, do this check of the wiring:
      <1> Remove the ACARS CMU. This is the task: ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801.
      <2> Remove CDU-1. This is the task: FMCS Control Display Unit (CDU) Removal, AMM TASK 34-61-01-000-802.
      <3> Do a check for an open circuit between these pins of connector D10727A of the ACARS CMU and connector D2181 of CDU-1 (WDM 23-27-32):

ACARS CMU D10727B	CDU-1 D2181
E15	14
F15	15

      <4> If you find a problem with the wiring, then do these steps:
         <a> Repair the wiring.

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<b> Re-install the ACARS CMU. This is the task: ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801.

<c> Re-install CDU-1. This is the task: FMCS Control Display Unit (CDU) - Installation, AMM TASK 34-61-01-400-802.

d) If the ACARS-SYSTEM CONFIG page does not print, replace the ACARS CMU. These are the tasks:

- ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801
- ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

7) If the <ACARS prompt did not show on CDU-2 and still does not show on CDU-2, do these steps:

a) Attempt to access a Digital Flight Data Acquisition Unit (DFDAU) screen that shows the status of the ARINC 429 data bus input from the CMU.

b) If the status shows that the data bus is active, do this check of the wiring:

<1> Remove the ACARS CMU. This is the task: ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801.

<2> Remove CDU-2. This is the task: FMCS Control Display Unit (CDU) Removal, AMM TASK 34-61-01-000-802.

<3> Do a check for an open circuit between these pins of connector D10727A of the ACARS CMU and connector D2221 of CDU-2 (WDM 23-27-33):

ACARS CMU	**CDU-2**
D10727B	**D2221**
E12	14
F12	15

<4> If you find a problem with the wiring, then do these steps:

<a> Repair the wiring.

<b> Re-install the ACARS CMU. This is the task: ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801.

<c> Re-install CDU-2. This is the task: FMCS Control Display Unit (CDU) - Installation, AMM TASK 34-61-01-400-802.

c) If the status shows that the data bus is inactive, replace the ACARS CMU. These are the tasks:

- ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801
- ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

——— END OF TASK ———

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816. ACARS Input Problem - Fault Isolation

A. Description

(1) The ground station reports that downlink messages from the ACARS system are incomplete

B. Possible Causes

(1) ACARS Communications Management Unit (CMU-1), M2127

(2) Flight Management Computer 1 (FMC-1), M1175

(3) Flight Management Computer 2 (FMC-2), M1632 (if installed)

(4) Digital Flight Data Acquisition Unit (DFDAU), M675

(5) Proximity Switch Electronics Unit (PSEU), M2061

(6) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

D. Related Data

(1) SSM 23-27-33, 23-27-35, 23-27-36, 23-27-38

(2) WDM 23-27-33, 23-27-35, 23-27-36, 23-27-38

E. Initial Evaluation

(1) If the data reported missing from the downlink messages is from flight crew manual entries, make sure the flight crew enters their data correctly.

(2) If the data reported missing from the downlink messages is OOOI data (Out, Off, On, In), do the Missing OOOI Data - Fault Isolation Procedure below.

(3) If the data reported missing from the downlink messages is from a LRU that talks to the ACARS CMU over a ARINC-429 data bus, do the Missing ARINC Data - Fault Isolation Procedure below.

F. Missing OOOI Data - Fault Isolation Procedure

(1) On a CDU, access the ACARS-SENSORS page.

(a) Do the Out-Off-On-In (OOOI) Sensor Test subtask of the ACARS System Test (AMM TASK 23-27-00-730-806-010). If one of the sensors fails the OOOI Sensor Test, do the fault isolation procedure for the failed sensor:

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1) PSEU Entry and Galley Service Door Monitored Problem - Fault Isolation, 52-10 TASK 801 for entry and galley doors.

2) PSEU Cargo Door Monitored Problem - Fault Isolation, 52-30 TASK 803 for cargo doors.

3) Parking Brake Switch Problem - Fault Isolation, 32-09 TASK 806 for the parking brake.

4) Nose Landing Gear Air/Ground Sensor Fault - Fault Isolation, 32-09 TASK 802 for the air/ground sensor.

(b) If all of the sensors passed the OOOI Sensor Test, do these steps:

1) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

2) Wait 30 seconds.

3) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

4) If the fault continues, install the ACARS software. This is the task: ACARS Software Installation with a Portable Data Loader (PDL), AMM TASK 23-27-33-470-801 or ACARS Communications Management Unit Software Installation with an Airborne Data Loader (ADL), AMM TASK 23-27-33-470-802 or ACARS Communications Management Unit Software Installation with an Enhanced Airborne Data Loader (eADL), AMM TASK 23-27-33-470-805.

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5) If the fault continues after the software is installed, replace the ACARS CMU. These are the tasks:

ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801

ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

G. Missing ARINC Data - Fault Isolation Procedure

(1) On a CDU, access the ACARS-PERIPHERALS screen, pages 1, 2 and 3.

(a) If the state of each connected input LRU (except for MCDU and PRINTER) shows “PRESENT” under the CONNECT prompt and “OK” under the STAT prompt, do the fault isolation procedure for the failed source LRU:

1) DFDAU - DFDAU Internal Failure - Fault Isolation, 31-31 TASK 802

2) FMC - FMC Hardware/Software Faults - Fault Isolation, 34-61 TASK 807

OMR 805 POST SB 737-23-1330

3) HFDR-1 - HF Communication System - BITE Procedure, 23-11 TASK 801

4) HFDR-2 - HF Communication System - BITE Procedure, 23-11 TASK 801

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

5) VHF-3 - VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801

(b) If the state of one of the connected input LRUs (except for MCDU and PRINTER) shows "ABSENT" under the CONNECT prompt, do these steps:

1) Cycle the power for the source LRU that shows “ABSENT”.

2) If the fault continues, replace the source LRU.

3) If the fault continues after source LRU replacement, examine the ARINC 429 data bus wiring between the source LRU and the ACARS CMU.

a) DFDAU - WDM 23-27-33 and WDM 31-35-02

b) FMC-1 - WDM 23-27-32 and WDM 34-61-25

c) FMC-2 (if installed) - WDM 23-27-33 and WDM 34-61-25

OMR 805 POST SB 737-23-1330

d) HFDR-1 - WDM 23-27-36

e) HFDR-2 - WDM 23-27-36

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

f) VHF-3 - VHF Communication Transceiver System - BITE Procedure, 23-12 TASK 801

——— END OF TASK ———

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OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

818. ACARS Call Annunciation Problem - Fault Isolation

A. Description

(1) The flight compartment chime does not sound for an ACARS uplink message.

(2) The VHF-C lamp on the SELCAL control panel does not come on for an ACARS uplink message.

(3) Pushing the VHF-C switchlight on the SELCAL control panel does not acknowledge the ACARS uplink message (VHF-C CALL lamp does not go out).

B. Possible Causes

(1) ACARS Communications Management Unit (CMU-1), M2127

(2) SELCAL Control Panel, P8-66

(3) Aural Warning Unit Module, M315

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6

Row	Col	Number	Name
OMR 802 POST SB 737-23-1334			
E	8	C00743	ACARS MU DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	7	C00744	ACARS MU AC
OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334			
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

D. Related Data

(1) SSM 23-22-11

(2) SSM 23-27-31

(3) WDM 23-22-11

(4) WDM 23-27-31

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E. Initial Evaluation

(1) Have an ACARS ground station send an uplink message to the airplane on VHF-3.

(a) If the SELCAL control panel VHF-3 switch lamp comes on and the flight compartment chime sounds, continue with step (2).

(b) If the VHF-3 switch lamp does not come on, but the chime sounds, do the Call Indicator Problem - Fault Isolation Procedure below.

(c) If the VHF-3 switch lamp comes on, but the chime does not sound, do the Flight Compartment Chime Problem - Fault Isolation Procedure below.

(2) On the SELCAL control panel, push and release the VHF-3 switch lamp.

(a) If the VHF-3 switch lamp goes out, then there was an intermittent fault.

(b) If the VHF-3 switch lamp does not go out, do the Call Reset Problem - Fault Isolation Procedure below.

F. Call Indicator Problem - Fault Isolation Procedure

(1) Do the test for a defective VHF-3 lamp on the SELCAL control panel. This is the task: Master Dim and Test - Operational Test, AMM TASK 33-18-00-710-802

(a) If the VHF-3 lamp tests OK, continue with step (2).

(b) If a defective VHF-3 lamp is found, replace the bulb in the switch lamp.

(c) If the fault continues after bulb replacement, replace the SELCAL Control Panel. These are the tasks:

SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801

SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801

(2) Have an SELCAL ground station make a call to the airplane on VHF-3.

(a) If the VHF-3 switch lamp on the SELCAL control panel does not come on, replace the SELCAL Control Panel. These are the tasks:

SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801

SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801

(b) If the VHF-3 switch lamp on the SELCAL control panel comes on, then continue.

(3) Do these steps:

(a) Examine the circuit from pin F9 of ACARS CMU connector D10727B to pin N of the SELCAL control panel connector D551 for an open wire (WDM 23-22-11 and WDM 23-27-31).

1) Repair the problems that you find.

G. Call Reset Problem - Fault Isolation Procedure

(1) On a CDU, access the ACARS-DISC IN page 3.

(2) Push and release the SELCAL control panel VHF-3 switch lamp more than once while watching the state of the GO-AHEAD input (MP09C).

(a) If the state of the pin alternated between “0” (ground) and “1” (open) as the switch was pushed and released, then there was an intermittent fault.

(b) If the state of the pin remained at “0” (ground) as the switch was pushed and released, do these steps:

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OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334 (Continued)

1) Disconnect plug D551 from the SELCAL control panel.
   a) If the state of the pin changes to “1” (open), replace the SELCAL Control Panel. These are the tasks:

      SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801

      SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801

   b) If the state of the pin remains at “0” (ground), examine the wire from pin F of SELCAL Control Panel connector D551 to pin C9 of the ACARS CMU connector DB10727B for a short to ground (WDM 23-22-11 and WDM 23-27-31).

      <1> Repair the problems that you find.

(c) If the state of the pin remained at “1” (open) as the switch was pushed and released, do these steps:
1) Disconnect plug D551 from the SELCAL control panel.
2) Connect a jumper wire between pins F and S of the SELCAL control panel connector D551.
   a) If the state of the pin changes to “0” (ground), replace the SELCAL Control Panel. These are the tasks:

      SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801

      SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801

   b) If the state of the pin remains at “1” (open), examine the wire from pin F of SELCAL Control Panel connector D551 to pin C9 of the ACARS CMU connector DB10727B for an open wire (WDM 23-22-11 and WDM 23-27-31).

      <1> Repair the problems that you find.

H. Flight Compartment Chime Problem - Fault Isolation Procedure

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6

Row	Col	Number	Name
OMR 802 POST SB 737-23-1334			
E	8	C00743	ACARS MU DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334			
E	9	C01500	CMU/ACARS DC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	9	C01500	CMU/ACARS DC (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805 POST SB 737-23-1330 (Continued)

OMR 809, 810, 812, 813; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(2) Do a BITE test of the Aural Warning. This is the task: Aural Warning Module BITE Test, AMM TASK 31-51-00-740-801.

(a) If the test results are satisfactory, continue with step (2).

(b) If the test results are not satisfactory, replace the aural warning module. These are the tasks:

Aural Warning Module Removal, AMM TASK 31-51-04-000-801

Aural Warning Module Installation, AMM TASK 31-51-04-400-801

(3) Do these steps:

(a) Examine the circuit from pin G9 of ACARS CMU connector D10727A to pin 8 of the aural warning module connector D940 for an open wire (WDM 23-22-11 and WDM 23-27-31).

1) Repair the problems that you find.

——— END OF TASK ———

OMR 815, 816, 818-826, 901-999

820. ACARS Call Annunciation Problem - Fault Isolation

A. Description

(1) Use this task when the flight compartment chime does not sound for an ACARS uplink message.

(2) Use this task when the ACARS message does not show on the Engine Display for an ACARS uplink message.

B. Possible Causes

(1) Software (CMU or DEU)

(2) ACARS Communications Management Unit (CMU-1), M2127

(3) Display Electronic Unit (DEU), M1808 or DEU, M1809

(4) Aural Warning Unit Module, M315

(5) Wiring

C. Initial Evaluation

(1) Have an ACARS ground station send an uplink message to the airplane on VHF-3.

(a) If the ACARS message shows on the Engine Display and the flight compartment chime sounds, then there was an intermittent fault.

(b) If the ACARS message does not show on the Engine Display, then do the ACARS Message Annunciation Problem fault isolation procedure below.

(c) If the chime does not sound, then do the ACARS Message Chime Problem fault isolation procedure below.

D. ACARS Message Annunciation Problem - Fault Isolation Procedure

(1) Do a check to make sure the CMU has the correct software installed. This is the task:

ACARS Communications Management Unit (CMU) Software Configuration Check, AMM TASK 23-27-33-700-802

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OMR 815, 816, 818-826, 901-999 (Continued)

(2) Do a check to make sure the DEU has the correct software installed (Display Electronic Unit Installation, AMM TASK 31-62-21-400-801).

(3) If the fault continues, replace the CMU, M2127. These are the tasks:

ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801

ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

(4) If the fault continues, replace the DEU, M1808 or M1809. These are the tasks:

Display Electronic Unit Removal, AMM TASK 31-62-21-000-801

Display Electronic Unit Installation, AMM TASK 31-62-21-400-801

(5) If the fault continues, examine the wiring between the CMU and the DEU (WDM 23-27-38), (WDM 31-62-15, WDM 31-62-25).

E/E Compartment	
CMU, M2127	**DEU-1, M1808**
D10727B	**D3973D**
J10	J15
K10	K15

(a) If the resistance check does not show continuity, repair the problems with the wiring.

E. ACARS Message Chime Problem - Fault Isolation Procedure

(1) Do a check to make sure the CMU has the correct software installed. This is the task:

ACARS Communications Management Unit (CMU) Software Configuration Check, AMM TASK 23-27-33-700-802

(2) Do a check to make sure the DEU has the correct software installed (Display Electronic Unit Installation, AMM TASK 31-62-21-400-801).

(3) If the fault continues, replace the Aural Warning Module, M315.

Aural Warning Module Removal, AMM TASK 31-51-04-000-801

Aural Warning Module Installation, AMM TASK 31-51-04-400-801

(4) If the fault continues, replace the DEU, M1808 or M1809.

Display Electronic Unit Removal, AMM TASK 31-62-21-000-801

Display Electronic Unit Installation, AMM TASK 31-62-21-400-801

(5) If the fault continues, examine the wiring between the CMU and the DEU (WDM 23-27-38).

E/E Compartment	
CMU, M2127	**DEU-1, M1808**
D10727B	**D3973D**
J10	J15
K10	K15

(a) If the resistance check does not show continuity, repair the problems with the wiring.

(6) If the fault continues, examine the wiring between the DEU and the Aural Warning Module (WDM 23-22-11 and WDM 31-62-41).

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OMR 815, 816, 818-826, 901-999 (Continued)

	E/E Compartment
	Aural Warning
DEU-1, M1808	**Module, M315**
D3973A	**D940**
E2	8

(a) If the resistance check does not show continuity, repair the problems with the wiring.

——— END OF TASK ———

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

825. ACARS Message APM WARN - Fault Isolation

A. Description

(1) Use this task when the scratchpad on the Aircraft Communications Addressing and Reporting System (ACARS) pages shows the message “APM WARN” while on the ground.

B. Possible Causes

(1) ACARS Communications Management Unit (CMU).

(2) Aircraft Personality Module (APM).

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

D. Related Data

(1) (SSM 23-27-31)

(2) (SSM 23-27-39)

(3) (WDM 23-27-31)

(4) (WDM 23-27-39)

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E. Initial Evaluation

(1) If the APM WARN message does not show in the CDU scratchpad, then there was an intermittent fault.

(2) If the APM WARN message shows in the CDU scratchpad, then do the applicable fault isolation procedure below.

F. ACARS Message APM WARN - Fault Isolation Procedure

(1) Prepare to cycle the circuit breakers. Do the steps that follow:

(a) Make sure that the circuit breakers are open for a minimum of 10 seconds.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(c) Measure the time.

(d) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC
E	9	C01500	CMU/ACARS DC

These circuit breakers are inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR 805 POST SB 737-23-1330			
E	8	C01483	CMU-1 AC (INOP)
E	9	C01500	CMU/ACARS DC (INOP)

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(e) If the APM WARN message does not show in the CDU scratchpad, then there was an intermittent fault.

(f) If the APM WARN message still shows in the CDU scratchpad, then continue.

(2) Do the ACARS - Operational Test. This is the task:ACARS - Operational Test, AMM TASK 23-27-00-740-816-010

(a) If the test fails, replace the ACARS Communications Management Unit (CMU) 1, M2127. These are the tasks:

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ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801

ACARS Communications Management Unit (CMU) Installation, AMM TASK 23-27-33-420-801

(b) If the test passes and the APM WARN message still shows in the CDU scratchpad, then continue.

(3) Do this check of the wiring between the CMU and the APM.

(a) Remove the CMU. This is the task: ACARS Communications Management Unit (CMU) Removal, AMM TASK 23-27-33-020-801

(b) Remove the APM. This is the task: ACARS Airplane Personality Module (APM) Removal, AMM TASK 23-27-35-020-801

OMR 701-799 POST SB 737-23-1330

(c) Use the table that follows to do continuity checks between the pins of the APM (M2374) connector D11411 and the CMU (M2127) connector D10727A.

M2374 D11411	M2127 D10727A
1	H1
2	E1
3	C1
4	K1
5	D1
6	A2
7	A1
8	B1
9	J1

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 802 POST SB 737-23-1334; OMR 805, 807 POST SB 737-23-1330

(d) Use the table that follows to do continuity checks between the pins of the APM (M2502) connector D11813 and the CMU (M2127) connector D10727A.

M2502 D11813	M2127 D10727A
1	H1
2	E1
3	C1
4	K1
5	D1
6	A2
7	A1
8	B1
9	J1

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OMR 809

(e) Use the table that follows to do continuity checks between the pins of the APM (M2502) connector D11813 and the CMU (M2127) connector D10727A.

M2502 D11813	M2127 D10727A
1	H1
2	E1
3	C1
4	K1
5	D1
6	A2
7	A1
8	B1
9	J1

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 805, 807 POST SB 737-23-1330; OMR 802 POST SB 737-23-1334

(f) If you find a problem with the wiring, then repair it as necessary.

(g) If the wiring checks good, then continue.

(4) Replace the APM module. This is the task:

ACARS Airplane Personality Module (APM) Installation, AMM TASK 23-27-35-420-801

G. Repair Confirmation

(1) If the APM WARN fault message does not show, then you corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. SELCAL Call Switch Problem - Fault Isolation

A. Description

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) The call switch on SELCAL control panel, P8-66 does not operate correctly.

OMR 815, 816, 818-826, 901-999

(2) A call switch on Audio control panel, does not operate correctly.

OMR ALL

B. Possible Causes

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) SELCAL control panel, P8-66.

OMR 815, 816, 818-826, 901-999

(2) Audio control panel, P8-6 (captain's), P8-7 (first officer's) or P5-15 (first observer's).

OMR ALL

(3) SELCAL decoder, M25

(4) Wiring problem

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

D. Related Data

(1) (SSM 23-22-11)

(2) (WDM 23-22-11)

E. Initial Evaluation

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Do this check of the SELCAL control panel:

(a) Push the call switch to reset for the applicable HF switch or VHF switch.

1) If the call light goes off for the applicable HF switch or VHF switch, then there was an intermittent fault.

2) If the call light does not go off for the applicable HF switch or VHF switch, then do the Fault Isolation Procedure below.

OMR 815, 816, 818-826, 901-999

(2) Do this check of the Audio control panel:

(a) Push the PTT for the applicable HF switch or VHF switch to reset the Call Light.

1) If the call light goes off for the applicable HF switch or VHF switch, then there was an intermittent fault.

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OMR 815, 816, 818-826, 901-999 (Continued)

2) If the call light does not go off for the applicable HF switch or VHF switch, then do the Fault Isolation Procedure below.

OMR ALL

F. Fault Isolation Procedure

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Replace the SELCAL control panel.

These are the tasks:

SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801,

SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.

(a) If the call switch does reset correctly, then you corrected the fault.

(b) If the call switch does not reset correctly, then continue.

OMR 815, 816, 818-826, 901-999

(2) Replace the Audio control panel.

These are the tasks:

Audio Control Panel Removal, AMM TASK 23-51-02-000-801,

Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) If the call switch resets correctly, then you corrected the fault.

(b) If the call switch does not reset correctly, then continue.

OMR ALL

(3) Replace the SELCAL decoder unit.

These are the tasks:

SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801,

SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.

(a) If the call switch does reset correctly, then you corrected the fault.

(b) If the call switch does not reset correctly, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(4) Do this check of the wiring:

(a) Remove the SELCAL control panel (SCP), P8-66. To remove it, do this task: SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801.

(b) Remove the SELCAL decoder unit, M00025. To remove it, do this task: SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801.

(c) Do a wiring check between these pins of connector D551 for the SELCAL control panel and connector D2555B for the SELCAL decoder unit: Refer to WDM 23-22-11 for details.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

FUNCTION KEY	SCP CONNECTOR		SELCAL DECODER CONNECTOR
HF1 (if applicable)	**D551**		**D2555B**
	pin G	. . . . .	pin D1
HF2 (if applicable)	**D551**		**D2555B**
	pin H	. . . . .	pin A2
VHF1	**D551**		**D2555B**
	pin D	. . . . .	pin A1
VHF2	**D551**		**D2555B**
	pin E	. . . . .	pin B1
VHF3 (if applicable)	**D551**		**D2555B**
	pin F	. . . . .	pin C1

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the SELCAL control panel. To install it, do this task: SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.
4) Re-install the SELCAL decoder unit. To install it, do this task: SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.
5) If the call switch does reset correctly, then you corrected the fault.

OMR 815, 816, 818-826, 901-999

(5) Do this check of the wiring:

(a) Remove the Remote Electronics Unit. To remove it, do this task: AMM TASK 23-51-01-000-801.

(b) Remove the SELCAL decoder unit, M00025. To remove it, do this task: SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801.

(c) Do a wiring check between the pins of connector D2501A for the Remote Electronics Unit (REU), and connector D2555B for the SELCAL decoder unit: Refer to WDM 23-22-11 for details.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

FUNCTION KEY	REU CONNECTOR	SELCAL DECODER CONNECTOR
HF1 (if applicable)	**D2501A** pin C10	**D2555B** pin C7
HF2 (if applicable)	**D2501A** pin J7	**D2555B** pin D7
VHF1	**D2501A** pin H6	**D2555B** pin D6
VHF2	**D2501A** pin A9	**D2555B** pin A7
VHF3 (if applicable)	**D2501A** pin F7	**D2555B** pin B7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the Remote Electronics Unit. To install it, do this task: AMM TASK 23-51-01-400-801.
4) Re-install the SELCAL decoder unit. To install it, do this task: SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.
5) If the call switch does reset correctly, then you corrected the fault.

OMR ALL

——— END OF TASK ———

802. SELCAL Call Light Problem - Fault Isolation

A. Description

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) The call light on the SELCAL control panel, P8-66 does not operate correctly.

OMR 815, 816, 818-826, 901-999

(2) The mic/call light on the Audio Control Panel does not operate correctly.

OMR ALL

B. Possible Causes

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) SELCAL control panel, P8-66

OMR 815, 816, 818-826, 901-999

(2) Audio Control Panel, captain's, first officer's, or observer's.

OMR ALL

(3) SELCAL decoder, M25

(4) Wiring problem

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

D. Related Data

(1) (SSM 23-22-11)

(2) (WDM 23-22-11)

E. Initial Evaluation

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Push the call light.

(a) If the light comes on, then do these steps:

1) Push the call light again.

a) If the call light goes off, then there was an intermittent fault.

b) If the call light stays on, then, do this task: SELCAL Call Switch Problem - Fault Isolation, 23-28 TASK 801.

(b) If the light does not come on, then do the Fault Isolation Procedure below.

OMR 815, 816, 818-826, 901-999

(2) Push the mic/call light.

(a) If the light comes on, then do these steps:

1) Push the mic call light again.

a) If the mic/call light goes off, then there was an intermittent fault.

b) If the mic/call light stays on, then, do this task: SELCAL Call Switch Problem - Fault Isolation, 23-28 TASK 801.

(b) If the light does not come on, then do the Fault Isolation Procedure below.

OMR ALL

F. Fault Isolation Procedure

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Replace the SELCAL control panel.

These are the tasks:

SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801,

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.

(a) If the call light does reset correctly, then you corrected the fault.

(b) If the call light does not reset correctly, then continue.

OMR 815, 816, 818-826, 901-999

(2) Replace the Audio Control Panel.

These are the tasks:

AMM TASK 23-51-02-000-801,

AMM TASK 23-51-02-400-801.

(a) If the call light does reset correctly, then you corrected the fault.

(b) If the call light does not reset correctly, then continue.

OMR ALL

(3) Replace the SELCAL decoder unit.

These are the tasks:

SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801,

SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.

(a) If the call switch does reset correctly, then you corrected the fault.

(b) If the call switch does not reset correctly, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(4) Do this check of the wiring:

(a) Remove the SELCAL control panel, P8-66. To remove it, do this task: SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801.

(b) Remove the SELCAL decoder unit, M00025. To remove it, do this task: SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801.

(c) Do a wiring check between these pins of connector D551 for the SELCAL control panel and connector D2555B for the SELCAL decoder unit: Refer to WDM 23-22-11 for details.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

FUNCTION KEY	SCP CONNECTOR		SELCAL DECODER CONNECTOR
HF1 (if applicable)	**D551**		**D2555B**
	pin P	. . . . .	pin C7
HF2 (if applicable)	**D551**		**D2555B**
	pin R	. . . . .	pin D7
VHF1	**D551**		**D2555B**
	pin L	. . . . .	pin D6
VHF2	**D551**		**D2555B**
	pin M	. . . . .	pin A7
VHF3 (if applicable)	**D551**		**D2555B**
	pin N	. . . . .	pin B7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the SELCAL control panel. To install it, do this task: SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.
4) Re-install the SELCAL decoder unit. To install it, do this task: SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.
5) If the call switch does reset correctly, then you corrected the fault.

OMR 815, 816, 818-826, 901-999

(5) Do this check of the wiring:

(a) Remove the Audio Control Panel. To remove it, do this task: Audio Control Panel Removal, AMM TASK 23-51-02-000-801.

(b) Remove the SELCAL decoder unit, M00025. To remove it, do this task: SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801.

(c) Do a wiring check between the pins of connector D141 for the Audio Control Panel (observer's) and connector D2555B for the SELCAL decoder unit: Refer to WDM 23-22-11 for details.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

	FUNCTION KEY	
	ACP CONNECTOR	**SELCAL DECODER CONNECTOR**
HF1 (if applicable)	**D141**	**D2555B**
	pin 4	pin C7
HF2 (if applicable)	**D141**	**D2555B**
	pin 5	pin D7
VHF1	**D141**	**D2555B**
	pin 1	pin D6
VHF2	**D141**	**D2555B**
	pin 2	pin A7
VHF3 (if applicable)	**D141**	**D2555B**
	pin 3	pin B7

(d) Do the wiring check for the captain's ACP and the first officer's ACP. Refer to WDM 23-22-11 for ACP connector numbers.

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the connectors.

3) Re-install the Audio Control Panel. To install it, do this task: Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

4) Re-install the SELCAL decoder unit. To install it, do this task: SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.

5) If the call switch does reset correctly, then you corrected the fault.

OMR ALL

——— END OF TASK ———

803. Aural Warning Unit Chimes Problem - Fault Isolation

A. Description

(1) The chime on the aural warning unit module, M315 does not operate correctly for calls on VHF or HF.

B. Possible Causes

(1) Remote electronics unit, M1353

(2) Aural warning unit, M315

(3) Wiring problem

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

D. Related Data

(1) (SSM 23-22-11)

(2) (WDM 23-22-11)

E. Initial Evaluation

(1) Initiate the chime in the aural warning unit, M315.

(a) If the chime comes on, then there was an intermittent fault.

(b) If the chime does not come on, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the aural warning unit, M315.

These are the tasks:

Aural Warning Module Removal, AMM TASK 31-51-04-000-801,

Aural Warning Module Installation, AMM TASK 31-51-04-400-801.

(a) If the chime operates correctly, then you corrected the fault.

(b) If the chime does not operate correctly, then continue.

(2) Replace the remote electronics unit, M1353.

These are the tasks:

Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801,

Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) If the chime operates correctly, then you corrected the fault.

(b) If the chime does not operate correctly, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) Do this check of the wiring:

(a) Remove the SELCAL control panel, P8-66. To remove it, do this task: SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801.

(b) Remove the aural warning unit, M315. To remove it, do this task: Aural Warning Module Removal, AMM TASK 31-51-04-000-801.

(c) Do a wiring check between these pins of connector D551 for the SELCAL control panel and connector D940 for the aural warning unit:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

	SCP CONNECTOR		AURAL WARN UNIT CONNECTOR
HF1 (if applicable)	**D551**		**D940**
	pin P	. . . . .	pin 8
HF2 (if applicable)	**D551**		**D940**
	pin R	. . . . .	pin 8
VHF1	**D551**		**D940**
	pin L	. . . . .	pin 8
VHF2	**D551**		**D940**
	pin M	. . . . .	pin 8
VHF3 (if applicable)	**D551**		**D940**
	pin N	. . . . .	pin 8

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the SELCAL control panel. To install it, do this task: SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.
4) Re-install the aural warning unit. To install it, do this task: Aural Warning Module Installation, AMM TASK 31-51-04-400-801.
5) If the chime operates correctly, then you corrected the fault.

OMR 815, 816, 818-826, 901-999

(4) Do this check of the wiring:

(a) Remove the Audio Selector Panel P5-15, P8-6, and P8-7. To remove it, do this task: AMM TASK 23-51-02-000-801.

(b) Remove the aural warning unit, M315. To remove it, do this task: AMM TASK 31-51-04-000-801.

(c) Do a wiring check between these pins of connector D141 for the Audio control panel and connector D940 for the aural warning unit:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

	ASP CONNECTOR		AURAL WARN UNIT CONNECTOR
HF1 (if applicable)	**D141** pin 4	. . . . .	**D940** pin 8
HF2 (if applicable)	**D141** pin 5	. . . . .	**D940** pin 8
VHF1	**D141** pin 1	. . . . .	**D940** pin 8
VHF2	**D141** pin 2	. . . . .	**D940** pin 8
VHF3 (if applicable)	**D141** pin 3	. . . . .	**D940** pin 8

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the Audio Selector Panel P5-15, P8-6, and P8-7. To remove it, do this task: AMM TASK 23-51-02-400-801.
4) Re-install the aural warning unit. To install it, do this task: AMM TASK 31-51-04-400-801.
5) If the chime operates correctly, then you corrected the fault.

OMR ALL

——— END OF TASK ———

804. SELCAL Operation Problem - Fault Isolation

A. Description

(1) The SELCAL system does not operate correctly for VHF or HF calls.

B. Possible Causes

(1) SELCAL decoder unit, M25

(2) Wiring problem

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

D. Related Data

(1) (SSM 23-22-11)

(2) (WDM 23-22-11)

E. Fault Isolation Procedure

(1) Replace the SELCAL decoder unit.

These are the tasks:

SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801,

SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.

(a) If the call switch does reset correctly, then you corrected the fault.

(b) If the call switch does not reset correctly, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(2) Do this check of the wiring:

(a) Remove the SELCAL control panel, P8-66. To remove it, do this task: SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801.

(b) Remove the SELCAL decoder unit, M25. To remove it, do this task: SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801.

(c) Do a wiring check between these pins of connector D551 for the SELCAL control panel and connector D2555B for the SELCAL decoder unit:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

	FUNCTION KEY	
	SCP CONNECTOR	**SELCAL DECODER CONNECTOR**
HF1 (if applicable)	**D551** pin P	**D2555B** pin C7
HF2 (if applicable)	**D551** pin R	**D2555B** pin D7
VHF1	**D551** pin L	**D2555B** pin D6
VHF2	**D551** pin M	**D2555B** pin A7
VHF3 (if applicable)	**D551** pin N	**D2555B** pin B7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the SELCAL control panel. To install it, do this task: SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801.
4) Re-install the SELCAL decoder unit. To install it, do this task: SELCAL Decoder Installation, AMM TASK 23-28-11-420-801.
5) If the chime operates correctly, then you corrected the fault.

OMR 815, 816, 818-826, 901-999

(3) Do this check of the wiring:

(a) Remove the Audio Selector Panel P5-15, P8-6, and P8-7. To remove it, do this task: AMM TASK 23-51-02-000-801.

(b) Remove the SELCAL decoder unit, M25. To remove it, do this task: AMM TASK 23-28-11-020-801.

(c) Do a wiring check between the pins of connector D141 for the Audio Selector Panel, and connector D2555B for the SELCAL decoder unit:

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OMR 815, 816, 818-826, 901-999 (Continued)

	FUNCTION KEY	
	AUDIO SELECTOR PANEL	**SELCAL DECODER CONNECTOR**
HF1 (if applicable)	**D141** pin 4	**D2555B** pin C7
HF2 (if applicable)	**D141** pin J7	**D2555B** pin D7
VHF1	**D141** pin 1	**D2555B** pin D6
VHF2	**D141** pin 2	**D2555B** pin A7
VHF3 (if applicable)	**D141** pin	**D2555B** pin B7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connectors.
3) Re-install the Audio Selector Panel. To install it, do this task: AMM TASK 23-51-02-400-801.
4) Re-install the SELCAL decoder unit. To install it, do this task: AMM TASK 23-28-11-420-801.
5) If the call switch does reset correctly, then you corrected the fault.

——— END OF TASK ———

OMR ALL

805. All SELCAL Call Lights On - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(a) All call lights on the Selective Call System (SELCAL) Control Panel, P8-66 come ON.

OMR 815, 816, 818-826, 901-999

(b) All mic/call lights on all the Audio Control Panel (ACP) come ON.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Possible Causes

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) SELCAL Control Panel, P8-66

OMR 815, 816, 818-826, 901-999

(2) Captain ACP, P8-6

(3) First Officer ACP, P8-7

(4) Observer ACP, P5-15

OMR ALL

(5) SELCAL Decoder, M25

C. Initial Evaluation

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Push the call light on the SELCAL Control Panel, P8-66.

(a) If all call lights go OFF, then there was an intermittent fault.

(b) If all call lights stay ON, then do the Fault Isolation Procedure below.

OMR 815, 816, 818-826, 901-999

(2) Push the mic/call light on the applicable ACP.

(a) If all mic/call lights go OFF, then there was an intermittent fault.

(b) If all mic/call lights stay ON, then do the Fault Isolation Procedure below.

OMR ALL

D. Fault Isolation Procedure

(1) Replace the SELCAL Decoder, M25. These are the tasks:

- SELCAL Decoder Unit Removal, AMM TASK 23-28-11-020-801
- SELCAL Decoder Installation, AMM TASK 23-28-11-420-801

(a) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(2) Replace the SELCAL Control Panel, P8-66. These are the tasks:

- SELCAL Control Panel - Removal, AMM TASK 23-28-21-000-801
- SELCAL Control Panel - Installation, AMM TASK 23-28-21-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(3) Replace the applicable ACP. These are the tasks:

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

E. Repair Confirmation

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Look at the call lights on the SELCAL Control Panel, P8-66.

(a) If all call lights go OFF, you corrected the problem.

(b) If all call lights do not go OFF, then continue the Fault Isolation Procedure at the subsequent step.

OMR 815, 816, 818-826, 901-999

(2) Look at the mic/call lights on the applicable ACP.

(a) If all mic/call lights go OFF, you corrected the problem.

(b) If all mic/call lights do not go OFF, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Passenger Address Does Not Operate from Flight Deck - Fault Isolation

A. Description

(1) No, intermittent, or poor quality audio can be heard from the passenger address (PA) speakers when the flight crew makes a PA announcement.

B. Possible Causes

(1) Microphone

(2) Wiring

(3) Control Wheel PTT switch.

(4) Passenger address amplifier, M63

(5) Remote electronic unit (REU), M1353.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) (SSM 23-31-11)

(2) (SSM 23-31-12)

(3) (SSM 23-31-13)

(4) (SSM 23-31-14)

(5) (SSM 23-51-11)

(6) (SSM 23-51-21)

(7) (SSM 23-51-31)

(8) (WDM 23-31-01)

(9) (WDM 23-31-02)

(10) (WDM 23-31-03)

(11) (WDM 23-31-04)

(12) (WDM 23-51-11)

(13) (WDM 23-51-21)

(14) (WDM 23-51-31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Fault Isolation Procedure

(1) Do this check of the microphone:

(a) Use a microphone in the flight deck, other than where the problem was reported, to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on all of the applicable speakers.

(b) If you can hear the announcement on the speakers, then replace the microphone where the problem was reported.

1) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not satisfactory, then continue.

(c) If you cannot hear the announcement on the speakers, or the announcement is intermittent, then continue.

(2) Do these steps to examine and repair the wires to the applicable microphone:

(a) For the PA microphone at the aft end of the control stand, use the 23-31 SSM and WDM listed in Related Data To examine these wires:

1) Examine and repair the wires between the connector D2639B at the passenger address amplifier and the connector D6001 in the control stand.

a) If you found a problem, then do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not OK, then continue.

(b) For other flight deck microphones, use the 23-51 SSM and WDM listed in Related Data to examine these wires:

NOTE: Other microphones are the Captains's or First Officer's hand microphone, boom microphone, oxygen mask microphone or the Observer seat microphones.

1) Remove the remote electronic unit (REU). To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

2) Examine and repair the wires between the connector D2501B at the REU and the applicable microphone jack:

NOTE: Airplanes without active noise suppression may use a three wire or four wire microphone connector. Airplanes with active noise suppression may use a three wire, four wire, or five wire microphone connector.

(c) If you found and repaired a wire problem, then do these steps:

1) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

2) If it is necessary, re-install the applicable microphone.

3) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not satisfactory, then continue.

(d) If no wire problem was found, then re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(3) Do this check of the Control Wheel PTT switch:

(a) Press and latch the control wheel PTT switch in the interphone position.

(b) Use a microphone in the flight deck to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on all of the applicable speakers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If you can hear the announcement on the speakers, do the Repair Confirmation at the end of this task.

3) If you cannot hear the announcement on the speakers, or the announcement is intermittent, then continue.

OMR 810, 812, 813

(c) Do the following steps:

1) Remove the latching balls from the control wheel PTT switches.

These are the tasks:

Control Wheel PTT Switch Removal, AMM TASK 23-51-04-000-801

Control Wheel PTT Switch Installation, AMM TASK 23-51-04-400-801

2) Replace the remote electronic unit, M1353.

NOTE: REU part number 10-62090-112 must be replaced with REU part number 10-2090-113.

These are the tasks:

Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801

Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

OMR ALL

(4) Replace the passenger address amplifier, M63.

These are the tasks:

Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801,

Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(5) Replace the remote electronic unit, M1353.

These are the tasks:

Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801,

Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Do this test of the passenger address system:

(a) Set the PA microphone selector switch on the audio control panel (ACP) to ON.

1) Make sure its light comes on.

(b) Push and hold the R/T - I/C switch on the ACP in the R/T position.

(c) Speak into the microphone.

1) Make sure you can hear the announcement on all of the applicable speakers.

(d) If you can hear the announcement on the speakers, then you corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

802. Passenger Address Volume Problem - Fault Isolation

A. Description

(1) The audio heard from the passenger address speakers is too loud or too quiet.

(2) The audio heard on the attendant and lavatory speakers is too loud or too quiet.

B. Possible Causes

(1) Master gain on PA amplifier

(2) PA Gain on Remote Electronic Unit (REU), M1353

(3) Passenger address (PA) amplifier, M63

(4) Wiring

(5) Remote electronic unit (REU), M1353

C. Related Data

(1) (SSM 73-22-31)

(2) (WDM 23-31-01)

(3) (WDM 23-31-02)

(4) (WDM 23-31-03)

(5) (WDM 23-31-04)

(6) (WDM 23-51-11)

(7) (WDM 23-51-21)

(8) (WDM 23-51-31)

D. Fault Isolation Procedure

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999; AIRPLANES WITH PA P/N 622-5342-101

(1) Adjust the master gain on the PA amplifier:

(a) Set and hold the test mode switch on the PA amplifier front panel to LEVEL.

(b) Turn the MASTER GAIN screw on the front panel clockwise or counterclockwise to the desired level.

NOTE: To get the maximum undistorted audio, turn the master gain screw clockwise until the front panel LEDs show 70.7.

(c) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

OMR ALL

(2) Replace the passenger address (PA) amplifier, M63.

These are the tasks:

Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801,

Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this check of the wiring:

(a) Remove the remote electronic unit (REU). To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel to get access to junction box J22:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

(c) Disconnect electrical connector D10918 from the engine 1 running relay, R564.

(d) Open this access panel to get access to junction box J24:

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

(e) Disconnect electrical connector D10916 from the engine 2 running relay, R563.

(f) Do a continuity check between these pins:

	REU CONNECTOR	RELAY CONNECTOR
ENGINE 1 RUNNING RELAY (R564)	**D2501A** pin F5	**D10918** pin A3
ENGINE 2 RUNNING RELAY (R563)	**D2501A** pin K5	**D10916** pin A3

(g) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.
2) Re-connect electrical connector D10918 to R564.
3) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

4) Re-connect electrical connector D10916 to R563.
5) Close this access panel:

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

6) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.
7) Do the Repair Confirmation at the end of this task.
   a) If the Repair Confirmation is not OK, then continue.

(h) If there is continuity, then do these steps and continue:

1) Re-connect electrical connector D10918 to R564.
2) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Re-connect electrical connector D10916 to R563.

4) Close this access panel:

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

(4) Install a new remote electronic unit (REU), M1353. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

E. Repair Confirmation

(1) Do this test of the passenger address system:

(a) Set the PA microphone selector switch on the audio control panel (ACP) to ON.

1) Make sure its light comes on.

(b) Push and hold the R/T - I/C switch on the ACP in the R/T position.

(c) Speak into the microphone.

1) Make sure you can hear the announcement from the passenger address speakers.

(d) If you can hear the announcement on the speakers, then you corrected the fault.

——— END OF TASK ———

803. Chime Does Not Go OFF - Fault Isolation

A. Description

(1) The chime in the Passenger Cabin does not go OFF.

B. Fault Isolation Procedure

(1) Replace the Passenger Address (PA) Amplifier, M63. These are the tasks:

- Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801
- Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801

OMR 807

(a) If the chime operates correctly, then you corrected the problem.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999; AIRPLANES WITH PA P/N 622-5342-101

(b) Do these steps to make sure that you corrected the problem:

1) On the PA Amplifier front panel, put the OPERATE/TONE Switch to the TONE position.

a) Make sure that you can hear a high chime sound on the PA Speakers.

2) Put the OPERATE/TONE Switch to the OPERATE position.

a) Make sure that the chime does not sound.

3) If you cannot hear the chime sound, then you corrected the problem.

OMR ALL

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

804. Cabin Chime Does Not Operate - Fault Isolation

A. Description

(1) The chime in the Passenger Cabin does not sound when one or more of these conditions occurs:

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(a) Passenger call button on the Passenger Service Unit (PSU) is pushed

OMR ALL

(b) Lavatory Call Button is pushed

(c) NO SMOKING Switch is set to the ON position

(d) FASTEN BELTS Switch is set to the ON position.

B. Possible Causes

(1) PA Amplifier

(2) Wiring

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) PSU

OMR ALL

(4) Lavatory Call Switch

(5) No Smoking Switch, S45

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(6) No power to the No Smoking Relay

(7) No Smoking Relay, R25

OMR ALL

(8) Fasten Belts Switch, S46

(9) No power to the Fasten Belts Relay

(10) Fasten Belts Relay, R26

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT
OMR 807			
C	8	C01708	SUPERNUMERARY LAVATORY RTN TO SEAT
C	9	C01707	SUPERNUMERARY FASTEN SEAT BELT
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Related Data

(1) WDM 23-31-01

(2) WDM 33-25-11

(3) WDM 33-27-11

(4) SSM 23-31-11

(5) SSM 33-25-11

(6) SSM 33-27-11

E. Initial Evaluation

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999; AIRPLANES WITH PA P/N 622-5342-101

(1) Do this test of the PA Amplifier:

(a) Put the OPERATE/TONE Switch on the front panel of the PA Amplifier to the TONE position.

1) Make sure you can hear a high chime sound on the Attendant, Lavatory, and PSU speakers.

(b) Put the OPERATE/TONE Switch to the OPERATE position.

(c) If you do not hear a high chime on the speakers, then do the Fault Isolation Procedure - No Chime below.

(d) If you hear a high chime on the speakers, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Do this test of the passenger call:

(a) Push the Attendant Call Button on the PSU.

1) Make sure you hear a high chime on the Attendant, Lavatory, and PSU speakers.

(b) If you do not hear a high chime on the speakers, then do the Fault Isolation Procedure - Passenger Call below.

(c) If you hear a high chime on the speakers, then continue.

OMR ALL

(3) Do this test of the Lavatory Call:

(a) Push the Call Button in the Lavatory.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure you hear a high chime on the Attendant, Lavatory, and PSU speakers.

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OMR 807

2) Make sure you hear a high chime on the Attendant and Lavatory speakers.

OMR ALL

(b) If you do not hear a high chime on the speakers, then do the Fault Isolation Procedure - Lavatory Call below.

(c) If you hear a high chime on the speakers, then continue.

(4) Do this test of the NO SMOKING Switch:

(a) Set the NO SMOKING Switch, on the P5 Overhead Panel in the Flight Compartment, to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

2) Make sure you hear a low chime on the Attendant and Lavatory speakers.

OMR ALL

(b) If you do not hear a low chime on the speakers, then do the Fault Isolation Procedure - No Smoking Chime below.

(c) If you hear a low chime on the speakers, then continue.

(5) Do this test of the FASTEN BELTS Switch:

(a) Set the FASTEN BELTS Switch, on the P5 Overhead Panel in the Flight Compartment, to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

2) Make sure you hear a low chime on the Attendant and Lavatory speakers.

OMR ALL

(b) If you do not hear a low chime on the speakers, then do the Fault Isolation Procedure - Fasten Belts Chime below.

(c) If you hear a low chime on the speakers, then there was an intermittent fault.

F. Fault Isolation Procedure - No Chime

(1) Replace the PA Amplifier. These are the tasks:

- Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801
- Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801

OMR 807

(a) If the chime operates correctly, then you corrected the problem.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999; AIRPLANES WITH PA P/N 622-5342-101

(b) Do these steps to make sure that you corrected the problem:

1) On the PA Amplifier front panel, put the OPERATE/TONE Switch to the TONE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that you can hear a high chime sound at the Attendant, Lavatory, and PSU speakers.

2) Put the OPERATE/TONE Switch to the OPERATE position.

a) If you can hear a chime, then you corrected the problem.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

G. Fault Isolation Procedure - Passenger Call

(1) Do this check of the wiring:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

(b) Remove the PSU. This is the task: Passenger Service Unit (PSU) - Removal, AMM TASK 25-23-61-000-801 or Passenger Service Unit - Removal, AMM TASK 25-23-61-000-804.

(c) Remove the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(d) Do a continuity check as follows:

PA AMP, M63 D2639B	PSU CONNECTOR
pin A11	pin 2

1) If there is not continuity, then do these steps:

a) Repair the wiring.

b) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

c) Re-install the PSU. This is the task: Passenger Service Unit (PSU) - Installation, AMM TASK 25-23-61-400-801 or Passenger Service Unit - Installation, AMM TASK 25-23-61-400-804.

d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

e) Push the Attendant Call Button on the PSU.

<1> Make sure you hear a chime over the Attendant, Lavatory, and PSU speakers.

f) If you hear a chime over the speakers, then you corrected the problem.

g) If you did not hear a chime over the speakers, then continue.

2) If there is continuity, then do these steps:

a) Install a new PSU. This is the task: Passenger Service Unit (PSU) - Installation, AMM TASK 25-23-61-400-801 or Passenger Service Unit - Installation, AMM TASK 25-23-61-400-804.

b) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

c) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

d) Push the Attendant Call Button on the PSU.

<1> Make sure you hear a chime over the Attendant, Lavatory, and PSU speakers.

e) If you hear a chime over the speakers, then you corrected the problem.

H. Fault Isolation Procedure - Lavatory Call

(1) Do this test of the Lavatory Call:

(a) Push the Call Button in the Lavatory.

1) Make sure the Amber Light on the Exit locator Signs comes ON.

(b) If the light does not come ON, then do these steps:

1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

2) Replace the Lavatory Call Switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) To replace it, do this task: Lavatory Call Light - Light/Switch Replacement, AMM TASK 33-27-00-960-802.

3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

4) Push the Call Button in the Lavatory.

a) Make sure you hear a high chime on the Attendant, Lavatory, and PSU speakers.

5) If you hear a high chime on the speakers, then you corrected the problem.

6) If you do not hear a high chime on the speakers, then continue.

(c) If the light comes ON, then continue.

(2) Do this check of the wiring (WDM 33-27-11):

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

(b) Disconnect applicable connector from the Lavatory Module.

(c) Remove the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(d) Do a continuity check as follows:

PA AMP, M63 D2639B		LAV MODULE CONNECTOR
pin A11	. . . . .	pin A2

1) If there is not continuity, then do these steps:

2) Repair the wiring.

3) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

4) Re-connect connector to the Lavatory Module.

5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

6) Push the Lavatory Call Button.
   a) Make sure you hear a chime over the Attendant, Lavatory, and PSU speakers.
7) If you hear a chime over the speakers, then you corrected the problem.

OMR ALL

I. Fault Isolation - No Smoking Chime

(1) Do this check of the NO SMOKING Switch (WDM 33-25-11):

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

(b) Disconnect connector D1620 from the No Smoking Relay, R25 in the P6 Panel.

(c) Set the NO SMOKING Switch, on the P5 Overhead Panel in the Flight Compartment, to the ON position.

(d) Do a continuity check as follows:

NO SMOKING
SIGN RELAY,
R25
D1620
pin X2 . . . . . GND

1) If there is not continuity, then do these steps:
   a) Replace the NO SMOKING Switch, S45.
   b) Make sure the NO SMOKING Switch is in the OFF position.
   c) Re-connect connector D1620 to the No Smoking Relay, R25 in the P6 Panel.
   d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 807 (Continued)

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C01706	FWD CABIN SIGN CONT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

e) Set the NO SMOKING Switch to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<1> Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<2> Make sure you hear a low chime on the Attendant and Lavatory speakers.

OMR ALL

f) If you hear a chime on the speakers, then you corrected the problem.

g) If you did not hear a chime on the speakers, then do these steps:

<1> Set the NO SMOKING Switch to the OFF position.

<2> Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

<3> Repair the wiring between pin X2 of connector D1620 on the No Smoking Relay, R25 and the NO SMOKING Switch, S45.

<4> Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

<5> Set the NO SMOKING Switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<6> Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

h) If you hear a chime on the speakers, then you corrected the problem.

2) If there is continuity, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Do this check for power to the No Smoking Relay (WDM 33-25-11):

(a) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

(b) Do a voltage check at these pins of connector at the No Smoking Relay, R25 and Structure Ground:

NO SMOKING SING RELAY, R25 D1620

pin X1 . . . . . GND
pin B2 . . . . . GND

1) If there is not voltage, then do these steps:

a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

b) Repair the wiring between the No Smoking Relay, R25 and this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

c) Re-connect electrical connector D1620 to the No Smoking Relay, R25 in the P6 Panel.

d) Set the NO SMOKING Switch to the OFF position.

e) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

f) Set the NO SMOKING Switch to the ON position.

<1> Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

g) If you hear a chime on the speakers, then you corrected the problem.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If there is voltage, then continue.

OMR ALL

(3) Do this check of the wiring (WDM 33-25-11):

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

(b) Remove the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(c) Do a continuity check as follows:

		NO SMOKING SIGN RELAY,
PA AMP, M63		**R25**
D2639B		**D1620**
pin D11	. . . . .	pin B1

1) If there is not continuity, then do these steps:

a) Repair the wiring.

b) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

c) Re-connect connector D1620 to the No Smoking Relay, R25 in the P6 Panel.

d) Set the NO SMOKING Switch to the OFF position.

e) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

f) Set the NO SMOKING Switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<1> Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<2> Make sure you hear a chime on the Attendant and Lavatory speakers.

OMR ALL

g) If you hear a chime on the speakers, then you corrected the problem.

2) If there is continuity, then continue.

a) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Replace the No Smoking Relay:

(a) Replace the No Smoking Relay, R25.

1) Re-connect connector D1620 to the No Smoking Relay, R25 in the P6 Panel.

2) Set the NO SMOKING Switch to the OFF position.

3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

4) Set the NO SMOKING Switch to the ON position.

a) Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

5) If you hear a chime on the speakers, then you corrected the problem.

OMR ALL

J. Fault Isolation - Fasten Belts Chime

(1) Do this check of the FASTEN BELTS Switch (WDM 33-25-11):

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

(b) Disconnect connector D1622 from the Fasten Belts Relay, R26 in the P6 Panel.

(c) Set the FASTEN BELTS Switch, on the P5 Overhead Panel in the Flight Compartment, to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Do a check for continuity as follows:

SEAT BELT
RELAY, R26
D1622
pin X2 . . . . . GND

1) If there is not continuity, then do these steps:
   a) Replace the FASTEN BELTS Switch, S46.
   b) Make sure the FASTEN BELTS Switch is in the OFF position.
   c) Re-connect connector D1622 to the Fasten Belts Relay, R26 in the P6 Panel.
   d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

   e) Set the FASTEN BELTS Switch to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<1> Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<2> Make sure you hear a low chime on the Attendant and Lavatory speakers.

OMR ALL

   f) If you hear a chime on the speakers, then you corrected the problem.
   g) If you did not hear a chime on the speakers, then do these steps:
      <1> Set the FASTEN BELTS Switch to the OFF position.
      <2> Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

<3> Repair the wiring between pin X2 of connector D1622 on the Fasten Belts Relay, R26 and the FASTEN BELTS Switch, S46.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<4> Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

<5> Set the FASTEN BELTS Switch to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<6> Make sure you hear a low chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<7> Make sure you hear a low chime on the Attendant and Lavatory speakers.

OMR ALL

h) If you hear a chime on the speakers, then you corrected the problem.

2) If there is continuity, then continue.

(2) Do this check for power to the Fasten Belts Relay (WDM 33-25-11):

(a) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

(b) Do a voltage check at these pins of connector at the Fasten Belts Relay, R26 and Structure Ground:

SEAT BELT RELAY, R26 D1622		**VOLTAGE**
pin X1	GND	28V DC
pin B2	GND	28V DC

1) If there is not voltage, then do these steps:

a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 807 (Continued)

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	11	C01706	FWD CABIN SIGN CONT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

b) Repair the wiring between the Fasten Belts Relay, R26 and these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name

OMR 807

D	11	C01706	FWD CABIN SIGN CONT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

c) Re-connect connector D1622 to the Fasten Belts Relay, R26 in the P6 Panel.

d) Set the FASTEN BELTS Switch to the OFF position.

e) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name

OMR 807

D	11	C01706	FWD CABIN SIGN CONT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

OMR ALL

f) Set the FASTEN BELTS Switch to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<1> Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<2> Make sure you hear a chime on the Attendant and Lavatory speakers.

OMR ALL

g) If you hear a chime on the speakers, then you corrected the problem.

2) If there is voltage, then continue.

(3) Do this check of the wiring (WDM 33-25-11):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

(b) Remove the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(c) Do a continuity check as follows:

PA AMP, M63		**SEAT BELT RELAY, R26**
D2639B		**D1622**
pin C11	. . . . .	pin B1

1) If there is not continuity, then do these steps:
   a) Repair the wiring.
   b) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.
   c) Re-connect connector D1622 to the Fasten Belts Relay, R26 in the P6 Panel.
   d) Set the FASTEN BELTS Switch to the OFF position.
   e) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

   f) Set the FASTEN BELTS Switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

<1> Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

<2> Make sure you hear a chime on the Attendant and Lavatory speakers.

OMR ALL

g) If you hear a chime on the speakers, then you corrected the problem.

2) If there is continuity, then do this step and continue:

a) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

(4) Replace the Fasten Belts Relay:

(a) Replace the Fasten Belts Relay, R26.

1) Re-connect connector D1622 to the Fasten Belts Relay, R26 in the P6 Panel.

2) Set the FASTEN BELTS Switch to the OFF position.

3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
OMR 807			
D	11	C01706	FWD CABIN SIGN CONT
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	11	C00442	PASSENGER CABIN SIGNS PASS SIGN CONT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
OMR ALL			
D	4	C00082	COMMUNICATIONS PA AMPL BAT

4) Set the FASTEN BELTS Switch to the ON position.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

a) Make sure you hear a chime on the Attendant, Lavatory, and PSU speakers.

OMR 807

b) Make sure you hear a chime on the Attendant and Lavatory speakers.

OMR ALL

5) If you hear a chime on the speakers, then you corrected the problem.

——— END OF TASK ———

805. Passenger Address System Problem - Fault Isolation

A. Description

(1) No audio, intermittent, or poor quality audio is heard from the PA Amplifier Speakers during an announcement.

B. Possible Causes

(1) Attendant's Handset

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) FWD (AFT) Cabin Attendant's Control Panel, P13 (P14)

OMR ALL

(3) Wiring

(4) PA Amplifier, M63

(5) Remote Electronics Unit (REU), M1353

OMR 701-799, 802, 805, 809, 810, 812, 813

(6) Pre-Recorded Announcement Machine (PRAM), M1276

OMR ALL

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT

OMR 701-799, 802, 805, 809, 810, 812, 813

C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

OMR ALL

D	4	C00082	COMMUNICATIONS PA AMPL BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name

OMR 807

C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC (INOP)

OMR ALL

D. Related Data

(1) SSM 23-31-11

(2) SSM 23-31-12

(3) SSM 23-31-13

(4) SSM 23-31-14

(5) WDM 23-31-01

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(6) WDM 23-31-02

(7) WDM 23-31-03

(8) WDM 23-31-04

(9) WDM 23-31-06

E. Fault Isolation Procedure

OMR 701-799, 802, 805, 809, 810, 812, 813

(1) Open, wait a minimum of 15 seconds, then close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

(2) Do this check of the Attendant's Handset:

(a) Use an Attendant's Handset, other than where the problem was reported, to make an announcement over the Passenger Address System (PAS).

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure you can hear the announcement on the Attendant, PSU, and Lavatory Speakers.

OMR 807

2) Make sure you can hear the announcement on the Attendant and Lavatory speakers.

OMR ALL

(b) If you can hear the announcement on the speakers, then replace the Attendant's Handset where the problem was reported. These are the tasks:

- Attendant Handset Removal, AMM TASK 23-42-01-000-801
- Attendant Handset Installation, AMM TASK 23-42-01-400-803

1) Do the Repair Confirmation at the end of this task.

(c) If you cannot hear the announcement on the speakers, or the announcement is intermittent, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Replace the applicable Cabin Attendant Control Panel (ACP). These are the tasks:

- Attendant's Panel and Components Removal, AMM TASK 25-25-11-000-801 or Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802
- Attendant's Panel and Components Installation, AMM TASK 25-25-11-400-801 or Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

(4) Do this check of the wiring between the PA Amplifier and the Attendant's Handset:

(a) Remove the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Use the WDM 23-31-01 to identify connector and pins of the Lavatory for the interface between the ACP and PA Amplifier connector D2639B, pin A2 and pin B2.

OMR ALL

(c) Disconnect the applicable Lavatory connector(s).

(d) Do a continuity check between the PA Amplifier and the ACP.

1) If there is no continuity between the pins, then do these steps:

a) Repair the wiring.

b) Re-connect the Lavatory connector(s).

c) Re-install the PA Amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If there is continuity between the pins, then do this step and continue:

a) Re-connect the Lavatory connector(s).

(5) Replace the PA Amplifier, M63. These are the tasks:

- Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801
- Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the REU, M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 809, 810, 812, 813

(7) Replace the PRAM, M1276. These are the tasks:

- Pre-Recorded Announcement Machine (PRAM) Removal, AMM TASK 23-31-07-000-801
- Pre-Recorded Announcement Machine (PRAM) Installation, AMM TASK 23-31-07-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Do a wiring check between the PRAM and the PA Amplifier (Look at INPUT 1 PTT signal (SSM 23-31-12).

NOTE: The PRAM has an effect on the PA Announcements made from the handsets, but not on the Flight Deck Handset. The PAS gives the highest priority for input 1 (pilot input). If the Push-To-Talk (PTT) 1 is pushed, all PA Handsets and PRAM Messages and Boarding Music will be overridden.

(a) If you find a problem, repair the wiring.

(b) Do the Repair Confirmation at the end of this task.

OMR ALL

F. Repair Confirmation

(1) Do this check of the PAS:

(a) Set the Attendant's Handset to the PA Mode.

(b) Push the PTT Button on the Attendant's Handset.

(c) Speak into the Attendant's Handset.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure that you can hear the announcement on the other Attendant, PSU, and Lavatory Speakers.

OMR 807

2) Make sure that you can hear the announcement on the other Attendant and Lavatory Speakers.

OMR ALL

(d) If you can hear the announcement on the speakers, then you corrected the problem.

(e) If you cannot hear the announcement on the speakers, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. Passenger Address Speaker Problem - Fault Isolation

A. Description

(1) No, intermittent, or poor quality audio can be heard on one of these passenger address speakers during an announcement:

(a) Attendant

(b) Lavatory

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) Passenger service unit.

OMR ALL

B. Possible Causes

(1) Speaker

(2) Wiring

(3) Remote electronic unit, M1353

C. Related Data

(1) SSM 23-31-11

(2) SSM 23-31-12

(3) SSM 23-31-13

(4) SSM 23-31-14

(5) WDM 23-31-01

(6) WDM 23-31-02

(7) WDM 23-31-03

(8) WDM 23-31-04

(9) WDM 23-31-06

D. Initial Evaluation

(1) Do this test of the passenger address system:

(a) Use a microphone in the flight deck to make an announcement on the passenger address system.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) Make sure you can hear the announcement on the attendant, lavatory, and passenger service unit speakers.

OMR 807

2) Make sure you can hear the announcement on the attendant and lavatory speakers.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) If you cannot hear the announcement on both forward attendant speakers or both aft attendant speakers, then do the Fault Isolation Procedure - Both Attendant Speakers below.

OMR ALL

(c) If you cannot hear the announcement, or the quality of the audio is bad, on an attendant speaker, then do the Fault Isolation Procedure - One Attendant Speaker below.

(d) If you cannot hear the announcement, or the quality of the audio is bad, on a lavatory speaker, then do the Fault Isolation Procedure - Lavatory Speaker below.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) If you cannot hear the announcement, or the quality of the audio is bad, on a passenger service unit (PSU) speaker, then do the Fault Isolation Procedure - PSU Speaker below.

OMR ALL

(f) If you hear the announcement on all speakers, then there was an intermittent fault.

E. Fault Isolation Procedure - Both Attendant Speakers

(1) Replace the remote electronic unit (REU), M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Use a microphone in the flight deck to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on both attendant's speakers.

(b) If you heard the announcement on both attendant's speakers, then you corrected the fault.

(c) If you did not hear the announcement on the attendant's speaker, then continue.

(2) Do this check of the wiring:

(a) Remove the remote electronic unit (REU), M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Remove the passenger address (PA) amplifier, M63. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(c) Remove the attendant speakers. This is the task: Attendant Speaker - Removal, AMM TASK 23-31-03-000-801.

NOTE: Record the wiring terminal connections. When you re-install or replace the attendant speakers, put the electrical wires to the same terminals.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Do a check for continuity between connector D2501A of the REU, or connector D2639B of the PA amplifier, and the attendant speakers:

Table 201/23-31-00-993-801

ATTENDANT SPEAKER	PA OR REU CONNECTOR		ATTENDANT SPEAKER
RIGHT FORWARD ATTENDANT'S SPEAKER (M1464)	D2501A		M1464
	pin H2	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire
LEFT FORWARD ATTENDANT'S SPEAKER (M157)	D2501A		M157
	pin H2	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire
RIGHT AFT ATTENDANT'S SPEAKER (M1212)	D2501A		M1212
	pin C1	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire
LEFT AFT ATTENDANT'S SPEAKER (M1213)	D2501A		M1213
	pin C1	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire

OMR ALL

(e) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.
2) Re-install the speakers. This is the task: Attendant Speaker - Installation, AMM TASK 23-31-03-400-801.

   NOTE: Make sure that the electrical connectors are connected to the same terminals that they were removed from.

3) Re-install the PA amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.
4) Re-install the REU. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.
5) Use a microphone in the flight deck to make an announcement over the passenger address system.
   a) Make sure you can hear the announcement on both attendant's speakers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

6) If you heard the announcement on both attendant's speakers, then you corrected the fault.

F. Fault Isolation Procedure - One Attendant Speaker

(1) Replace the attendant speaker. These are the tasks:

- Attendant Speaker - Removal, AMM TASK 23-31-03-000-801
- Attendant Speaker - Installation, AMM TASK 23-31-03-400-801

(a) Use a microphone in the flight deck to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on the attendant's speaker.

(b) If you heard the announcement on the attendant's speaker, then you corrected the fault.

(c) If you did not hear the announcement on the attendant's speaker, then continue.

(2) Do this check of the wiring.

(a) Remove the remote electronic unit (REU), M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Remove the passenger address (PA) amplifier, M63. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(c) Remove the attendant speakers. This is the task: Attendant Speaker - Removal, AMM TASK 23-31-03-000-801.

NOTE: Record the wiring terminal connections. When you re-install or replace the attendant speakers, put the electrical wires to the same terminals.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Do a check for continuity between connector D2501A of the REU, or connector D2639B of the PA amplifier, and the attendant speakers:

Table 202/23-31-00-993-803

ATTENDANT SPEAKER	PA OR REU CONNECTOR		ATTENDANT SPEAKER
RIGHT FORWARD ATTENDANT'S SPEAKER (M1464)	D2501A		M1464
	pin H2	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire
LEFT FORWARD ATTENDANT'S SPEAKER (M157)	D2501A		M157
	pin H2	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire
RIGHT AFT ATTENDANT'S SPEAKER (M1212)	D2501A		M1212
	pin C1	----------	1K/2K/4K terminal wire
	D2639B		

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 202/23-31-00-993-803 (Continued)

ATTENDANT SPEAKER	PA OR REU CONNECTOR		ATTENDANT SPEAKER
	pin B15	----------	COM terminal wire
LEFT AFT ATTENDANT'S SPEAKER (M1213)	D2501A		M1213
	pin C1	----------	1K/2K/4K terminal wire
	D2639B		
	pin B15	----------	COM terminal wire

OMR ALL

(e) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.

2) Re-install the speakers. This is the task: Attendant Speaker - Installation, AMM TASK 23-31-03-400-801.

NOTE: Make sure that the electrical connectors are connected to the same terminals that they were removed from.

3) Re-install the PA amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

4) Re-install the REU. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

5) Use a microphone in the flight deck to make an announcement over the passenger address system.

a) Make sure you can hear the announcement on both attendant's speakers.

6) If you heard the announcement on both attendant's speakers, then you corrected the fault.

G. Fault Isolation Procedure - Lavatory Speaker

(1) Replace the lavatory speaker. These are the tasks:

- Lavatory Speaker Removal, AMM TASK 23-31-05-000-801
- Lavatory Speaker Installation, AMM TASK 23-31-05-400-801

(a) Use a microphone in the flight deck to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on the lavatory speaker.

(b) If you heard the announcement on the lavatory speaker, then you corrected the fault.

(c) If you did not hear the announcement on the lavatory speaker, then continue.

(2) Do this check of the wiring.

(a) Remove the passenger address (PA) amplifier, M63. To remove it, do this task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(b) Disconnect electrical connector D3732 from the left forward lavatory module.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) Disconnect electrical connector D3738 from the left aft lavatory module.

(d) Disconnect electrical connector D3740 from the right aft lavatory module.

OMR 901-999

(e) Disconnect electrical connector D12596 from the left aft lavatory module.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(f) Do a check for continuity between the PA amplifier and the lavatory module connectors:

Table 203/23-31-00-993-806

LAVATORY SPEAKER	PA AMP CONNECTOR		LAVATORY MODULE CONNECTOR
LEFT FORWARD LAVATORY SPEAKER	D2639B		D3732
	pin A15	----------	pin 13
	pin B15	----------	pin 14
LEFT AFT LAVATORY SPEAKER	D2639B		D3738
	pin A15	----------	pin 13
	pin B15	----------	pin 14
RIGHT AFT LAVATORY SPEAKER	D2639B		D3740
	pin A15	----------	pin 13
	pin B15	----------	pin 14
OMR 901-999			
LEFT AFT LAVATORY SPEAKER	D2639B		D12596
	pin A15	----------	pin 13
	pin B15	----------	pin 14

OMR ALL

(g) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.

2) Re-connect electrical connector D3732 to the left forward lavatory module.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

3) Re-connect electrical connector D3738 to the left aft lavatory module.

4) Re-connect electrical connector D3740 to the right aft lavatory module.

OMR 901-999

5) Re-connect electrical connector D12596 to the left aft lavatory module.

OMR ALL

6) Re-install the PA amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

7) Use a microphone in the flight deck to make an announcement over the passenger address system.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure you can hear the announcement on the lavatory speaker.

8) If you heard the announcement on the lavatory speaker, then you corrected the fault.

H. Fault Isolation Procedure - PSU Speaker

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Replace the PSU speaker.

(a) These are the tasks:

1) Passenger Service Unit (PSU) Speaker - Removal, AMM TASK 23-31-02-000-801

2) Passenger Service Unit (PSU) Speaker - Installation, AMM TASK 23-31-02-400-801

(b) Use a microphone in the flight deck to make an announcement over the passenger address system.

1) Make sure you can hear the announcement on the PSU speaker.

(c) If you heard the announcement on the PSU speaker, then you corrected the fault.

(d) If you did not hear the announcement on the PSU speaker, then continue.

OMR 807

(2) Do this check of the wiring.

(a) Remove the passenger address (PA) amplifier, M63. This is the task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(b) Remove the ASU or LSU speaker. This is the task: Attendant Speaker - Removal, AMM TASK 23-31-03-000-801 or Lavatory Speaker Removal, AMM TASK 23-31-05-000-801.

(c) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.

2) Re-install the ASU or LSU speaker. This is the task: Attendant Speaker - Installation, AMM TASK 23-31-03-400-801 or Lavatory Speaker Installation, AMM TASK 23-31-05-400-801.

3) Re-install the PA amplifier. This is the task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

4) Use a microphone in the flight deck to make an announcement over the passenger address system.

a) Make sure you can hear the announcement on the PSU speaker.

5) If you heard the announcement on the PSU speaker, then you corrected the fault.

——— END OF TASK ———

OMR 701-799, 802, 805, 809, 810, 812, 813

807. Boarding Music Problem - Fault Isolation

A. Description

(1) The pre-recorded announcement machine (PRAM) does not operate, or the quality of the audio is bad.

B. Possible Causes

(1) Pre-recorded announcement machine (PRAM), M1276 or M1966

(2) Passenger address (PA) amplifier, M63

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Initial Evaluation

(1) Do a test of the boarding music operation:

(a) Push the ENT button on the PRAM.

(b) Push the MUSIC button.

(c) Push '1' on the PRAM to select channel 1.

(d) Push the START button.

(e) Turn the volume control to the middle position.

1) Make sure you hear clear music on all of the PSU, lavatory, and PA speakers.

(f) Turn the volume control clockwise.

1) Make sure the volume level increases at all of the speakers.

(g) Turn the volume control counterclockwise.

1) Make sure the volume level decreases at all of the speakers.

(h) Turn the volume control to the middle position.

(i) Push the STOP button on the tape reproducer.

1) Make sure the boarding music stops.

(j) If the boarding music system did not pass any of the steps above, then do the Fault Isolation Procedure below.

(k) If the boarding music system operated correctly, then there was an intermittent fault.

E. Fault Isolation Procedure

(1) Do this test of the boarding music operation:

(a) Set the forward attendant's handset to the PA mode.

(b) Push the PTT button on the forward attendant's handset.

(c) Make an announcement from the forward attendant's handset.

1) Make sure you can hear the announcement on the aft attendant, lavatory, and passenger service unit speakers.

(d) If you heard an announcement on the speakers, then do these steps:

1) Replace the pre-recorded announcement machine (PRAM), M1276 or M1966.

These are the tasks:

Pre-Recorded Announcement Machine (PRAM) Removal, AMM TASK 23-31-07-000-801,

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813 (Continued)

Pre-Recorded Announcement Machine (PRAM) Installation, AMM TASK 23-31-07-400-801.

2) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not satisfactory, then continue.

(e) If you did not hear an announcement on the speakers, then do these steps:

1) Replace the passenger address (PA) amplifier, M63.

These are the tasks:

Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801,

Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

2) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not satisfactory, then continue.

(2) Do this check of the wiring:

(a) Remove the passenger address (PA) amplifier, M63. To remove it, do this task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

(b) Remove the pre-recorded announcements machine (PRAM). To remove it, do this task: Pre-Recorded Announcement Machine (PRAM) Removal, AMM TASK 23-31-07-000-801.

(c) Do a check for continuity between these pins of connector D2639B on the PA amplifier, and connector D1507 of the pre-recorded announcement/boarding music tape reproducer:

D2639B		D1507
pin D5	. . . . .	pin 1
pin A5	. . . . .	pin 18
pin B5	. . . . .	pin 17

(d) If there is not continuity between the pins, then do these steps:

1) Repair the wiring.

2) Re-install the PRAM. To install it, do this task: Pre-Recorded Announcement Machine (PRAM) Installation, AMM TASK 23-31-07-400-801.

3) Re-install the PA amplifier. To install it, do this task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

4) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Do a test of the boarding music operation:

(a) Push the ENT button on the PRAM.

(b) Push the MUSIC button.

(c) Push '1' on the PRAM to select channel 1.

(d) Push the START button.

(e) Turn the volume control to the middle position.

1) Make sure you hear clear music on all of the PSU, lavatory, and PA speakers.

(f) Turn the volume control clockwise.

1) Make sure the volume level increases at all of the speakers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813 (Continued)

(g) Turn the volume control counterclockwise.

1) Make sure the volume level decreases at all of the speakers.

(h) Turn the volume control to the middle position.

(i) Push the STOP button on the tape reproducer.

1) Make sure the boarding music stops.

(j) If the boarding music system operated correctly, then you corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Passenger Video Monitor Problem - Fault Isolation

A. Description

(1) This procedure is for video monitors that have one of these problems:

(a) The video monitor has been damaged.

(b) Video monitor picture quality is unacceptable.

B. Fault Isolation Procedure

(1) Replace the video monitor.

These are the tasks:

PSU-Mounted LCD Monitor Removal, AMM TASK 23-32-13-000-801,

PSU-Mounted LCD Monitor Installation, AMM TASK 23-32-13-400-801.

——— END OF TASK ———

802. Passenger Video Monitor Does Not Operate - Fault Isolation

A. Description

(1) One or several video monitors do not operate.

B. Possible Causes

OMR 805, 809, 810, 812, 813

(1) VSCU programming

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Video monitor

(3) Video distribution unit (VDU)

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	5	C01450	ENT VID CONT CENTER DC
C	7	C01451	ENT VID CONT CENTER AC
D	5	C01452	VIDEO 1
D	6	C01453	VIDEO 2
D	7	C01454	VIDEO 3
D	8	C01455	VIDEO 4

D. Related Data

(1) (WDM 23-32-3X,-4X,-5X,-6X)

(2) (SSM 23-32-3X,-4X,-5X,-6X)

E. Initial Evaluation

(1) Do this test of the passenger video system:

(a) Push the power button on the video system control unit (VSCU) to the on position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make the applicable selections to display the video from VR 1 on all cabin video monitors.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR 1.

(c) Make sure the applicable video monitors lower from the stowed position.

(d) Make sure that the video from VR 1 on all of the video monitors.

(e) If you see video from VR 1 on all but one video monitor, then do the Fault Isolation Procedure - Single Video Monitor, below.

(f) If you see the video from VR 1 on all but one row of the monitors or all but two adjacent monitors, then do the Fault Isolation Procedure - Single Row/Adjacent Video Monitors, below.

(g) If you do not see video from VR 1 on the forward zone of video monitors, then do the Fault Isolation Procedure - Forward Monitors, below.

(h) If you do not see video from VR 1 on the aft zone of video monitors, then do the Fault Isolation Procedure - Aft Monitors, below.

(i) If you do not see video from VR 1 on several video monitors, then do the Fault Isolation Procedure - Multiple Video Monitors, below.

(j) If you see video from VR 1 on all video monitors, then there was an intermittent fault.

F. Fault Isolation Procedure - Single Video Monitor

NOTE: You must do the Initial Evaluation before doing this procedure.

OMR 805, 809, 810, 812, 813

(1) Do a check of the video system configuration. To check the configuration, do this task: Video System Configuration, AMM TASK 23-32-02-800-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Move the affected video monitor to the location of a known good monitor:

NOTE: If the affected video monitor is the forward (retractable) LCD monitor, then skip to the next subtask.

(a) These are the procedures to move the affected monitor. These are the tasks: PSU-Mounted LCD Monitor Removal, AMM TASK 23-32-13-000-801PSU-Mounted LCD Monitor Installation, AMM TASK 23-32-13-400-801.

(b) Repeat the initial evaluation.

(c) If you do not see the video on the affected monitor then re-install the known good monitor in it's original location:

1) These are the procedures to re-install the monitor. These are the tasks: PSU-Mounted LCD Monitor Removal, AMM TASK 23-32-13-000-801PSU-Mounted LCD Monitor Installation, AMM TASK 23-32-13-400-801.

(d) Install a replacement video monitor at the affected monitor's original location.

1) This is the procedure to install the monitor. Do this task: PSU-Mounted LCD Monitor Installation, AMM TASK 23-32-13-400-801.

(e) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Do these steps to check the video monitor connectors for damage:

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(a) Disconnect the electrical connectors from the affected video monitor.

(b) Make sure that there is no damage to the electrical connectors or the pins.

(c) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Reconnect the electrical connectors to the video monitor.

3) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(4) Do these steps to check the video distribution unit (VDU):

(a) Remove the video distribution unit (VDU) for the affected video monitor (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(b) Make sure that there is no damage to the electrical connectors or the pins.

(c) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the VDU. To install it, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

3) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation was not OK, then continue.

(d) If there is no damage to the connectors or pins, then do these steps:

1) Install a new VDU in the affected VDU's original location. To install it, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

2) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(5) Do this check of the wiring:

(a) Remove the affected video monitor. To remove it, do this task: PSU-Mounted LCD Monitor Removal, AMM TASK 23-32-13-000-801.

(b) Remove the VDU for the affected monitor (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(c) Do a check for continuity of all wiring between the video monitor and the VDU (WDM 23-32-XX).

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the VDU. To install it, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

3) Re-install the video monitor. To install it, do this task: PSU-Mounted LCD Monitor Installation, AMM TASK 23-32-13-400-801.

4) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Single Row/Adjacent Video Monitors

NOTE: You must do the Initial Evaluation before doing this procedure.

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OMR 805, 809, 810, 812, 813

(1) Do a check of the video system configuration. To check the configuration, do this task: Video System Configuration, AMM TASK 23-32-02-800-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Replace the video distribution unit (VDU) for the affected video monitors.

These are the tasks:

Video Distribution Unit Removal, AMM TASK 23-32-03-000-801,

Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Do this check of the wiring:

(a) Remove the VDU for the affected video monitors (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(b) Remove the VDU for the forward video monitors (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(c) Make sure that there is no damage to the electrical connectors or the pins on both VDUs.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install both VDUs. To install them, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

3) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(e) If there is no damage to the connectors or pins, then do these steps:

1) Do a check for continuity of the wiring between the VDUs.

2) If there is not continuity then do these steps:

a) Repair the wiring.

b) Re-install both VDUs. To install them, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

c) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - Forward Monitors

OMR 805, 809, 810, 812, 813

(1) Do a check of the video system configuration. To check the configuration, do this task: Video System Configuration, AMM TASK 23-32-02-800-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

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OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Replace the first video distribution unit (VDU) of the forward zone.

These are the tasks:

Video Distribution Unit Removal, AMM TASK 23-32-03-000-801,

Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Do this check of the wiring:

(a) Remove the first VDU, M1934. To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(b) Remove the video system control unit (VSCU). To remove it, do this task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Do a check for continuity of the wiring between the VSCU and the VDU.

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the VSCU. To install it, do this task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VDU. To install it, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

4) Do the Repair Confirmation at the end of this task.

I. Fault Isolation Procedure - Aft Monitors

OMR 805, 809, 810, 812, 813

(1) Do a check of the video system configuration. To check the configuration, do this task: Video System Configuration, AMM TASK 23-32-02-800-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Replace the first video distribution unit (VDU) of the aft zone.

These are the tasks:

Video Distribution Unit Removal, AMM TASK 23-32-03-000-801,

Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Do this check of the wiring:

(a) Remove the first VDU of the aft zone. To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.

(b) Remove the video system control unit (VSCU). To remove it, do this task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Do a check for continuity of the wiring between the VSCU and the VDU.

(d) If there is not continuity, then do these steps:

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1) Repair the wiring.
2) Re-install the VSCU. To install it, do this task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.
3) Re-install the VDU. To install it, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.
4) Do the Repair Confirmation at the end of this task.

J. Fault Isolation Procedure - Multiple Video Monitors

OMR 805, 809, 810, 812, 813

(1) Do a check of the video system configuration. To check the configuration, do this task: Video System Configuration, AMM TASK 23-32-02-800-801.
   (a) Do the Repair Confirmation at the end of this task.
      1) If the Repair Confirmation is not OK, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Replace the video distribution unit (VDU) for the affected video monitor closest to the front of the airplane (WDM 23-32-XX).

These are the tasks:

Video Distribution Unit Removal, AMM TASK 23-32-03-000-801,

Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.

   (a) Do the Repair Confirmation at the end of this task.
      1) If the Repair Confirmation is not OK, then continue.

(3) Do this check of the wiring:
   (a) Remove the VDU for the affected video monitor closest to the front of the airplane (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.
   (b) Remove the VDU for the video monitor in front of the affected video monitors (WDM 23-32-XX). To remove it, do this task: Video Distribution Unit Removal, AMM TASK 23-32-03-000-801.
   (c) Do a check for continuity of the wiring between both VDUs.
   (d) If there is not continuity, then do these steps:
      1) Repair the wiring.
      2) Re-install both VDUs. To install them, do this task: Video Distribution Unit Installation, AMM TASK 23-32-03-400-801.
      3) Do the Repair Confirmation at the end of this task.

K. Repair Confirmation

(1) Do this test of the passenger video system:
   (a) Make the applicable selections to display the video from VR 1 on the cabin monitors.

   NOTE: You will make the selections on the VSCU and if necessary, on the VR 1.

      1) Make sure the applicable video monitors lower from the stowed position.
   (b) Look for the video display on all of the video monitors.
   (c) If you see the video on all of the video monitors then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

803. Video Reproducer Problems - Fault Isolation

A. Description

(1) A Video Reproducer (VR) has one of these problems:

(a) VR does not turn on.

(b) VR controls do not operate.

B. Possible Causes

(1) Video Reproducers

(2) Connectors or pins

(3) Video System Control Unit (VSCU), M2065

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	5	C01450	ENT VID CONT CENTER DC
C	7	C01451	ENT VID CONT CENTER AC

D. Related Data

(1) WDM 23-32-11

(2) SSM 23-32-11

E. Initial Evaluation

(1) Do this test of the VR:

(a) Push the power button on the VSCU to the on position.

1) Make sure the power light on the VR comes on.

(b) If the light on the VR does not come on, then do the Fault Isolation Procedure - VR Does Not Turn On below.

(c) If the light does come on, then continue.

(d) Put a media source (tape, Compact Disc (CD), or DVD) into the VR if necessary.

(e) Make the applicable selections to display the video from VR 1 on the VSCU preview monitor.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR 1.

1) Make sure that you see video from the VR on the VSCU preview monitor.

(f) If the VR plays but you do not see video from the VR on the VSCU preview monitor, then do the Fault Isolation Procedure - No Video From VR below.

(g) If the VR plays but you do not see video from the VR on the monitors, then do the Fault Isolation Procedure - No Video From VR below.

(h) If the VR does not play, or any other attempt to control the media source (tape, CD, or DVD) does not work, then do the Fault Isolation Procedure - VR Controls Do Not Operate below.

(i) If you see video from the VR on the VSCU preview monitor, then there was an intermittent fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) If you see video from the VR on the monitors, then there was an intermittent fault.

F. Fault Isolation Procedure - VR Does Not Turn On

OMR 701-799, 802

(1) Do this check for power:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	7	C01451	ENT VID CONT CENTER AC

(c) Do a check for 115V AC on pins 1 and 2 of the VR connector.

(d) If you do not find the 115V AC, then do these steps:

1) Examine and repair the wiring between the VSCU and the VR.

(e) If you found and repaired a wire problem, then do these steps:

1) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

2) Do the Repair Confirmation at the end of this procedure.

(f) If the 115V AC is OK, then do a check for 28V DC on pins 3 and 4 of the VR connector.

(g) If there is not 28V DC, then do these steps:

1) Repair the wiring between the VSCU and the VR.

2) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

3) Do the Repair Confirmation at the end of this procedure.

(h) If there is 28V DC, then continue.

OMR 805, 809, 810, 812, 813

(2) Do this check for power:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	7	C01451	ENT VID CONT CENTER AC

(c) Do a check for 115V AC on pins 3 and 4 of the VR connector.

(d) If there is not 115V AC, then do these steps:

1) Repair the wiring between the VSCU and the VR.

2) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

3) Do the Repair Confirmation at the end of this procedure.

(e) If there is 115V AC, then continue.

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OMR 701-799, 802, 805, 809, 810, 812, 813

(3) Move the VR to another location:

(a) Remove a working VR from another location. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Install the affected VR at the working VR's location. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

(c) Make the selections to show video from the affected VR, in the new location, on the VSCU preview monitor.

1) Make sure you see video from the VR on the VSCU preview monitor.

(d) If you do not see video from the VR on the VSCU preview monitor, then do these steps:

1) Remove the affected VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

2) Re-install the working VR into its original location. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

3) Install a new VR into the affected VR's original location. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(e) If you do see video from the VR on the VSCU preview monitor, then continue.

(4) Do these steps to check the VR and VSCU connectors for damage:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the VSCU. This is the task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Make sure that there is no damage to the electrical connectors or the pins.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(e) If there is no damage to the connectors or pins, then continue.

(5) Do this check of the wiring:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the VSCU. This is the task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Do a check for continuity of the wiring between the VR and the VSCU.

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

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2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(e) If there is continuity, then continue.

(6) Replace the VSCU, M2065. These are the tasks:

- Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001
- Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - No Video From VR

(1) Do this check of the wiring:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the VSCU. This is the task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Make sure that there is no damage to the electrical connectors or the pins.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(e) If there is no damage to the connectors or pins, then do these steps:

OMR 805, 809, 810, 812, 813

	VR	
	VR Connector	**VSCU Connector**
VR 1	**D10659**	**D10657B**
	pin 63	pin 9
	pin 64	pin 10
VR 2 (if installed)	**D10661**	**D10657B**
	pin 63	pin 16
	pin 64	pin 17
VR 3 (if installed)	**D10663**	**D10627B**
	pin 63	pin 24
	pin 64	pin 25

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OMR 701-799, 802

(g) Do a check for continuity between these pins of the VR and the VSCU:

	VR	
	VR Connector	VSCU Connector
VR 1	D10659	D10667
	pin 30	pin 1
	pin 30	pin 1
VR 2 (if installed)	D10661	D10667
	pin 30	pin 2
	pin 30	pin 2

OMR 701-799, 802, 805, 809, 810, 812, 813

(h) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(i) If there is continuity, then continue.

(2) Replace the VR. These are the tasks:

- Video Reproducer Removal, AMM TASK 23-32-01-000-801
- Video Reproducer Installation, AMM TASK 23-32-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the VSCU. These are the tasks:

- Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001
- Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001

(a) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - VR Controls Do Not Operate

(1) Do this check of the connectors:

(a) Remove the VR. This is the task: Video Reproducer Removal, AMM TASK 23-32-01-000-801.

(b) Remove the VSCU. This is the task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(c) Make sure that there is no damage to the electrical connectors or the pins.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

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3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(e) If there is no damage to the connectors or pins, then continue.

(2) Do this check of the wiring:

OMR 805, 809, 810, 812, 813

(a) Do a check for continuity between these pins of the VR and the VSCU:

	VR	
	VR Connector	**VSCU Connector**
VR 1	**D10659**	**D10657B**
	pin 54	pin 55
	pin 53	pin 56
VR 2 (if installed)	**D10661**	**D10657B**
	pin 54	pin 57
	pin 53	pin 58
VR 3 (if installed)	**D10663**	**D10657B**
	pin 54	pin 59
	pin 53	pin 60

OMR 701-799, 802

(b) Do a check for continuity between these pins of the VR and the VSCU:

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OMR 701-799, 802 (Continued)

	VR	
	VR Connector	**VSCU Connector**
VR 1	**D10659**	**D10673**
	pin 6	pin 38
	pin 7	pin 38
VR 1	**D10659**	**D10675**
	pin 19	pin 1
	pin 20	pin 2
VR 2 (If installed)	**D10661**	**D10673**
	pin 6	pin 38
	pin 7	pin 38
VR 2 (If installed)	**D10661**	**D10675**
	pin 19	pin 4
	pin 20	pin 5

OMR 701-799, 802, 805, 809, 810, 812, 813

(c) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the VSCU. This is the task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

3) Re-install the VR. This is the task: Video Reproducer Installation, AMM TASK 23-32-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(3) Replace the VR. These are the tasks:

- Video Reproducer Removal, AMM TASK 23-32-01-000-801
- Video Reproducer Installation, AMM TASK 23-32-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the VSCU. These are the tasks:

- Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001
- Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001

(a) Do the Repair Confirmation at the end of this task.

I. Repair Confirmation

(1) Do this test of the VR:

(a) Push the power button on the VSCU to the on position.

(b) Put a media source (tape, CD, or DVD) into the VR if necessary.

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(c) Make the applicable selections to display the video from VR 1 on the VSCU preview monitor.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR 1.

1) Make sure that you see video from the VR on the VSCU preview monitor.

(d) If you see video from the VR on the VSCU preview monitor, then you corrected the problem.

(e) If test was not satisfactory, then continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

804. Video Reproducer Video Quality Problem - Fault Isolation

A. Description

(1) The quality of the video from a video reproducer (VR) is not acceptable.

B. Fault Isolation Procedure

(1) Replace the video reproducer.

These are the tasks:

Video Reproducer Removal, AMM TASK 23-32-01-000-801,

Video Reproducer Installation, AMM TASK 23-32-01-400-801.

——— END OF TASK ———

805. Video Reproducer Audio Quality Problem - Fault Isolation

A. Description

(1) The quality of the audio from a video reproducer (VR) is not acceptable.

B. Possible Causes

(1) Video reproducers

(2) Audio multiplexer, M1956

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	7	C01451	ENT VID CONT CENTER AC
C	8	C01456	ENTERTAINMENT AUDIO

D. Related Data

(1) (WDM 23-32-31, 23-34-01)

(2) (SSM 23-32-11, 23-34-01)

E. Initial Evaluation

(1) Do this test of the video reproducer:

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

(b) Push the power button on the video system control unit (VSCU) to the on position.

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(c) Connect headphones to the headphone jack on the VSCU.

(d) Put a media source (tape, CD, or DVD) into the VR.

(e) Make the applicable selections to display the video from the VR on all passenger video monitors, the VSCU preview monitor, and the audio on the PA system.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR.

1) Make sure you hear clear audio on the headphones connected to the VSCU.

2) Make sure you hear clear audio on the passenger address (PA) speakers.

(f) If you do not hear clear audio on the headphones connected to the VSCU, do the Fault Isolation Procedure - VR Problem below.

(g) If you hear clear audio on the headphones connected to the VSCU, but do not hear clear audio on the PA speakers, then do the Fault Isolation Procedure - Multiplexer Problem below.

(h) If you hear clear audio on the headphones connected to the VSCU and on the PA speakers, then there was an intermittent problem.

F. Fault Isolation Procedure - VR Problem

NOTE: You must do the Initial Evaluation in this task before you do this procedure.

(1) Move the VR to a different location:

(a) Remove the affected VR. To remove it, (AMM TASK 23-32-01-000-801).

(b) Remove a working VR. To remove it, (AMM TASK 23-32-01-000-801).

(c) Install the affected VR in the working VR's original location. To install it, (AMM TASK 23-32-01-400-801).

(d) Make the applicable selections to display the video from VR 1 on the VSCU preview monitor.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR 1.

(e) Make sure you hear clear audio on the headphones connected to the VSCU.

(f) If you do not hear clear audio on the headphones connected to the VSCU, then do these steps:

1) Remove the affected VR. To remove it, (AMM TASK 23-32-01-000-801).

2) Re-install the working VR to its original location. To install it, (AMM TASK 23-32-01-400-801).

3) Install a new VR in the affected VR's original location. To install it, (AMM TASK 23-32-01-400-801).

4) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(g) If you do hear clear audio on the headphones connected to the VSCU, then do these steps and then continue:

1) Remove the affected VR. To remove it, (AMM TASK 23-32-01-000-801).

2) Re-install the working VR in its original location. To install it, (AMM TASK 23-32-01-400-801).

(2) Do this check of the wiring:

(a) Remove the video system control unit (VSCU). To remove it, (AMM TASK 23-32-02-000-801-001).

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(b) Do a check of the wiring between the VR and the VSCU (WDM 23-32-31).

(c) If there is a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the VSCU. To install it, (AMM TASK 23-32-02-400-801-001).

3) Re-install the VR. To install it, (AMM TASK 23-32-01-400-801).

4) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Multiplexer Problem

(1) Replace the audio multiplexer, M1956.

These are the tasks:

Audio Multiplexer Removal, AMM TASK 23-34-02-000-801,

Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the video system control unit (VSCU), M2065. To remove it, do this task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(b) Remove the audio multiplexer, M1956. To remove it, do this task: Audio Multiplexer Removal, AMM TASK 23-34-02-000-801.

(c) Do a check for continuity of the wiring between the VSCU and the audio multiplexer (WDM 23-32-31).

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the audio multiplexer, M1956. To install it, (AMM TASK 23-34-02-400-801).

3) Re-install the VSCU, M2065. To install it, (AMM TASK 23-32-02-400-801-001).

4) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Do this test of the video reproducer (VR):

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

(b) Push the power button on the video system control unit (VSCU) to the on position.

(c) Connect headphones to the headphone jack on the VSCU.

(d) Put a media source (tape, CD, or DVD) into the VR.

(e) Make the applicable selections to display the video from the VR on the VSCU preview monitor.

NOTE: You will make the selections on the VSCU and if it is necessary, on the VR.

1) Make sure you hear clear audio on the headphones connected to the VSCU.

2) Make sure you hear clear audio on the PA speakers.

(f) If you hear clear audio on the headphones connected to the VSCU and on the PA speakers, then you corrected the fault.

END OF TASK

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806. Airshow Does Not Operate - Fault Isolation

A. Description

(1) The airshow system does not operate.

B. Possible Causes

OMR 701-799, 802

(1) Airshow compact disc

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Connectors and pins

(3) Digital interface unit (DIU), M1499

(4) Video system control unit (VSCU), M2065

(5) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	5	C01450	ENT VID CONT CENTER DC

D. Related Data

(1) (WDM 23-32-21)

E. Initial Evaluation

(1) Do this test of the airshow system:

(a) Push the power button on the video system control unit (VSCU) to the on position.

(b) Make the selections on the VSCU to show video from the airshow system on the VSCU preview monitor.

1) Make sure you see video from the airshow system on the VSCU preview monitor.

(c) If you do not see video from the airshow system, then do the Fault Isolation Procedure below.

(d) If you do see video from the airshow system, then there was an intermittent fault.

F. Fault Isolation Procedure

OMR 701-799, 802

(1) Do this check for a compact disc (CD):

(a) Push the button on the digital interface unit (DIU) to eject the CD caddy.

1) Make sure a CD is installed in the CD caddy.

(b) If there is no CD installed in the CD caddy, then do these steps:

1) Install an airshow CD into the caddy.

2) Put the CD caddy into the DIU.

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OMR 701-799, 802 (Continued)

3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	5	C01450	ENT VID CONT CENTER DC

4) Wait at least 10 seconds.

5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	5	C01450	ENT VID CONT CENTER DC

6) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(c) If there is a CD installed, then continue.

OMR 701-799, 802, 805, 809, 810, 812, 813

(2) Do this check of the connectors and pins:

(a) Remove the VSCU. To remove it, do this task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(b) Remove the DIU. To remove it, do this task: Digital Interface Unit (DIU) Removal, AMM TASK 23-32-06-000-804.

(c) Check the connectors and pins for damage.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the DIU. To install it, do this task: Digital Interface Unit (DIU) Installation, AMM TASK 23-32-06-400-801.

3) Re-install the VSCU. To install it, do this task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

4) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(e) If there is no damage to the connectors or pins, do these steps and continue:

1) Re-install the DIU. To install it, do this task: Digital Interface Unit (DIU) Installation, AMM TASK 23-32-06-400-801.

2) Re-install the VSCU. To install it, do this task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

(3) Replace the DIU. These are the tasks:

- Digital Interface Unit (DIU) Removal, AMM TASK 23-32-06-000-804
- Digital Interface Unit (DIU) Installation, AMM TASK 23-32-06-400-801

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(4) Replace the VSCU. These are the tasks:

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- Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001
- Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(5) Do this check of the wiring:

(a) Remove the VSCU. To remove it, do this task: Video System Control Unit Removal, AMM TASK 23-32-02-000-801-001.

(b) Remove the DIU. To remove it, do this task: Digital Interface Unit (DIU) Removal, AMM TASK 23-32-06-000-804.

(c) Do a check for continuity of the wiring between the DIU and the VSCU (WDM 23-32-21).

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the DIU. To install it, do this task: Digital Interface Unit (DIU) Installation, AMM TASK 23-32-06-400-801.

3) Re-install the VSCU. To install it, do this task: Video System Control Unit Installation, AMM TASK 23-32-02-400-801-001.

4) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this test of the airshow system:

(a) Push the power button on the video system control unit (VSCU) to the on position.

(b) Make the selections on the VSCU to show video from the airshow system on the VSCU preview monitor.

1) Make sure you see video from the airshow system on the VSCU preview monitor.

(c) If you see video from the airshow system, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Damaged Passenger Control Unit - Fault Isolation

A. Description

(1) The passenger control unit (PCU) at a seat has been damaged or is not controllable.

B. Fault Isolation Procedure

(1) Replace the passenger control unit (PCU).

These are the tasks:

Passenger Control Unit Removal, AMM TASK 23-34-11-000-801,

Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

——— END OF TASK ———

802. Seat Audio Problems - Fault Isolation

A. Description

(1) The passenger entertainment audio cannot be heard, or is of unacceptable quality, at one or many seats.

B. Possible Causes

(1) Audio system configuration

(2) Passenger control unit (PCU)

(3) Wiring

(4) Seat electronic box (SEB)

(5) Audio multiplexer (AMUX), M1956

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	10	C01584	ENTERTAINMENT AUDIO DC
C	8	C01456	ENTERTAINMENT AUDIO

OMR 701-799, 802

C	10	C01101	ENTERTAINMENT MUX

OMR 701-799, 802, 805, 809, 810, 812, 813

D. Related Data

(1) (WDM 23-34-01, 23-34-02)

E. Initial Evaluation

OMR 805, 809, 810, 812, 813

(1) Do this check of the audio BITE:

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

(b) Push the BITE and MODE SELECT buttons on the audio Built-In-Test Equipment (BITE) panel at the same time.

(c) Enter 8131 when prompted for a PASS CODE.

(d) Push the MODE SELECT button.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 809, 810, 812, 813 (Continued)

(e) Wait for the BITE test to complete.

(f) If the BITE panel shows the message -NO LAYOUT DATA- then do the Fault Isolation Procedure - No Layout Data below.

(g) If the BITE panel shows the message SEAT XXX PCU ERROR, then do the Fault Isolation Procedure - Seat PCU Error below.

(h) If the BITE panel shows the message LT/RT XX-YY NO CONCT then do the Fault isolation Procedure - Line Fault below.

(i) If the BITE panel shows the message LT/RT XX-YY NOT CNTC then do the Fault isolation Procedure - Multiple Line Faults below.

(j) If the BITE panel shows the message L/R FWD/AFT LINE 1/2 ERR then do the Fault Isolation Procedure - No Communication In Column below.

(k) If the BITE panel shows the message L/R FWD/AFT LAYOUT ERR, SEAT XX LT PCU X ERR, or COL X RW XX PCU X ERR then do the Fault Isolation Procedure - Faulty Seat Layout below.

(l) If the BITE panel shows the message BITE OK PUSH ENT KEY then there was an intermittent fault.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(2) Do this test of the passenger entertainment system:

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

OMR 701-799, 802; AIRPLANES WITH CD PLAYERS

(b) Install Compact Disc (CD)s in the Audio Entertainment Player (AEP). To install them, do this task: Compact Disc (CD) Installation, AMM TASK 23-34-01-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Connect headphones to the jack at an affected seat, another seat in the same seat group, another seat in the same column (left or right), and another seat in the opposite column.

(d) Put the channel selector switch on each Passenger Control Unit (PCU) to 7.

1) Make sure you hear audio from the AEP on the headphones.

(e) If you do not hear audio on one seat, but you do hear audio on all others, then do the Fault Isolation Procedure - One Seat below.

(f) If you do not hear audio on the seats in the same seat group, but you do hear audio on the other seats, then do the Fault Isolation Procedure - Seat Group below.

(g) If you do not hear audio on the seats in the same column, but you do hear audio on the seat on the opposite column, then do the Fault Isolation Procedure - Column below.

(h) If you do not hear audio on any seat, then do the Fault Isolation Procedure - All Seats below.

(i) If you heard audio on all seats, then there was an intermittent fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 809, 810, 812, 813

F. Fault Isolation Procedure - No Layout Data

(1) Download BITE panel EEPROM data to the Audio Multiplexer (AMUX). To do this, do this task: Audio Multiplexer Memory Update, AMM TASK 23-34-03-700-801.

(a) Do the Repair Confirmation at the end of this task.

(2) Do a check of the left and right Seat Electronics Box (SEB) CCT CB on the AMUX, M1956.

(a) If the left and right SEB CCT CB are open then close them.

(b) Do the Repair Confirmation at the end of this task.

(3) Replace the AMUX.

These are the tasks:

- Audio Multiplexer Removal, AMM TASK 23-34-02-000-801
- Audio Multiplexer Installation, AMM TASK 23-34-02-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Seat PCU Error

(1) Replace the PCU at seat XXX (as shown in the BITE message).

These are the tasks:

- Passenger Control Unit Removal, AMM TASK 23-34-11-000-801
- Passenger Control Unit Installation, AMM TASK 23-34-11-400-801

(a) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - Line Fault

(1) Do these steps to check the SEB connectors for damage:

(a) Remove the SEB at row XX LT (left) or RT (right) as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(b) Remove the SEB at seat YY LT (left) or RT (right) as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(c) Make sure that there is no damage to the electrical connectors or pins on any of the SEB connectors.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install both SEBs. To install them, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Do the Repair Confirmation at the end of this task.

(e) If there is no damage to the connectors or pins, then continue.

(2) Do this check of the wiring:

(a) If necessary, remove the SEBs at rows XX and LT (left) or RT (right) as shown in the BITE message. To remove them, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(b) Do a check for continuity of the wiring between the SEBs (WDM 23-34-01).

(c) If there is not continuity, then do these steps:

1) Repair the wiring.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 809, 810, 812, 813 (Continued)

2) Re-install both SEBs. To install them, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Do the Repair Confirmation at the end of this task.

(d) If there is continuity, then do this step and continue:

1) Re-install the SEB at row YY. To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

(3) Install a new SEB at row XX. To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

(a) Do the Repair Confirmation at the end of this task.

I. Fault Isolation Procedure - Multiple Line Faults

(1) Do these steps to check the seat electronic box connectors for damage:

(a) Remove the SEB at row XX LT (left) or RT (right) as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(b) Remove the SEB at seat YY LT (left) or RT (right) as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(c) Make sure that there is no damage to the electrical connectors or pins on any of the SEB connectors.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install both SEBs. To install them, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Do the Repair Confirmation at the end of this task.

(e) If there is no damage to the connectors or pins, then continue.

(2) Do this check of the wiring:

(a) If necessary, remove the SEBs at rows XX and YY LT (left) or RT (right) as shown in the BITE message. To remove them, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(b) Do a check for continuity of the wiring between the SEBs (WDM 23-34-01).

(c) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install both SEBs. To install them, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Do the Repair Confirmation at the end of this task.

(3) Do this check of the seat layout data:

(a) Push the MODE SELECT button on the BITE panel.

1) Make sure the correct CONFIG number is shown on the BITE panel display.

(b) If an incorrect CONFIG number is shown, then do these steps:

1) Download BITE panel EEPROM data to the AMUX. To do this, do this task: Audio Multiplexer Memory Update, AMM TASK 23-34-03-700-801.

2) Do the Repair Confirmation at the end of this task.

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OMR 805, 809, 810, 812, 813 (Continued)

J. Fault Isolation Procedure - No Communication In Column

(1) Do a check of the left and right SEB CCT CB on the AMUX, M1956.

(a) If the left and right SEB CCT CB are open then close them.

(b) Do the Repair Confirmation at the end of this task.

(2) Do these steps to check the electrical connectors for damage:

(a) Remove the AMUX, M1956. To remove it, do this task: Audio Multiplexer Removal, AMM TASK 23-34-02-000-801.

(b) Remove the first SEB in the L (left) or R (right) FWD (forward) or AFT column as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(c) Make sure that there is no damage to the electrical connectors or pins on any of the audio AMUX or SEB connectors.

(d) If there is damage to the connectors or pins, then do these steps:

1) Repair the connectors or pins.

2) Re-install the SEB. To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Re-install the AMUX, M1956. To install it, do this task: Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

4) Do the Repair Confirmation at the end of this task.

(e) If there is no damage to the connectors or pins, then continue.

(3) Do this check of the wiring:

(a) If necessary, remove the AMUX, M1956. To remove it, do this task: Audio Multiplexer Removal, AMM TASK 23-34-02-000-801.

(b) If necessary, remove the first SEB in the L (left) or R (right) FWD (forward) or AFT column as shown in the BITE message. To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

(c) Do a check for continuity of the wiring between the AMUX and the SEB (WDM 23-34-01).

(d) If there is not continuity, then do these steps:

1) Repair the wiring.

2) Re-install the SEB. To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

3) Re-install the AMUX, M1956. To install it, do this task: Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

4) Do the Repair Confirmation at the end of this task.

(4) Do this check of the seat layout data:

(a) Push the MODE SELECT button on the BITE panel.

1) Make sure the correct CONFIG number is shown on the BITE panel display.

(b) If an incorrect CONFIG number is shown, then do these steps:

1) Download BITE panel EEPROM data to the AMUX. To do this, do this task: Audio Multiplexer Memory Update, AMM TASK 23-34-03-700-801.

2) Do the Repair Confirmation at the end of this task.

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OMR 805, 809, 810, 812, 813 (Continued)

(5) Install a new SEB. To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

(a) Do the Repair Confirmation at the end of this task.

K. Fault Isolation Procedure - Faulty Seat Layout

(1) Do this check of the seat layout data:

(a) Push the MODE SELECT button on the BITE panel.

1) Make sure the correct CONFIG number is shown on the BITE panel display.

(b) If an incorrect CONFIG number is shown, then do these steps:

1) Download BITE panel EEPROM data to the AMUX. To do this, do this task: Audio Multiplexer Memory Update, AMM TASK 23-34-03-700-801.

2) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802

L. Fault Isolation Procedure - One Seat

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(1) Move the PCU to another location:

(a) Remove the PCU at the affected seat.

OMR 701-799, 802

1) To remove it, do this task: Passenger Control Unit Removal, AMM TASK 23-34-11-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Remove a PCU at an unaffected seat.

OMR 701-799, 802

1) To remove it, do this task: Passenger Control Unit Removal, AMM TASK 23-34-11-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Install the PCU from the affected seat at the unaffected seat location.

OMR 701-799, 802

1) To install it, do this task: Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(d) Connect headphones to the jack at the unaffected seat.

(e) Put the channel selector switch on the PCU to 7.

1) Make sure you hear audio from the AEP on the headphones.

(f) If you do not hear audio at the unaffected seat location, then do these steps:

1) Install a new PCU in the affected seat location.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

a) To install it, do this task: Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

2) Re-install the PCU from the unaffected seat to its original location.

OMR 701-799, 802

a) To install it, do this task: Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

3) Do the Repair Confirmation at the end of this task.

(g) If you do hear audio at the unaffected seat location, re-install the PCU to the affected seat and continue.

OMR 701-799, 802

1) To install it, do this task: Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(2) Replace the SEB for the affected seat.

OMR 701-799, 802

(a) These are the tasks:

- Seat Electronics Box Removal, AMM TASK 23-34-04-000-801
- Seat Electronics Box Installation, AMM TASK 23-34-04-400-801

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring:

(a) Remove the PCU for the affected seat.

OMR 701-799, 802

1) To remove it, do this task: Passenger Control Unit Removal, AMM TASK 23-34-11-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Remove the SEB for the affected seat.

OMR 701-799, 802

1) To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Do a check of the wiring between the PCU and the SEB (WDM 23-34-02).

(d) If there is a problem with the wiring, do these steps:

1) Repair the wiring.

2) Re-install the SEB.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

a) To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

3) Re-install the PCU.

OMR 701-799, 802

a) To install it, do this task: Passenger Control Unit Installation, AMM TASK 23-34-11-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

4) Do the Repair Confirmation at the end of this task.

M. Fault Isolation Procedure - Seat Group

(1) Replace the SEB for the affected seat group.

OMR 701-799, 802

(a) These are the tasks:

- Seat Electronics Box Removal, AMM TASK 23-34-04-000-801
- Seat Electronics Box Installation, AMM TASK 23-34-04-400-801

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring:

(a) Remove the SEB for the affected row.

OMR 701-799, 802

1) To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Remove the SEB for the row in front of the affected row.

OMR 701-799, 802

1) To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Do a check of the wiring between the SEBs (WDM 23-34-02).

(d) If there is a problem with the wiring, do these steps:

1) Repair the wiring.

2) Re-install both SEBs.

OMR 701-799, 802

a) To install them, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

3) Do the Repair Confirmation at the end of this task.

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OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL (Continued)

N. Fault Isolation Procedure - Column

(1) Do a check of the left and right SEB CCT CB on the AMUX, M1956.

(a) If the left and right SEB CCT CB are open then close them.

(b) Do the Repair Confirmation at the end of this task.

(2) Replace the AMUX, M1956.

OMR 701-799, 802

(a) These are the tasks:

- Audio Multiplexer Removal, AMM TASK 23-34-02-000-801
- Audio Multiplexer Installation, AMM TASK 23-34-02-400-801

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring:

(a) Remove the SEB in the first row of the affected column.

OMR 701-799, 802

1) To remove it, do this task: Seat Electronics Box Removal, AMM TASK 23-34-04-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(b) Remove the AMUX, M1956.

OMR 701-799, 802

1) To remove it, do this task: Audio Multiplexer Removal, AMM TASK 23-34-02-000-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Do a check of the wiring between the SEB and the AMUX (WDM 23-34-02).

(d) If there is a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the AMUX.

OMR 701-799, 802

a) To install it, do this task: Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

3) Re-install the SEB.

OMR 701-799, 802

a) To install it, do this task: Seat Electronics Box Installation, AMM TASK 23-34-04-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

4) Do the Repair Confirmation at the end of this task.

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OMR 701-799, 802

O. Fault Isolation Procedure - All Seats

(1) Do this check of the audio entertainment system configuration:

(a) Make the selections on the Video System Control Unit (VSCU) to show the VSCU configuration.

1) Make sure the VSCU Configuration display shows YES next to AMUX.

(b) If the status of AMUX is NO, then do these steps:

1) Select AMUX and then push the SELECT button.

a) Make sure the status of AMUX changes to YES.

2) Do the Repair Confirmation at the end of this test.

(c) If the status of AMUX is YES, then continue.

(d) Make the selections on the VSCU to do the AMUX INITIALIZATION.

1) Make sure the VSCU shows the correct number of SEBs in each column.

(e) If the number of SEBs is correct, then do the Repair Confirmation at the end of this task.

(f) If the number of SEBs is not correct, then continue.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(2) Do a check of the left and right SEB CCT CB on the AMUX, M1956.

(a) If the left and right SEB CCT CB are open then close them.

(b) Do the Repair Confirmation at the end of this task.

(3) Do this check for power:

(a) Remove the AMUX, M1956.

OMR 701-799, 802

1) To remove it, do this task: Audio Multiplexer Removal, AMM TASK 23-34-02-000-801.

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	10	C01584	ENTERTAINMENT AUDIO DC
C	8	C01456	ENTERTAINMENT AUDIO
C	10	C01101	ENTERTAINMENT MUX

(c) Do a check for 115V AC on pins 2 and 4 on connector D10527C of the AMUX.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(d) If there is 115V AC on the pins, then do these steps:

1) Install a new multiplexer.

OMR 701-799, 802

a) To install it, do this task: Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

2) Do the Repair Confirmation at the end of this task.

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OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL (Continued)

(e) If there is not 115V AC on the pins, then re-install the AMUX and continue.

OMR 701-799, 802

1) To install it, do this task: Audio Multiplexer Installation, AMM TASK 23-34-02-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(4) Replace the audio entertainment multiplexer (AEM), M1891.

(a) Do the Repair Confirmation at the end of this task.

(5) Replace the power relay mux/AEP, R660.

(a) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 809, 810, 812, 813

P. Repair Confirmation

OMR 805, 809, 810, 812, 813

(1) Do this check of the audio BITE:

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the ON position.

(b) Push the BITE and MODE SELECT buttons on the audio BITE panel at the same time.

(c) Enter 8131 when prompted for a PASS CODE.

(d) Push the MODE SELECT button.

(e) Wait for the BITE test to complete.

(f) If the BITE panel shows the message BITE OK PUSH ENT KEY then you corrected the fault.

(g) If the BITE panel does not show the message BITE OK PUSH ENT KEY, continue the Fault Isolation Procedure at the subsequent step.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(2) Do this test of the passenger entertainment system:

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the ON position.

OMR 701-799, 802; AIRPLANES WITH CD PLAYERS

(b) Install CDs in the AEP. To install them, do this task: Compact Disc (CD) Installation, AMM TASK 23-34-01-400-801.

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL

(c) Connect headphones to all affected areas.

(d) Put the channel selector switch on each PCU to 7.

1) Make sure you hear audio from the AEP on the headphones.

(e) If you heard audio on all seats, then you corrected the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802; AIRPLANES WITHOUT THE AUDIO BITE PANEL (Continued)

(f) If you cannot hear audio on all seats, continue the Fault Isolation Procedure at the subsequent step.

OMR 701-799, 802, 805, 809, 810, 812, 813

——— END OF TASK ———

803. Seat Audio Does Not Turn Off - Fault Isolation

A. Description

(1) The ENTERTAINMENT IFE button on the forward attendant's panel does not turn off the passenger entertainment system.

B. Fault Isolation Procedure

(1) Replace the audio entertainment module on the forward attendant's panel, M1891.

——— END OF TASK ———

804. Damaged Audio Entertainment Player - Fault Isolation

A. Description

(1) The audio entertainment player (AEP) has been damaged.

B. Fault Isolation Procedure

(1) Replace the audio entertainment player (AEP).

These are the tasks:

Audio Entertainment Player Removal, AMM TASK 23-34-01-000-803,

Audio Entertainment Player Installation, AMM TASK 23-34-01-400-803.

——— END OF TASK ———

805. Audio Entertainment Player Does Not Operate - Fault Isolation

A. Description

(1) There is no audio from the AEP on any channel of all passenger control units.

B. Possible Causes

(1) Audio entertainment player (AEP), M01955

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	10	C01584	ENTERTAINMENT AUDIO DC
C	8	C01456	ENTERTAINMENT AUDIO

OMR 701-799, 802

C	10	C01101	ENTERTAINMENT MUX

OMR 701-799, 802, 805, 809, 810, 812, 813

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Initial Evaluation

(1) Do this test of the audio entertainment player (AEP):

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

OMR 701-799, 802, 805, 809, 810, 812, 813; AIRPLANES WITH CD PLAYERS

(b) Install compact discs (CDs) in the AEP. To install them, do this task: Compact Disc (CD) Installation, AMM TASK 23-34-01-400-801.

OMR 701-799, 802, 805, 809, 810, 812, 813

(c) Connect headphones to the headphone jack at a seat.

(d) Put the channel selector switch on the passenger control unit (PCU) to 7.

(e) Adjust the volume control to a comfortable level.

1) Make sure you hear audio from the AEP on the headphones.

(f) If you do not hear audio from the AEP, then do the Fault Isolation Procedure below.

(g) If you hear audio from the AEP then there was an intermittent fault.

E. Fault Isolation Procedure

(1) Replace the audio entertainment player, M01955. These are the tasks:

- Audio Entertainment Player Removal, AMM TASK 23-34-01-000-803
- Audio Entertainment Player Installation, AMM TASK 23-34-01-400-803

(a) Push the ENTERTAINMENT IFE button on the forward attendant's panel to the on position.

OMR 701-799, 802, 805, 809, 810, 812, 813; AIRPLANES WITH CD PLAYERS

(b) Install compact discs (CDs) in the AEP. To install them, do this task: Compact Disc (CD) Installation, AMM TASK 23-34-01-400-801.

OMR 701-799, 802, 805, 809, 810, 812, 813

(c) Connect headphones to the headphone jack at a seat.

(d) Put the channel selector switch on the passenger control unit (PCU) to 7.

(e) Adjust the volume control to a comfortable level.

1) Make sure you hear audio from the AEP on the headphones.

(f) If you hear audio from the AEP then you corrected the fault.

——— END OF TASK ———

806. Seat Electronics Box/Seat to Seat Cable - Fault Isolation

A. Description

(1) If all the PCU display 1-18 for channel selections there may be a problem with an SEB or the Seat to Seat Cables.

(2) If there is a problem with the SEB or the seat to seat cables it can damage the communication circuitry in the AMUX.

B. Possible Causes

(1) Seat Electronics Box (SEB)

(2) Seat to Seat Cables

(3) Audio Multiplexer (AMUX), M1956

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	10	C01584	ENTERTAINMENT AUDIO DC
C	8	C01456	ENTERTAINMENT AUDIO

OMR 701-799, 802

C	10	C01101	ENTERTAINMENT MUX

OMR 701-799, 802, 805, 809, 810, 812, 813

D. Related Data

(1) (WDM 23-34-02)

E. Fault Isolation Procedure - Channels 1-18 displayed on PCU

(1) If only one seat group displays channels 1-18, replace the SEB at that seat group or the SEB located under the seat group in front of the affected seats.

(a) Do the Repair Confirmation at the end of the task.

(2) Do a check of the wiring:

(a) Turn OFF the In-Flight Entertainment (IFE) power switch on the forward attendant's panel.

(b) Pull and collar all applicable IFE circuit breakers.

(c) Remove the sidewall disconnect for each column (WDM 23-34-02).

(d) Do the following steps to check the wiring:

1) Check the resistance between pin 4 and pin 5 on the sidewall disconnect for each column.

a) Resistance between these pins should be approximately 4.2 Kilo Ohms.

2) Measure the resistance in all 8 pins in each sidewall disconnect. Use the following table for approximate pin measurements.

PIN	PIN	Ohmmeter Reading
4	5	4.2 Kilo Ohms
14	15	4.5 Ohms
5	10	2.2 Kilo Ohms
4	10	2.2 Kilo Ohms
1	2	88 Ohms

(e) If the resistance between pin 4 and 5 is not 4.2 Kilo Ohms or there is a short between the other pins do the following:

1) Move the terminating plug forward until the resistance reads 4.2 Kilo Ohm.

2) The short between the pins disappears.

(f) You have now isolated the problem and need to replace the defective seat to seat cable (WDM 23-34-02) or the SEB (AMM PAGEBLOCK 23-34-04/401).

1) Do the Repair Confirmation at the end of the task.

(g) Do a check of the left and right SEB CCT CB on the AMUX, M1956.

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1) If the left and right SEB CCT CB are open then close them.

a) Do the Repair Confirmation at the end of this task.

(h) If there are no defects you can replace the AMUX (AMM PAGEBLOCK 23-34-02/401) and reattach all sidewall disconnects (WDM 23-34-02).

1) Do the Repair Confirmation at the end of the task.

(i) Reset all applicable IFE circuit breakers.

(j) Turn ON the IFE power switch at the first attendant's front panel.

OMR 805, 809, 810, 812, 813

(3) Update AMUX memory (AMM PAGEBLOCK 23-34-03/201).

(a) Do the Repair Confirmation at the end of the task.

OMR 701-799, 802, 805, 809, 810, 812, 813

(4) If all PCU's display channel 1-12 you are finished.

(5) If after replacing the AMUX the PCU's still display channels 1-18 check the following:

NOTE: If the problem was not fixed prior to installing the new MUX, it may become damaged and need replacement again.

(a) BITE Panel.

1) Replace if necessary.

(b) Check wiring between the AMUX and side wall disconnect.

(c) Check wiring between BITE Panel and AMUX.

F. Repair Confirmation

(1) Do a check of the PCU display.

(a) If the PCUs display channel 1-12, then you corrected the problem.

(b) If the PCUs display channel 1-18, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

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801. Service Interphone Connection Problem - Fault Isolation

A. Description

(1) The service interphone does not connect to the flight interphone or other service interphone locations.

B. Possible Causes

(1) Remote Electronic Unit (REU), M1353

(2) SERVICE INTERPHONE switch, S50

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) (SSM 23-41-11)

(2) (WDM 23-41-11)

E. Initial Evaluation

(1) Do these steps to prepare the service interphone system for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the ON position.

(b) Connect a boom mic/headset to the pilot's station in the flight deck.

(c) Connect a boom mic/headset to the service interphone jack at the external power panel near the nose wheel well area.

(d) Push the SERV INT microphone selector switch on the pilot's audio control panel (ACP).

(e) Push the volume control for the SVC microphone selector switch and turn to the middle position.

(2) Do this check of the service interphone system:

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly from the headset.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make sure you can hear the voice clearly on the pilot's headset.

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(g) If the voice is not heard on either headset, then do the Fault Isolation Procedure below.

(h) If the voice is heard on both headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do these steps.

(1) Replace the Remote Electronic Unit (REU), M1353.

These are the tasks:

Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801,

Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Replace the SERVICE INTERPHONE switch, S50, on the P5 overhead panel.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Do this check of the wiring for open circuit:

(a) Remove the REU, M1353 from the E4-1 shelf. To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Remove the SERVICE INTERPHONE switch, S50, from the P5 overhead panel.

(c) Do a check for an open circuit between these pins of the SERVICE INTERPHONE switch, S50, and connector D2501B of the REU:

S50		**D2501B**
pin 2	. . . . .	pin D6

(d) Remove the EE rack service interphone jack, D6025.

(e) Do a check for an open circuit between these pins of the SERVICE INTERPHONE switch, S50, and the EE rack service interphone jack, D6025:

S50		**D6025**
pin 1	. . . . .	pin R

(f) Do a check for an open circuit between these pins of connector D2501B of the REU and the EE rack service interphone jack, D6025:

D2501B		**D6025**
pin D7	. . . . .	pin S
pin E7	. . . . .	pin T

(g) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

3) Re-install the SERVICE INTERPHONE switch.

4) Re-install the EE rack service interphone jack.

5) Do the Repair Confirmation at the end of this task:

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a) If the Repair Confirmation is not OK, then continue.

(4) Do this check for short circuit:

(a) Remove the REU, M1353 from the E4-1 shelf. To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Make sure the SERVICE INTERPHONE switch, S50, is at the ON position

(c) Do a check for short circuit at connector D2501B:

D2501B		D2501B
Pin D6	. . . . .	D7
Pin E7	. . . . .	D7

(d) If a short circuit is identified, then do these steps:

1) Inspect and isolate each jack and wiring for the source of short circuit such as corrosion, fluid contamination or spring contacts touching the jeck body. Refer to WDM 23-41-11.

NOTE: Inspect jacks D6017, D6011 and D6057 before you check the other jacks. Due to their location, they are more likely to have short circuit problems.

2) Repair or replace the damaged jack or wiring

(e) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(f) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the service interphone system:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do this check.

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly on the headset.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make sure you can hear the voice clearly on the pilot's headset.

(g) If the voice is heard on both headsets, then you corrected the fault.

——— END OF TASK ———

802. Service Interphone Audio Problem at One Jack - Fault Isolation

A. Description

(1) The audio heard at one of the service interphone jacks is unsatisfactory.

B. Possible Causes

(1) Service Interphone Jack

(2) Wiring

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) (SSM 23-41-11)

(2) (WDM 23-41-11)

E. Initial Evaluation

(1) Do this check of the service interphone jack:

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly on the headset at the applicable service interphone jack.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make sure you can hear the voice clearly on the pilot's headset.

(g) If the voice is not heard on either headset, then do the Fault Isolation Procedure below.

(h) If the voice is heard on both headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do these steps.

(1) Replace the applicable jack.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the REU from the E4-1 shelf. To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Do a check for an open circuit between these pins of connector D2501B of the REU and the applicable service interphone jack:

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	JACK LOCATION	
	REU CONNECTOR	JACK CONNECTOR
EXTERNAL POWER RECEPTACLE PANEL	D2501B pin D7	D6007 pin S
	D2501B pin E7	D6007 pin T

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	JACK LOCATION	
	REU CONNECTOR	JACK CONNECTOR
EE RACK	D2501B	D6025
	pin D7	pin S
	D2501B	D6025
	pin E7	pin T
APU	D2501B	D6017
	pin D7	pin S
	D2501B	D6017
	pin E7	pin T
AFT ENTRY LIGHT PANEL	D2501B	D6015
	pin D7	pin S
	D2501B	D6015
	pin E7	pin T
LEFT WHEEL WELL	D2501B	D6011
	pin D7	pin S
	D2501B	D6011
	pin E7	pin T
RIGHT WHEEL WELL	D2501B	D6057
	pin D7	pin S
	D2501B	D6057
	pin E7	pin T
RIGHT WING REFUELING SLAT	D2501B	D6013
	pin D7	pin S
	D2501B	D6013
	pin E7	pin T

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

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3) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the service interphone system:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do this check.

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly on the headset at the applicable service interphone jack.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make you can hear the voice clearly on the pilot's headset.

(g) If the voice is heard on both headsets, then you corrected the fault.

——— END OF TASK ———

803. Service Interphone Audio Problem at All Jacks - Fault Isolation

A. Description

(1) The audio heard at all of the service interphone jacks is unsatisfactory.

B. Possible Causes

(1) Remote Electronic Unit (REU), M1353

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) (SSM 23-41-11)

(2) (WDM 23-41-11)

E. Initial Evaluation

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Do these steps to prepare the service interphone system for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the ON position.

(b) Connect a boom mic/headset to the pilot's station in the flight deck.

(c) Connect a boom mic/headset to the EE rack service interphone jack.

(d) Push the SVC microphone selector switch on the pilot's audio control panel (ACP).

(e) Push the volume control for the SVC microphone selector switch and turn to the middle position.

OMR 815, 816, 818-826, 901-999

(2) Do these steps to prepare the service interphone system for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the ON position.

(b) Connect a boom mic/headset to the pilot's station in the flight deck.

(c) Connect a boom mic/headset to the EE rack service interphone jack.

(d) Push the CABIN microphone selector switch on the pilot's audio control panel (ACP).

(e) Push the volume control for the CABIN microphone selector switch and turn to the middle position.

OMR ALL

(3) Do this check of the service interphone system:

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly on the headset at the EE rack service interphone jack.

NOTE: Repeat this step at several of the service interphone jack locations to make sure the problem occurs at all jacks.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make sure you can hear the voice clearly on the pilot's headset.

(g) If the voice is not heard on either headset, from each service interphone jack you tried, then do the Fault Isolation Procedure below.

(h) If the voice is heard on both headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do these steps.

(1) Replace the Remote Electronic Unit (REU), M1353.

These are the tasks:

Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801,

Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the REU from the E4-1 shelf. To remove it, do this task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Remove the EE rack service interphone jack, D6025.

(c) Do a check for an open circuit between these pins of connector D2501B of the REU and the EE rack service interphone jack, D6025:

D2501B		D6025
pin D7	. . . . .	pin S
pin E7	. . . . .	pin T

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the REU. To install it, do this task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

3) Re-install the EE rack service interphone jack, D6025.

4) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the service interphone system:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do these steps.

(a) Push the PTT switch on the pilot's control wheel to the INT position.

(b) Speak into the pilot's boom microphone.

(c) Make sure the ground crew can hear the voice clearly on the headset at the EE rack service interphone jack.

(d) Push the switch to the center (OFF) position.

(e) Have the ground crew speak into the boom microphone.

(f) Make sure you can hear the voice clearly on the pilot's headset.

(g) If the voice is heard on both headsets, then you corrected the fault.

——— END OF TASK ———

804. Attendant Handset Problem - Fault Isolation

A. Description

(1) The audio heard at an Attendant's Handset is unsatisfactory.

(2) If the Attendant's Handset has problems with a PA Announcement, refer to Passenger Address System Problem - Fault Isolation, 23-31 TASK 805.

B. Possible Causes

(1) Attendant's Handset, M74 (M75)

NOTE: 737BCF does not have a M75 installed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) FWD (AFT) Cabin Attendant's Control Panel, P13 (P14)

NOTE: 737BCF does not have a P14 installed.

(3) Remote Electronics Unit (REU), M1353

(4) Wiring

OMR 815, 816, 818-826, 901-999

(5) Inline Filter M2931 or M2932

OMR ALL

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) WDM 23-41-11

(2) WDM 23-42-11

(3) SSM 23-41-11

(4) SSM 23-42-11

OMR 815, 816, 818-826, 901-999

(5) WDM 23-42-12

OMR ALL

E. Initial Evaluation

(1) Do this check of the Attendant's Handset:

(a) Lift the handset from the cradle.

(b) Push the button for the attendant on the handset keypad.

(c) Make sure that you hear a chime from the PA Speakers and a Master Call Light at the Attendant's Station comes ON.

(d) Make voice communication to the other attendant's station.

1) If the sound you hear on the Attendant's Handset is satisfactory, then there was an intermittent fault.

2) If the sound you hear on the Attendant's Handset is not satisfactory, do the Fault Isolation Procedure below.

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F. Fault Isolation Procedure

(1) Make sure that the connection between the handset and cord is correct.

NOTE: To connect a cord to the handset, push the cord connector into the handset connector until you hear an audible click.

(2) Replace the Attendant's Handset/Cord Assembly, M74 (M75). These are the tasks:

- Attendant Handset Removal, AMM TASK 23-42-01-000-801
- Attendant Handset Cord Removal, AMM TASK 23-42-01-000-802
- Attendant Handset Installation, AMM TASK 23-42-01-400-803
- Attendant Handset Cord Installation, AMM TASK 23-42-01-400-802

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the applicable Cabin Attendant's Control Panel, P13 (P14). These are the tasks:

- Attendant's Panel and Components Removal, AMM TASK 25-25-11-000-801 or Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802
- Attendant's Panel and Components Installation, AMM TASK 25-25-11-400-801 or Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the REU, M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(5) Replace the applicable inline filter M2931 or M2932. These are the tasks:

- Inline Filter Removal, AMM TASK 23-42-04-000-801
- Inline Filter Installation, AMM TASK 23-42-04-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(6) Do this check of the wiring:

(a) Remove the Attendant's Handset. This is the task: Attendant Handset Removal, AMM TASK 23-42-01-000-801.

(b) Remove the REU, M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(c) Do a check for an open circuit between the Attendant's Handset, M74 (M75), and REU, M1353 as follows (WDM 23-41-11, WDM 23-42-11):

P-A		D2501B
pin C5	. . . . .	pin G8
pin C6	. . . . .	pin F7
pin D5	. . . . .	pin G9

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

2) Re-install the Attendant's Handset. This is the task: Attendant Handset Installation, AMM TASK 23-42-01-400-803.

3) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

4) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(7) Do this check of the wiring:

(a) Remove the attendant handset, M75. This is the task: Attendant Handset Removal, AMM TASK 23-42-01-000-801.

(b) Remove the inline filter, 2931. This is the task: Inline Filter Removal, AMM TASK 23-42-04-000-801.

(c) Do a check for an open circuit between the attendant handset and the inline filter as follows (WDM 23-42-12):

J1	D11815
C5	4
D5	2
C6	3

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the Attendant's Handset, M75. This is the task: Attendant Handset Installation, AMM TASK 23-42-01-400-803.

3) Re-install the inline filter, 2931. This is the task: Inline Filter Installation, AMM TASK 23-42-04-400-801.

4) Do the Repair Confirmation at the end of this task.

(8) Do this check of the wiring:

(a) Remove the inline filter, 2931. This is the task: Inline Filter Removal, AMM TASK 23-42-04-000-801.

(b) Remove the REU, M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801

(c) Do a check for an open circuit between the inline filter and the REU as follows (WDM 23-42-12):

D11815	D2501B
13	G8
7	G9
6	F7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the inline filter, 2931. This is the task: Inline Filter Installation, AMM TASK 23-42-04-400-801.

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OMR 815, 816, 818-826, 901-999 (Continued)

3) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

4) Do the Repair Confirmation at the end of this task.

(9) Do this check of the wiring:

(a) Remove the attendant handset, M74. This is the task: Attendant Handset Removal, AMM TASK 23-42-01-000-801.

(b) Remove the inline filter, 2932. This is the task: Inline Filter Removal, AMM TASK 23-42-04-000-801.

(c) Do a check for an open circuit between the attendant handset and the inline filter as follows (WDM 23-42-12):

J1		D13363
15	. . . . .	3
16	. . . . .	2
14	. . . . .	4

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the Attendant's Handset, M74. This is the task: Attendant Handset Installation, AMM TASK 23-42-01-400-803.

3) Re-install the inline filter, 2932. This is the task: Inline Filter Installation, AMM TASK 23-42-04-400-801.

4) Do the Repair Confirmation at the end of this task.

(10) Do this check of the wiring:

(a) Remove the inline filter, 2932 This is the task: Inline Filter Removal, AMM TASK 23-42-04-000-801.

(b) Remove the REU, M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801

(c) Do a check for an open circuit between the inline filter and the REU as follows (WDM 23-42-12):

D13363		D2501B
3	. . . . .	G8
2	. . . . .	G9
4	. . . . .	F7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the inline filter, 2932 This is the task: Inline Filter Installation, AMM TASK 23-42-04-400-801.

3) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

4) Do the Repair Confirmation at the end of this task.

OMR ALL

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G. Repair Confirmation

(1) On the Handset Keypad, push the button for the attendant

(a) Make sure that you hear a chime from the PA Speakers and a Master Call Light at the Attendant's Station comes ON.

(2) Make voice communication to the other Attendant's Station.

(a) If the sound you hear on the Attendant's Handset is satisfactory, then you corrected the problem.

(b) If the sound you hear on the Attendant's Handset is not satisfactory, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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801. Attendant Control Panel (ACP) - BITE Procedure

A. General

OMR 815, 816, 818-826, 901-999

(1) This procedure provides the system test for the Attendant Control Panels (ACPs). The system test is performed at the forward ACP.

(a) For the system test, the forward ACP will initiate BITE test on both the forward and the aft ACPs. The BITE test will check the ACPs for internal faults and their communication status. The forward ACP will then query the light LRU's for fault information. If there are ACP or lighting faults, the ACP will show the faults at the end of the test.

(2) The procedure also provides the task to view faults that are generated during data loading.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) This BITE procedure is not applicable for airplanes without Boeing Sky Interior.

OMR ALL

B. System Test Procedure

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) NA

OMR 815, 816, 818-826, 901-999

(2) At the forward ACP, do the steps that follow to start the system test:

(a) In the ACP main menu column, push the Maintenance button.

(b) In the Maintenance menu column, push the System Test button.

(c) In the control window, push the Start Test button.

NOTE: When the system test operates, the System Test and Start Test buttons are active, the other buttons become disabled and the status bar will display the approximate percent complete. When the test completes, the System Test and the Start Test buttons remain active and other buttons will be available for selection.

(d) Check the result window for the system test faults.

NOTE: The result window will show the faults with their maintenance message numbers and corrective actions. If there are no faults, a message will be displayed to indicate that there are no system test faults.

(e) To save the fault data on a diskette, refer to: Attendant Control Panel (ACP) - Saving Fault Data, AMM TASK 23-42-03-970-802.

(f) To correct the fault, do the actions recommended on the screen or use the fault maintenance message number and refer to the table at the end of this task to find the fault isolation task for the applicable maintenance message.

C. Data Load Fault Data Review Procedure

(1) Do the steps that follow to view the Data Load Faults:

NOTE: These faults are generated during data loading, user will see the warning messages about the faults.

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OMR 815, 816, 818-826, 901-999 (Continued)

(a) In the Maintenance menu column, push the Fault Data button.

NOTE: After you push the Fault Data button, four buttons will appear in the control window: System Test Faults, Data Load Faults, Save All and Exit. The result window will show instructions for user to view or save the fault data.

(b) Push the Data Load Faults button in the control window.

(c) Check the result window for the data load faults.

NOTE: The result window will show the faults with their maintenance message numbers and corrective actions. If there are no faults, a message will be displayed to indicate that there are no data load faults.

(d) To save the fault data on a diskette, refer to: Attendant Control Panel (ACP) - Saving Fault Data, AMM TASK 23-42-03-970-802.

(e) To correct the faults, do the actions recommended on the screen or using the fault maintenance message number and refer to the table at the end of this task to find the fault isolation task for the applicable maintenance messages.

OMR ALL

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11001 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11002 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11003 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11004 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11005 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11006 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11007 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 808
ACP	23-11008 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 809
ACP	23-11010 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11011 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11012 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807
ACP	23-11013 INTERNAL ACP FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11014 INTERNAL ACP FAULT DETECTED IN FORWARD AND AFT ACPS	23-42 TASK 809
ACP	23-11015 <FORWARD | AFT> WORK LIGHT FAULT DETECTED	23-42 TASK 811
ACP	23-11016 <FORWARD | AFT> MDCD LIGHT FAULT DETECTED	23-42 TASK 812
ACP	23-11017 28 VDC POWER FAULT DETECTED IN <FORWARD | AFT> ACP	23-42 TASK 814
ACP	23-11018 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11019 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11020 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11021 INTERNAL ACP FAULT DETECTED	23-42 TASK 808
ACP	23-11022 COMMUNICATION WITH AFT ACP LOST	23-42 TASK 815
ACP	23-11023 CABIN TEMPERATURE FAULT DETECTED	23-42 TASK 816
ACP	23-11024 <FORWARD | AFT> THRESHOLD LIGHT FAULT DETECTED	23-42 TASK 813
ACP	23-11025 INTERNAL ACP FAULT DETECTED	23-42 TASK 807
ACP	23-11026 LAVATORY INOPERATIVE	23-42 TASK 817
ACP	23-11027 CLEAN CHECK SENSORS	23-42 TASK 817
ACP	23-11028 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11029 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11030 LIGHT <COLUMN>-<ADDRESS> EXTRA RESPONSE DURING ADDRESSING	23-42 TASK 810
ACP	23-11031 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 810
ACP	23-11032 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 810
ACP	23-11033 LIGHT <COLUMN>-<ADDRESS> INCORRECT STANDARD SCENES DETECTED	23-42 TASK 810
ACP	23-11034 LIGHT <COLUMN>-<ADDRESS> INCORRECT CUSTOM SCENES DETECTED	23-42 TASK 810
ACP	23-11035 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 1	23-42 TASK 819
ACP	23-11036 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 2	23-42 TASK 819
ACP	23-11037 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 3	23-42 TASK 819

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11038 NO SYSTEM TEST LRU RESPONSES DETECTED ON COLUMN 4	23-42 TASK 819
ACP	23-11039 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 1-<X>	23-42 TASK 818
ACP	23-11040 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 2-<X>	23-42 TASK 818
ACP	23-11041 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 3-<X>	23-42 TASK 818
ACP	23-11042 SYSTEM TEST RS-485 CABLE BREAK DETECTED AT LRU 4-<X>	23-42 TASK 818
ACP	23-11043 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ADDRESSING	23-42 TASK 820
ACP	23-11044 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ADDRESSING	23-42 TASK 844
ACP	23-11045 LIGHT <COLUMN>-<ADDRESS> EXTRA RESPONSE DURING ADDRESSING	23-42 TASK 821
ACP	23-11046 LIGHT <COLUMN>-<ADDRESS> INCORRECT LIGHT TYPE DETECTED	23-42 TASK 822
ACP	23-11047 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING ZONING	23-42 TASK 823
ACP	23-11048 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING ZONING	23-42 TASK 823
ACP	23-11049 LIGHT <COLUMN>-<ADDRESS> INCORRECT ZONE DETECTED	23-42 TASK 824
ACP	23-11050 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11051 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11052 LIGHT <COLUMN>-<ADDRESS> FAILED STANDARD SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11053 LIGHT <COLUMN>-<ADDRESS> NO RESPONSE DURING CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11054 LIGHT <COLUMN>-<ADDRESS> OUT OF SEQUENCE RESPONSE DURING CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11055 LIGHT <COLUMN>-<ADDRESS> FAILED CUSTOM SCENE DOWNLOAD	23-42 TASK 823
ACP	23-11056 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 1	23-42 TASK 825
ACP	23-11057 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 2	23-42 TASK 825

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	23-11058 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 3	23-42 TASK 825
ACP	23-11059 NO DATA LOAD LRU RESPONSES DETECTED ON COLUMN 4	23-42 TASK 825
ACP	23-11060 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 1	23-42 TASK 826
ACP	23-11061 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 2	23-42 TASK 826
ACP	23-11062 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 3	23-42 TASK 826
ACP	23-11063 DATA LOAD RS-485 CABLE BREAK DETECTED ON COLUMN 4	23-42 TASK 826
ACP	23-11064 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11065 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11066 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11067 INTERNAL ACP FAULT DETECTED	23-42 TASK 827
ACP	23-11068 COMMUNICATIONS WITH AFT ACP LOST	23-42 TASK 828
ACP	23-11069 LSAP CONFIGURATION INVALID	23-42 TASK 829
ACP	23-11070 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11071 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11072 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11073 INTERNAL ACP FAULT DETECTED	23-42 TASK 830
ACP	23-11074 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11075 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11076 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11077 DISK FAULT DETECTED	23-42 TASK 831
ACP	23-11078 INCOMPATIBLE DISK DETECTED	23-42 TASK 832
ACP	23-11079 INCOMPATIBLE DISK DETECTED	23-42 TASK 832
ACP	23-11080 COMMUNICATIONS WITH ADL LOST	23-42 TASK 833
ACP	33-11001 Light <column>-<address> Component ID Error	33-20 TASK 817
ACP	33-11002 Light <column>-<address> Calibration Data CRC Error	33-20 TASK 817
ACP	33-11003 Light <column>-<address> Firmware Version Disagree Error	33-20 TASK 817
ACP	33-11004 Light <column>-<address> Power Supply Error	33-20 TASK 817
ACP	33-11005 Light <column>-<address> Temperature Sensor Error	33-20 TASK 817
ACP	33-11006 Light <column>-<address> RAM Check Error	33-20 TASK 817

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
ACP	33-11007 Light <column>-<address> Slave Token Error	33-20 TASK 817
ACP	33-11008 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11009 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11010 Light <column>-<address> Slave Communication Error	33-20 TASK 817
ACP	33-11011 Light <column>-<address> LED Wrap Data	33-20 TASK 817
ACP	33-11012 Light <column>-<address> Zone/Address Data CRC Error	33-20 TASK 818
ACP	33-11013 Light <column>-<address> Zone Address Disagree	33-20 TASK 818
ACP	33-11014 Light <column>-<address> Standard Scene CRC Error	33-20 TASK 818
ACP	33-11015 Light <column>-<address> Custom Scene CRC Error	33-20 TASK 818
ACP	33-11016 Light <column>-<address> Watchdog Timer Error	33-20 TASK 817
ACP	33-11017 Light <column>-<address> Master Token Timeout Error	33-20 TASK 819
ACP	33-11018 Light <column>-<address> Master Token Release Error	33-20 TASK 819
ACP	33-11019 Light <column>-<address> Loss of communication	33-20 TASK 819

END OF TASK

802. Flight Deck to Attendant, Call System Does Not Operate - Fault Isolation

A. Description

(1) The pink attendant call light on the Exit Locator Signs does not come on and the call chime does not sound when the flight crew presses the ATTEND button on the P5 forward overhead panel.

B. Possible Causes

(1) Attendant Call Switch, S36

(2) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

D. Related Data

(1) (SSM 23-42-11)

(2) (SSM 23-43-11)

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(3) (WDM 23-42-11)

(4) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the attendant's CALL light:

(a) Press the ATTEND button on the P5 forward overhead panel.

1) Make sure the pink call light on the forward and aft Exit Locator Signs comes on.

2) Make sure you hear a chime in the passenger cabin.

(b) Press the button for reset on the attendant's handset.

(c) If the pink call lights do not come on and the chime does not sound, then do the Fault Isolation Procedure below.

(d) If the pink call lights go on and the chime sounds, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the Attendant's Call switch, S36, on the P5 forward overhead panel.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do these checks of the wiring:

(a) Remove the Exit Locator Sign, L1086 (forward) or L1223 (aft).

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(b) Remove the handset logic control card from the attendant's panel.

OMR 815, 816, 818-826, 901-999

(c) Remove the attendant control panel. To remove it, do this task: Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(d) Do a check for an open circuit between connector D3160 (forward) or D10180 (aft), of the Exit Locator Sign, and connector D11068 (forward) or D11466 (aft) of the lavatory module. For correct pins, refer to WDM 23-42-11.

OMR 815, 816, 818-826, 901-999

(e) Do a check for an open circuit between connector D3160 (forward) or D10180 (aft), of the Exit Locator Sign, and connector D14368 (forward attendant control panel) or D14376 (aft attendant control panel). For correct pins, refer to WDM 23-42-11.

OMR ALL

(f) Remove the Attendant's Call switch, S36, from the P5 forward overhead panel.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(g) Do a check for an open circuit between connector D40170J of the attendant's call switch, S36, and connector D11068 of the forward lavatory module. For correct pins, refer to WDM 23-42-11.

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OMR 815, 816, 818-826, 901-999

(h) Do a check for an open circuit between connector D40170J of the attendant's call switch, S36, and connector D14368 of the forward attendant control panel. For correct pins, refer to WDM 23-42-11.

OMR ALL

(i) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the Attendant's Call switch.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

3) Re-install the handset logic control card.

OMR 815, 816, 818-826, 901-999

4) Re-install the attendant control panel. Do this task: Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802.

OMR ALL

5) Re-install the Exit Locator Sign.

6) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the attendant's CALL light:

(a) Press the ATTEND button on the P5 forward overhead panel.

1) Make sure the pink call light on the forward and aft Exit Locator Signs comes on.

2) Make sure you hear a chime in the passenger cabin.

(b) Press the button for reset on the attendant's handset.

(c) If the pink call lights go on and the chime sounds, then you corrected the fault.

——— END OF TASK ———

803. Attendant Call Chime Does Not Operate - Fault Isolation

A. Description

(1) The chime from the Passenger Address system does not come on, but the call lights come on when the flight crew presses the ATTENDANT button on the P5 forward overhead panel.

B. Possible Causes

(1) Passenger Address Amplifier, M63

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

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D. Related Data

(1) (SSM 23-42-11)

(2) (SSM 23-43-11)

(3) (WDM 23-42-11)

(4) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the call chime:

(a) Press either the ATTEND button on the P5 forward overhead panel, or the button for the attendant on the attendant handset.

1) Make sure you hear a chime in the passenger cabin.

(b) Press the button for reset on the attendant's handset.

(c) If the chime does not sound, then do the Fault Isolation Procedure below.

(d) If the chime sounds, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the Passenger Address Amplifier, M63.

These are the tasks:

Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801,

Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the Passenger Address Amplifier, M63. To remove it, do this task: Passenger Address (PA) Amplifier - Removal, AMM TASK 23-31-01-000-801.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(b) Remove the handset logic control card from the attendant's panel.

OMR 815, 816, 818-826, 901-999

(c) Remove the attendant control panel. To remove it, do this task: Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(d) Do a check for an open circuit between the Passenger Address Amplifier M63, and connector D11068 (forward) or D11466 (aft) of the lavatory module. Refer to WDM 23-42-11.

OMR 815, 816, 818-826, 901-999

(e) Do a check for an open circuit between the Passenger Address Amplifier M63, and connector D14368 (forward attendant control panel) or D14376 (aft attendant control module). Refer to WDM 23-42-11.

OMR ALL

(f) If there is an open circuit, then do these steps:

1) Repair the wiring.

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2) Re-install the Passenger Address Amplifier. To install it, do this task: Passenger Address (PA) Amplifier - Installation, AMM TASK 23-31-01-400-801.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

3) Re-install the handset logic control card.

OMR 815, 816, 818-826, 901-999

4) Re-install the attendant control panel. Do this task: Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802.

OMR ALL

5) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the attendant's CALL light:

(a) Press either the ATTEND button on the P5 forward overhead panel, or the button for the attendant on the attendant handset.

1) Make sure you hear a chime in the passenger cabin.

(b) Press the button for reset on the attendant's handset.

(c) If you heard a chime, then you corrected the fault.

——— END OF TASK ———

804. Flight Compartment Call Chime Does Not Operate - Fault Isolation

A. Description

(1) The chime from the Aural Warning Module does not come on, but the CALL light on the P5 forward overhead panel comes on when the ground crew presses the PILOT CALL button on the P19 external power panel or an attendant presses the button for the pilot on the attendant handset.

B. Possible Causes

(1) Aural Warning Module, M315

(2) Handset, M74 or M75

(3) PILOT CALL switch, S33

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

D. Related Data

(1) SSM 23-42-11

(2) SSM 23-43-11

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(3) WDM 23-42-11

(4) WDM 23-43-11

E. Initial Evaluation

(1) Do this check of the call chime:

(a) Press the PILOT CALL button on the P19 External Power panel and the button for the pilot on the forward and aft handsets.

(b) Make sure you hear a chime in the flight deck.

(c) If you do not hear a chime when you push the button for the pilot on one handset but you hear the chime for the other handset, then do the Fault Isolation Procedure - Handset.

(d) If you do not hear a chime when you push the button for the pilot on the two handsets but you hear a chime when you push the PILOT CALL button on the P19 External Power panel, then do the Fault Isolation Procedure - Handset Wiring.

(e) If you hear a chime when you push the button for the pilot on one of the handsets but you do not hear a chime when you push the PILOT CALL button on P19 panel, then do the Fault Isolation Procedure - Switch.

(f) If you do not hear a chime when you push the button for the pilot on the two handsets and the PILOT CALL button on P19 panel, then do the Fault Isolation Procedure - Aural Warning Module.

F. Fault Isolation Procedure - Handset

(1) Replace the defective handset, M74 or M75. These are the tasks:

- Attendant Handset Removal, AMM TASK 23-42-01-000-801
- Attendant Handset Installation, AMM TASK 23-42-01-400-803

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the defective handset. To remove it, do this task: Attendant Handset Removal, AMM TASK 23-42-01-000-801

(b) Do a check between the handset logic control card and the terminator box SM9 (WDM 23-42-11):

Handset Logic Control Card Handset Connector	**Terminator Box SM9 SM9**
pin E13	SM9

(c) If you find problems with the wiring, do these steps:

1) Repair the wiring.

2) Re-install the handset. To install it, do this task: Attendant Handset Installation, AMM TASK 23-42-01-400-803.

3) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Fault Isolation Procedure - Handset Wiring

(1) Do a check of the wiring from the handsets to the Aural Warning Module.

(a) Remove the defective handset. To remove it, do this task: Attendant Handset Removal, AMM TASK 23-42-01-000-801

(b) Remove the Aural Warning Module. To remove it, do this task: Aural Warning Module Removal, AMM TASK 31-51-04-000-801

(c) Do a check of the wiring between the handset logic control card and the Aural Warning Module (WDM 23-42-11).

Handset Logic Control Card Handset Connector	Aural Warning Module D940
pin E13	pin 7

(d) If you find problems with the wiring, do these steps:

1) Repair the wiring.

2) Re-install the handset. To install it, do this task: Attendant Handset Installation, AMM TASK 23-42-01-400-803.

3) Re-install the Aural Warning Module. To install it, do this task: Aural Warning Module Installation, AMM TASK 31-51-04-400-801.

4) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - Switch

(1) Replace the PILOT CALL switch, S33 on the P19 External Power panel.

(a) Do the Repair Confirmation at the end of this task.

I. Fault Isolation Procedure - Aural Warning Module

(1) Replace the Aural Warning Module, M315. These are the tasks:

- Aural Warning Module Removal, AMM TASK 31-51-04-000-801
- Aural Warning Module Installation, AMM TASK 31-51-04-400-801

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Do this check of the wiring:

(a) Remove the Aural Warning Module. To remove it, do this task: Aural Warning Module Removal, AMM TASK 31-51-04-000-801.

(b) Press and hold the PILOT CALL switch, S33, on the P19 External Power panel or the button for the pilot on the attendant's handset.

(c) Do a check for 28 VDC at the Aural Warning Module connector D940, pin 7 (pin 6 ground).

(d) Release the PILOT CALL switch or button for the pilot on the attendant handset.

(e) If there is not 28 VDC, then do these steps:

1) Repair the wiring.

2) Re-install the Aural Warning Module. To install it, do this task: Aural Warning Module Installation, AMM TASK 31-51-04-400-801.

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3) Do the Repair Confirmation at the end of this task.

J. Repair Confirmation

(1) Do this check of the call chime:

(a) Press either the PILOT CALL button on the P19 External Power panel, or the button for the pilot on the attendant handset.

(b) Make sure you hear a chime in the flight deck.

(c) If the chime sounds, then you corrected the fault.

——— END OF TASK ———

805. Attendant to Flight Deck, Call System Does Not Operate - Fault Isolation

A. Description

(1) The Cabin Crew cannot communicate with the flight deck or the other Crew Stations with the Crew Call and Interphone System.

B. Possible Causes

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) FWD (AFT) Cabin Attendant's Control Panel, P13 (P14)

OMR ALL

(2) Magnetic Actuated Reset Switch

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

D. Related Data

(1) WDM 23-42-11

(2) WDM 23-43-11

(3) SSM 23-42-11

(4) SSM 23-43-11

E. Initial Evaluation

(1) Do this check of the Attendant's Handset:

(a) Lift the handset from the cradle.

(b) Push the button for the attendant on the Handset Keypad.

(c) Make sure that you hear a chime from the PA Speakers and the Pink Attendant Call Light at the other Attendant's Station comes ON.

(d) Make voice communication to the other Attendant's Station.

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1) If you hear the voice on the Attendant's Handset, then there was an intermittent fault.

2) If you do not hear the voice on the Attendant's Handset, do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Replace the applicable Cabin Attendant's Control Panel. These are the tasks:

- Attendant's Panel and Components Removal, AMM TASK 25-25-11-000-801 or Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802
- Attendant's Panel and Components Installation, AMM TASK 25-25-11-400-801 or Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

(2) Do this check of the Cradle Reset Magnet:

(a) Remove the Handset Handle Cradle. This is the task: Attendant Handset Cradle Removal, AMM TASK 23-42-01-000-803.

(b) Remove the retainer.

(c) Repair the Cradle Reset Magnet. This is the task: Attendant Handset Cradle Installation, AMM TASK 23-42-01-400-801

(d) Re-install the retainer.

(e) Re-install the Handset Cradle.

(f) Put the Handset in the cradle.

(g) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the handset:

(a) Lift the handset from the Handle Cradle.

(b) Push the button for the attendant on the Handset Keypad.

1) Make sure that you hear a chime from the PA Speakers and the Pink Attendant Call Light at the other Attendant's Station comes ON.

(c) Make voice communication to the other Attendant's Station.

1) If you hear the voice on the Attendant's Handset, then you corrected the problem.

2) If you do not hear the voice on the Attendant's Handset, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. Attendant To Flight Deck, Captain's Call Light Does Not Operate - Fault Isolation

A. Description

(1) The CALL light on the forward overhead panel does not come on, but the call chime sounds when the PILOT button is pushed on an attendant's handset.

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B. Possible Causes

(1) Master Dim and Test switch, S3

(2) Captain's Call Light, L19

(3) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

D. Related Data

(1) (SSM 23-42-11)

(2) (SSM 23-43-11)

(3) (WDM 23-42-11)

(4) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the captain's CALL light:

(a) Set the Master Dim and Test switch, S3, on the P1 main instrument panel, to BRT.

(b) Press and hold the button for the pilot on the attendant's handset.

1) Make sure the CALL light on the P5 forward overhead panel goes on.

(c) Release the button on the attendant's handset.

(d) If the CALL light does not go on, then do the Fault Isolation Procedure below.

(e) If the CALL light goes on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this test of the Captain's CALL light, L19:

(a) Set the Master Dim and Test switch, S3, on the P1 main instrument panel, to TEST.

(b) If the CALL light does not go on, then replace the Captain's CALL light.

1) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(c) If the CALL light goes on, then continue.

(2) Do this check of the wiring:

(a) Remove the Captain's CALL light, L19, from the P5 forward overhead panel.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(b) Remove the handset logic control card from the attendant's panel.

OMR 815, 816, 818-826, 901-999

(c) Remove the attendant control panel. To remove it, do this task: Attendant's Panel with LCD Touch Panel - Removal, AMM TASK 25-25-11-000-802

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OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(d) Do a check for an open circuit between connector D40170J of the Captain's CALL light, L19, and connector D11068 (forward) or D11466 (aft) of the lavatory module. For correct pins, refer to WDM 23-42-11.

OMR 815, 816, 818-826, 901-999

(e) Do a check for an open circuit between connector D40170J at the Captain's CALL light, L19, and connector D14368 (forward attendant control panel) or D14376 (aft attendant control panel). For correct pins, refer to WDM 23-42-11.

OMR ALL

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

2) Re-install the handset logic control card.

OMR 815, 816, 818-826, 901-999

3) Re-install the attendant control panel. Do this task: Attendant's Panel with LCD Screen - Installation, AMM TASK 25-25-11-400-802.

OMR ALL

4) Re-install the Captain's CALL light.

5) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the Captain's CALL light:

(a) Press and hold the button for the pilot on the attendant's handset.

1) Make sure the CALL light on the P5 forward overhead panel goes on.

(b) Release the button on the attendant's handset.

(c) If the CALL light goes on, then you corrected the fault.

——— END OF TASK ———

OMR 815, 816, 818-826, 901-999

807. Internal ACP Fault In Forward/Aft ACP - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11001
(b) 23-11002
(c) 23-11003
(d) 23-11004
(e) 23-11010
(f) 23-11011
(g) 23-11012
(h) 23-11013
(i) 23-11025

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Attendant Control Panel (ACP)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

808. Internal ACP Fault - Data Loading - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11005

(b) 23-11006

(c) 23-11007

(d) 23-11018

(e) 23-11019

(f) 23-11020

(g) 23-11021

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Data load

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

C. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

D. Fault Isolation Procedure

(1) Do ADL to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault.

——— END OF TASK ———

809. Internal ACP Fault - HW Config - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11008

(b) 23-11014

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Attendant Control Panel (ACP)

C. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

D. Fault Isolation Procedure

(1) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

810. Light Column Address Fault - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11028

(b) 23-11029

(c) 23-11030

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(d) 23-11031

(e) 23-11032

(f) 23-11033

(g) 23-11034

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Data load

C. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

D. Fault Isolation Procedure

(1) Set the cabin/utility switch on the overhead panel P5 to OFF and ON position to cycle power to the light LRU's.

(2) Do the system test, refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do ACP to LRU or Phase 2 Data Load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault.

END OF TASK

811. Fwd/Aft Work Light Fault - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11015.

(2) The fault is generated by the system test.

B. Possible Causes

(1) Wiring to the Fwd/Aft Work Light

(2) Attendant Control Panel (ACP)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

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D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do the wiring check and correct problems (if applicable) for the Work Light. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(5) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(6) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

812. Fwd/Aft MDCD Light Fault - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11016.

(2) The fault is generated by the system test.

B. Possible Causes

(1) Wiring to the Main Deck Cargo Door (MDCD) Light

(2) Attendant Control Panel (ACP)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

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OMR 815, 816, 818-826, 901-999 (Continued)

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do the wiring check and correct problems (if applicable) for the MDCD Light. Refer to: Cargo/Service Compartment Lighting Problem - Fault Isolation, 33-30 TASK 801.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(5) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(6) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

813. Fwd/Aft Threshold Light Fault - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11024.

(2) The fault is generated by the system test.

B. Possible Causes

(1) Wiring to the Fwd/Aft Threshold Light

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

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(3) Do the wiring check and correct problems (if applicable) for the Threshold Light. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault.

——— END OF TASK ———

814. 28 Vdc Power Fault in Fwd/Aft ACP - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11017.

(2) The fault is generated by the system test.

B. Possible Causes

(1) 28 Vdc Power

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) If the fault shows for both forward and aft ACP's, do this check for 28 Vdc at the circuit breaker:

(a) Do a check for 28 Vdc at the load terminal of the circuit breaker C01936 ATTENDANT PANEL to structure ground.

(b) If there is not 28 Vdc at the load terminal, replace the circuit breaker.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(3) Do a check for wiring from the circuit breaker C01936 to the applicable ACP and fix problems.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

815. Communication with Aft ACP Lost - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11022.

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OMR 815, 816, 818-826, 901-999 (Continued)

(2) The fault is generated by the system test.

B. Possible Causes

(1) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to:Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do a check for the wiring between the forward ACP and the aft ACP and fix problems if you find them. Refer to :WDM 23-42-XX.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

816. Cabin Temperature Fault - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11023.

(2) The fault is generated by the system test.

B. Possible Causes

(1) Data Load

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do data load for the CDB software. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do data load for the OPS software. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault.

——— END OF TASK ———

817. Lavatory Inoperative / Clean Check Sensors - Fault Isolation

A. Description

(1) This task is for the maintenance messages 23-11026 and 23-11027.

(2) The fault is generated by the system test.

B. Possible Causes

(1) The Lavatory Logic Control Module

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do a check of the Lavatory Logic Control Module and correct problems (if applicable). Refer to: LOGIC CONTROL MODULE - MAINTENANCE PRACTICES, AMM 38-33-03/201.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

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(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do a check of wiring to the Lavatory Logic Control Module and correct problems (if applicable). Refer to: WDM 38-3X.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

END OF TASK

818. RS-485 Cable Break - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11039

(b) 23-11040

(c) 23-11041

(d) 23-11042

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Wiring connection

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do a check for wiring connection and correct problems (if applicable) for the light(s) indicated on the ACP. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

819. No LRU Responses - Fault Isolation

A. Description

(1) This task is for the following maintenance messages:

(a) 23-11035

(b) 23-11036

(c) 23-11037

(d) 23-11038

(2) The fault(s) is generated by the system test.

B. Possible Causes

(1) Wiring connection at the light

(2) Wiring connection at the forward ACP

(3) Forward ACP

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do a wiring connection check and correct problems (if applicable) at the specified light. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(2) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Do a wiring connection check and correct problems (if applicable) at the forward ACP. Refer to: WDM 23-42-XX.

(4) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(5) Replace the forward ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(6) Do the system test. Refer to: Attendant Control Panel (ACP) - BITE Procedure, 23-42 TASK 801.

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(a) If the maintenance message does not show, then you corrected the fault.

END OF TASK

820. LRU Address Fault - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11043

(2) The fault(s) is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) The light LRU

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. This is the task: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do a check and correct problems (if applicable) for the light LRU. This is the task: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(2) Do the ACP to LRU or Phase 2 data load. This is the task: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

END OF TASK

821. LRU Address - Extra Response - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) 23-11045

(2) The fault is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) The Configuration Database (CDB) software

(2) The light LRU

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OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Make sure that the CDB part number is correct.

(a) If the CDB part number is not correct, do the data load again with the correct CDB. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the CDB part number is correct, then continue.

(2) Make sure the CDB matches the airplane configuration.

(a) If the CDB does not match the airplane configuration, do the ACP to LRU or Phase 2 data load again with the correct CDB that matches the airplane configuration. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the CDB matches the airplane configuration, then continue.

(3) Do a check for wiring connection and correct problems for the light LRU. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(4) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

822. LRU Address - Incorrect Light - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) 23-11046

(2) The fault is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) The Configuration Database (CDB) software

(2) The light LRU

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Make sure that the CDB part number is correct.

(a) If the CDB part number is not correct, do the data load again with the correct CDB. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the CDB part number is correct, then continue.

(2) Make sure the CDB matches the airplane configuration.

(a) If the CDB does not match the airplane configuration, do the ACP to LRU or Phase 2 data load again with the correct CDB that matches the airplane configuration. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the CDB matches the airplane configuration, then continue.

(3) Make sure the LRU light part number is correct.

(a) If the LRU light part number is not correct, replace it with the correct light. Refer to chapter 33 for the correct procedure.

(4) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

823. LRU Address Fault - Zone/Scene Download - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11047

(b) 23-11048

(c) 23-11050

(d) 23-11051

(e) 23-11052

(f) 23-11053

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(g) 23-11054

(h) 23-11055

(2) The fault(s) is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) The light LRU

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Set the cabin/utility switch on the overhead panel P5 to OFF and ON position to cycle power to the light LRU's.

(2) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault message does not show, then you corrected the fault. If the message shows, then continue.

(3) Replace the light LRU.

(4) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

824. LRU Address - Incorrect Zone - Fault Isolation

A. Description

(1) This task is for the maintenance message:

(a) 23-11049

(2) The fault is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) The Configuration Database (CDB)

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OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Make sure that the CDB part number is correct.

(a) If the CDB part number is not correct, do the data load again with the correct CDB. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the CDB part number is correct, then continue.

(2) Make sure the CDB matches the airplane configuration.

(a) If the CDB does not match the airplane configuration, do the ACP to LRU or Phase 2 data load again with the correct CDB that matches the airplane configuration. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

825. No Data Load LRU Responses - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11056

(b) 23-11057

(c) 23-11058

(d) 23-11059

(2) The fault(s) is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) Wiring

(2) The forward ACP

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OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Set the cabin/utility switch on the overhead panel P5 to OFF and ON position to cycle power to the light LRU's.

(2) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(3) Do a wiring connection check and correct problems (if applicable) at the specified light. Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(4) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(5) Replace the forward ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(6) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

826. Data Load RS-485 Cable Break - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11060

(b) 23-11061

(c) 23-11062

(d) 23-11063

(2) The fault(s) is generated during the ACP to LRU or Phase 2 data load.

B. Possible Causes

(1) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to LRU or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Set the cabin/utility switch on the overhead panel P5 to OFF and ON position to cycle power to the light LRU's.

(2) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(3) Do a wiring connection check and correct problems (if applicable) at the specified light(s). Refer to: Passenger Compartment Lighting Problem - Fault Isolation, 33-20 TASK 801.

(4) Do the ACP to LRU or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

827. Aft ACP Internal Fault - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11064

(b) 23-11065

(c) 23-11066

(d) 23-11067

(2) The fault is generated during the ACP to ACP or Phase 2 data load.

B. Possible Cause

(1) The aft ACP

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

D. Initial Evaluation

(1) Do the ACP to ACP or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the ACP to ACP or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(3) Replace the aft ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(4) Do the ACP to ACP or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

828. Communication with Aft ACP Lost - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11068.

(2) The fault is generated during the ACP to ACP or Phase 2 data load.

B. Possible Causes

(1) Wiring

(2) The ACP's

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to ACP or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows on the screen, then continue.

E. Fault Isolation Procedure

(1) Do a wiring check between the forward and the aft ACP's. Refer to: WDM 23-42-XX.

(a) Correct the problems that you find (if applicable).

(2) Do the ACP to ACP or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

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OMR 815, 816, 818-826, 901-999 (Continued)

(3) Replace the aft ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(4) Do the ACP to ACP or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(5) Replace the forward ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(6) Do the ACP to ACP or Phase 2 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

829. LSAP Configuration Invalid - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11069.

(2) The fault is generated during the ACP to ACP or Phase 2 data load.

B. Possible Causes

(1) Incorrect LSAP

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ACP to ACP or Phase 2 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Make sure the LSAP part number is correct. If it is not correct, do the ACP to ACP or Phase 2 data load again with the correct LSAP part number. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

830. Forward ACP Internal Fault - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11070

(b) 23-11071

(c) 23-11072

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(d) 23-11073

(2) The fault is generated during the ADL to ACP or Phase 1 data load.

B. Possible Causes

(1) The forward ACP

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ADL to ACP or Phase 1 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(2) Do the ADP to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(3) Replace the forward ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(4) Do the ADP to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

831. LSAP Disk Fault - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11074

(b) 23-11075

(c) 23-11076

(d) 23-11077

(2) The fault is generated during the ADL to ACP or Phase 1 data load.

B. Possible Causes

(1) The LSAP diskette

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OMR 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ADL to ACP or Phase 1 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Check the LSAP diskette, make sure that it is not physically broken. If it is broken, replace it with another diskette (same LSAP part number).

(2) Do the ADL to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

832. Incompatible Disk Fault - Fault Isolation

A. Description

(1) This task is for the maintenance messages:

(a) 23-11078

(b) 23-11079

(2) The fault is generated during the ADL to ACP or Phase 1 data load.

B. Possible Causes

(1) Incorrect LSAP

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ADL to ACP or Phase 1 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

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OMR 815, 816, 818-826, 901-999 (Continued)

E. Fault Isolation Procedure

(1) Make sure that the LSAP diskette part number is correct. If it is not correct, replace it with the diskette with correct LSAP part number.

(2) Do the ADL to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

833. Communications with ADL Lost - Fault Isolation

A. Description

(1) This task is for the maintenance message 23-11080.

(2) The fault is generated during the ADL to ACP or Phase 1 data load.

B. Possible Causes

(1) Incorrect ADL set up

(2) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Do the ADL to ACP or Phase 1 data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Make sure that the ADL selector switch is in the correct position.

(a) If the ADL selector switch is not in the correct position, reset the switch and do the data load again. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

1) If the fault does not show, then you corrected the fault.

(b) If the ADL selector switch is correct, then continue.

(2) Open and close the circuit breaker C01936 to cycle the power to the ACP's.

(3) Do the ADL to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault. If the fault shows, then continue.

(4) Do a wiring check from the ACP to the ADL. Refer to: WDM 23-42-XX

(5) Do the ADL to ACP or Phase 1 data load. Refer to: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

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OMR 815, 816, 818-826, 901-999 (Continued)

(a) If the fault does not show, then you corrected the fault.

——— END OF TASK ———

839. LED on the ACP Switch Assembly (ISA) is ON - Fault Isolation

A. Description

(1) When the LED on the ISA is on steadily (not flashing), this indicates an over temperature condition and the ACP is not operational.

(2) When the LED on the ISA is on and flashing, this indicates an internal critical fault with the ACP and the ACP is shut down.

B. Possible Causes

(1) Ambient temperature is out of range

(2) ACP internal fault

C. Initial Evaluation

(1) If the LED is on and not flashing (indicating that ambient temperature is out of range), no maintenance action is necessary. The ACP will become operational when the temperature is in range (32°F (0°C) to122°F (50°C)).

(2) If the LED is flashing, there is an internal fault with the ACP. Do the fault isolation procedure that follows.

D. Fault Isolation Procedure

(1) Do the steps that follow when the LED is flashing:

(a) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(b) If the LED on the ISA does not turn on, then you corrected the fault.

——— END OF TASK ———

840. Attendant Control Panel - Touch Screen Inoperative

A. Description

(1) When the ACP Touch Screen is inoperative, it does not respond to any touching inputs.

B. Possible Causes

(1) The ACP Touch Screen is locked up.

(2) The Touch Screen Display function is not operative.

C. Initial Evaluation

(1) To activate the Touch Screen, touch the two opposite corners of the screen sequentially.

(a) If the Touch Screen responds to inputs then you corrected the problem.

(b) If the Touch Screen does not respond to any touching inputs, do the fault isolation procedure that follows.

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OMR 815, 816, 818-826, 901-999 (Continued)

D. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

E. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP.

(a) Touch the two opposite corners of the Touch Screen to activate the Touch Screen

(b) If the Touch screen responds to inputs, then there was an intermittent fault.

(c) If the Touch Screen does not respond to touching inputs, then continue.

(2) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(a) If the touch screen operates, then you corrected the fault.

——— END OF TASK ———

841. Amber LED Light on ACP Switch Assembly does not come on during operational test

A. Description

(1) After power-up, the amber LED at the left corner of the ACP should turn on for about 2 seconds and then turn off. If the LED does not turn on at power-up, it might be inoperative.

B. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

C. Fault Isolation Procedure

(1) Open and close the circuit breaker C01936 to cycle the power to the ACP.

(a) Check the LED again during the first 2 seconds of power-up.

(b) If the LED light turns on during the first 2 seconds of power-up and then turn off, then there was an intermittent fault.

(c) If the LED light does not turn on during the first 2 seconds of power-up, then continue.

(2) Replace the applicable ACP. Refer to:AMM PAGEBLOCK 25-25-11/401.

(a) If the LED light turns on during the first 2 seconds of power-up and then turn off, then you corrected the fault.

——— END OF TASK ———

844. LRU Address Fault - Out of Sequence

A. Description

(1) This task is for maintenance message:

(a) 23–11044

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(2) The fault(s) is generated during Phase 2 Data Load.

(3) During Phase 2 Data Load, the error text “responded out of sequence” will appear in red color next to the light address and location.

(4) This error message is only shown on ACP OPS V5. This means that there were retries on the data bus during data load.

B. Possible Causes

(1) The light LRU

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

D. Initial Evaluation

(1) Set the cabin/utility switch on the overhead panel P5 to OFF and ON position to cycle power to the light LRU's.

(2) Do the Phase 2 data load again. This is the task: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then there was an intermittent fault.

(b) If the fault shows, then continue.

E. Fault Isolation Procedure

(1) Do an Operational Check of the lights. This is the task: Passenger Compartment Lights - Operational Test, AMM TASK 33-20-00-710-804-001.

(2) Do an ACP Configuration Check. This is the task: Attendant Control Panel (ACP) Configuration Check, AMM TASK 23-42-03-700-801.

(a) If all the BSI lights are responding to ACP commands normally and there are no other errors from the Phase 2 Data Load and the Configuration Check, then the Phase 2 Data load was successful. Corrective action for the “Out of Sequence” error can be deferred.

NOTE: The “Out of Sequence” error is due to a higher than tolerance impedance value on the light assembly data bus. The Phase 2 Data Load can still be successful even with this error.

(3) Replace the light LRU.

(a) Replace the applicable light(s) that responded out of sequence.

(b) Do this task Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

F. Repair Confirmation

(1) Do the Phase 2 data load again. This is the task: Attendant Control Panel (ACP) Software Loading, AMM TASK 23-42-03-470-801.

(a) If the fault does not show, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Ground Crew Call Horn Does Not Operate - Fault Isolation

A. Description

(1) The ground crew call horn does not sound in the nose wheel well area.

B. Possible Causes

(1) Ground Crew Call Horn, M53

(2) Ground Crew Call Switch, S32

(3) Wiring

(4) No power to the Ground Crew Call Horn

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

D. Related Data

(1) (SSM 23-43-11)

(2) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the ground crew call horn:

(a) Push and hold the GRD CALL switch on the P5 forward overhead panel.

1) Make sure that you hear the ground crew call horn in the nose wheel well area.

(b) Release the GRD CALL switch.

(c) If the ground crew call horn does not sound, then do the Fault Isolation Procedure below.

(d) If the ground crew call horn sounds, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this check for electrical power to the ground crew call horn:

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

(b) Remove the Ground Crew Call Horn, M53, from the nose wheel well area. To remove it, do this task: Ground Crew Call Horn Removal, AMM TASK 23-43-02-000-801.

(c) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

(d) Push and hold the GRD CALL switch on the P5 forward overhead panel.

(e) Do a check for 28 VDC across the Ground Crew Call Horn wires (one 28 VDC, the other ground).

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(f) Release the GRD CALL switch.

(g) If there is 28 VDC, then do these steps:

1) Install a new Ground Crew Call Horn. To install it, do this task: Ground Crew Call Horn Installation, AMM TASK 23-43-02-420-801.

2) Do the Repair Confirmation at the end of this task.

(h) If there is not 28 VDC, then continue.

(2) Replace the Ground Crew Call Switch, S32, on the P5 forward overhead panel.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not OK, then continue.

(3) Repair the wiring between the ground crew call horn, M53, and the circuit breaker, C73, on the P18-3 circuit breaker panel.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the ground crew call horn:

(a) Push and hold the GRD CALL switch on the P5 forward overhead panel.

1) Make sure that you hear the ground crew call horn in the nose wheel well area.

(b) Release the GRD CALL switch.

(c) If the ground crew call horn sounds, then you corrected the fault.

——— END OF TASK ———

802. Ground Crew Call Horn Operates Continuously - Fault Isolation

A. Description

(1) The ground crew call horn stays on continuously.

B. Possible Causes

(1) Air Data Inertial Reference Unit (ADIRU), M1749 (left) or M1752 (right)

(2) Ground crew call switch, S32

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC

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F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

D. Related Data

(1) (SSM 23-43-11)

(2) (WDM 23-43-11)

E. Fault Isolation Procedure

(1) Do this task: Ground Crew Call Horn Sounds - Fault Isolation, 34-21 TASK 829.

(a) If the ground crew call horn goes off, then you corrected the fault.

(b) If you still hear the ground crew call horn, then continue.

(2) Replace the Ground Crew Call Switch, S32, on the P5 forward overhead panel.

(a) Do this check of the ground crew call horn:

1) Push and hold the GRD CALL switch on the P5 forward overhead panel.

a) Make sure you hear the ground crew call horn in the nose wheel well area.

2) Release the GRD CALL switch.

a) Make sure that the ground crew call horn goes off.

3) If the ground crew call horn sounds and goes off, then you corrected the fault.

——— END OF TASK ———

803. Ground Crew to Flight Deck, Call System Does Not Operate - Fault Isolation

A. Description

(1) The CALL light on the forward overhead panel does not come on and the Aural Warning Unit does not chime when the ground crew pushes the PILOT CALL button on the External Power Panel.

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B. Possible Causes

(1) PILOT CALL Switch, S33

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

D. Related Data

(1) (SSM 23-43-11)

(2) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the captain's CALL light:

(a) Press and hold the PILOT CALL button on the P19 External Power panel.

1) Make sure the CALL light on the P5 forward overhead panel goes on.

2) Make sure that you hear a chime in the flight deck.

(b) Release the PILOT CALL button.

(c) If the CALL light does not go on and the chime does not sound, then do the Fault Isolation Procedure below.

(d) If the CALL light goes on and the chime sounds, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the PILOT CALL switch, S33, on the P19 External Power panel.

(a) Do this check of the captain's CALL light:

1) Press and hold the PILOT CALL button on the P19 External Power panel.

a) Make sure the CALL light on the P5 forward overhead panel goes on.

b) Make sure that you hear a chime in the flight deck.

2) Release the PILOT CALL button.

3) If the CALL light goes on and you hear a chime, then you corrected the fault.

——— END OF TASK ———

804. Ground Crew to Flight Deck, Call Light Does Not Operate - Fault Isolation

A. Description

(1) The CALL light on the forward overhead panel does not come on, but the call chime sounds when the ground crew pushes the PILOT CALL button on the External Power Panel.

B. Possible Causes

(1) Master Dim and Test Switch, S3

(2) Captain's Call Light, L19

(3) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

D. Related Data

(1) (SSM 23-43-11)

(2) (WDM 23-43-11)

E. Initial Evaluation

(1) Do this check of the captain's CALL light:

(a) Set the Master Dim and Test switch, S3, on the P1 main instrument panel, to BRT.

(b) Push and hold the PILOT CALL button, S33, on the P19 External Power panel.

1) Make sure the CALL light on the P5 forward overhead panel goes on.

(c) Release the PILOT CALL button.

(d) If the CALL light does not go on, then do the Fault Isolation Procedure below.

(e) If the CALL light goes on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this test of the Captain's CALL light, L19:

(a) Set the Master Dim and Test switch, S3, on the P1 main instrument panel, to TEST.

(b) If the CALL light does not go on, then replace the Captain's CALL light.

1) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is not OK, then continue.

(c) If the CALL light goes on, then continue.

(2) Do this check of the wiring:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT

(b) Remove the Captain's CALL light, L19, from the P5 forward overhead panel.

(c) Remove the PILOT CALL switch, S33, from the P19 External Power panel.

(d) Do a check for an open circuit between these pins of the Captain's CALL light, L19, and the PILOT CALL switch, S33:

L19		S33
pin 1	. . . . .	pin C

(e) If you find a problem with the wiring, then do these steps:

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1) Repair the wiring.
2) Re-install the PILOT CALL switch.
3) Re-install the Captain's CALL light.
4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT

5) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the captain's CALL light:
   (a) Press and hold the PILOT CALL switch, S33, on the P19 External Power panel.
      1) Make sure the captain's CALL light, L19, on the P5 forward overhead panel goes on.
   (b) Release the PILOT CALL switch.
   (c) If the CALL light goes on, then you corrected the fault.

——— END OF TASK ———

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801. Microphone/Headset Problem - Fault Isolation

A. Description

(1) A flight interphone hand microphone, headset, or boom microphone/headset does not operate.

B. Possible Causes

(1) Hand microphone, headset, or boom microphone/headset

(2) Microphone, headset, or boom microphone/headset jack

(3) Audio Control Panel (ACP), P8-6 (captain's), P8-7 (first officer's), or P5-15 (observer's)

(4) Oxygen Mask stowage panels.

(5) Remote Electronic Unit (REU), M1353

(6) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the flight interphone for initial evaluation and for fault isolation:

OMR 701-799, 802, 805, 807, 809, 812, 813, 815, 816, 818-826, 901-999

(a) Make sure the AURAL WARN MUTE switch on the front panel of the REU is not in the “Mute” position.

NOTE: The switch is in “Mute” position when it is horizontal to the bottom edge of the front panel.

OMR ALL

(b) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the OFF position.

(c) Connect a microphone and headset to all flight interphone stations.

(d) Push the flight microphone selector switch on all Audio Control Panels (ACPs). The flight microphone selector switch is labelled FLT, FLT INT, or INT.

(e) Push the volume control for the flight microphone selector switch to on.

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(2) Do this check of the flight interphone:

(a) Push and hold the test switch on the affected station's Audio Control Panel in the R/T or RADIO position. The switch is labelled R/T - I/C or RADIO - INT.

(b) Speak into the affected station's microphone.

1) Make sure you can hear the voice clearly from the other stations' headset.

(c) Release the test switch.

(d) If the voice is not heard on all headsets, then do the Fault Isolation Procedure below.

(e) If the voice is heard on all headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the affected hand microphone, headset, or boom microphone/headset.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Replace the affected station's hand microphone, headset, or boom microphone/headset jack.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Replace the affected Audio Control Panel (ACP):

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(4) Replace the Remote Electronic Unit (REU), M1353:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(5) Use the SSM and WDM listed in Related Data to do a check of the wires from the remote electronics unit:

(a) Remove the affected station's hand microphone, headset, or boom microphone/headset jack.

(b) Remove the REU, M1353 from the E4-1 shelf (AMM TASK 23-51-01-000-801).

(c) Examine and repair the wires between the REU connector D2501B and the microphone, headset, or boom microphone/headset jack connector.

1) If you found and repaired a wire problem, then do these steps:

a) Re-install the REU (AMM TASK 23-51-01-400-801).

b) If it is necessary, re-install the hand microphone, headset, or boom microphone/headset jack.

c) Do the Repair Confirmation at the end of this task.

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G. Repair Confirmation

(1) If it is necessary, do these steps to prepare the flight interphone for Repair Confirmation:

(a) Make sure that the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the OFF position.

(b) Make sure that a microphone and headset is connected at all flight interphone stations.

(c) Makes sure that these steps are completed:

1) Push the flight microphone selector switch on all Audio Control Panels (ACPs). The flight microphone selector switch is labelled FLT, FLT INT, or INT.

2) Push the volume control for the flight microphone selector switch to on.

(2) Do this check of the flight interphone:

(a) Push and hold the test switch on the affected station's Audio Control Panel in the R/T or RADIO position. The switch is labelled R/T - I/C or RADIO - INT.

(b) Speak into the affected station's microphone.

1) Make sure you can hear the voice clearly from the other stations' headset.

(c) Release the test switch.

(d) If the voice is heard on all headsets, then you have corrected the problem.

——— END OF TASK ———

802. Damaged Microphone/Headset - Fault Isolation

A. Fault Isolation Procedure

(1) Replace the affected hand microphone, headphone, or boom microphone/headset.

——— END OF TASK ———

803. Flight Interphone Problem At All Stations - Fault Isolation

A. Description

(1) The flight interphone does not operate at any flight deck station.

B. Possible Causes

(1) Remote Electronics Unit (REU), M1353

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

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D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the flight interphone for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the OFF position.

(b) Connect a boom mic/headset to both pilots' boom microphone/headset jacks in the flight deck.

(c) Push the flight microphone selector switch on all Audio Control Panels (ACPs). The flight microphone selector switch is labelled FLT, FLT INT, or INT.

(d) Push the volume control for the flight microphone selector switch to on.

(2) Do this check of the flight interphone:

(a) Push and hold the PTT switch on the affected pilot's control wheel to the MIC position.

(b) Speak into the boom microphone.

1) Make sure you can hear the voice clearly from the other pilot's headset.

(c) Release the PTT switch on the control wheel.

(d) Push and hold the PTT switch on the other pilot's control wheel to the MIC position.

(e) Speak into the boom microphone.

1) Make sure you can hear the voice clearly from the other pilot's headset.

(f) Release the PTT switch on the control wheel.

(g) If the voice is not heard on either headset, then do the Fault Isolation Procedure below.

(h) If the voice is heard on both headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the Remote Electronic Unit (REU), M1353:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do this check of the flight interphone:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do this check.

1) Push and hold the PTT switch on the pilot's control wheel to the MIC position.

2) Speak into the boom microphone/headset.

a) Make sure you can hear the voice clearly from the other pilot's headset.

3) Release the PTT switch on the control wheel.

4) Push and hold the PTT switch on the other pilot's control wheel to the MIC position.

5) Speak into the boom microphone.

a) Make sure you can hear the voice clearly from the other pilot's headset.

6) Release the PTT switch on the control wheel.

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7) If the voice is heard on both headsets, then you corrected the fault.

——— END OF TASK ———

804. Flight Interphone Problems At One Station - Fault Isolation

A. Description

(1) The flight interphone does not operate at one station. The other stations operate correctly.

B. Possible Causes

(1) Push-To-Talk (PTT) switch, S519 (captain's) or S520 (first officer's)

OMR 805, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Glareshield PTT switch, S1091 (captain's) or S989 (first officer's)

OMR ALL

(3) Audio Control Panel (ACP), P8-6 (captain's), P8-7 (first officer's), or P5-15 (observer's)

(4) Remote Electronics Unit (REU), M1353

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the flight interphone for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 overhead panel, to the OFF position.

(b) Connect a microphone and headset to jacks at all flight interphone stations in the flight deck.

(c) Push the flight microphone selector switch on all ACPs. The flight microphone selector switch is labelled FLT, FLT INT, or INT.

(d) Push the volume control for the flight microphone selector switch to on.

(2) Do this check of the flight interphone:

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(a) Push and hold the test switch on the affected station's ACP in the R/T or RADIO. The switch is labelled R/T - I/C or RADIO - INT.

(b) Speak into the affected station's microphone.

1) Make sure that you can hear the voice clearly from the other stations' headset.

(c) Release the test switch.

(d) If the voice is not heard on all headsets, then do the Fault Isolation Procedure below.

(e) If the voice is heard on all headsets, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the PTT switch, S519 (captain's) or S520 (first officer's):

- Control Wheel PTT Switch Removal, AMM TASK 23-51-04-000-801
- Control Wheel PTT Switch Installation, AMM TASK 23-51-04-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

OMR 805, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Replace the glareshield PTT switch, S1091 (captain's) or S989 (first officer's):

- Glareshield PTT Switch Removal, AMM TASK 23-51-05-000-801
- Glareshield PTT Switch - Installation, AMM TASK 23-51-05-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

OMR ALL

(3) Replace the affected ACP:

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(4) Replace the REU, M1353:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) If it is necessary, do these steps to prepare the flight interphone for Repair Confirmation:

(a) Make sure that the SERVICE INTERPHONE switch, S50, on the P5 overhead panel, to the OFF position.

(b) Make sure that a microphone and headset is connected at all flight interphone stations.

(c) Makes sure that these steps are completed:

1) Push the flight microphone selector switch on all ACPs. The flight microphone selector switch is labelled FLT, FLT INT, or INT.

2) Push the volume control for the flight microphone selector switch to on.

(2) Do this check of the flight interphone:

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(a) Push and hold the test switch on the affected station's ACP in the R/T or RADIO position. The switch is labelled R/T - I/C or RADIO - INT.

(b) Speak into the affected station's microphone.

1) Make sure that you can hear the voice clearly from the other stations' headset.

(c) Release the test switch.

(d) If the voice is heard on all headsets, then you have corrected the problem.

END OF TASK

805. Captain's Flight Interphone Speaker Problem - Fault Isolation

A. Description

(1) Sound is not heard from the captain's flight interphone speaker.

B. Possible Causes

(1) Captain's Flight Interphone Speaker, M77

(2) Captain's Audio Control Panel, P8-6

(3) Remote Electronic Unit (REU), M1353

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the flight interphone for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the OFF position.

(b) Connect a microphone to the first officer's microphone jack.

(c) Set all audio control panels (ACPs) to these conditions:

1) Push all audio monitor switches to off.

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2) Push the flight microphone selector switch to on. The flight microphone selector switch is labelled FLT, FLT INT, or INT.
   a) Make sure its light comes on.
3) Push the volume control for the flight microphone selector switch. The flight microphone selector switch is labelled FLT, FLT INT, or INT.
   a) Make sure its light comes on.
4) Turn the volume control for the flight microphone selector switch to the middle position. The flight microphone selector switch is labelled FLT, FLT INT, or INT.

(d) Set the captain's ACP to these conditions:
1) Push the SPKR volume control switch to on.
   a) Make sure its light comes on.
2) Turn the SPKR volume control switch to the middle position or to the volume level you are comfortable with.

(2) Do this check of the captain's flight interphone speaker:
(a) Push and hold the PTT switch on the first officer's control wheel to the MIC position.
(b) Speak into the first officer's microphone.
(c) Make sure you can hear the voice clearly from the captain's flight interphone speaker.
(d) If the voice is not heard on the captain's flight interphone speaker, then do the Fault Isolation Procedure below.
(e) If the voice is heard on the captain's flight interphone speaker, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the captain's flight interphone speaker:
- Flight Interphone Speaker Removal, AMM TASK 23-51-03-000-802
- Flight Interphone Speaker Installation, AMM TASK 23-51-03-000-804.

(a) Do the Repair Confirmation at the end of this task.
(b) Do the Repair Confirmation at the end of this task.
1) If the Repair Confirmation is not OK, then continue.

(2) Replace the captain's Audio Control Panel (ACP):
- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.
1) If the Repair Confirmation is not OK, then continue.

(3) Replace the Remote Electronic Unit (REU):
- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.
1) If the Repair Confirmation is not OK, then continue.

(4) Do this check of the wiring:
(a) Remove the REU, M1353 from the E4-1 shelf (AMM TASK 23-51-01-000-801).
(b) Remove the captain's flight interphone speaker (AMM TASK 23-51-03-000-802).

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(c) Do a check for an open circuit between connector D2501B, pin D8 of the REU, M1353, and connector D127, pin 1 of the captain's flight interphone speaker, M77:

(d) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the captain's flight interphone speaker (AMM TASK 23-51-03-000-804).

3) Re-install the REU (AMM TASK 23-51-01-400-801).

4) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the captain's flight interphone speaker:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do this check.

(a) Push and hold the PTT switch on the first officer's control wheel to the MIC position.

(b) Speak into the first officer's microphone.

(c) Make sure you can hear the voice clearly from the captain's flight interphone speaker.

(d) If the voice is heard on the captain's flight interphone speaker, then you corrected the fault.

——— END OF TASK ———

806. First Officer's Flight Interphone Speaker Problem - Fault Isolation

A. Description

(1) Sound is not heard from the first officer's flight interphone speaker.

B. Possible Causes

(1) Remote Electronic Unit (REU), M1353

(2) First Officer's Audio Control Panel, P8-6

(3) First Officer's Flight Interphone Speaker, M78

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

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(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the flight interphone for initial evaluation and for fault isolation:

(a) Set the SERVICE INTERPHONE switch, S50, on the P5 Overhead Panel to the OFF position.

(b) Connect a microphone to the captain's microphone jack.

(c) Set all audio control panels (ACPs) to these conditions:

1) Push all audio monitor switches to off.

2) Push the flight microphone selector switch to on. The flight microphone selector switch is labelled FLT, FLT INT, or INT.

a) Make sure its light comes on.

3) Push the volume control for the flight microphone selector switch. The flight microphone selector switch is labelled FLT, FLT INT, or INT.

a) Make sure its light comes on.

(d) Set the first officer's ACP to these conditions:

1) Push the SPKR volume control switch to on.

a) Make sure its light comes on.

2) Turn the SPKR volume control switch to the middle position or to the volume level you are comfortable with.

(2) Do this check of the first officer's flight interphone speaker:

(a) Push and hold the PTT switch on the captain's control wheel to the MIC position.

(b) Speak into the captain's microphone.

(c) Make sure you can hear the voice clearly from the first officer's flight interphone speaker.

(d) If the voice is not heard on the first officer's flight interphone speaker, then do the Fault Isolation Procedure below.

(e) If the voice is heard on the first officer's flight interphone speaker, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the first officer's Audio Control Panel (ACP):

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(2) Replace the Remote Electronic Unit (REU):

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(3) Replace the first officer's flight interphone speaker:

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- Flight Interphone Speaker Removal, AMM TASK 23-51-03-000-802
- Flight Interphone Speaker Installation, AMM TASK 23-51-03-000-804.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not OK, then continue.

(4) Do this check of the wiring:

(a) Remove the REU, M1353 from the E4-1 shelf (AMM TASK 23-51-01-000-801).

(b) Remove the first officer's flight interphone speaker (AMM TASK 23-51-03-000-802).

(c) Do a check for an open circuit between connector D2501B, pin C11 of the REU, M1353, and connector D129, pin 1 of the first officer's flight interphone speaker, M78:

(d) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the first officer's flight interphone speaker (AMM TASK 23-51-03-000-804).

3) Re-install the REU (AMM TASK 23-51-01-400-801).

4) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the first officer's flight interphone speaker:

NOTE: You must do the steps to prepare for fault isolation that are in the Initial Evaluation before you can do this check.

(a) Push and hold the PTT switch on the captain's control wheel to the MIC position.

(b) Speak into the captain's microphone.

(c) Make sure you can hear the voice clearly from the first officer's flight interphone speaker.

(d) If the voice is heard on the first officer's flight interphone speaker, then you corrected the fault.

——— END OF TASK ———

807. Audio Control Panel Problems - Fault Isolation

A. Description

(1) The audio control panel does not operate correctly at one station. The other stations operate correctly.

B. Possible Causes

(1) Audio Control Panel (ACP), P8-6 (captain's), P8-7 (first officer's), or P5-15 (observer's)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do these steps to prepare the audio control panel for initial evaluation and for fault isolation:

(a) Visually examine the audio control panel.

1) Identify the light or switch that does not operate correctly.

F. Fault Isolation Procedure

(1) Replace the affected Audio Control Panel (ACP):

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the audio control panel:

(a) Operate the light or switch that did not operate correctly.

(b) If the light or switch now operates correctly, then you corrected the fault.

——— END OF TASK ———

808. Captain`s (First Officer`s, Observer`s) Oxygen Mask Communication Problems - Fault Isolation

A. Description

(1) The Captain`s (First Officer`s, Observer`s) oxygen mask communication does not operate correctly.

B. Possible Causes

(1) Oxygen Mask, M524 (M525, M526)

(2) Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7)

(3) Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15)

(4) REU, M1353

(5) Wiring

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	2	C00412	INSTR XFR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Related Data

(1) SSM 23-51-11

(2) WDM 23-51-11

(3) WDM 23-51-21

(4) WDM 23-51-31

E. Initial Evaluation

(1) Do a test for the applicable oxygen mask microphone: Oxygen Mask Microphone - Operational Test, AMM TASK 23-51-00-710-803.

(a) If the oxygen mask microphone operate correctly, then there was an intermittent fault.

(b) If the oxygen mask microphone does not operate correctly, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Oxygen Mask, M524 (M525, M526). These are the tasks:

- Oxygen Mask/Regulator Removal, AMM TASK 35-12-85-000-802 or Crew and Supernumerary Oxygen Mask/Regulator Removal, AMM TASK 35-12-85-000-806
- Oxygen Mask/Regulator Installation, AMM TASK 35-12-85-400-802 or Crew and Supernumerary Oxygen Mask/Regulator Installation, AMM TASK 35-12-85-400-806.

(a) Do the Repair Confirmation at the end of this task.

(2) Do this wiring check (WDM 23-51-11, WDM 23-51-21, WDM 23-51-31):

(a) Use Wiring Diagram Manual (WDM) to identify the pins of connector D6049 (D6051, D6053) and pins of connector D2501B for the interface between Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7) and REU, M1353.

(b) Remove the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Removal, AMM TASK 35-12-85-000-801.

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(c) Remove the REU, M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(d) Do a wiring check between pins of connector D6049 (D6051, D6053) at the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel and pins of connector D2501B at the E4-1 Shelf.

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Installation, AMM TASK 35-12-85-400-801.

3) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

4) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(f) If you do not find a problem with the wiring, then do this step and continue:

1) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(g) If you do not find a problem with the wiring, then do this step and continue:

1) Re-install the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Installation, AMM TASK 35-12-85-400-801.

OMR 815, 816, 818-826, 901-999

(3) Do this wiring check (WDM 23-51-11, WDM 23-51-21, WDM 23-51-31):

(a) Remove the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Removal, AMM TASK 35-12-85-000-801.

(b) Remove the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). This is the task: Audio Control Panel Removal, AMM TASK 23-51-02-000-801.

(c) Do a wiring check as follows:

CAPTAIN	
CAPTAINS OXYGEN MASK STOWAGE PNL, P21-7 D6049	**CAPT AUDIO CONTROL PANEL, P8-6 D133**
pin E	pin E

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FIRST OFFICER

F/O OXYGEN MASK STOWAGE PNL, P23-7 D6051	**F/O AUDIO CONTROL PANEL, P8-7 D137**
pin E	pin E

OBSERVER

OXYGEN MASK STOWAGE PNL, P61-7 D6053	**OBS AUDIO CONTROL PANEL, P5-15 D139**
pin E	pin E

(d) Do a continuity check to make sure that this pin of connector D6049 (D6051, D6053) has continuity to the Structure Ground as follows:

CAPTAIN

CAPTAINS OXYGEN MASK STOWAGE PNL, P21-7 D6049	
pin F	GND

FIRST OFFICER

F/O OXYGEN MASK STOWAGE PNL, P23-7 D6051	
pin F	GND

OBSERVER

OXYGEN MASK STOWAGE PNL, P61-7 D6053	
pin F	GND

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-install the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Installation, AMM TASK 35-12-85-400-801.

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OMR 815, 816, 818-826, 901-999 (Continued)

3) Re-install the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). This is the task: Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then do this step and continue:

1) Re-install the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). This is the task: Oxygen Mask Stowage Box Installation, AMM TASK 35-12-85-400-801.

OMR ALL

(4) Do this wiring check (WDM 23-51-11, WDM 23-51-21, WDM 23-51-31):

(a) Remove the REU, M1353. This is the task: Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801.

(b) Remove the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). This is the task: Audio Control Panel Removal, AMM TASK 23-51-02-000-801.

(c) Do a wiring check as follows:

CAPTAIN

CAPT AUDIO CONTROL PANEL, P8-6 D133		**REMOTE ELECTRONIC UNIT, M1353 D2501B**
pin A	. . . . .	pin K1
pin B	. . . . .	pin K2

FIRST OFFICER

F/O AUDIO CONTROL PANEL, P8-6 D137		**REMOTE ELECTRONIC UNIT, M1353 D2501B**
pin A	. . . . .	pin K6
pin B	. . . . .	pin K7

OBSERVER

OBS AUDIO CONTROL PANEL, P5-15 D139		**REMOTE ELECTRONIC UNIT, M1353 D2501B**
pin A	. . . . .	pin K11
pin B	. . . . .	pin K12

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Re-install the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). This is the task: Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wiring, then do these steps and continue:

a) Re-install the REU, M1353. This is the task: Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

b) Re-install the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). This is the task: Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(5) Replace the Captain`s (First Officer`s, Observer`s) Oxygen Mask Stowage Panel, P21-7 (P23-7, P61-7). These are the tasks:

- Oxygen Mask Stowage Box Removal, AMM TASK 35-12-85-000-801
- Oxygen Mask Stowage Box Installation, AMM TASK 35-12-85-400-801.

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the Captain`s (First Officer`s, Observer`s) ACP, P8-6 (P8-7, P5-15). These are the tasks:

- Audio Control Panel Removal, AMM TASK 23-51-02-000-801
- Audio Control Panel Installation, AMM TASK 23-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

(7) Replace the REU, M1353. These are the tasks:

- Remote Electronics Unit (REU) Removal, AMM TASK 23-51-01-000-801
- Remote Electronics Unit (REU) Installation, AMM TASK 23-51-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do a test for the applicable oxygen mask microphone: Oxygen Mask Microphone - Operational Test, AMM TASK 23-51-00-710-803.

(a) If the observed fault symptom is gone, then you corrected a problem.

(b) If the observed fault symptom stays, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Voice Recorder Monitor Jack Signal Problem - Fault Isolation

A. Description

(1) No sound is heard through the headset/headphone.

B. Possible Causes

(1) Headset/headphone

(2) Voice recorder unit, M383

(3) Voice recorder control panel, P5-7

(4) Wiring or connector.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 815, 816, 818-826, 901-999			
D	6	C01948	VOICE RCDR/RIPS
OMR 701-799, 802, 805, 807, 809, 810, 812, 813			
D	7	C00107	VOICE RCDR

OMR ALL

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
OMR 815, 816, 818-826, 901-999			
D	7	C00107	VOICE RCDR (INOP)

OMR ALL

D. Related Data

(1) (SSM 23-71-11)

(2) (WDM 23-71-01)

E. Initial Evaluation

(1) Connect the headphone, STD-1390 to the voice recorder control panel at the pilot's overhead panel, P5.

(2) Set the volume control switches on the audio control panels (ACP) to the off position.

(3) Set the CVR AUTO/ON switch to the ON position.

NOTE: Both engines must be off for at least five minutes.

(a) Make sure that the CVR AUTO/ON switch stays in the ON position.

(4) Make sure that you can hear the flight deck conversation in the headphone.

(5) Push and hold the TEST switch on the voice recorder control panel.

(a) Make sure that you can hear a tone in the headphone.

1) If you hear a tone in the headphone, then continue.

2) If you do not hear a tone in the headphone, then do the Fault Isolation procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(6) Release the TEST switch.

(7) Speak into the microphones for the captain, the first officer, and the first observer, and the area microphone on the P5 overhead panel.

(a) If you hear your voice on all four channels through the headphone at the voice recorder control panel, then there was an intermittent fault.

(b) If you can not hear your voice on all four channels through the headphone at the voice recorder control panel, then do the Fault Isolation Procedure below.

(8) Set the CVR AUTO/ON switch back to the AUTO position.

F. Fault Isolation Procedure

(1) Replace the voice recorder, M383.

These are the tasks:

Voice Recorder Removal, AMM TASK 23-71-11-000-801,

Voice Recorder Installation, AMM TASK 23-71-11-400-801.

(a) Push and hold the TEST switch on the voice recorder control panel.

(b) If you hear a modulated sound through the headphone at the voice recorder control panel, then you corrected the fault.

(c) If you do not hear a modulated sound through the headphone at the voice recorder control panel, then continue.

(d) Release the TEST switch.

(2) Replace the voice recorder control panel, P5-7.

These are the tasks:

Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801,

Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.

(a) Push and hold the TEST switch on the voice recorder control panel.

(b) If you hear a modulated sound through the headphone at the voice recorder control panel, then you corrected the fault.

(c) If you do not hear a modulated sound through the headphone at the voice recorder control panel, then continue.

(d) Release the TEST switch.

(3) Do this check of the wiring:

(a) Remove the voice recorder control panel, P5-7. To remove it, do this task: Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801.

(b) Do a wiring check between these pins of connector D179, at the P5 overhead panel, and connector D177, at the voice recorder rack.

D179		D177
pin B	. . . . .	pin 13
pin C	. . . . .	pin 14
pin L	. . . . .	pin 47
pin K	. . . . .	pin 45
pin V	. . . . .	pin 5
pin W	. . . . .	pin 6

(c) If you find a problem with the wiring, then do these steps:

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1) Repair the wiring.
2) Re-install the voice recorder unit, M383. To install it, do this task:Voice Recorder Installation, AMM TASK 23-71-11-400-801.
3) Re-install the voice recorder control panel, P5-7. To install it, do this task:Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.
4) Push the TEST switch on the voice recorder control panel for 3 to 5 seconds.
5) If you hear a modulated sound through the headphone at the voice recorder control panel, then you corrected the fault.

——— END OF TASK ———

804. Voice Recorder STATUS Light or TEST Light Problem - Fault Isolation

A. Description

(1) When you push the TEST Button on the Cockpit Voice Recorder (CVR) Control Panel for approximately one half second, the STATUS or TEST Light blinks, flickers, dims or does not come ON momentarily.

B. Possible Causes

(1) Voice Recorder Unit, M383

(2) Voice Recorder Control Panel, P5-7

OMR 815, 816, 818-826, 901-999

(3) Recorder Independent Power Supply (RIPS) Battery Pack

(4) RIPS, M2862

(5) CVR Dispatch Relay, R1069

(6) Circuit Breaker, C1537

OMR ALL

(7) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

OMR 815, 816, 818-826, 901-999

D	6	C01948	VOICE RCDR/RIPS

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

D	7	C00107	VOICE RCDR

OMR ALL

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name

OMR 815, 816, 818-826, 901-999

D	7	C00107	VOICE RCDR (INOP)

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Related Data

(1) SSM 23-71-11

OMR 815, 816, 818-826, 901-999

(2) SSM 23-27-35

OMR ALL

(3) WDM 23-71-01

OMR 815, 816, 818-826, 901-999

(4) WDM 23-27-35

OMR ALL

E. Initial Evaluation

OMR 815, 816, 818-826, 901-999

(1) Do this task: ACARS - Operational Test, AMM TASK 23-27-00-740-816-010 to make sure that Aircraft Communications Addressing And Reporting System (ACARS) is serviceable.

NOTE: The ACARS must be serviceable with the correct software installed in the ACARS Communications Management Unit (CMU) in order to get the operational test of the Voice Recorder System to pass.

OMR ALL

(2) Set the CVR AUTO/ON Switch to the ON position.

NOTE: Both engines must be off for at least 5 minutes.

(a) Make sure that the CVR AUTO/ON Switch stays in the ON position.

(3) Push the TEST Switch, on the P5-7 Voice Recorder Control Panel, for approximately one half second.

(a) If the STATUS or TEST Light comes ON momentarily, then there was an intermittent problem.

(b) If the STATUS or TEST Light blinks, flickers, dims or does not come ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

OMR 815, 816, 818-826, 901-999

(1) Do this check of the wiring (WDM 23-71-01):

(a) At the maintenance port connector D10101, use a digital/analog multimeter, COM-1793, to do a continuity check at pin 1.

1) If pin 1 does not read continuity to ground (resistance is open or infinity), then repair the wiring.

a) Do the Repair Confirmation at the end of this task.

2) If pin 1 shows continuity to ground (resistance is approximately zero (0) Ω, then continue.

3) Attach the Black (Negative) lead of the multimeter to Pin 1 of connector D10101 as a ground reference when taking measurements in the subsequent steps.

(b) At the maintenance port connector D10101, use a digital/analog multimeter, COM-1793, set to the DIODE position and do a continuity check between pin 1 and pin 6.

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OMR 815, 816, 818-826, 901-999; AIRPLANES WITH RIPS PART NUMBER 100-3020-400

1) If a continuity check between pin 1 and pin 6 does not read a value between 400 to 700 millivolts (ground condition), then replace the Battery Pack of the RIPS, M2862. This is the task: Recorder Independent Power Supply (RIPS) Battery - Replacement, AMM TASK 23-71-22-960-801.
   a) Supply electrical power. This is the task: Supply Electrical Power, AMM TASK 24-22-00-860-811.
      <1> Wait for 15 (fifteen) minutes after you supply electrical power to the airplane.
      NOTE: If you do not wait fifteen minutes, it is possible the value you see will not be satisfactory because the battery is at a low state of charge.
   b) Do the Repair Confirmation at the end of this task.
2) If continuity check between pin 1 and pin 6 reads a value between 400 to 700 millivolts (ground state), then continue.

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH RIPS PART NUMBER 100-3010-401

3) If a continuity check between pin 1 and pin 6 does not read a value less than 700 millivolts (ground condition), then replace the Battery Pack of the RIPS, M2862. This is the task: Recorder Independent Power Supply (RIPS) Battery - Replacement, AMM TASK 23-71-22-960-801.
   a) Supply electrical power. This is the task: Supply Electrical Power, AMM TASK 24-22-00-860-811.
      <1> Wait for 15 (fifteen) minutes after you supply electrical power to the airplane.
      NOTE: If you do not wait fifteen minutes, it is possible the value you see will not be satisfactory because the battery is at a low state of charge.
   b) Do the Repair Confirmation at the end of this task.
4) If continuity check between pin 1 and pin 6 reads a value less than 700 millivolts (ground state), then continue.

OMR 815, 816, 818-826, 901-999

(c) At the maintenance port connector D10101, use a digital/analog multimeter, COM-1793, to do a continuity check between pin 1 and pin 8.
   1) If a continuity check between pin 1 and pin 8 does not read a value < 100 Ω, then do the steps that follow:
      a) Do a wiring check between these pins of connector D177 at the Voice Recorder Rack and connector D10727B at the CMU-1 (WDM 23-27-35):

VOICE RECORDER, M383 D177	**CMU-1, M2127 D10727B**
pin 33	pin D15
pin 34	pin C15

      b) If you find a problem with the wiring, then do these steps:

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OMR 815, 816, 818-826, 901-999 (Continued)

<1> Repair the wiring.

<2> At the maintenance port connector D10101, use a digital/analog multimeter, COM-1793, to do a continuity check between pin 1 and pin 8.

<a> If a continuity check between pin 1 and pin 8 does not read a value < 100 Ω, then go to the step below that looks at the front panel of the Voice Recorder Unit, M383.

<b> If continuity check between pin 1 and pin 8 reads a value < 100 Ω, then continue.

2) If continuity check between pin 1 and pin 8 reads a value < 100 Ω, then continue.

(d) At the maintenance port connector D10101, use a digital/analog multimeter, COM-1793, to do a continuity check between pin 1 and pin 12.

1) If a continuity check between pin 1 and pin 12 reads a value < 100 Ω, then replace the Voice Recorder Control Panel, P5-7. These are the tasks:

- Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801
- Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801

2) If continuity check between pin 1 and pin 12 does not read a value < 100 Ω, then continue.

OMR ALL

(2) Look at the Front Panel of the Voice Recorder Unit, M383.

(a) If the BITE Indicator comes ON and stays ON, then replace the Voice Recorder Unit, M383. These are the tasks:

- Voice Recorder Removal, AMM TASK 23-71-11-000-801
- Voice Recorder Installation, AMM TASK 23-71-11-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If the BITE Indicator does not come ON, then continue.

OMR 815, 816, 818-826, 901-999

(3) Replace the Battery Pack of the RIPS, M2862. This is the task: Recorder Independent Power Supply (RIPS) Battery - Replacement, AMM TASK 23-71-22-960-801.

(a) Supply electrical power. This is the task: Supply Electrical Power, AMM TASK 24-22-00-860-811.

1) Wait for 15 (fifteen) minutes after you supply electrical power to the airplane.

NOTE: If you do not wait fifteen minutes, it is possible the value you see will not be satisfactory because the battery is at a low state of charge.

(b) Do the Repair Confirmation at the end of this task.

(4) Replace the RIPS, M2862. These are the tasks:

- Recorder Independent Power Supply - Removal, AMM TASK 23-71-22-000-801
- Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(5) Replace the P5-7 Voice Recorder Control Panel. These are the tasks:

- Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801
- Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(6) Replace the CVR Dispatch Relay, R1069 on the J11 Panel as follows (WDM 23-71-01):

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(b) Replace the CVR Dispatch Relay, R1069.

(c) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(d) Do the Repair Confirmation at the end of this task.

OMR ALL

(7) Do this check of the wiring (WDM 23-71-01):

(a) Remove the Voice Recorder Unit, M383. This is the task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.

(b) Remove the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801.

(c) Do a wiring check as follows:

VOICE RECORDER CONT PNL, P5-7 D179		**VOICE RECORDER, M383 D177**
pin E	. . . . .	pin 16

1) If you find the problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.

c) Re-install the Voice Recorder Unit, M383. This is the task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find the problem with the wiring, then do these steps and continue:

a) Re-install the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.

b) Re-install the Voice Recorder Unit, M383. This is the task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999

(8) Do this voltage check (WDM 23-71-01):

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(b) Remove the CVR Dispatch Relay, R1069 from the J11 Panel.

(c) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(d) Do a voltage check as follows:

RELAY-CVR DISPATCH, R1069 D13187		**VOLTAGE**
pin 7	GND	28 VDC

1) If the correct voltage is not present, then do this check of the wiring:

a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

b) Do a wiring check as follows:

RELAY-CVR DISPATCH, R1069 D13187	**VOICE RCDR RELAY, C1537**
pin 7	term =L

<1> If you find a problem with the wiring, then do these steps:

<a> Repair the wiring.

<b> Re-install the CVR Dispatch Relay, R1069.

<c> Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

<d> Do the Repair Confirmation at the end of this task.

<2> If you do not find a problem with the wiring, then do these steps:

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<a> Replace this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

<b> Do the Repair Confirmation at the end of this task.

2) If the correct voltage is present, then continue.

(9) Do this check of the wiring (WDM 23-71-01):

(a) Remove the Voice Recorder Unit, M383. This is the task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.

(b) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(c) Do a wiring check as follows:

RELAY-CVR DISPATCH, R1069 D13187		**VOICE RECORDER, M383 D177**
pin 2	. . . . .	pin 15
pin 3	. . . . .	pin 31

1) If you find the problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the CVR Dispatch Relay, R1069.

c) Re-install the Voice Recorder Unit, M383. This is the task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

e) Do the Repair Confirmation at the end of this task.

2) If you do not find the problem with the wiring, then do this step and continue:

a) Re-install the Voice Recorder Unit, M383. This is the task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

(10) Do this check of the wiring (WDM 23-71-01):

(a) Remove the RIPS, M2862. This is the task: Recorder Independent Power Supply - Removal, AMM TASK 23-71-22-000-801.

(b) Do a wiring check as follows:

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OMR 815, 816, 818-826, 901-999 (Continued)

RELAY-CVR DISPATCH, R1069 D13187	**RIPS, M2862 D13185**
pin 5	pin 10

1) If you find the problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the CVR Dispatch Relay, R1069.
   c) Re-install the RIPS, M2862. This is the task: Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find the problem with the wiring, then do this step and continue:
   a) Re-install the CVR Dispatch Relay, R1069.

(11) Do this check of the wiring (WDM 23-71-01):
   (a) Remove the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801.
   (b) Do a wiring check as follows:

VOICE RECORDER CONT PNL, P5-7 D179	**RIPS, M2862 D13185**
pin D	pin 11

1) If you find the problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.
   c) Re-install the RIPS, M2862. This is the task: Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801.
   d) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(12) Do this check of the wiring (WDM 23-71-01):
   (a) Remove the Voice Recorder Unit, M383. This is the task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.
   (b) Remove the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801.
   (c) Do a wiring check as follows:

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OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

VOICE RECORDER CONT PNL, P5-7 D179	VOICE RECORDER, M383 D177
pin D	pin 15
pin E	pin 16

1) If you find the problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the P5-7 Voice Recorder Control Panel. This is the task: Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.
   c) Re-install the Voice Recorder Unit, M383. This is the task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.
   d) Do the Repair Confirmation at the end of this task.

OMR ALL

G. Repair Confirmation

(1) Push the TEST switch, on the P5-7 Voice Recorder Control Panel, for approximately one half second.
   (a) If the STATUS or TEST Light comes ON momentarily, then you corrected the fault.
   (b) If the STATUS or TEST Light does not come ON momentarily, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

805. CVR System Operation Problem (Airplanes With AUTO/ON Switch) - Fault Isolation

A. Description

(1) Cockpit voice recorder (CVR) system does not operate correctly.

B. Possible Causes

(1) Voice recorder unit, M383
(2) Voice recorder control panel, P5-7
(3) CVR AUTO/ON switch, P5-88

OMR 815, 816, 818-826, 901-999

(4) Recorder Independent Power Supply, M2862

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(5) CVR sense engine run relay, R779

OMR 815, 816, 818-826, 901-999

(6) MOAO RIPS power cutoff relay, R1112

OMR ALL

(7) CVR switch latching relay, R780
(8) Engine 1 running relay, R737
(9) Engine 2 running relay, R738

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(10) Wiring problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

OMR 815, 816, 818-826, 901-999

D	6	C01948	VOICE RCDR/RIPS

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

D	7	C00107	VOICE RCDR

OMR ALL

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name

OMR 815, 816, 818-826, 901-999

D	7	C00107	VOICE RCDR (INOP)

OMR ALL

D. Related Data

(1) (SSM 23-71-11)

(2) (WDM 23-71-01)

E. Initial Evaluation

(1) Do these steps to do the Initial Evaluation:

OMR 815, 816, 818-826, 901-999

(a) Do this task: ACARS - Operational Test, AMM TASK 23-27-00-740-816-010 to make sure that ACARS system is serviceable.

OMR ALL

(b) Connect the headphone, STD-1390 to the HEADPHONE/HEADSET jack on the voice recorder control panel.

(c) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

(d) Set the left engine start lever to the IDLE position.

1) Make sure that the right engine start lever is in the OFF position.

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2) Wait approximately 5 minutes.

NOTE: The CDS will energize the engine running relay 5 minutes after you put the start lever in the IDLE position.

3) Make sure that the CVR AUTO/ON switch moves from the ON position to the AUTO position.

(e) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light comes on once on the CVR control panel.

(f) If the test is not satisfactory, then make a note that there is a problem with the left engine/CVR operation and continue.

(g) Set the left engine start lever to the OFF position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

1) Wait approximately 5 minutes.

NOTE: The voice recorder will continue to operate for 5 minutes after both engines have stopped running, or engine running simulation has been discontinued.

OMR 815, 816, 818-826, 901-999

2) Wait approximately 10 minutes.

NOTE: The voice recorder will continue to operate for 10 minutes after both engines have stopped running, or engine running simulation has been discontinued.

OMR ALL

(h) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

(i) Set the right engine start lever to the IDLE position.

1) Wait approximately 5 minutes.

NOTE: The CDS will energize the engine running relay 5 minutes after you put the start lever in the IDLE position.

2) Make sure that the CVR AUTO/ON switch moves from the ON position to the AUTO position.

(j) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light comes on once on the CVR control panel.

(k) If the test was not satisfactory, then make a note that there is a problem with the right engine/CVR operation and continue.

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(l) Set the right engine start lever to the OFF position.

(m) Make sure that the left and right engine start levers are each set to the off position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

1) Wait approximately 5 minutes.

OMR 815, 816, 818-826, 901-999

2) Wait approximately 10 minutes.

OMR ALL

(n) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

2) If the CVR AUTO/ON switch does not stay in the ON position, then go to the Fault Isolation Procedure - CVR AUTO/ON Switch Operation Problem, Switch Does Not Stay ON.

(o) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light comes on once on the CVR control panel.

(p) If the test was not satisfactory, then make a note that there is a problem with the CVR AUTO/ON switch operation and continue.

(2) If the CVR did not operate correctly, then do these steps:

(a) If the CVR did not operate correctly for all of the conditions above, then do the Fault Isolation Procedure - CVR System Operation Problem.

(b) If there was a problem with the left and right engine/CVR operation, then do the Fault Isolation Procedure - Left and Right Engine/CVR Operation Problem.

(c) If there was a problem with the left engine/CVR operation, then do the Fault Isolation Procedure - Left Engine/CVR Operation Problem.

(d) If there was a problem with the right engine/CVR operation, then do the Fault Isolation Procedure - Right Engine/CVR Operation Problem.

(e) If there was a problem with the CVR AUTO/ON switch operation, then do the Fault Isolation Procedure - CVR AUTO/ON Switch Operation Problem.

(f) If there was a problem with the CVR AUTO/ON switch operation where the switch did not stay ON, then do the Fault Isolation Procedure - CVR AUTO/ON Switch Operation Problem, Switch Does Not Stay ON.

(3) If the CVR did operate correctly, then there was an intermittent fault.

F. Fault Isolation Procedure - CVR System Operation Problem

(1) Replace the voice recorder, M383.

These are the tasks:

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Voice Recorder Removal, AMM TASK 23-71-11-000-801,

Voice Recorder Installation, AMM TASK 23-71-11-400-801.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(2) Replace the voice recorder control panel, P5-7.

These are the tasks:

Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801,

Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) Do this check of the wiring to the voice recorder:

(a) Remove the voice recorder, M383. To remove it, do this task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.

(b) Set one of the two engine start levers to the IDLE position.

(c) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	7	C00107	VOICE RCDR

(d) Do a check for 115 VAC between pin 2 and pin 3 (ground) of connector D177 for the voice recorder.

(e) If there is not 115 VAC between pin 2 and pin 3 of connector D177, then do these steps:

1) Do a check for 115 VAC between the load terminal of VOICE RCDR circuit breaker, C107 and structure ground.

2) If there is not 115 VAC at the load terminal, then do these steps:

a) Replace the following circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	7	C00107	VOICE RCDR

b) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	7	C00107	VOICE RCDR

c) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Do the Repair Confirmation at the end of this task.

3) If there is 115 VAC at the load terminal, then do these steps:

a) Repair the wiring between pin 2 of D177 and the circuit breaker, C107.

b) Do the Repair Confirmation at the end of this task.

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OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

(f) If there is 115 VAC between pin 2 and pin 3 of connector D177, then continue.

OMR 815, 816, 818-826, 901-999

(4) Do this check of the wiring to the voice recorder:

(a) Remove the voice recorder, M383. To remove it, do this task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.

(b) Set one of the two engine start levers to the IDLE position.

(c) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS

(d) Do a check for 28 VDC between pin 9 and pin 3 (ground) of connector D177 for the voice recorder.

(e) If there is not 28 VDC between pin 9 and pin 3 of connector D177, then do these steps:

1) Do a check for 28 VDC between the load terminal of VOICE RCDR circuit breaker, C1948 and structure ground.

2) If there is not 28 VDC at the load terminal, then do these steps:

a) Replace the following circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS

b) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS

c) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Do the Repair Confirmation at the end of this task.

3) If there is 28 VDC at the load terminal, then do these steps:

a) Do a check for 28 VDC between pin A2 and pin X2 (ground) of connector D13401 at the MOAO RIPS power cutoff relay, R1112.

b) If there is 28 VDC at pin A2 of MOAO RIPS power cutoff relay, R1112, then do these steps:

<1> Replace the MOAO RIPS power cutoff relay, R1112.

<a> Do the Repair Confirmation at the end of this task.

<b> If the Repair Confirmation is not satisfactory, then continue.

<2> Repair the wiring between pin A1 of MOAO RIPS power cutoff relay, R1112 and pin 9 of the voice recorder, M383.

<a> Do the Repair Confirmation at the end of this task.

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<b> If the Repair Confirmation is not satisfactory, then continue.

c) If there is not 28 VDC at pin A2 of MOAO RIPS power cutoff relay, R1112, then do these steps:

<1> Do a check for 28 VDC between pin 4 and structure ground of connector D13185 at the RIPS, M2862.

<2> If there is 28 VDC input to the RIPS, M2862, then do these steps:

<a> Replace the battery pack of the RIPS, M2862.

This is the task:

Recorder Independent Power Supply (RIPS) Battery - Replacement, AMM TASK 23-71-22-960-801

<b> Wait for 15 minutes to continue next the step.

NOTE: It is necessary to wait 15 minutes for the battery to charge up after the power supplied to the airplane. If you do not wait for 15 minutes, it is possible that the test will not be satisfactory.

<c> Do the Repair Confirmation at the end of this task.

<d> If the Repair Confirmation is not satisfactory, then continue.

<e> Replace the RIPS unit, M2862.

These are the tasks:

Recorder Independent Power Supply - Removal, AMM TASK 23-71-22-000-801,

Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801.

<f> Do the Repair Confirmation at the end of this task.

<g> If the Repair Confirmation is not satisfactory, then continue.

<h> Repair the wiring between pin 8 of connector D13185 at the RIPS, M2862 and pin A2 of connector D13401 at relay, R1112.

<i> Do the Repair Confirmation at the end of this task.

<3> If there is no 28 VDC input to the RIPS, M2862, then do these steps:

<a> Repair the wiring between pin 4 of connector D13185 and the circuit breaker, C1948.

<b> Do the Repair Confirmation at the end of this task.

(f) If there is 28 VDC between pin 9 and pin 3 of connector D177, then continue.

OMR ALL

(5) Do this check of the wiring to the CVR AUTO/ON switch:

(a) Disconnect connector D11125 from the CVR AUTO/ON switch.

(b) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

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(c) Do a check for 28 VDC between pin 4 of connector D11125 and structure ground.

(d) If there is not 28 VDC at pin 4 of connector D11125, then do these steps:

1) Do a check for 28 VDC between the load terminal of VOICE RCDR RELAY circuit breaker, C1537 and structure ground.

2) If there is not 28 VDC at the load terminal of VOICE RCDR RELAY circuit breaker C1537, then do these steps:

a) Replace the circuit breaker C1537.

b) Re-connect connector D11125 to the CVR AUTO/ON switch.

c) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Do the Repair Confirmation at the end of this task.

3) If there is 28 VDC at the load terminal of VOICE RCDR RELAY circuit breaker C1537, then do these steps:

a) Repair the wiring between pin 4 of connector D11125 and the circuit breaker, C1537.

b) Re-connect connector D11125 to the CVR AUTO/ON switch.

c) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

d) Do the Repair Confirmation at the end of this task.

(e) If there is 28 VDC at pin 4 of the CVR AUTO/ON switch, then re-connect connector D11125 to the CVR AUTO/ON switch and continue.

(6) Do this check of the wiring between the voice recorder control panel and the voice recorder:

(a) Remove the voice recorder control panel, P5-7. To remove it, do this task: Voice Recorder Control Panel Removal, AMM TASK 23-71-12-000-801.

(b) Do a wiring check between these pins of connector D179, at the P5 overhead panel, and connector D177, for the voice recorder.

D179		D177
pin A	. . . . .	pin 12
pin E	. . . . .	pin 16
pin D	. . . . .	pin 15
pin W	. . . . .	pin 6
pin V	. . . . .	pin 5
pin C	. . . . .	pin 14
pin L	. . . . .	pin 47
pin K	. . . . .	pin 45
pin B	. . . . .	pin 13
pin F	. . . . .	pin 17

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.

3) Re-install the voice recorder control panel, P5-7. To install it, do this task: Voice Recorder Control Panel Installation, AMM TASK 23-71-12-400-801.

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4) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Left and Right Engine/CVR Operation Problem

(1) Set one of the two engine start levers to the IDLE position.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the PSEU BITE test shows an ENGINE RUN fault, then go to the fault isolation task for the applicable maintenance message to correct the fault.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(b) If the PSEU BITE test does not show an ENGINE RUN fault, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) Replace the CVR sense engine run relay, R779.

NOTE: R779 is in the J22 junction box.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

OMR 815, 816, 818-826, 901-999

(4) Replace the MOAO RIPS power cutoff relay, R1112.

NOTE: R1112 is in the J11 junction box.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(5) Do this check of the wiring:

(a) Remove the CVR sense engine run relay, R779.

(b) Do a check for 115 VAC at pin A2 of connector D11073 for the relay.

(c) If there is not 115 VAC at pin A2 of R779, then do these steps:

1) Do a check for 115 VAC between the load terminal of VOICE RECORDER circuit breaker, C107 and structure ground.

2) If there is not 115 VAC at the load terminal of VOICE RECORDER circuit breaker C107, then do these steps:

a) Replace the circuit breaker C107.

b) Re-install the CVR sense engine run relay, R779.

c) Do the Repair Confirmation at the end of this task.

3) If there is 115 VAC at the load terminal of VOICE RECORDER circuit breaker C107, then do these steps:

a) Repair the wiring between pin A2 of connector D11073 and the circuit breaker.

b) Re-install the CVR sense engine run relay, R779.

c) Do the Repair Confirmation at the end of this task.

(d) If there is 115 VAC at pin A2 of connector D11073, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999

(6) Do this check of the wiring:

(a) Remove the MOAO RIPS power cutoff relay, R1112.

(b) Do a check for 28 VDC at pin A2 of connector D13401 for the relay, R1112.

(c) If there is not 28 VDC at pin A2 of connector D13401 at relay, R1112, then do these steps:

1) Replace the RIPS, M2862.

These are the tasks:

Recorder Independent Power Supply - Removal, AMM TASK 23-71-22-000-801,

Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

2) Do a check for 28 VDC between the load terminal of VOICE RECORDER circuit breaker, C1948 and structure ground.

a) If there is not 28 VDC at the load terminal of VOICE RECORDER circuit breaker C1948, then do these steps:

<1> Replace the circuit breaker C1948.

<2> Do the Repair Confirmation at the end of this task.

<3> If the Repair Confirmation is not satisfactory, then continue.

b) If there is 28 VDC at the load terminal of VOICE RECORDER circuit breaker C1948, then continue.

3) Examine the wiring between the RIPS, M2862 and the MOAO RIPS power cutoff relay, R1112:

a) Do a continuity check of the wiring:

RIPS, M2862	**Relay, R1112**
D13185	**D13401**
pin 8	pin A2

b) Repair the wiring problem that you find.

c) Do the Repair Confirmation at the end of this task.

d) If the Repair Confirmation is not satisfactory, then continue.

4) Examine the wiring between the RIPS, M2862 and the VOICE RCDR/RIPS circuit breaker, C1948:

a) Do a continuity check of the wiring:

RIPS, M2862	**CB, C1948**
D13185	**C1948**
pin 4	pin Load Terminal

b) Do a continuity check of the wiring:

c) Repair the wiring problem that you find.

d) Do the Repair Confirmation at the end of this task.

e) If the Repair Confirmation is not satisfactory, then continue.

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OMR 815, 816, 818-826, 901-999 (Continued)

(d) If there is 28 VDC at pin A2 of connector D13401, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(7) Do this check of the wiring:

(a) Do a check for 28 VDC between pin X1 and pin X2 (ground) of connector D11073.

(b) If there is not 28 VDC between pin X1 and pin X2 of connector D11073, then do these steps:

1) Do a check for 28 VDC at the load terminal of the VOICE RCDR RELAY circuit breaker, C1537.

2) If there is not 28 VDC at C1537, then do these steps:

a) Replace the circuit breaker C1537.

b) Re-install the CVR sense engine run relay, R779.

c) Do the Repair Confirmation at the end of this task.

3) If there is 28 VDC at the load terminal of the VOICE RCDR RELAY circuit breaker, C1537, then do these steps:

a) Repair the wiring between pin X1 of connector D11073 and the circuit breaker.

b) Re-install the CVR sense engine run relay, R779.

c) Do the Repair Confirmation at the end of this task.

(c) If there is 28 VDC from X1 to X2 of connector D11073, then continue.

OMR 815, 816, 818-826, 901-999

(8) Do this check of the wiring:

(a) Do a check for 28 VDC between pin X1 and pin X2 (ground) of connector D13401 at relay, R1112.

(b) If there is not 28 VDC between pin X1 and pin X2 of connector D13401, then do these steps:

1) Repair the wiring between pin X1 of connector D13401 at relay, R1112 and pin 8 of connector D13185 at the RIPS, M2862.

2) Do the Repair Confirmation at the end of this task.

(c) If there is 28 VDC from X1 to X2 of D13401, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(9) Do this check of the ground circuit for the CVR sense engine run relay, R779:

(a) Make sure pin C1 of connector D11073 goes to ground.

(b) If pin C1 goes does not go to ground, then do these steps:

1) Repair the wiring between pin C1 of connector D11073 and structure ground.

a) Re-install the CVR sense engine run relay, R779.

b) Do the Repair Confirmation at the end of this task.

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OMR 815, 816, 818-826, 901-999

(10) Do this check of the ground circuit for the MOAO RIPS power cutoff relay, R1112:

(a) Make sure pin C1 of connector D13401 at relay, R1112 goes to ground.

(b) If pin C1 goes does not go to ground, then do these steps:

1) Replace the CVR Manual On/Auto Off switch.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

2) Repair the wiring between pin C1 of connector D13401 and pin 3 of CVR Manual On/Auto Off switch.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

3) Repair the wiring between pin 1 of CVR Manual On/Auto Off switch and structure ground.

a) Do the Repair Confirmation at the end of this task.

OMR ALL

H. Fault Isolation Procedure - Left Engine/CVR Operation Problem

(1) Replace the engine 1 running relay, R737.

NOTE: R737 is in the J22 junction box.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(2) Do this check of the wiring:

(a) Remove the engine 1 running relay, R737.

(b) Make sure that pin C2 of connector D12538 for the relay R737, goes to ground.

(c) If pin C2 of connector D12538 does not go to ground, then do these steps:

1) Repair the wiring between pin C2 of connector D12538 and structure ground.

2) Re-install the engine 1 running relay, R737.

3) Do the Repair Confirmation at the end of this task.

(d) If pin C2 of connector D12538 goes to ground, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) Do this check of the wiring between the CVR sense engine run relay, R779 and the engine 1 running relay, R737:

(a) Remove the CVR sense engine run relay, R779.

NOTE: R779 is in the J22 junction box.

(b) Do a check for an open circuit between these pins of connector D11073 for relay R779 and connector D12538 for relay R737:

D11073		D12538
pin C1	. . . . .	pin C1

1) Repair the wiring.

2) Re-install the engine 1 running relay, R737.

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OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

3) Re-install the CVR sense engine run relay, R779.

4) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(4) Do this check of the wiring between the MOAO RIPS power cutoff relay, R1112 and the engine 1 running relay, R737:

NOTE: R1112 is in the J11 junction box and R737 is in the J22 junction box.

(a) Do a continuity check of wiring between these pins of connector D13401 for relay R1112 and connector D12538 for relay R737:

Relay, R1112		**Relay, R737**
D13401		**D12538**
pin C1	. . . . .	pin C1

1) Repair the wiring problem that you find.

2) Do the Repair Confirmation at the end of this task.

OMR ALL

I. Fault Isolation Procedure - Right Engine/CVR Operation Problem

(1) Replace the engine 2 running relay, R738.

NOTE: R738 is in the J24 junction box.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(2) Do this check of the wiring:

(a) Remove the engine 2 running relay, R738.

(b) Make sure that pin C2 of connector D12540 for the relay R738, goes to ground.

(c) If pin C2 of connector D12540 does not go to ground, then do these steps:

1) Repair the wiring between pin C2 of connector D12540 and structure ground.

2) Re-install the engine 2 running relay, R738.

3) Do the Repair Confirmation at the end of this task.

(d) If pin C2 of connector D12540 goes to ground, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(3) Do this check of the wiring between the CVR sense engine run relay, R779 and the engine 2 running relay, R738:

(a) Remove the CVR sense engine run relay, R779.

NOTE: R779 is in the J22 junction box.

(b) Do a check for an open circuit between these pins of connector D11073 for relay R779 and connector D12540 for relay R738:

D11073		**D12540**
pin C1	. . . . .	pin C1

1) Repair the wiring.

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OMR 701-799, 802, 805, 807, 809, 810, 812, 813 (Continued)

2) Re-install the engine 2 running relay, R738.
3) Re-install the CVR sense engine run relay, R779.
4) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(4) Do this check of the wiring between the MOAO RIPS power cutoff relay, R1112 and the engine 2 running relay, R738:

NOTE: R1112 is in the J11 junction box and R787 is in the J24 junction box.

(a) Do a continuity check of the wiring between these pins of connector D13401 for relay R1112 and connector D12540 for relay R738:

Relay, R1112		**Relay, R787**
D13401		**D12540**
pin C1	. . . . .	pin C1

1) Repair the wiring problem that you find.
2) Do the Repair Confirmation at the end of this task.

OMR ALL

J. Fault Isolation Procedure - CVR AUTO/ON Switch Operation Problem

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) Do this check for 115 VAC to the CVR AUTO/ON switch:
(a) Disconnect connector D11125 from the CVR AUTO/ON switch, P5-88.
(b) Do a check for 115 VAC at pin 1, of connector D11125.
(c) If there is not 115 VAC at pin 1 of connector D11125, then do these steps:
1) Do a check for 115 VAC at the load terminal of circuit breaker, C107.
2) If there is 115 VAC at the load terminal of circuit breaker C107, then do these steps:
a) Repair the wiring between pin 1 of connector D11125 and the circuit breaker.
b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.
c) Do the Repair Confirmation at the end of this task.
3) If there is not 115 VAC at the load terminal of circuit breaker C107, then do these steps:
a) Replace the circuit breaker, C107.
b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.
c) Do the Repair Confirmation at the end of this task.
(d) If there is 115 VAC at pin 1 of connector D11125, then continue.

OMR 815, 816, 818-826, 901-999

(2) Do this check for 28 VDC to the CVR AUTO/ON switch:
(a) Disconnect connector D11125 from the CVR AUTO/ON switch, P5-88.
(b) Do a check for 28 VDC at pin 4 of connector D11125.
(c) If there is not 28 VDC at pin 4 of connector D11125, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

1) Do a check for 28 VDC at the load terminal of circuit breaker, C1537.
2) If there is 28 VDC at the load terminal of circuit breaker C1537, then do these steps:
   a) Repair the wiring between pin 4 of connector D11125 and the circuit breaker.
   b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.
   c) Do the Repair Confirmation at the end of this task.
3) If there is not 28 VDC at the load terminal of circuit breaker C1537, then do these steps:
   a) Replace the circuit breaker, C1537.
   b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.
   c) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 4 of connector D11125, then continue.

OMR ALL

(3) Replace the CVR AUTO/ON switch, P5-88.
   (a) Do the Repair Confirmation at the end of this task.
   (b) If the Repair Confirmation is not satisfactory, then continue.

OMR 815, 816, 818-826, 901-999

(4) Replace the MOAO RIPS power cutoff relay, R1112.
   (a) Do the Repair Confirmation at the end of this task.
   (b) If the Repair Confirmation is not satisfactory, then continue.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(5) Do this check of the wiring between the CVR AUTO/ON switch and the voice recorder:
   (a) Disconnect connector D11125 from the CVR AUTO/ON switch, P5-88.
   (b) Remove the voice recorder, M383. To remove it, do this task: Voice Recorder Removal, AMM TASK 23-71-11-000-801.
   (c) Do a check of the wiring between these pins of connector D11125 for the CVR AUTO/ON switch and connector D177 for the voice recorder:

D11125		D177
pin 3	. . . . .	pin 2

   (d) Repair the wiring.
   (e) Re-install the voice recorder, M383. To install it, do this task: Voice Recorder Installation, AMM TASK 23-71-11-400-801.
   (f) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.
   (g) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

(6) Do this check of the wiring between the CVR AUTO/ON switch and the voice recorder:
   (a) Do a check of the wiring between these pins of connector D13401 for the MOAO RIPS power cutoff relay, R1112 and connector D11125 for the CVR AUTO/ON switch:

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	CVR AUTO/ON
Relay, R1112	**switch**
D13401	**D11125**
pin C1	pin 3

(b) Do a check of the wiring between these pins of connector D13401 for the MOAO RIPS power cutoff relay, R1112 and connector D177 for the voice recorder:

	Voice Recorder,
Relay, R1112	**M383**
D13401	**D177**
pin A1	pin 9

(c) Do a check of the wiring between these pins of connector D13401 for the MOAO RIPS power cutoff relay, R1112 and connector D13185 for the RIPS, M2862:

Relay, R1112	**RIPS, M2862**
D13401	**D13185**
pin A2	pin 8
pin X1	pin 8

(d) Repair the wiring problem that you find.

(e) Do the Repair Confirmation at the end of this task.

OMR ALL

K. Fault Isolation Procedure - CVR AUTO/ON Switch Operation Problem, Switch Does Not Stay ON

(1) Do this check for 28 VDC at the CVR AUTO/ON switch:

(a) Make sure that the engines have been off for at least 5 minutes.

(b) Disconnect connector D11125 from the CVR AUTO/ON switch, P5-88.

(c) Do a check for 28 VDC at pin 4 of connector D11125.

(d) If there is not 28 VDC at pin 4 of connector D11125, then do these steps:

1) Do a check for 28 VDC from the load terminal of VOICE RCDR RELAY circuit breaker C1537 to structure ground.

2) If there is not 28 VDC at the load terminal of VOICE RCDR RELAY circuit breaker C1537, then do these steps:

a) Replace the circuit breaker, C1537.

b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.

c) Do the Repair Confirmation at the end of this task.

3) If there is 28 VDC at the load terminal of circuit breaker C1537, then do these steps:

a) Repair the wiring between pin 4 of connector D11125 and the circuit breaker.

b) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.

c) Do the Repair Confirmation at the end of this task.

(e) If there is 28 VDC at pin 4 of connector D11125, then continue.

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(2) Do this check of the CVR AUTO/ON switch:

(a) Make sure that pin 5 of connector D11125 for the CVR AUTO/ON switch goes to ground.

(b) If pin 5 of connector D11125 goes to ground, then do these steps:

1) Replace the CVR AUTO/ON switch, P5-88.

2) Do the Repair Confirmation at the end of this task.

(c) If pin 5 of connector D11125 does not go to ground, then re-connect connector D11125 to the CVR AUTO/ON switch and continue.

(3) Do this check for 28 VDC at the CVR switch latching relay, R780:

(a) Remove the CVR switch latching relay, R780.

NOTE: R780 is in the J22 junction box.

(b) Do a check for 28 VDC between pin X1 of connector D11131 for the relay, R780 and structure ground.

(c) If there is not 28 VDC at pin X1 of connector D11131, then do these steps:

1) Repair the wiring between these pins of connector D11131 for the relay, R780 and load terminal of the VOICE RCDR RELAY circuit breaker, C1537:

Relay, R780		**CB, C1537**
D11131		**C1537**
pin X1	. . . . .	pin Load Terminal

2) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin X1 of D11131, then continue.

(4) Do this check of the wiring between the CVR switch latching relay, R780 and the engine 2 running relay, R738:

(a) Make sure that pin X2 of D11131 goes to ground.

(b) If pin X2 of D11131 does not go to ground, then do these steps:

1) Replace the engine 2 running relay, R738.

2) Re-install the CVR latching relay, R780.

3) Do the Repair Confirmation at the end of this task.

4) If the Repair Confirmation is not satisfactory, then do these steps:

a) Remove the engine 2 running relay, R738.

b) Remove the CVR switch latching relay, R780.

c) Do a wiring check between these pins of connector D12540 for relay R738 and connector D11131 for relay R780:

D12540		**D11131**
pin C3	. . . . .	pin X2

d) Repair the problem in the wiring.

e) Re-install the engine 2 running relay, R738.

f) Re-install the CVR switch latching relay, R780.

g) Do the Repair Confirmation at the end of this task.

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(c) If pin X2 of D11131 goes to ground, then continue.

(5) Do this check of the wiring between the CVR switch latching relay, R780 and the engine 1 running relay, R737:

(a) Make sure that pin A2 of D11131 goes to ground.

(b) If pin A2 of D11131 does not go to ground, then do these steps:

1) Replace the engine 1 running relay, R737.

2) Re-install the CVR latching relay, R780.

3) Do the Repair Confirmation at the end of this task.

4) If the Repair Confirmation is not satisfactory, then do these steps:

a) Remove the engine 1 running relay, R737

b) Remove the CVR switch latching relay, R780.

c) Do a wiring check between these pins of connector D12538 for relay R737 and connector D11131 for relay R780:

D12538		D11131
pin C3	. . . . .	pin A2

d) Repair the problem in the wiring.

e) Re-install the engine 1 running relay, R737.

f) Re-install the CVR switch latching relay, R780.

g) Do the Repair Confirmation at the end of this task.

5) If pin A2 of D11131 goes to ground, then continue.

(6) Do this check of the CVR switch latching relay, R780:

(a) Do a check for 28 VDC between pin A1 of D11131 and structure ground.

(b) If there is not 28 VDC at pin A1 of D11131, then do these steps:

1) Disconnect connector D11125 from the CVR AUTO/ON switch, P5-88.

2) Do a check of the wiring between these pins of connector D11125 and connector D11131:

D11125		D11131
pin 5	. . . . .	pin A1

3) Repair the wiring.

4) Re-install the CVR switch latching relay, R780.

5) Re-connect connector D11125 to the CVR AUTO/ON switch, P5-88.

6) Do the Repair Confirmation at the end of this task.

(c) If there is 28 VDC at pin A1 of D11131 then do these steps:

1) Install a new CVR switch latching relay, R780.

2) Do the Repair Confirmation at the end of this task.

L. Repair Confirmation

(1) Do this check of the CVR operation:

(a) Connect the headphone, STD-1390 to the HEADPHONE/HEADSET jack on the voice recorder control panel.

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(b) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

(c) Set the left engine start lever to the IDLE position.

1) Make sure that the right engine start lever is in the OFF position.

2) Wait approximately 5 minutes.

NOTE: The CDS will energize the engine running relay 5 minutes after you put the start lever in the IDLE position.

3) Make sure that the CVR AUTO/ON switch moves from the ON position to the AUTO position.

(d) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light on the CVR control panel comes on once.

(e) Set the left engine start lever to the OFF position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

1) Wait approximately 5 minutes.

NOTE: The voice recorder will continue to operate for 5 minutes after both engines have stopped running, or engine running simulation has been discontinued.

OMR 815, 816, 818-826, 901-999

2) Wait approximately 10 minutes.

NOTE: The voice recorder will continue to operate for 5 minutes after both engines have stopped running, or engine running simulation has been discontinued.

OMR ALL

(f) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

(g) Set the right engine start lever to the IDLE position.

1) Wait approximately 5 minutes.

NOTE: The CDS will energize the engine running relay 5 minutes after you put the start lever in the IDLE position.

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2) Make sure that the CVR AUTO/ON switch moves from the ON position to the AUTO position.

(h) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light on the CVR control panel comes on once.

(i) Set the right engine start lever to the OFF position.

(j) Make sure that the left and right engine start levers are each set to the off position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

1) Wait approximately 5 minutes.

OMR 815, 816, 818-826, 901-999

2) Wait approximately 10 minutes.

OMR ALL

(k) Set the CVR AUTO/ON switch (P5 overhead panel) to the ON position.

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

NOTE: Both engines must be off for at least 5 minutes.

OMR 815, 816, 818-826, 901-999

NOTE: Both engines must be off for at least 10 minutes.

OMR ALL

1) Make sure that the CVR AUTO/ON switch stays in the ON position.

(l) Push the TEST switch on the CVR control panel (P5 panel) for one half second and release it.

1) Make sure that you hear a test tone in the headphone.

2) Make sure that the light on the CVR control panel comes on once.

(m) Set the CVR AUTO/ON switch back to the AUTO position.

(n) If the CVR operates correctly, then you corrected the fault.

——— END OF TASK ———

OMR 815, 816, 818-826, 901-999

806. RIPS Maintenance Status Check Failed - Fault Isolation

A. Description

(1) Use this task when the results of the RIPS Maintenance Status Check at the RIPS Maintenance Port are not satisfactory.

NOTE: The RIPS Maintenance Status Check in the Aircraft Maintenance Manual (AMM) is a part of the RIPS System Test task. It is also a part of the RIPS Maintenance Discrete Check task.

B. Possible Causes

(1) RIPS Battery

(2) RIPS Unit, M2862

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OMR 815, 816, 818-826, 901-999 (Continued)

C. Related Data

(1) SSM 23-71-11

(2) WDM 23-71-01

D. Initial Evaluation

(1) Use a digital/analog multimeter, COM-1793 and make sure it is set to the DIODE position.

(2) Do a check for continuity at the RIPS Maintenance Port connector D10101 between pin 1 and pin 6 (WDM 23-71-01).

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH RIPS PART NUMBER 100-3020-400

(a) If you see a value between 400 to 700 millivolts (ground state), do the No Fault Status Check Within Limits Fault Isolation Procedure.

(b) If you see a value that is not between 400 to 700 millivolts (ground state), do the No Fault Status Check Not Within Limits Fault Isolation Procedure.

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH RIPS PART NUMBER 100-3010-401

(c) If a continuity check between pin 1 and pin 6 does not read a value less than 700 millivolts (ground condition), do the No Fault Status Check Not Within Limits Fault Isolation Procedure.

(d) If continuity check between pin 1 and pin 6 reads a value less than 700 millivolts (ground state), do the No Fault Status Check Within Limits Fault Isolation Procedure.

OMR 815, 816, 818-826, 901-999

E. No Fault Status Check Within Limits - Fault Isolation Procedure

(1) Replace the RIPS Battery. This is the task: Recorder Independent Power Supply (RIPS) Battery - Replacement, AMM TASK 23-71-22-960-801.

(a) Supply electrical power. This is the task: Supply Electrical Power, AMM TASK 24-22-00-860-811.

1) Wait for 15 (fifteen) minutes after you supply electrical power to the airplane before you do the tests that follow.

NOTE: If you do not wait fifteen minutes, it is possible the value you see will not be satisfactory because the battery is at a low state of charge.

(b) Use a digital/analog multimeter, COM-1793 and set it to the DIODE position.

(c) Do a check for continuity at connector D10101 between pin 1 and pin 10 (WDM 23-71-01).

1) If you see a value of “OL” (out-of-limit), then you corrected the problem.

2) Install the protective dust-cap cover on the Maintenance Port (Connector D10101).

F. No Fault Status Check Not Within Limits - Fault Isolation Procedure

(1) Replace the RIPS Unit, M2862. These are the tasks:

- Recorder Independent Power Supply - Removal, AMM TASK 23-71-22-000-801
- Recorder Independent Power Supply - Installation, AMM TASK 23-71-22-400-801

(a) Supply electrical power. This is the task: Supply Electrical Power, AMM TASK 24-22-00-860-811.

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OMR 815, 816, 818-826, 901-999 (Continued)

1) Wait for 15 (fifteen) minutes after you supply electrical power to the airplane before you do the tests that follow.

NOTE: If you do not wait fifteen minutes, it is possible the value you see will not be satisfactory because the battery is at a low state of charge.

(b) Use a digital/analog multimeter, COM-1793 and set it to the DIODE position.

(c) Do a check for continuity at connector D10101 between pin 1 and pin 10 (WDM 23-71-01).

1) If you see a value of “OL” (out-of-limit), then you corrected the problem.

2) Install the protective dust-cap cover on the Maintenance Port (Connector D10101).

——— END OF TASK ———

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807. No Surveillance Video For One Camera - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) There is no Surveillance Video for Camera 1.

(b) There is no Surveillance Video for Camera 2.

(c) There is no Surveillance Video for Camera 3.

B. Possible Causes

(1) Video Camera:

- Camera 1, M3002, or
- Camera 2, M3026, or
- Camera 3, M3004

(2) Video Interface Unit (VIU):

- VIU, M3001 or
- VIU, M3029

(3) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

D. Related Data

(1) WDM 23-70-11

E. Initial Evaluation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the MFD shows Surveillance Video for all three cameras, then there was an intermittent fault.

(b) If the MFD does not show Surveillance Video for one camera only, and the airplane has VIU P/N M3029 installed, then do the “Fault Isolation Procedure - Airplanes with VIU, M3029” below.

F. Fault Isolation Procedure - Airplanes with VIU, M3029

(1) Make sure that there is Input Power from the VIU to the applicable Camera 1 (2, 3), M3002 (M3026, M3004) as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(b) Disconnect the applicable CAM 1 (2, 3) connector D12325 (D11423, D12329) (WDM 23-70-11).

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(c) Close this circuit breaker:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(d) Do a voltage check at the applicable CAM 1 (2, 3) connector as follows (WDM 23-70-11):

CAMERA 1

CONNECTOR D12325		CONNECTOR D12325	VOLTAGE
pin D	. . . . .	pin G	+12V DC
pin E	. . . . .	pin G	-12V DC

CAMERA 2

CONNECTOR D11423		CONNECTOR D11423	VOLTAGE
pin D	. . . . .	pin G	+12V DC
pin E	. . . . .	pin G	-12V DC

CAMERA 3

CONNECTOR D12329		CONNECTOR D12329	VOLTAGE
pin D	. . . . .	pin G	+12V DC
pin E	. . . . .	pin G	-12V DC

1) If the voltage check is as specified above, then replace the applicable CAM 1 (2, 3), M3002 (M3026, M3004). These are the tasks:
   - Video Camera - Removal, AMM TASK 23-75-02-000-804
   - Video Camera - Installation, AMM TASK 23-75-02-400-804

   a) Do the Repair Confirmation at the end of this task.

2) If the voltage check is not as specified above, then replace the VIU, M3029. These are the tasks:
   - Video Interface Unit - Removal, AMM TASK 23-75-10-000-801
   - Video Interface Unit - Installation, AMM TASK 23-75-10-400-801

   a) Do the Repair Confirmation at the end of this task.

(2) Do this wiring check (WDM 23-70-11):

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(b) Disconnect the applicable CAM 1 (2, 3) connector D12325 (D11423, D12329).

(c) Disconnect connector D12859 from the VIU.

(d) Examine the wiring between the applicable CAM 1 (2, 3) and the VIU, M3029 as follows:

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CAMERA 1

CAM 1 D12325	**VIU D12859**
pin A	pin A
pin B	pin C-
pin D	pin C
pin E	pin B

CAM 1 D12325	
pin G	GD11100-ST

CAMERA 2

CAM 2 D11423	**VIU D12859**
pin A	pin E
pin B	pin D
pin D	pin G
pin E	pin F

CAM 2 D11423	
pin G	GD3734-ST

CAMERA 3

CAM 3 D12329	**VIU D12859**
pin A	pin J
pin B	pin H
pin D	pin L
pin E	pin K

CAM 3 D12329	
pin G	GD3924-ST

1) If you find a problem with the wiring do these steps:
   a) Repair the wiring.
   b) Reconnect the applicable CAM 1 (2, 3) connector D12325 (D11423, D12329).
   c) Reconnect connector D12859 to the VIU.
   d) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the video shows correctly on the Common Display System (CDS) from all three cameras, then you corrected the problem.

(b) If the video does not show correctly on the CDS from all three cameras, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

808. Distorted Surveillance Video for One Camera - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Surveillance Video shows, but the display is dark for one camera.

(b) Lines show on the Video Display for one camera.

(c) Surveillance Video is jittery or jumpy or the picture appears to roll.

(d) Other Surveillance Video Display problems for one camera.

B. Possible Causes

(1) Video Camera:

- Camera 1, M3002, or
- Camera 2, M3026, or
- Camera 3, M3004

(2) VIU:

- VIU, M3001 or
- VIU, M3029

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	12	C01373	DISPLAY CTR LWR

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

D. Related Data

(1) WDM 23-70-11

(2) WDM 31-62-42

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the MFD shows Surveillance Video correctly for all three cameras, then there was an intermittent fault.

NOTE: Only the Lower-Center MFD can show Flight Deck Entry Video.

The Inboard Left, and Inboard Right MFD Units are connected to FDEVSS.

However to successfully show video, the optional video-capable MFDs are required.

(b) If Surveillance Video shows for all three cameras, but on one camera the video is dark, the display is jumpy, lines show or there are other display problems then do these steps:

1) Make a record of the Camera Control Panel (CP) position (R-C-L) with the problem.

NOTE: Position C on the Control Panel, M3000, is Camera 1, M3002 on the Wiring Diagram Manual (WDM).

Position L on the Control Panel is Camera 2, M3026.

Position R is Camera 3, M3004.

NOTE: Only the Lower-Center MFD can show Flight Deck Entry Video.

The Inboard Left, and Inboard Right MFD Units are connected to FDEVSS.

However to successfully show video, the optional video-capable MFDs are required.

(c) If the Surveillance Video on one camera is dark, the display is jumpy, lines show or there are other display problems, and the airplane has VIU P/N M3029 installed, then do the "Fault Isolation Procedure - Airplanes with VIU, M3029" below.

F. Fault Isolation Procedure - Airplanes with VIU, M3029

(1) Examine the camera and lens.

(a) Make sure that the lens is not covered, and the view is not blocked.

(2) Replace the applicable CAM 1 (2, 3), M3002 (M3026, M3004). These are the tasks:

- Video Camera - Removal, AMM TASK 23-75-02-000-804
- Video Camera - Installation, AMM TASK 23-75-02-400-804

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the VIU, M3029. These are the tasks:

- Video Interface Unit - Removal, AMM TASK 23-75-10-000-801
- Video Interface Unit - Installation, AMM TASK 23-75-10-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Do this wiring check (WDM 23-70-11):

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(b) Disconnect the applicable CAM 1 (2, 3) connector D12325 (D11423, D12329).

(c) Disconnect connector D12859 from the VIU.

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(d) Examine the wiring between the applicable CAM 1 (2, 3) and the VIU, M3029 as follows:

CAMERA 1

CAM 1 D12325	**VIU D12859**
pin A	pin A
pin B	pin C-

CAM 1 D12325	
pin G	GD11100-ST

CAMERA 2

CAM 2 D11423	**VIU D12859**
pin A	pin E
pin B	pin D

CAM 2 D11423	
pin G	GD3734-ST

CAMERA 3

CAM 3 D12329	**VIU D12859**
pin A	pin J
pin B	pin H

CAM 3 D12329	
pin G	GD3924-ST

1) If you find a problem with the wiring do these steps:
   a) Repair the wiring.
   b) Reconnect the applicable CAM 1 (2, 3) connector D12325 (D11423, D12329).
   c) Reconnect connector D12859 to the VIU.
   d) Do the Repair Confirmation at the end of this task.

(5) Replace the VIU, M3001. These are the tasks:
- Video Interface Unit - Removal, AMM TASK 23-75-10-000-801
- Video Interface Unit - Installation, AMM TASK 23-75-10-400-801

(a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the video shows correctly on the CDS from all three cameras, then you corrected the problem.

(b) If the video does not show correctly on the CDS from all three cameras, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

809. Surveillance Video Problems for All Cameras - Fault Isolation

A. Description

(1) This task is for this (these) Observed Fault(s):

(a) There is no video for all three Video Surveillance Cameras.

B. Possible Causes

(1) VIU:

- VIU, M3001 or
- VIU, M3029

(2) Camera CP, M3000

(3) Wiring

(4) Impedance Transformer:

- Impedance Transformer, T3022 or
- Impedance Transformer, T3029

NOTE: The Transformer is located between the Lower Center Display Unit, N00190 and the Video Surveillance System (WDM 23-70-11, WDM 31-62-42).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	12	C01373	DISPLAY CTR LWR

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
A	8	C01627	CABIN UTIL RLY PWR

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

D. Related Data

(1) WDM 23-70-11

(2) WDM 31-62-42

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the Multi-Function Display (MFD) shows Surveillance Video for all three cameras correctly, then there was an intermittent fault.

(b) If the MFD does not show Surveillance Video for all three cameras, and the airplane has VIU P/N M3001 installed, then do the “Fault Isolation Procedure - Airplanes with VIU, M3001” below.

(c) If the MFD does not show Surveillance Video for all three cameras, and the airplane has VIU P/N M3029 installed, then do the “Fault Isolation Procedure - Airplanes with VIU, M3029” below.

OMR 815, 816, 818-826, 901-999; Airplanes with VIU M3001

F. Fault Isolation Procedure - Airplanes with VIU, M3001

(1) On the Electric Meters, Battery and Galley Power Module, P5-13, make sure that the In-Flight Entertainment (IFE)/PASS SEAT Switch is set to ON.

(2) Make sure that the VIU has power as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(b) Disconnect connector D12323 from the VIU, M03001 (WDM 23-70-11).

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(d) Do a check for 115V Alternating Current (AC) on connector D12323 between pins A and F (WDM 23-70-11).

(e) If you do not find 115V AC at the VIU connector, then do the steps that follow:

1) Examine and repair the wiring between the VIU, M3001, and circuit breaker C01641 (WDM 23-70-11).

2) Reconnect connector D12323 to the VIU.

3) Do the Repair Confirmation at the end of this task.

(f) If you find 115V AC at the VIU connector, then continue.

(3) Replace the VIU. These are the tasks: .

- Video Interface Unit - Removal, AMM TASK 23-75-10-000-801
- Video Interface Unit - Installation, AMM TASK 23-75-10-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Impedance Transformer, T3022 (or T3029) for the Lower Center MFD (WDM 23-70-11, WDM 31-62-41).

(a) Do the Repair Confirmation at the end of this task.

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OMR 815, 816, 818-826, 901-999; Airplanes with VIU M3001 (Continued)

(5) Do a continuity check of the camera CP circuitry at the VIU connector as follows (WDM 23-70-11):

(a) Disconnect connector D12321 from the VIU, M3001.

(b) Do a check to make sure that each position of the 3-position Rotary Switch, on the Camera Control Panel, has continuity to the VIU.

NOTE: You must set the Rotary Switch to the specified position to complete each of the three circuits.

CP CAM SEL-POS L

VIU Connector D12321	VIU Connector D12321	RESISTANCE
pin X	pin HH (RTN)	CONTINUITY

CP CAM SEL-POS C

VIU Connector D12321	VIU Connector D12321	RESISTANCE
pin U	pin HH (RTN)	CONTINUITY

CP CAM SEL-POS R

VIU Connector D12321	VIU Connector D12321	RESISTANCE
pin R	pin HH (RTN)	CONTINUITY

(c) Do a continuity check of the DSPL (Display) push-button on the Camera CP from the Video Switch (VS) connector:

NOTE: To complete the circuit, you must push DSPL on the Camera Control Panel. The position of the Rotary Switch does not affect this check.

CP DSPL SEL

VIU Connector D12321	VIU Connector D12321	RESISTANCE
pin M	pin GG (RTN)	CONTINUITY (with DSPL Pushed)

(d) If the continuity checks are not as specified above, then do these steps:

1) Replace the Camera CP. These are the tasks:

- Flight Deck Camera Select Panel - Removal, AMM TASK 23-75-03-000-802
- Flight Deck Camera Select Panel - Installation, AMM TASK 23-75-03-400-802

a) Do the Repair Confirmation at the end of this task.

2) Do the continuity checks again.

a) If you still find a problem, then repair the wiring.

b) Reconnect connector D12321 to the VIU.

c) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999; Airplanes with VIU M3029

G. Fault Isolation Procedure - Airplanes with VIU, M3029

(1) On the Electric Meters, Battery and Galley Power Module, P5-13, make sure that the IFE/PASS SEAT Switch is set to ON.

(2) Make sure the VIU has power as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(b) Disconnect connector D12865 from the VIU, M3029 (WDM 23-70-11).

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
B	1	C01641	SURVEILLANCE CAMERA

(d) Do a check for 115V AC on connector D12865 between pins A and F (WDM 23-70-11).

(e) If you do not find 115V AC at the VIU connector, then do the steps that follow:

1) Examine and repair the wiring between the VIU, M3029, and circuit breaker C01641 (WDM 23-70-11).

2) Reconnect connector D12865 to the VIU.

3) Do the Repair Confirmation at the end of this task.

(f) If you find 115V AC at the VIU connector, then continue.

(3) Replace the VIU. These are the tasks: .

- Video Interface Unit - Removal, AMM TASK 23-75-10-000-801
- Video Interface Unit - Installation, AMM TASK 23-75-10-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Impedance Transformer, T3022 (or T3029) for the Lower Center MFD (WDM 23-70-11, WDM 31-62-42).

(a) Do the Repair Confirmation at the end of this task.

(5) Do a continuity check of the Camera CP circuitry at the VIU connector as follows (WDM 23-70-11):

(a) Disconnect connector D12863 from the VIU, M3029.

(b) Do a check to make sure that each position of the 3-position Rotary Switch, on the Camera CP, has continuity to the VIU.

NOTE: You must set the Rotary Switch to the specified position to complete each of the three circuits.

CP CAM SEL-POS L

VIU Connector D12863		VIU Connector D12863	RESISTANCE
pin X	. . . . .	pin HH (RTN)	CONTINUITY

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999; Airplanes with VIU M3029 (Continued)

CP CAM SEL-POS C

VIU Connector D12863	VIU Connector D12863	RESISTANCE
pin U	pin HH (RTN)	CONTINUITY

CP CAM SEL-POS R

VIU Connector D12863	VIU Connector D12863	RESISTANCE
pin R	pin HH (RTN)	CONTINUITY

(c) Do a continuity check of the DSPL (Display) push-button on the Camera CP from the VS connector:

NOTE: To complete the circuit, you must push DSPL on the Camera Control Panel. The position of the Rotary Switch does not affect this check.

CP DSPL SEL

VIU Connector D12863	VIU Connector D12863	RESISTANCE
pin M	pin GG (RTN)	CONTINUITY (with DSPL Pushed)

(d) If the continuity checks are not as specified above, then do these steps:

1) Replace the Camera CP. These are the tasks:
   - Flight Deck Camera Select Panel - Removal, AMM TASK 23-75-03-000-802
   - Flight Deck Camera Select Panel - Installation, AMM TASK 23-75-03-400-802

   a) Do the Repair Confirmation at the end of this task.

2) Do the continuity checks again.

   a) If you still find a problem, then repair the wiring.

   b) Reconnect connector D12863 to the VIU.

   c) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818-826, 901-999

H. Repair Confirmation

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, AMM TASK 23-75-00-730-804.

(a) If the MFD shows Surveillance Video for all three cameras correctly, then you corrected the problem.

(b) If the MFD does not show Surveillance Video for all three cameras, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 810, 812, 813, 815, 816, 818-824, 901-999

802. Flight Compartment Outlet Unit Does Not Operate - Fault Isolation

A. Description

(1) The Outlet Unit (OU) in the flight compartment does not provide power to a Portable Electronic Device (PED).

NOTE: Make sure that the PED operates normally at another location before doing this test.

B. Possible Causes

(1) Outlet Unit

(2) In-Seat Power Supply

(3) Circuit Breaker

(4) Wiring problem

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
A	3	C01643	FD - PC POWER

D. Related Data

(1) WDM 25-29-11

E. Initial Evaluation

(1) Make sure that power is applied to the Flight Compartment PC power system.

(a) Test the system again by cycling the applicable circuit breaker.

(b) Do this task on the applicable OU: Flight Compartment PC Power System - Operational Test, AMM TASK 23-82-00-710-801.

(c) If the OU provides power to the PED, there was an intermittent fault.

(d) If one OU is operational, do the fault isolation task below for “One OU is Operational”.

(e) If both OUs are not operational, do the fault isolation task below for “Both OUs are Not Operational”.

F. One OU is Operational - Fault Isolation Procedure

(1) Replace the applicable OU. These are the tasks:

- Outlet Unit Removal, AMM TASK 23-82-01-000-801
- Outlet Unit Installation, AMM TASK 23-82-01-400-801.

(a) If the OU is operational, then you corrected the fault.

(b) If the problem continues, examine the wiring between the ISPS and the OU.

1) If you find a problem, repair or replace the applicable wiring.

2) If the OU is operational, then you corrected the fault.

G. Both OUs are Not Operational - Fault Isolation Procedure

(1) Replace the ISPS. These are the tasks:

- In-Seat Power Converter Removal, AMM TASK 23-82-03-000-801
- In-Seat Power Converter Installation, AMM TASK 23-82-03-400-801.

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OMR 810, 812, 813, 815, 816, 818-824, 901-999 (Continued)

(a) If the OUs are operational, then you corrected the fault.

(b) If the OUs are not operational, replace the circuit breaker.

(c) If the OUs are operational, then you corrected the fault.

——— END OF TASK ———

OMR 825, 826

803. Flight Compartment Outlet Unit Has No Power Or USB Problem

A. Description

(1) The Outlet Unit (OU) in the flight compartment does not provide power to a PED or all PC power outlets do not have power.

B. Possible Causes

(1) CAPT (F/O) PC power outlet , M3249 (M3250)

(2) In-Seat Power Supply, M3248

(3) Circuit Breaker, C1643

(4) Wiring problem

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
A	3	C01643	FD - PC POWER

D. Related Data

(1) WDM 25-29-11

E. Initial Evaluation

(1) Do this task: Flight Compartment PC Power System - Operational Test, AMM TASK 23-82-00-710-801.

(a) If both OUs are operational, there was an intermittent fault.

(b) If one OU is operational, do the fault isolation task below for “One OU is Operational”.

(c) If both OUs are not operational, do the fault isolation task below for “Both OUs are Not Operational”.

F. One OU is Operational - Fault Isolation Procedure

(1) If the LED indication on the power outlet module does not operate in any condition, do this check of the voltage at the in-seat power supply:

NOTE: You must measure the outlet voltage to know if the power outlet module operates correctly.

(a) Disconnect connector D16740 from the in-seat power supply, M3248.

(b) Do a check for voltage between pin 1 of connector D16740 and ground (WDM 25-29-11). The correct voltage is 115 VAC.

(c) If the correct voltage is not present, then do this check of the voltage at the FD - PC POWER circuit breaker:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 825, 826 (Continued)

1) Get access to the FD - PC POWER circuit breaker, C1643.
2) Do a check for voltage between terminal =L of the circuit breaker and ground (WDM 25-29-11). The correct voltage is 115 VAC.
3) If the correct voltage is not present, then do these steps:
   a) Re-connect connector D16740 to the in-seat power supply, M3248.
   b) Replace the FD - PC POWER circuit breaker, C1643.
   c) Do the Repair Confirmation at the end of this task.
4) If the correct voltage is present, then do this check of the wiring:
   a) Do a wiring check as follows (WDM 25-29-11):

ISPS D16740		Circuit Breaker C1643
pin 1	. . . . .	term = L

   b) If you find a problem with the wiring, then do these steps:
      <1> Repair the wiring.
      <2> Re-connect connector D16740 to the in-seat power supply, M3248.
      <3> Do the Repair Confirmation at the end of this task.

(d) If the correct voltage is present, then continue.

(e) Re-connect connector D16740 to the in-seat power supply, M3248.

(2) Do a check of the wiring between the in-seat power supply, M3248 and the applicable CAPT (F/O) PC power outlet , M3249 (M3250) (WDM 25-29-11).

(a) If you find a problem with the wiring, then do these steps:
   1) Repair the wiring.
   2) Do the Repair Confirmation at the end of this task.

(3) Replace the ISPS, M3248. These are the tasks:

- In-Seat Power Supply Removal, AMM TASK 23-82-02-000-801
- In-Seat Power Supply Installation, AMM TASK 23-82-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the applicable CAPT (F/O) PC power outlet , M3249 (M3250). These are the tasks:

- Outlet Unit Removal, AMM TASK 23-82-01-000-801
- Outlet Unit Installation, AMM TASK 23-82-01-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Both OUs are Not Operational - Fault Isolation Procedure

(1) Do this check of the voltage at the in-seat power supply:

(a) Disconnect connector D16740 from the in-seat power supply, M3248.

(b) Do a check for voltage between pin 1 of connector D16740 and ground (WDM 25-29-11). The correct voltage is 115 VAC.

(c) If the correct voltage is not present, then do this check of the wiring:

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OMR 825, 826 (Continued)

1) Open this circuit breaker:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
A	3	C01643	FD - PC POWER

2) Do a wiring check as follows (WDM 25-29-11):

ISPS		Circuit Breaker
D16740		C1643
pin 1	. . . . .	term = L

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D16740 to the in-seat power supply, M3248.
   c) Close this circuit breaker:

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
A	3	C01643	FD - PC POWER

   d) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect connector D16740 to the in-seat power supply, M3248.
   b) Replace the FD - PC POWER circuit breaker, C1643.
   c) Do the Repair Confirmation at the end of this task.

(d) If the correct voltage is present, then continue.

(2) Replace the ISPS, M3248. These are the tasks:
- In-Seat Power Supply Removal, AMM TASK 23-82-02-000-801
- In-Seat Power Supply Installation, AMM TASK 23-82-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Do this task: Flight Compartment PC Power System - Operational Test, AMM TASK 23-82-00-710-801.
   (a) If both OUs are operational, then you corrected the fault.
   (b) If one OU is not operational, continue the Fault Isolation Procedure at the subsequent step.
   (c) If both OUs are not operational, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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